String Theory I and II

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0 Introduction

The key idea of string theory is to replace fundamental (i.e. point-like) particles by strings. While the propagation of point particles in spacetime gives rise to 1-dimensional worldlines, strings sweep out 2-dimensional worldsheets, see figure 1.

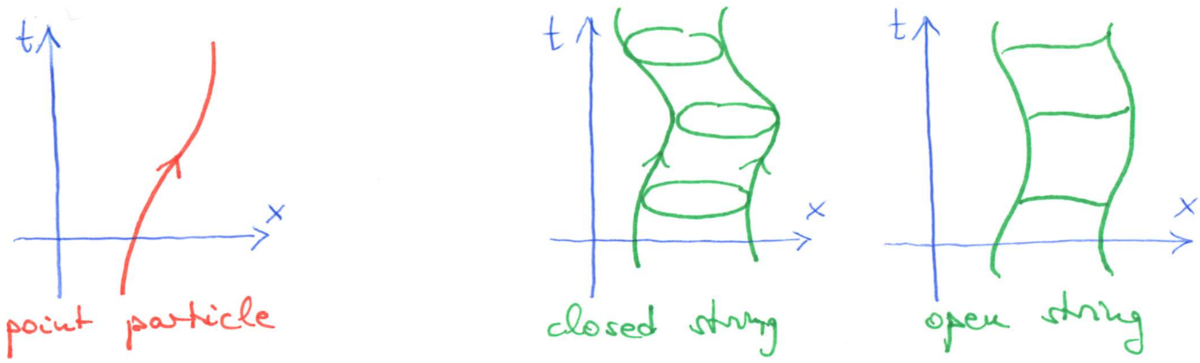

Figure 1: Worldlines of point particles versus worldsheets of strings.

Roughly speaking, different point-particle species (quarks, leptons, gauge bosons, gravitons, higher-spin excitations, etc.) are then reinterpreted as different vibration modes of the strings.

- Massless closed-string excitations include a spin-2 particle subject to diffeomorphism invariance. These properties uniquely identify it as a graviton since general relativity (along with its general coordinate invariance) is the unique theory of massless spin two! Hence, string theory is a theory of quantum gravity.

- Massless open-string excitations are identified as abelian or non-abelian gauge bosons. Hence, string theory gives rise to gauge interactions as we know them from the Standard Model.

- Both closed and open strings have an infinite tower of massive higher-spin excitations, where the maximal spin grows linearly with the mass squares.

0.1 Features and mysteries of string theory

- String theory is a perturbative theory of quantum gravity. Its loop diagrams are free of UV divergences which can be intuitively understood as follows: Vertices in Feynman diagrams contain several special interaction points whose lack of spatial extent can be viewed as causing UV- or short-distance divergences (in the integral over loop momenta). The analogues of Feynman diagrams in string perturbation theory are worldsheets of different topologies, where the interactions zones are smeared out (see figure 2) such that the UV behaviour is softened.

- Interacting open strings (in particular their loop diagrams) necessarily introduce closed strings. Hence, by the gauge bosons and gravitons among their massless excitations, string theory necessarily unifies gauge interactions with gravity!

- String theory "predicts" its spacetime dimensionality D through mathematical consistency conditions. We find $D = 26$ for the bosonic theory and $D = 10$ for supersymmetric strings. The observed

Figure 2: Ultraviolet-divergent Feynman integral (in dimensions $D > 8$) versus ultraviolet-finite worldsheet diagram in one-loop closed-string amplitude.

$D = 4$ spacetime can be accommodated with the ten dimensions of superstrings via compactified (i.e. very small) extra dimensions as well as extra ingredients such as D branes.

- String theory geometrizes gauge groups (say the $SU(3) \times SU(2) \times U(1)$ of the Standard Model), interaction strengths and numbers of generations (the three generations of quarks and leptons in the Standard Model), for instance through the number of intersection points of D branes. Also, string theory induces or exploits creative generalizations of geometry.

- Even though the theory of superstrings is unique after taking dualities into account, it has a large number of solutions. Depending on the way of counting these solutions, one can arrive a large numbers of vacua such as 10^{500}. In spite of this huge landscape, it is very hard to reproduce the Standard Model features.

- Consistency requires spacetime supersymmetry, an extension of Poincaré symmetry (Lorentz transformations plus translations) that mixes bosons and fermions. More specifically, strings require supersymmetry at high energies (e.g. the Planck scale $\sqrt{\frac{\hbar c}{G_{\rm N}}} \sim 10^{19}$ GeV). This is by no means in conflict with the non-observation of supersymmetry at LHC (as of fall 2020) since the electroweak or LHC-accessible scales are so much lower (in the range of 10^4 GeV).

- In contrast to the more than 20 free parameters of the Standard Model, string theory only requires a single fundamental parameter as an input. This "string scale" can either be incorporated into a mass- or length scale, related by $M_{\rm string} = \frac{1}{\ell_{\rm string}}$. Alternatively, one speaks of the inverse string tension or "Regge slope" $\alpha' = \ell^2_{\rm string}$. The coupling constant in string perturbation theory (which determines the relative weight of different worldsheet topologies) in turn is dynamically determined by the vacuum expectation value (VEV) of a massless scalar field in the closed-string spectrum, the so-called dilaton. Also, geometric data such as the size of extra dimensions can be traced back to VEVs of scalar fields.

- String theory is notoriously difficult to test experimentally. This difficulty is a common theme to any candidate theory of quantum gravity since higher-derivative corrections to general relativity are very hard to measure. On the one hand, there are considerations of an extremely low string scale $M_{\rm string} \sim 10^4 - 10^6$ GeV (based on large extra dimensions) which might give rise to envision string signatures at LHC. On the other hand, there is a wide range of possibilities (across at least 15 orders of magnitude) on where to expect the string scale.

- String theory benefits from and inspires beautiful mathematics:

* Mirror symmetry between Calabi-Yau manifolds (certain compactification geometries for 6 real or 3 complex spacetime dimensions) that maps different pieces of topological information to each other and which is inspired by a duality between so-called type IIA and type IIB superstring theories
    * The low-energy or α'-expansion of string amplitudes introduces certain special numbers and functions that are under active investigation in both particle physics (in the context of Feynman integrals) and pure mathematics (number theory and algebraic geometry): multiple zeta values and polylogarithms (tree-level amplitudes) and their elliptic generalizations (one-loop amplitudes). There are many open questions on the number-theoretic content of higher-loop amplitudes.

- String theory reveals surprising connections between gauge theory and gravity.
    * At weak coupling, various examples of scattering amplitude in perturbative gravity can be assembled from squares of building blocks in gauge-theory amplitudes. Such amplitude relations are sometimes phrased as gravity = (gauge theory)^2 and have a clean string-theory derivation at tree level due to the relation between open- and closed-string amplitudes. Similar relations among field-theory amplitudes are observed at the level of the loop integrand and have a similar string-theory origin from the chiral-splitting approach to string loop amplitudes.
    * According to Maldacena's AdS/CFT correspondence, string theory in AdS_{d+1} spacetimes is dual to conformal gauge theories on their d-dimensional boundary. This correspondence is also known as the gauge-gravity duality. It relates the weak- and strong-coupling regimes of the two sides and often allows to infer the result of prohibitively hard computations in one theory from more accessible ones in the dual theory. The AdS/CFT correspondence is a nice realization of the holographic principle, an idea in quantum gravity stating that the information on a higher-dimensional system may be encoded in a lower-dimensional one.

- Still, our understanding of string theories is rather incomplete,
    * Apart from the notable exception of the AdS/CFT correspondence for certain boundary conditions, string perturbation theory has not yet been derived from a non-perturbative formulation of string theory.
    * D branes are hypersurfaces in spacetime where the endpoints of open strings are confined. D branes must be dynamical in a non-perturbative formulation of string theory, but they become infinitely heavy at weak coupling. Hence, the dynamics of D branes is invisible to string perturbation theory (similar to massive degrees of freedom that have been integrated out)

0.2 A brief history

- In the late 60's, string theory was born out of an attempt to study hadron spectra and interactions. In particular, the so-called Veneziano amplitude (1968) – a particularly crossing symmetric expression for a candidate amplitude for hadron scattering – is often referred to as the first equation of string theory.

- In 1974, Scherk and Schwarz suggested to interpret the massless spin-2 excitation of closed strings as a graviton. Prior to this proposal, the spin-2 excitation was viewed as a disturbing feature of strings, but thanks to the new viewpoint of Scherk and Schwarz, string theory was ready to be interpreted as a candidate theory of quantum gravity.

- In 1984, Green and Schwarz showed that potential gauge anomalies of the type I superstring cancel for suitable choice of the gauge group. Prior to this remarkable discovery, superstring theories were under the reputation that anomalies would inevitably spoil their consistency. With the remarkable discovery of anomaly cancellation – sometimes referred to as the first superstring revolution – a broad range of researches started to work on string theory.

- In 1995, Witten explained how to relate a total of 5 known superstring theories through a web of dualities, triggering the second superstring revolution. While string theory was supposed to yield a unique theory of quantum gravity, a total of 5 different-looking incarnations of superstrings have been known (I, IIA, IIB, $het_{SO(32)}$, $het_{E_8\times E_8}$) that threatened the desired uniqueness. Thanks to Witten's web of dualities, these five theories were recognized as different phases of a single overarching theory dubbed "M theory".

- In 1998, Maldacena came up with the aforementioned AdS/CFT correspondence.

0.3 Outline of the courses string theory I and II

The 16 lectures of the "String theory I" course will cover

- Relativistic point particle vs closed strings
- Quantizing bosonic strings
- Conformal field theory
- Polyakov path integral and ghosts
- String perturbation theory and tree amplitudes

while "String theory II" is planned to cover the following topics

- Bosonic strings in background fields
- Basics of the RNS superstring
- String theory and spacetime supersymmetry
- Superstring scattering amplitudes
- String dualities and AdS/CFT

0.4 Literature

In "String theory I" and earlier parts of "String theory II", I will mainly follow

- David Tong, "String Theory", Part III lecture notes given in Cambridge, arXiv:0908.0333

- Ralph Blumenhagen, Dieter Lüst, Stefan Theisen, "Basic Concepts of String Theory", Springer 2013, ISBN 978-3-642-29496-9

and further resources will be mentioned in the "String theory II" course. Moreover, there is a large body of nice textbooks which I can recommend for complementary reading:

- Joseph Polchinski, "String theory I. An introduction to the bosonic string" and "String theory II. Superstring theory and beyond", Cambridge University Press 2005
- Michael Green, John Schwarz, Edward Witten, "Superstring theory I. Introduction" and "Superstring theory II. Loop amplitudes, anomalies & phenomenology", Cambridge University Press 1987
- Katrin Becker, Melanie Becker, John Schwarz, "String theory and M-theory, a modern introduction", Cambridge University Press 2006
- Elias Kiritsis, "String theory in a nutshell", Princeton University Press 2019
- Philippe Di Francesco, Pierre Mathieu, David Sénéchal, "Conformal field theory", Springer 1997
- Martin Ammon, Johanna Erdmenger, "Gauge / Gravity duality, foundations and applications", Cambridge University Press 2015

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1 Relativistic point particle vs closed strings

Unless otherwise specified, we will consider strings in a background[1] of D-dimensional Minkowski spacetime $\mathbb{R}^{1,D-1}$ with $t = X^0$ and a metric in "mostly plus" signature $\eta_{\mu\nu} = \text{diag}(-1, +1, \ldots, +1)$. Lorentz indices $\mu, \nu, \lambda, \ldots = 0, 1, \cdots, D-1$ will be taken from the middle of the Greek alphabet. Moreover, we will work in units where $c = \hbar = 1$.

1.1 Point-particle action = worldline length

We describe the propagation of a relativistic point particle through spacetime by a worldline as depicted in figure 3. The latter is parameterized through a "proper time" τ with $\dot{X}^\mu \equiv \frac{dX^\mu}{d\tau}$. A possible action functional to derive the equations of motion is furnished by the length of the worldline[2]

$$S_{old}[X] = -m \int d\tau \sqrt{-\frac{dX^\mu}{d\tau}\frac{dX^\nu}{d\tau}\eta_{\mu\nu}} \tag{1.1}$$

[1] Of course, string theory as a theory of quantum gravity is prepared for different backgrounds that can be attained by "coherent states" built from graviton excitations to be introduced later in this course. We will work with $\mathbb{R}^{1,D-1}$ as a "reference spacetime" that serves as a starting point for studying small and large fluctuations.

[2] As will be explained below, the square-root dependence of S_{old} on X^μ is inconvenient for quantization, so we will encounter an equivalent action S_{new} in the next subsections.

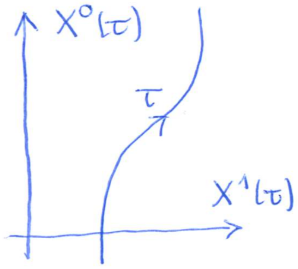


Figure 3: Two-dimensional projection of the worldline of a point particle.

It is invariant under monotonic reparametrizations

$$\tilde{\tau} = \tilde{\tau}(\tau) \to d\tilde{\tau} = \frac{d\tilde{\tau}}{d\tau} d\tau \ , \qquad \dot{X}^\mu = \frac{dX^\mu}{d\tilde{\tau}} \frac{d\tilde{\tau}}{d\tau} \tag{1.2}$$

so we have a cancellation of $\frac{d\tilde{\tau}}{d\tau}$ in

$$S_{old}[X] = -m \int d\tilde{\tau} \sqrt{-\frac{dX^\mu}{d\tilde{\tau}} \frac{dX^\nu}{d\tilde{\tau}} \eta_{\mu\nu}} \tag{1.3}$$

This reparametrization invariance reflects the fact that there is no physical meaning to the choice of the worldline coordinate τ. As a consequence, one out of D degrees of freedom of X^μ is fake: One can for instance gauge away X^0 by enforcing $X^0(\tilde{\tau}) = \tilde{\tau}$ via suitable choice of $\tilde{\tau}(\tau)$.

In the frame with $X^\mu = (t, \vec{x})$, we have (with velocity $\vec{v} = \frac{d\vec{x}}{dt}$)

$$\dot{X}^\mu = \left(\frac{dt}{d\tau}, \frac{dt}{d\tau} \frac{d\vec{x}}{dt} \right)^\mu = \frac{dt}{d\tau} (1, \vec{v})^\mu \tag{1.4}$$

and the action takes the following form

$$\begin{aligned} S_{old}[X] &= -m \int d\tau \sqrt{-\left(\frac{dt}{d\tau} \right)^2 (1, \vec{v})^\mu (1, \vec{v})^\nu \eta_{\mu\nu}} \\ &= -m \int dt \sqrt{1 - \vec{v} \cdot \vec{v}} \\ &= \int dt \left(-m + \frac{m}{2} \vec{v}^2 + O(\vec{v}^4) \right) \end{aligned} \tag{1.5}$$

For this choice of frame, $\vec{x}$ represent the $D-1$ dynamical degrees of freedom, while t is merely a parameter. In the non-relativistic limit $|\vec{v}| \ll 1$ performed in the last step, we recover the well-known expressions $E_{kin} = \frac{1}{2} m \vec{v}^2$ and $E_{pot} = mc^2$ in the Lagrangian (recall that we have set $c = 1$),

$$S_{non-rel} = \int dt L_{non-rel} \ , \qquad L_{non-rel} = E_{kin} - E_{pot} + O(\vec{v}^4) \tag{1.6}$$

As a manifestly Lorentz-invariant way to see that one of the degrees of freedom in $X^\mu(\tau)$ is unphysical, we observe that the canonical momenta

$$S_{old}[X] = \int dt L_{old} \ , \qquad p_\mu = \frac{\partial L_{old}}{\partial \dot{X}^\mu} = \frac{m \dot{X}_\mu}{\sqrt{-\dot{X}^\lambda \dot{X}^\rho \eta_{\rho\lambda}}} \tag{1.7}$$

obey an algebraic relation, the "mass-shell condition"

$$p_\mu p^\mu = -m^2 \tag{1.8}$$

Note that $X^\mu = X^\mu(\tau)$ are thought of as "worldline fields" or maps from the worldline to $\mathbb{R}^{1,D-1}$. To avoid cluttering, we will henceforth denote Lorentz-index contractions by a dot, i.e. $A^\mu B^\nu \eta_{\mu\nu} = A \cdot B$.

1.2 Lightning recap of functional differentiation

Since we will frequently need to compute variations of actions, this section aims to briefly review functional differentiation and fix our conventions. The key idea is to generalize $\frac{\partial x^\mu}{\partial x^\nu} = \delta^\mu_\nu$ to the continuum, i.e. to pass from the discrete choices of differentiation variables x^μ with $\mu = 0, 1, \cdots, D-1$ to continuous labels $\tau \in \mathbb{R}$ in the argument of $x^\mu \to f(\tau)$.

Given that the Dirac δ-distribution is the continuum limit of the Kronecker delta δ^μ_ν, we generalize $\frac{\partial}{\partial x^\nu}$ to a functional derivative $\frac{\delta}{\delta f(\tau')}$ subject to

$$\frac{\delta f(\tau)}{\delta f(\tau')} = \delta(\tau - \tau') \tag{1.9}$$

$$\int_a^b d\tau g(\tau)\delta(\tau - \tau') = g(\tau') \ , \qquad \text{provided that } \tau' \in (a, b) \tag{1.10}$$

The functional derivative is taken to commute with ordinary derivatives

$$\frac{\delta}{\delta f(\tau')} \frac{df(\tau)}{d\tau} = \frac{d}{d\tau} \frac{\delta f(\tau)}{\delta f(\tau')} = \frac{d}{d\tau} \delta(\tau - \tau') \tag{1.11}$$

where the derivative of the Dirac δ-distribution is defined via integration by parts

$$\begin{aligned} \int_a^b d\tau g(\tau) \frac{d}{d\tau} \delta(\tau - \tau') &= g(\tau)\delta(\tau - \tau')\big|_{\tau=a}^{\tau=b} - \int_a^b d\tau \frac{dg(\tau)}{d\tau} \delta(\tau - \tau') \\ &= -\frac{dg(\tau')}{d\tau'} \end{aligned} \tag{1.12}$$

The surface term $g(\tau)\delta(\tau - \tau')\big|_{\tau=a}^{\tau=b}$ is discarded under the assumption that $\tau' \neq a$ and $\tau' \neq b$.

When applied to the worldline action, the conventions for the functional derivative feature an additional Kronecker delta in $\frac{\delta X^\mu(\tau)}{\delta X^\nu(\tau')} = \delta^\mu_\nu \delta(\tau - \tau')$

$$\begin{aligned} \frac{\delta S_{old}}{\delta X^\mu(\tau')} &= -m \int d\tau \frac{-2\left(\frac{\delta}{\delta X^\mu(\tau')} \dot{X}^\nu(\tau)\right) \dot{X}_\nu(\tau)}{2\sqrt{-\dot{X}^2(\tau)}} \\ &= m \int d\tau \frac{\delta^\nu_\mu \frac{d}{d\tau} \delta(\tau - \tau') \dot{X}_\nu(\tau)}{\sqrt{-\dot{X}^2(\tau)}} \\ &= -m \frac{d}{d\tau'} \frac{\dot{X}_\mu(\tau')}{\sqrt{-\dot{X}^2(\tau')}} = -\frac{d}{d\tau} \frac{\partial L_{old}}{\partial \dot{X}^\mu}\Big|_{\tau=\tau'} \end{aligned} \tag{1.13}$$

More generally, with $S[X] = \int d\tau L(\tau)$, you recover the general form of the Euler-Lagrange equations that you know from your course on analytical mechanics

$$\frac{\delta S[X]}{\delta X^\mu(\tau)} = \frac{\partial L}{\partial X^\mu} - \frac{d}{d\tau} \frac{\partial L}{\partial \dot{X}^\mu} + \left(\frac{d}{d\tau}\right)^2 \frac{\partial L}{\partial \ddot{X}^\mu} - \dots \tag{1.14}$$

If the Lagrangian depends on higher derivatives beyond $X^\mu, \dot{X}^\mu, \ddot{X}^\mu$, the $\dots$ stand for an alternating sequence of higher derivatives $\left(\frac{d}{d\tau}\right)^n \frac{\partial L}{\partial((\frac{d}{d\tau})^n X^\mu)}$.

1.3 Quantization friendly point particle action

In the context of path-integral quantization, the square-root form $\sim \sqrt{-\dot{X}^2}$ of the worldsheet action S_{old} is highly inconvenient. This can be illustrated by comparison with a quadratic action

$$\mathcal{S}_{quad}[X] = \int d\tau X^\mu(\tau) D_{\mu\nu} X^\nu(\tau) \tag{1.15}$$

where the linear operator $D_{\mu\nu}$ may involve any number of τ-derivatives: The path integral

$$\int \mathcal{D}[X] e^{-\mathcal{S}_{quad}[X]} = \int \mathcal{D}[X] e^{-\int d\tau X^\mu(\tau) D_{\mu\nu} X^\nu(\tau)} \tag{1.16}$$

is simply the continuum-version of a Gaussian integral $\int_{-\infty}^{\infty} d^n x e^{-\vec{x}D\vec{x}} = \sqrt{\frac{\pi^n}{\det D}}$. However, the path integral constructed from the square-root containing worldsheet action

$$\int \mathcal{D}[X] e^{-\mathcal{S}_{old}[X]} = \int \mathcal{D}[X] e^{m \int d\tau \sqrt{-\dot{X}^2(\tau)}} \tag{1.17}$$

already causes major complications in its finite dimensional counterpart $\int_{-\infty}^{\infty} d^n x e^{-\sqrt{\vec{x}A\vec{x}}}$.

The square-root dependence of $\mathcal{S}_{old}[X]$ can be bypassed by introducing a worldline metric $e(\tau) \equiv \sqrt{-g_{\tau\tau}}$ or "einbein" as an extra worldline field. Starting from the new action

$$S_{new}[X, e] = \frac{1}{2} \int d\tau \left(\frac{1}{e} \dot{X}^2 - em^2 \right) \tag{1.18}$$

we can recover the old action S_{old} by inserting the equation of motion (e.o.m.) of the einbein

$$\frac{\delta S_{new}[X, e]}{\delta e(\tau)} = -\frac{1}{2e^2} (\dot{X}^2(\tau) + e^2 m^2) = 0 \tag{1.19}$$

into S_{new}. We denote the solution to $\frac{\delta S_{new}[X,e]}{\delta e(\tau)} = 0$ by $e_{class} = \sqrt{-\frac{\dot{X}^2}{m^2}}$ (the subscript referring to "classical configuration"):

$$\begin{aligned} S_{new}[X, e_{class}] &= \frac{1}{2} \int d\tau \left(\frac{1}{\sqrt{-\frac{\dot{X}^2}{m^2}}} \dot{X}^2 - \sqrt{-\frac{\dot{X}^2}{m^2}} m^2 \right) \\ &= -m \int d\tau \sqrt{-\dot{X}^2} = S_{old}[X] \end{aligned} \tag{1.20}$$

Similarly we can recover the e.o.m. $\frac{\delta S_{old}[X]}{\delta X^\mu(\tau)} = 0$ for X^μ from S_{new} at $e = e_{class}$,

$$\frac{\delta S_{new}[X, e]}{\delta X_\mu(\tau)} = -\frac{d}{d\tau} \frac{\dot{X}^\mu}{e}, \qquad \frac{\delta S_{new}[X, e_{class}]}{\delta X_\mu(\tau)} = -\frac{d}{d\tau} \frac{m\dot{X}^\mu}{\sqrt{-\dot{X}^2}} \tag{1.21}$$

Another new feature of S_{new} is that the mass-shell condition becomes an e.o.m.

$$\begin{aligned} p_\mu &= \frac{\partial L_{new}}{\partial \dot{X}^\mu} = \frac{\dot{X}_\mu}{e} \\ p_\mu p^\mu|_{e_{class}} &= \frac{X^\mu X_\mu}{e_{class}^2} = -m^2 \end{aligned} \tag{1.22}$$

The new action retains the reparametrization invariance of S_{old}. We shall consider infinitesimal one-dimensional diffeomorphisms $\tau \to \tilde{\tau} = \tau - \eta(\tau)$ and work to first order in $\eta(\tau)$. This is treated as an active coordinate transformation, where the coordinates remain fixed and the value of a field at τ is moved to $\tau - \eta(\tau)$, i.e. $\tilde{X}(\tilde{\tau}) = X(\tau)$. This implies the following infinitesimal transformation of X^μ (dropping higher derivatives of X^μ that are $O(\eta^2)$ and using $\tilde{X}^\mu(\tau) = \tilde{X}^\mu(\tilde{\tau} + \eta)$)

$$\delta X^\mu = \tilde{X}^\mu(\tau) - X^\mu(\tau) = \eta \frac{dX^\mu}{d\tau} \tag{1.23}$$

and the einbein is taken to transform as a density on the worldline

$$\delta e = \frac{d}{d\tau}(\eta e) = \eta \dot{e} + e\dot{\eta} \tag{1.24}$$

These infinitesimal transformations are easily checked to shift the action by a total derivative

$$\delta S_{new}[X,e] = \frac{1}{2}\int d\tau \frac{d}{d\tau}\Big(\frac{\eta}{e}\dot{X}^2 - \eta e m^2\Big) = 0 \tag{1.25}$$

which we drop by the assumption that the Lagrantian at $\tau \to \pm\infty$ falls off. Hence, the new action is classically equivalent to the old one, but easier to quantize (since its path integral is Gaussian). Moreover, S_{new} has a non-trivial massless limit $m \to 0$.

1.4 String action = worldsheet area

We shall generalize the discussion of the point particle to closed strings that are parametrized by a periodic spatial coordinate $\sigma \sim \sigma + 2\pi$. Closed strings sweep out two-dimensional "worldsheets" Σ, see figure 4, whose embedding in spacetime "target space" is defined by maps $X^\mu : \Sigma \to \mathbb{R}^{1,D-1}$ with

$$X^\mu(\tau, \sigma + 2\pi) = X^\mu(\tau,\sigma) \tag{1.26}$$

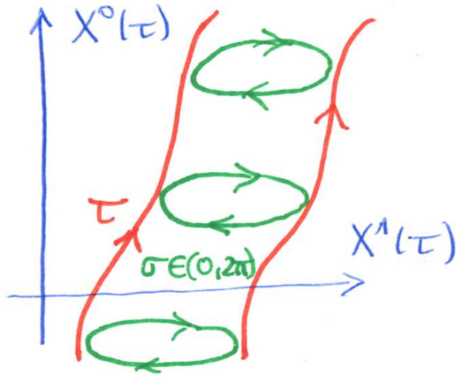


Figure 4: Two-dimensional projection of the worldsheet of a closed string.

By analogy with the worldline length as a point-particle action, a natural proposal for a string-generalization is the worldsheet area. This is known as the Nambu-Goto action

$$S_{NG}[X] = -T\int dA_{Mink} \tag{1.27}$$

In preparation for the area element dA_{Mink} tailored to $X^\mu(\tau,\sigma)$ in Minkowski spacetime, we first determine the Euclidean area element A_{Eucl} (with $\vec{X}(\tau,\sigma) \in \mathbb{R}^D$). As illustrated in figure 5, A_{Eucl} is computed from the area of a parallelogram spanned by the tangent vectors $d\vec{\ell}_\sigma = d\sigma \vec{X}'$ and $d\vec{\ell}_\tau = d\tau \dot{\vec{X}}$,

$$\begin{aligned} dA_{Eucl} &= |d\vec{\ell}_\sigma||d\vec{\ell}_\tau|\sin\theta \\ &= d^2\sigma |\vec{X}'||\dot{\vec{X}}|\sqrt{1-\cos^2\theta} \\ &= d^2\sigma\sqrt{|\dot{\vec{X}}|^2|\vec{X}'|^2 - (\dot{\vec{X}}\cdot\vec{X}')^2} \end{aligned} \tag{1.28}$$

where $d^2\sigma = d\tau d\sigma$ as well as $\vec{X}' = \frac{\partial \vec{X}}{\partial\sigma}$ and $\dot{\vec{X}} = \frac{\partial\vec{X}}{\partial\tau}$. Moreover, we used that $\sin\theta = \sqrt{1-\cos^2\theta}$ with $\cos\theta = \frac{\dot{\vec{X}}\cdot\vec{X}'}{|\dot{\vec{X}}||\vec{X}'|}$. Throughout this course, we use the following notation $\sigma^\alpha = (\tau,\sigma)$ for the worldsheet coordinates with indices $\alpha,\beta,\ldots = 0,1$ from the beginning of the Greek alphabet. Then, the argument of the square root can be related to the Euclidean version of the "induced metric"

$$\gamma_{\alpha\beta} = \frac{\partial X^\mu}{\partial\sigma^\alpha}\frac{\partial X^\nu}{\partial\sigma^\beta}\eta_{\mu\nu}\,, \qquad (\gamma_{\alpha\beta})_{Eucl} = \frac{\partial\vec{X}}{\partial\sigma^\alpha}\cdot\frac{\partial\vec{X}}{\partial\sigma^\beta} \tag{1.29}$$

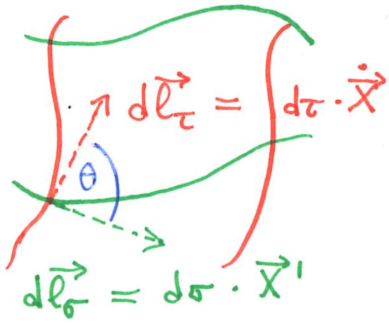


Figure 5: Two-dimensional projection of the worldsheet of a closed string.

More precisely, the Euclidean area element may be recognized as the determinant

$$dA_{Eucl} = d^2\sigma \sqrt{\det \gamma_{Eucl}} \tag{1.30}$$

which straightforwardly translates into its Minkowski version by inserting a minus sign into the square root to account for the signature of $\eta_{\mu\nu}$,

$$dA_{Mink} = d^2\sigma \sqrt{-\det \gamma} \tag{1.31}$$

Hence, the worldsheet area or Nambu-Goto action is given by

$$\begin{aligned} S_{NG}[X] &= -T \int d^2\sigma \sqrt{-\det \gamma} \\ &= -T \int d^2\sigma \sqrt{(\dot{X} \cdot X')^2 - |\dot{X}|^2 |X'|^2} \end{aligned} \tag{1.32}$$

The meaning of the normalization constant T can be understood by taking a "snapshot" of the string: In a frame $X^0 = \tau \times \text{const}$ and when the instantaneous kinetic energy is taken to vanish $\frac{d\vec{x}}{dt} = 0$, one can identify T as the string tension,

$$T = \frac{\text{potential energy}}{\text{spatial length}} = \text{tension} \tag{1.33}$$

We will often reexpress the tension via

$$T = \frac{1}{2\pi\alpha'} \tag{1.34}$$

where $\alpha' = \ell_{string}^2$ is known as the "Regge slope".

As will be shown in the homework problems, the e.o.m. of S_{NG} is the reparametrization invariant wave equation (with shorthand $\partial_\alpha = \frac{\partial}{\partial \sigma^\alpha}$)

$$\frac{\delta S_{NG}[X]}{\delta X_\mu} = T \partial_\alpha (\sqrt{-\det \gamma}\, \gamma^{\alpha\beta} \partial_\beta X^\mu) \tag{1.35}$$

In this course, we study fundamental strings without any thickness $L \to 0$. In absence of a thickness parameter L, there is no chance of adding a rigidity term $\sim L \int K^2$ to S_{NG}, where K is the extrinsic curvature. Other areas of physics involve non-fundamental strings (e.g. QCD, cosmology or magnetic flux tubes in superconductors). In these cases, $S_{NG}[X]$ is a universal part of the respective worldsheet action, but additional L-dependent terms specific to the context will appear.

1.5 Quantization friendly string action

Similar to the worldline length S_{old}, the Nambu-Goto action S_{NG} is plagued by a square-root dependence on the fields $X^\mu(\sigma)$ that complicates quantization[3]. We shall now study the worldsheet analogue of passing to a classically equivalent action $S_{old} \rightarrow S_{new}$ which is quadratic in the fields. This is accomplished by the Polyakov action S_P involving an independent worldsheet metric $h_{\alpha\beta}(\sigma)$ that generalizes the einbein $e(\tau)$ on the worldline (with shorthand $h = \det h$ and inverse $h^{\alpha\beta}h_{\beta\gamma} = \delta^\alpha_\gamma$):

$$S_P[X,h] = -\frac{1}{4\pi\alpha'}\int d^2\sigma\sqrt{-h}h^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} \tag{1.36}$$

From the worldsheet perspective, $S_P[X,h]$ describes D scalar fields $X^\mu(\sigma)$ coupled to two-dimensional gravity via $h^{\alpha\beta}(\sigma)$.

One can recover the Nambu-Goto action from S_P by inserting the e.o.m. of the metric $h_{\alpha\beta}$

$$\frac{\delta S_P[X,h]}{\delta h^{\alpha\beta}} = -\frac{\sqrt{-h}}{4\pi\alpha'}\left(\partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} - \frac{1}{2}h_{\alpha\beta}(h^{\gamma\delta}\partial_\gamma X\cdot\partial_\delta X)\right) \tag{1.37}$$

The classical configurations h^{class} solving $\frac{\delta S_p[X,h^{class}]}{\delta h^{\alpha\beta}} = 0$ are proportional to the induced metric $\gamma_{\alpha\beta} = \frac{\partial X^\mu}{\partial\sigma^\alpha}\frac{\partial X^\nu}{\partial\sigma^\beta}\eta_{\mu\nu}$, namely $h^{class}_{\alpha\beta} = F(\sigma)\gamma_{\alpha\beta}$, with scaling factor $F(\sigma) = \frac{2}{h^{\gamma\delta}\partial_\gamma X\cdot\partial_\delta X}$. When plugging back into S_P, the factors of $F(\sigma)$ are found to cancel (the inverse $\gamma^{\alpha\beta}$ yields $\gamma^{\alpha\beta}\gamma_{\alpha\beta} = 2$):

$$\begin{aligned} S_P[X,h^{class}] &= -\frac{1}{4\pi\alpha'}\int d^2\sigma\sqrt{-F^2\det\gamma}\,\frac{\gamma^{\alpha\beta}\gamma_{\alpha\beta}}{F} \\ &= -\frac{1}{2\pi\alpha'}\int d^2\sigma\sqrt{-\det\gamma} = S_{NG}[X] \end{aligned} \tag{1.38}$$

For later reference, we introduce the worldsheet energy momentum tensor

$$\begin{aligned} T_{\alpha\beta} &= \frac{4\pi}{\sqrt{-h}}\frac{\delta S_P[X,h]}{\delta h^{\alpha\beta}} \\ &= -\frac{1}{\alpha'}\left(\partial_\alpha X\cdot\partial_\beta X - \frac{1}{2}h_{\alpha\beta}(h^{\gamma\delta}\partial_\gamma X\cdot\partial_\delta X)\right) \end{aligned} \tag{1.39}$$

You may recognize this definition from the study of classical field theory in presence of gravity.

The e.o.m. for X^μ are given by

$$\frac{\delta S_P[X,h]}{\delta X^\mu} = -\partial_\alpha\frac{\partial L_p}{\partial(\partial_\alpha X^\mu)} = \frac{1}{2\pi\alpha'}\partial_\alpha(h^{\alpha\beta}\sqrt{-h}\partial_\beta X_\mu) \tag{1.40}$$

which resemble the e.o.m. from the Nambu-Goto action up to $h_{\alpha\beta} \leftrightarrow \gamma_{\alpha\beta}$. In fact, the classical configuration $h^{class}_{\alpha\beta} = F(\sigma)\gamma_{\alpha\beta}$ also maps the e.o.m. from S_P and S_{NG} to each other,

$$\begin{aligned} \frac{\delta S_P[X,h]}{\delta X^\mu} &= \frac{1}{2\pi\alpha'}\partial_\alpha(\sqrt{-F^2\det\gamma}\frac{\gamma^{\alpha\beta}}{F}\partial_\beta X_\mu) \\ &= \frac{1}{2\pi\alpha'}\partial_\alpha(\sqrt{-\det\gamma}\gamma^{\alpha\beta}\partial_\beta X_\mu) = \frac{\delta S_{NG}}{\delta X^\mu} \end{aligned} \tag{1.41}$$

which is another way of seeing the classical equivalence of S_P and S_{NG} found above. At a technical level, the classical equivalence stems from the fact that the scale factor $F(\sigma)$ drops out from both the action and the e.o.m.

The Polyakov action enjoys the following types of symmetries

[3] Here and below, the notation $X^\mu(\sigma)$ refers to the dependence of X^μ on both of $\sigma^\alpha = (\tau,\sigma)$ and does *not* indicate that X^μ is independent on τ.

- reparametrization / diffeomophism invariance, $\sigma^\alpha \to \tilde{\sigma}^\alpha = \sigma^\alpha - \eta^\alpha(\sigma)$ with infinitesimal $\eta(\sigma)$

$$\begin{aligned} \delta X^\mu &= \eta^\alpha \partial_\alpha X^\mu \\ \delta h_{\alpha\beta} &= \eta^\gamma \partial_\gamma h_{\alpha\beta} + h_{\gamma\alpha}\partial_\beta \eta^\gamma + h_{\gamma\beta}\partial_\alpha \eta^\gamma \end{aligned} \tag{1.42}$$

  which are the infinitesimal versions of

$$\begin{aligned} \tilde{X}^\mu(\tilde{\sigma}) &= X^\mu(\sigma) \\ \tilde{h}_{\alpha\beta}(\tilde{\sigma}) &= \frac{\partial \sigma^\gamma}{\partial \tilde{\sigma}^\alpha}\frac{\partial \sigma^\delta}{\partial \tilde{\sigma}^\beta} h_{\gamma\delta}(\sigma) \end{aligned} \tag{1.43}$$

- local Weyl invariance (note that this is *not* a diffeomorphism!)

$$\begin{aligned} \delta X^\mu(\sigma) &= 0 \\ \delta h_{\alpha\beta}(\sigma) &= 2\phi(\sigma) h_{\alpha\beta}(\sigma) \end{aligned} \tag{1.44}$$

  which is the infinitesimal version of $h_{\alpha\beta} \to \Omega^2(\sigma) h_{\alpha\beta}$ with $\Omega(\sigma) = e^{\phi(\sigma)}$. The σ-dependent scale factor $\Omega^2(\sigma)$ drops out in the same way as $F(\sigma)$ drops out of $S_P[X, h^{class}]$.

- The action also has a global Poincare invariance $X^\mu \to \Lambda^\mu{}_\nu X^\nu + c^\mu$, where $\Lambda \in SO(1, D-1)$ are Lorentz transformations in spacetime. Note, however, that the Lorentz indices $\mu, \nu, \ldots$ are internal from the worldsheet perspective in the same way as the flavor index of quarks in the Standard Model is "internal" from the spacetime perspective.

We have seen in the case of the worldline that reparametrization symmetry is crucial in order to identify the number of d.o.f. in X^μ. Similarly, the local symmetries of the worldsheet action are essential for the later counting of d.o.f. and must be preserved. Hence, one cannot modify S_P by extra terms that break diffeomorphism or Weyl invariance. This rules out the addition of a potential $S_V[X,h] = \int d^2\sigma \sqrt{-h} V(X)$ or a cosmological constant $S_\mu[h] = \mu \int d^2\sigma \sqrt{-h}$.

1.6 Fixing conformal gauge

As we will see, the symmetries of the Polyakov action S_P imply that we can locally gauge-fix the metric to $h_{\alpha\beta} \to \eta_{\alpha\beta} = \mathrm{diag}(-1, 1)$. As a symmetric "$2 \times 2$ matrix" $h_{\alpha\beta} = h_{\beta\alpha}$, the worldsheet metric may have up to three independent entries. The diffeomorphisms $\sigma^\alpha \to \tilde{\sigma}^\alpha(\sigma)$ with $\alpha = 0, 1$ can be used to locally fix two of them, e.g. to set the off-diagonal entries to zero and to relate the diagonal ones. This choice is known as

$$h_{\alpha\beta}(\sigma) = e^{2\phi(\sigma)} \eta_{\alpha\beta} \qquad \text{conformal gauge} \tag{1.45}$$

which can always be attained locally since the partial differential equations of the required $\sigma(\tilde{\sigma})$ locally have solutions. The conformal factor $e^{2\phi(\sigma)}$ in turn can be removed via Weyl transformation, leading to the announced Minkowski form $h_{\alpha\beta} \to \eta_{\alpha\beta}$.

Still, the Minkowski form of the worldsheet metric leaves residual gauge freedom: the subclass of diffeomorphisms that can be undone via Weyl transformation. This residual gauge freedom is analogous to the fact that Lorenz gauge $\partial_\mu A^\mu = 0$ in Maxwell electrodynamics is preserved by gauge transformations $A^\mu \to A^\mu + \partial^\mu \Lambda$ that obey the massless wave equation $\partial^2 \Lambda = 0$.

Conformal gauge brings the Polyakov action into the following form:

$$S_P[X, h_{\alpha\beta} = e^{2\phi}\eta_{\alpha\beta}] = -\frac{1}{4\pi\alpha'}\int d^2\sigma \partial_\alpha X \cdot \partial^\alpha X \equiv S_{CG}[X] \tag{1.46}$$

Its equation of motion is easily seen to be the free wave equation

$$\frac{\delta S_{CG}[X]}{\delta X^\mu} \sim \partial_\alpha \partial^\alpha X_\mu = (\partial_\sigma^2 - \partial_\tau^2) X_\mu = 0 \tag{1.47}$$

and string propagation $X^\mu(\sigma)$ in spacetime is governed by its solutions.

Even though the e.o.m. for X^μ in conformal gauge no longer contains an explicit reference to the worldsheet metric, the latter contributes an important constraint: We still have to impose the e.o.m. for $h_{\alpha\beta}$, setting $h_{\alpha\beta}$ to $\eta_{\alpha\beta}$ *after* the variation.

$$\begin{aligned} 0 = T_{\alpha\beta} &= \frac{4\pi}{\sqrt{-h}}\frac{\delta S_P[X,h]}{\delta h^{\alpha\beta}}|_{h_{\alpha\beta}\to\eta_{\alpha\beta}} \\ &= -\frac{1}{\alpha'}(\partial_\alpha X \cdot \partial_\beta X - \frac{1}{2}\eta_{\alpha\beta}\partial_\gamma X \partial^\gamma X) \end{aligned} \tag{1.48}$$

The energy-momentum tensor as defined in the previous subsection is easily checked to be traceless, $T^\alpha{}_\alpha = 0$, so the e.o.m. for $h_{\alpha\beta}$ translates into the following two component constraints:

$$\begin{aligned} -\alpha' T_{01} &= \dot{X} \cdot X' = \frac{\partial X^\mu}{\partial \sigma}\frac{\partial X_\mu}{\partial \tau} = 0 \\ -\alpha' T_{00} &= -\alpha' T_{11} = \frac{1}{2}(\dot{X}^2 + (X')^2) = 0 \end{aligned} \tag{1.49}$$

The availability of these extra constraints beyond the wave equation is analogous to $p^2 = -m^2$ on the worldline which follows from $\frac{\delta S_{new}[X,e]}{\delta e} = 0$.

1.7 Mode expansion

We conclude this introductory section by investigating the classical solutions to the free wave equation and the Fourier modes of the constraints $T_{\alpha\beta} = 0$. We exploit that, in two dimensions, the wave operator can be factorized by picking lightcone coordinates:

$$\sigma^\pm = \tau \pm \sigma \ , \qquad \partial_\pm = \frac{\partial}{\partial \sigma^\pm} = \frac{1}{2}(\partial_\tau \pm \partial_\sigma) \ , \qquad \partial_\alpha \partial^\alpha = -4\partial_+\partial_- \tag{1.50}$$

Hence, the e.o.m. for X^μ in conformal gauge and lightcone coordinates takes the form

$$\partial_+ \partial_- X_\mu = 0 \tag{1.51}$$

Its most general solution decomposes into two independent functions X_L, X_R of one variable each:

$$X^\mu(\sigma) = X^\mu_L(\sigma^+) + X^\mu_R(\sigma^-) \tag{1.52}$$

For closed strings, we additionally need to impose periodic boundary conditions $\sigma \sim \sigma + 2\pi$ which leads to a discrete Fourier expansion for both X_L and X_R:

$$\begin{aligned} X^\mu_L(\sigma^+) &= \frac{1}{2}x^\mu + \frac{\alpha'}{2}p^\mu \sigma^+ + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{\tilde{\alpha}^\mu_n}{n}e^{-in\sigma^+} \\ X^\mu_R(\sigma^-) &= \frac{1}{2}x^\mu + \frac{\alpha'}{2}p^\mu \sigma^- + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{\alpha^\mu_n}{n}e^{-in\sigma^-} \end{aligned} \tag{1.53}$$

The first two terms add up to $x^\mu + \alpha' p^\mu \tau$ and signal motion at constant momentum p^μ. Note that the periodic boundary conditions do not admit a linear term $\sim \sigma$ even though it solves the wave equation. The factors of $\frac{\alpha'}{2}$ reflecting the normalization of the Fourier modes α_n^μ and $\tilde{\alpha}_n^\mu$ are chosen for later convenience. We emphasize that the inverse string tension α' must not be confused with the Fourier modes $\alpha_n^\mu, \tilde{\alpha}_n^\mu$.

It remains to impose the constraints $T_{\alpha\beta} = 0$, using the conversions $\partial_\tau = \partial_+ + \partial_-$ and $\partial_\sigma = \partial_+ - \partial_-$. We calculate the independent entries of the worldsheet energy-momentum tensor to be

$$\begin{aligned} -\alpha' T_{01} &= (\partial_+ X)^2 - (\partial_- X)^2 = 0 \\ -\alpha' T_{00} &= (\partial_+ X)^2 + (\partial_- X)^2 = 0 \end{aligned} \tag{1.54}$$

i.e. $(\partial_+ X)^2$ and $(\partial_- X)^2$ have to vanish separately. The Fourier modes of these constraints take a particularly convenient form if we denote

$$\alpha_0^\mu = \tilde{\alpha}_0^\mu = \sqrt{\frac{\alpha'}{2}} p^\mu \tag{1.55}$$

i.e. the momentum p^μ can be viewed as the joint zero mode of $\partial_+ X^\mu$ and $\partial_- X^\mu$:

$$\begin{aligned} \partial_+ X^\mu = \partial_+ X_L^\mu &= \sqrt{\frac{\alpha'}{2}} \sum_{n \in \mathbb{Z}} \tilde{\alpha}_n^\mu e^{-in\sigma^+} \\ \partial_- X^\mu = \partial_- X_R^\mu &= \sqrt{\frac{\alpha'}{2}} \sum_{n \in \mathbb{Z}} \alpha_n^\mu e^{-in\sigma^-} \end{aligned} \tag{1.56}$$

Then, the constrains become

$$\begin{aligned} 0 = (\partial_+ X)^2 &= \frac{\alpha'}{2} \sum_{n \in \mathbb{Z}} \sum_{\ell \in \mathbb{Z}} \tilde{\alpha}_n \cdot \tilde{\alpha}_\ell e^{-i(n+\ell)\sigma^+} = \alpha' \sum_{n \in \mathbb{Z}} \tilde{L}_n e^{-in\sigma^+} \\ 0 = (\partial_- X)^2 &= \frac{\alpha'}{2} \sum_{n \in \mathbb{Z}} \sum_{\ell \in \mathbb{Z}} \alpha_n \cdot \alpha_\ell e^{-i(n+\ell)\sigma^+} = \alpha' \sum_{n \in \mathbb{Z}} L_n e^{-in\sigma^-} \end{aligned} \tag{1.57}$$

where the last step defines the Fourier modes of the energy-momentum tensor:

$$\begin{aligned} L_n &= \frac{1}{2} \sum_{m \in \mathbb{Z}} \alpha_{n-m} \cdot \alpha_m \\ \tilde{L}_n &= \frac{1}{2} \sum_{m \in \mathbb{Z}} \tilde{\alpha}_{n-m} \cdot \tilde{\alpha}_m \end{aligned} \tag{1.58}$$

Hence, our constraints are reorganized in Fourier space and read

$$L_n = \tilde{L}_n = 0 \ \forall \ n \in \mathbb{Z} \tag{1.59}$$

Note that the zero modes of the constraints already fix the mass spectrum: The constraint

$$L_0 = \frac{\alpha'}{4} p^2 + \sum_{n>0} \alpha_n \cdot \alpha_{-n} \tag{1.60}$$

together with $p^2 = -m^2_{class}$ yields the following expressions for the classical values of the mass square:

$$m^2_{class} = \frac{4}{\alpha'} \sum_{n>0} \alpha_n \cdot \alpha_{-n} = \frac{4}{\alpha'} \sum_{n>0} \tilde{\alpha}_n \cdot \tilde{\alpha}_{-n} \tag{1.61}$$

However, there will be an extra term in the quantum theory, once the Fourier modes $\alpha_n^\mu, \tilde{\alpha}_n^\mu$ are promoted to operators and no longer commute.

2 Quantizing closed bosonic string

In this section, we will apply canonical quantization / the operator formulation to the two-dimensional field theory on the worldsheet defined by the Polyakov action $S_P[X,h]$. As you know from your quantum-field-theory course, path integrals provide an alternative approach to quantization, and we will discuss them in part 5 of the course.

Gauge symmetries are delicate in quantization, and there are two ways to handle the diffeomorphism and Weyl gauge symmetry of the Polyakov action:

- In the next two subsection, we will *first* do covariant quantization and *then* use gauge fixing constraints, this is analogous to the Gupta-Bleuler quantization in QED.
- From the third subsection onwards, we will *first* solve the constraints and *then* quantize the distinct solutions; this is analogous to QED in Coulomb gauge which obscures Lorentz invariance in spacetime.

2.1 Opening line in covariant quantization

We define the canonical momentum conjugate to X^μ by the derivative w.r.t. τ (as opposed to the spatial coordinate σ). In conformal gauge $h_{\alpha\beta} \to \eta_{\alpha\beta}$, it reads

$$\begin{aligned} \Pi^\mu(\tau,\sigma) &= \frac{\partial L_{CG}}{\partial \dot{X}_\mu} = \frac{1}{2\pi\alpha'}\dot{X}^\mu \\ &= \frac{p^\mu}{2\pi} + \frac{1}{2\pi\sqrt{2\alpha'}}\sum_{n\neq 0}(\tilde{\alpha}^\mu_n e^{-in\sigma^+} + \alpha^\mu_n e^{-in\sigma^-}) \end{aligned} \tag{2.1}$$

where $\sigma^\pm = \tau \pm \sigma$. Since we have singled out ∂_τ over ∂_σ, Lorentz covariance on the worldsheet is broken (though Lorentz covariance in target space is preserved). We impose canonical equal-time commutators,

$$\begin{aligned} [X^\mu(\tau,\sigma), \Pi_\nu(\tau,\sigma')] &= i\delta^\mu_\nu \delta(\sigma-\sigma') \\ [X^\mu(\tau,\sigma), X^\nu(\tau,\sigma')] &= [\Pi_\mu(\tau,\sigma), \Pi_\nu(\tau,\sigma')] = 0 \end{aligned} \tag{2.2}$$

where $\delta(\sigma-\sigma')$ is the continuum extension of the Kronecker-delta familiar from $[x^\mu, p_\nu] = i\delta^\mu_\nu$ in quantum mechanics. As will be shown in the exercises, these canonical commutators are equivalent to the following commutation relations among the operators in the mode expansion:

$$\begin{aligned} [x^\mu, p_\nu] &= i\delta^\mu_\nu \\ [x^\mu, x^\nu] &= [p_\mu, p_\nu] = 0 \\ [\tilde{\alpha}^\mu_n, \tilde{\alpha}^\nu_m] &= [\alpha^\mu_n, \alpha^\nu_m] = n\delta_{n+m}\eta^{\mu\nu} \\ [\alpha^\mu_n, \tilde{\alpha}^\nu_m] &= 0 \end{aligned} \tag{2.3}$$

Both $\{\alpha^\mu_n,\ n\in\mathbb{Z}\}$ and $\{\tilde{\alpha}^\mu_n,\ n\in\mathbb{Z}\}$ comprise infinitely many copies of the algebra $[a,a^\dagger]=1$ of the quantum mechanical (QM) harmonic oscillator. If we define

$$a^\mu_n \equiv \frac{\alpha^\mu_n}{\sqrt{n}}\,, \qquad a^{\dagger\nu}_n \equiv \frac{\alpha^\nu_{-n}}{\sqrt{n}}\,, \qquad n>0 \tag{2.4}$$

then we have the normalization as in the quantum-mechanics course

$$[a^\mu_n, a^{\dagger\nu}_m] = \eta^{\mu\nu}\delta_{n,m}\,, \qquad [a^\mu_n, a^\nu_m] = [a^{\dagger\mu}_n, a^{\dagger\nu}_m] = 0\,, \qquad n,m>0 \tag{2.5}$$

The Hamiltonian of the QM harmonic oscillator is

$$H_{QM} = \frac{1}{2}(a^\dagger a + a a^\dagger) = a^\dagger a + \frac{1}{2} \tag{2.6}$$

where we normal-ordered the a and $a^\dagger$ to have the lowering operator a on the right and the raising operator $a^\dagger$ on the left. The term $+\frac{1}{2}$ is then identified as the zero-point energy, and the terminology of lowering / raising operators is motivated by

$$[H_{QM}, a^\dagger] = a^\dagger, \quad [H_{QM}, a] = -a \tag{2.7}$$

We can build a Fock space $|0\rangle, a^\dagger|0\rangle, a^\dagger a^\dagger|0\rangle, \ldots$ from some vacuum state $|0\rangle$, and the requirement that the spectrum of the Hamiltonian is bounded from below requires $a|0\rangle = 0$.

In the string-theory scenario with infinitely many pairs $a_n^\mu, a_n^{\dagger\nu}$, we take $\alpha_n^\mu \sim a_n^\mu$ with $n > 0$ as the lowering operators which annihilate the vacuum. The Hamiltonian follows from tracing out both the discrete labels $\mu = 0, 1, \ldots, D-1$ and the continuous one $\sigma \in (0, 2\pi)$ (see the exercises for some intermediate steps)

$$\begin{aligned} H &= \int_0^{2\pi} d\sigma \big(\Pi_\mu(\tau,\sigma)\dot{X}^\mu(\tau,\sigma) - L_{CG}(\tau,\sigma)\big) \\ &= \frac{1}{4\pi\alpha'} \int_0^{2\pi} d\sigma \big(\dot{X}^2 + (X')^2\big) = \ldots \\ &= \frac{\alpha'}{2}p^2 + \frac{1}{2}\sum_{n>0} \big(\alpha_{-n}\cdot\alpha_n + \alpha_n\cdot\alpha_{-n} + (\alpha \leftrightarrow \tilde{\alpha})\big) \end{aligned} \tag{2.8}$$

where we use normal ordering $\alpha_n \cdot \alpha_{-n} = \alpha_{-n}\cdot\alpha_n + D\cdot n$ (the D coming from $\eta_{\mu\nu}\eta^{\mu\nu}$)

$$H = \frac{\alpha'}{2}p^2 + \sum_{n=1}^{\infty}(\alpha_{-n}\cdot\alpha_n + \tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n) + D\sum_{n=1}^{\infty} n \tag{2.9}$$

Formally, this Hamiltonian has infinite zero-point energy. The role of $\alpha^\mu_{n<0}, \tilde{\alpha}^\mu_{n<0}$ as raising operators and $\alpha^\mu_{n>0}, \tilde{\alpha}^\mu_{n>0}$ as lowering operators is confirmed by

$$[H, \alpha_n^\mu] = -n\alpha_n^\mu\,, \qquad [H, \tilde{\alpha}_n^\mu] = -n\tilde{\alpha}_n^\mu \tag{2.10}$$

What goes beyond the QM analogy: There are additional x^μ, p_ν operators in the mode expansion of $X^\mu(\tau,\sigma)$. In contrast to the QM situation, they are not a combination of a's and $a^\dagger$'s. We take the vacuum $|0;p\rangle$ w.r.t. $\{\alpha_n, \tilde{\alpha}_n,\ n>0\}$ to be a p^μ eigenstate with the normalization

$$\langle p';0|0;p\rangle = \delta^D(p-p') \tag{2.11}$$

Then the lowering operators act as usual,

$$\alpha^\mu_{n>0}|0;p\rangle = \tilde{\alpha}^\mu_{n>0}|0;p\rangle = 0 \tag{2.12}$$

We can build the Fock space of physical states by raising operators $\alpha^\mu_{n<0}, \tilde{\alpha}^\mu_{n<0}$,

$$|\text{phys}\rangle \in \big\{\alpha_{-1}^{\mu_1}\alpha_{-1}^{\mu_2}\ldots\alpha_{-2}^{\nu_1}\ldots\tilde{\alpha}_{-1}^{\lambda_1}\tilde{\alpha}_{-1}^{\lambda_2}\ldots\tilde{\alpha}_{-2}^{\rho_1}\ldots|0;p\rangle\big\} \tag{2.13}$$

From the free Lorentz indices $\mu_i, \nu_i, \ldots$, one can identify the spectrum as an infinite tower of higher-spin excitations.

2.2 Constraints in covariant quantization

The discussion in the previous section leaves two loose ends up to now:

- How to implement the classical vanishing of the worldsheet energy momentum tensor
$$T_{\alpha\beta} = -\frac{1}{\alpha'}\left(\partial_\alpha X\cdot\partial_\beta X - \frac{1}{2}\eta_{\alpha\beta}\eta^{\gamma\delta}\partial_\gamma X\cdot\partial_\delta X\right) \tag{2.14}$$
in the quantum theory?

- The indefinite signature of Minkowski spacetime $\mathbb{R}^{1,D-1}$ gives rise to negative-norm states: By $(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$ and $\eta^{\mu\nu} = \mathrm{diag}(-1,+1,\dots,+1)$, the collection of states $\{\alpha_{-1}^\lambda\tilde{\alpha}_{-1}^\rho|0;p\rangle\}$ contains negative-norm representatives:
$$\langle p';0|\alpha_1^\mu\tilde{\alpha}_1^\nu\alpha_{-1}^\lambda\tilde{\alpha}_{-1}^\rho|0;p\rangle = \eta^{\mu\lambda}\eta^{\nu\rho}\delta^D(p-p') \tag{2.15}$$
If these negative-norm states are physical, then unitarity of the theory is violated.

In quantization of QED, the negative-norm states are removed by imposing the gauge-fixing constraints. In our string-theory setup, the removal of negative-norm states will ultimately come from the operator-valued version of the energy-momentum constraints $T_{\alpha\beta} = 0$. We shall pass to operator-valued Fourier modes $L_n, \tilde{L}_n$ in $\sigma^\pm = \tau\pm\sigma$ coordinates

$$\begin{aligned} T_{++} &= -\frac{1}{\alpha'}(\partial_+X)^2 = -\sum_{n\in\mathbb{Z}}\tilde{L}_n e^{-in\sigma^+} \\ T_{--} &= -\frac{1}{\alpha'}(\partial_-X)^2 = -\sum_{n\in\mathbb{Z}}L_n e^{-in\sigma^-} \end{aligned} \tag{2.16}$$

All the $L_n, \tilde{L}_n$ with $n\neq 0$ are unambiguous,

$$\begin{aligned} L_{n\neq 0} &= \frac{1}{2}\sum_{\ell\in\mathbb{Z}}\alpha_{n-\ell}\cdot\alpha_\ell \\ \tilde{L}_{n\neq 0} &= \frac{1}{2}\sum_{\ell\in\mathbb{Z}}\tilde{\alpha}_{n-\ell}\cdot\tilde{\alpha}_\ell \end{aligned} \tag{2.17}$$

since the composing operators $\alpha_{n-\ell}, \alpha_\ell$ and $\tilde{\alpha}_{n-\ell}, \tilde{\alpha}_\ell$ commute in these cases. For $L_0, \tilde{L}_0$, in turn, we have to pick an ordering convention and shall now choose normal ordering (lowering operators on the right)

$$\begin{aligned} L_0 &= \frac{\alpha_0^2}{2} + \sum_{n=1}^\infty \alpha_{-n}\cdot\alpha_n \\ \tilde{L}_0 &= \frac{\tilde{\alpha}_0^2}{2} + \sum_{n=1}^\infty \tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n \end{aligned} \tag{2.18}$$

where

$$\frac{\alpha_0^2}{2} = \frac{\tilde{\alpha}_0^2}{2} = \frac{\alpha'}{4}p^2 = -\frac{\alpha'}{4}m^2 \tag{2.19}$$

As a quantum version of the classical vanishing of $T_{\alpha\beta}$, one could consider the scenario where all the $L_n, \tilde{L}_n$ with $n\in\mathbb{Z}$ annihilate physical states. However, it will be sufficient to impose

$$L_{n>0}|\mathrm{phys}\rangle = \tilde{L}_{n>0}|\mathrm{phys}\rangle = 0 \tag{2.20}$$

since, by $(L_n)^\dagger = L_{-n}$, this ensures that the matrix elements of $\{L_n, \tilde{L}_n,\ n \in \mathbb{Z}\setminus\{0\}\}$ w.r.t. physical states $|\text{phys}\rangle$ and $|\text{phys}'\rangle$ vanish

$$\langle \text{phys}'|L_n|\text{phys}\rangle = 0 = \langle \text{phys}'|\tilde{L}_n|\text{phys}\rangle\,, \qquad n \neq 0 \tag{2.21}$$

In the simple example of the vacuum $|\text{phys}\rangle \to |0;p\rangle$, this is a consequence of the defining property of annihilation operators

$$\alpha^\mu_{n>0}|0;p\rangle = \tilde{\alpha}^\mu_{n>0}|0;p\rangle = 0 \tag{2.22}$$

The above normal-ordering conventions for the operators L_0 and $\tilde{L}_0$ were an ad-hoc choice. Hence, we should allow for some scalar offset $a \in \mathbb{C}$ when we impose the constraint

$$(L_0 - a)|\text{phys}\rangle = (\tilde{L}_0 - a)|\text{phys}\rangle = 0 \tag{2.23}$$

that corresponds to the zero mode of the energy-momentum tensor. It is instructive to see how the normal-ordering constant a affects the mass spectrum.

$$m^2|\text{phys}\rangle = \frac{4}{\alpha'}\left(-a + \sum_{n=1}^{\infty}\alpha_{-n}\cdot\alpha_n\right)|\text{phys}\rangle = \frac{4}{\alpha'}\left(-a + \sum_{n=1}^{\infty}\tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n\right)|\text{phys}\rangle \tag{2.24}$$

Our choice to pick the same offset a for both L_0 and $\tilde{L}_0$ implies the level-matching condition $(L_0 - \tilde{L}_0)|\text{phys}\rangle = 0$. In other words, physical closed-string states have the same oscillation numbers w.r.t. α^μ_n and $\tilde{\alpha}^\mu_n$. In the next subsections, we will settle a by Lorentz-invariance at quantum level.

2.3 Lightcone gauge

Now we are going to switch gears and impose $T_{\alpha\beta}$ constraints *before* quantization. The starting point is to recall the residual diffeo$\times$Weyl invariance that keeps the worldsheet metric $h_{\alpha\beta}(\sigma^\pm)$ in Minkowski form $\eta_{\alpha\beta}$,

$$h_{\alpha\beta}(\sigma^\pm) = \eta_{\alpha\beta} \overset{\text{certain diffeo's}}{\longrightarrow} e^{2\phi(\sigma^+,\sigma^-)}\eta_{\alpha\beta} \overset{\text{Weyl}}{\longrightarrow} \eta_{\alpha\beta} \tag{2.25}$$

In lightcone coordinates where $\eta_{\alpha\beta}d\sigma^\alpha d\sigma^\beta = -d\sigma^+ d\sigma^-$, the residual gauge invariance consists of two copies of *one-variable* diffeomorphisms $\sigma^+ \to \tilde{\sigma}^+(\sigma^+)$ and independently $\sigma^- \to \tilde{\sigma}^-(\sigma^-)$, where we admit no σ^- dependence to $\tilde{\sigma}^+(\sigma^+)$. When compared with the set of *two-variable* diffeomorphisms $\tilde{\sigma}^\alpha(\sigma^\beta)$, the residual gauge freedom is a set of measure zero! It affects the components of $T_{\alpha\beta}$ via

$$\partial_+ X^\mu \to \frac{\partial}{\partial\tilde{\sigma}^+}X^\mu = \frac{\partial\sigma^+}{\partial\tilde{\sigma}^+}\partial_+ X^\mu \tag{2.26}$$

For one given spacetime direction μ, one can independently enforce $\partial_+ X^\mu$ and $\partial_- X^\mu$ to match an arbitrary function with periodicity $\sigma \cong \sigma + 2\pi$: One simply has to dial $\sigma^+ \to \tilde{\sigma}^+(\sigma^+)$ such that the Jacobian compensates the deviation from the desired expression, i.e.: $\frac{\partial\sigma^+}{\partial\tilde{\sigma}^+} = \frac{\text{desired expression for } \partial_+ X^\mu}{\text{old version of } \partial_+ X^\mu}$.

However, this can only be done for a single spacetime direction, and this choice of μ direction breaks the manifest $SO(1, D-1)$ Lorentz invariance. We will do so by picking the following lightcone directions in the spacetime

$$X^\pm = \sqrt{\frac{1}{2}}(X^0 \pm X^{D-1}) \tag{2.27}$$

This is analogous to the lightcone coordinates $\sigma^{\pm} = \tau \pm \sigma$ on the worldsheet, except for the fact that spacetime has $D-2$ leftover or "transversal" directions (whose choice is an ad-hoc convention). We hope that the following discussion does not cause any confusion between the $\pm$ super- and subscripts for spacetime and worldsheet quantities ($X^{\pm}$ versus $\sigma^{\pm}$).

Now we exhaust the residual gauge symmetry $\sigma^{\pm} \to \tilde{\sigma}^{\pm}(\sigma^{\pm})$ to fix the "lightcone gauge"

$$\partial_+ X^+ = \partial_- X^+ = \frac{\alpha'}{2} p^+ \tag{2.28}$$

i.e. to enforce both $\partial_+ X^+$ and $\partial_- X^+$ to be the same constant. (Here, the superscript refers to the lightcone in spacetime, while the subscripts refer to the worldsheet lightcone $\frac{1}{2}(\partial_\tau \pm \partial_\sigma)$). For the undifferentiated X_L^+ and X_R^+, this still leaves the center-of-mass position x^+ as an integration constant

$$\begin{aligned} X_L^+ &= \frac{1}{2} x^+ + \frac{\alpha'}{2} p^+ \sigma^+ \\ X_R^+ &= \frac{1}{2} x^+ + \frac{\alpha'}{2} p^+ \sigma^- \end{aligned} \tag{2.29}$$

Let us now consider the classical worldsheet energy-momentum constraints

$$0 = \partial_\pm X \cdot \partial_\pm X = -2 \partial_\pm X^+ \partial_\pm X^- + \sum_{i=1}^{D-2} \partial_\pm X^i \partial_\pm X^i \tag{2.30}$$

With the above gauge-fixing of X^+, one obtains a simple expression when solving for the X^- component

$$\begin{aligned} \partial_+ X_L^- &= \frac{1}{\alpha' p^+} \sum_{i=1}^{D-2} \partial_+ X^i \partial_+ X^i \\ \partial_- X_R^- &= \frac{1}{\alpha' p^+} \sum_{i=1}^{D-2} \partial_- X^i \partial_- X^i \end{aligned} \tag{2.31}$$

where $p^+ \neq 0$. With the usual mode expansion for the directions $X^{i=1,2,\dots,D-2}$ outside the lightcone,

$$\alpha' p^- = \frac{1}{p^+} \sum_{i=1}^{D-2} \left(\frac{\alpha'}{2} p^i p^i + \sum_{n \neq 0} \alpha_n^i \alpha_{-n}^i \right) \tag{2.32}$$

$$\alpha_n^- = \sqrt{\frac{1}{2\alpha'}} \frac{1}{p^+} \sum_{k \in \mathbb{Z}} \sum_{i=1}^{D-2} \alpha_{n-k}^i \alpha_k^i \tag{2.33}$$

while x^- is left unconstrained (it does not enter the quantities $\partial_\pm X^-$ in the classical constraint). The same equations hold with $\alpha_n \leftrightarrow \tilde{\alpha}_n$ interchanged. This leads to the following lightcone formulation of the classical mass spectrum:

$$\begin{aligned} m_{class}^2 = 2p^+ p^- - \sum_{i=1}^{D-2} p^i p^i &= \frac{4}{\alpha'} \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i \\ &= \frac{4}{\alpha'} \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \tilde{\alpha}_{-n}^i \tilde{\alpha}_n^i \end{aligned} \tag{2.34}$$

2.4 Lightcone quantization

In a classical analysis, residual gauge freedom has been used to fix p^-, α_n^- and $\tilde{\alpha}_n^-$ in terms of p^+ and $p^i, \alpha_n^i, \tilde{\alpha}_n^i$ with $i = 1, 2, \dots, D-2$. When shall now perform the quantization only for the independent variables. By following the steps in the manifestly Lorentz-covariant quantization

$$\begin{aligned} [x^i, p^j] &= i\delta^{ij} \\ [x^-, p^+] &= -i \\ [\alpha^i_n, \alpha^j_k] &= [\tilde{\alpha}^i_n, \tilde{\alpha}^j_k] = n\delta^{ij}\delta_{n+k,0} \end{aligned} \tag{2.35}$$

This time, we have $2(D-2)$ instead of $2D$ towers of harmonic oscillators!

We still take the ground state $|0;p\rangle$ w.r.t. the dynamical oscillators $\{\alpha^i_n, \tilde{\alpha}^i_n,\ i = 1, 2, \ldots, D-2\}$ to be an eigenstate of $p^\mu = (p^+, p^-, p^i)$ subject to

$$\alpha^i_{n>0}|0;p\rangle = \tilde{\alpha}^i_{n>0}|0;p\rangle = 0 \tag{2.36}$$

One can still impose $[x^+, p^-] = -i$, and we have the following structure for the spectrum of physical states

$$\{|\text{phys}\rangle\} \in \{\alpha^{i_1}_{-1}\alpha^{i_2}_{-1}\ldots\alpha^{j_1}_{-2}\ldots\tilde{\alpha}^{k_1}_{-1}\tilde{\alpha}^{k_2}_{-1}\ldots\tilde{\alpha}^{l_1}_{-2}\ldots|0;p\rangle\} \tag{2.37}$$

Here, the matrix of inner products is positive definite since $\eta^{ij} = \delta^{ij} \geq 0$ for $i, j = 1, 2, \ldots, D-2$, so we have successfully removed the negative-norm states!

As in covariant quantization, there are ordering issues in p^- and m^2, so we need to once more prepare ourselves to an offset $a \in \mathbb{C}$ in the mass spectrum

$$\begin{aligned} m^2 = 2p^+p^- - \sum_{i=1}^{D-2} p^i p^i &= \frac{4}{\alpha'}\left(\sum_{i=1}^{D-2}\sum_{n=1}^{\infty} \alpha^i_{-n}\alpha^i_n - a\right) \\ &= \frac{4}{\alpha'}\left(\sum_{i=1}^{D-2}\sum_{n=1}^{\infty} \tilde{\alpha}^i_{-n}\tilde{\alpha}^i_n - a\right) \end{aligned} \tag{2.38}$$

where p^- contains $\sum_{n\neq 0}\alpha^i_n\alpha^i_{-n}$. The level-matching condition is

$$\sum_{i=1}^{D-2}\sum_{n=1}^{\infty}(\alpha^i_{-n}\alpha^i_n - \tilde{\alpha}^i_{-n}\tilde{\alpha}^i_n) = 0 \tag{2.39}$$

which is the only constraint on the oscillator numbers of $\alpha^{i_1}_{-1}\alpha^{i_2}_{-1}\ldots\alpha^{j_1}_{-2}\ldots\tilde{\alpha}^{k_1}_{-1}\tilde{\alpha}^{k_2}_{-1}\ldots\tilde{\alpha}^{l_1}_{-2}\ldots$ in the above schematic expression for physical states. We shall now discuss two approaches to fixing the normal-ordering constant a:

2.4.1 The physicist's way

The physicist's way of fixing a is to use Lorentz invariance, i.e. to impose that lightcone quantization does not introduce any violations of Lorentz invariance. We first note that the ground state $|0;p\rangle$ is a Lorentz scalar with $m^2 = -\frac{4a}{\alpha'}$. The first excited states $\alpha^i_{-1}\tilde{\alpha}^j_{-1}|0;p\rangle$ admitted by level matching obey $m^2 = \frac{4}{\alpha'}(1-a)$. There are $(D-2)^2$ such states, and we will explain that this counting and Lorentz invariance require a=1.

- We shall recall some basics on the representations of the Poincaré group (Lorentz transformations and translations). If $m^2 \neq 0$, we can find a reference frame $p^\mu = (m, 0, \ldots, 0)$ via Lorentz rotations, so massive particles form representations of the little group $SO(D-1)$ that stabilizes this choice of p^μ. When $m^2 = 0$ in turn, the best we can find is $p^\mu = (E, 0, \ldots, 0, E)$. Hence, massless particles form representations of the little group $SO(D-2)$ stabilizing the latter choice of p^μ.

- The $(D-2)^2$ states $\alpha^i_{-1}\tilde{\alpha}^j_{-1}|0;p\rangle$ only fit into representations of $SO(D-2)$, not into those of $SO(D-1)$. This follows from the fact that the candidate representations of $SO(D-1)$ have dimensions $\frac{1}{2}D(D-1)-1$ (symmetric and traceless two-tensor), $\frac{1}{2}(D-1)(D-2)$ (antisymmetric two tensor) and 1 (trace). Thus, the state must be massless to ensure that the relevant little group is $SO(D-2)$,

$$m^2\alpha^i_{-1}\tilde{\alpha}^j_{-1}|0;p\rangle \sim (1-a) = 0 \tag{2.40}$$

- The massless states $\alpha^i_{-1}\tilde{\alpha}^j_{-1}|0;p\rangle$ do not yet form an *irreducible* representation of the Lorentz group. We can disentangle the trace and the antisymmetric part, and find three irreducibles (i.e. particle species) at the massless level of the bosonic string

$$\alpha^i_{-1}\tilde{\alpha}^j_{-1} = \underbrace{\alpha^{(i}_{-1}\tilde{\alpha}^{j)}_{-1} - \frac{1}{D-2}\delta^{ij}\alpha^k_{-1}\tilde{\alpha}^k_{-1}}_{\text{massless spin 2} \,\Rightarrow\, \text{graviton}} \oplus \underbrace{\alpha^{[i}_{-1}\tilde{\alpha}^{j]}_{-1}}_{\text{“B-field”}} \oplus \underbrace{\frac{1}{D-2}\delta^{ij}\alpha^k_{-1}\tilde{\alpha}^k_{-1}}_{\text{dilaton}}$$

  As will be discussed later on, the vacuum expectation value of the dilaton will set the string coupling constants which weights different orders in string perturbation theory.

- For the above choice $a = 1$, the ground state has negative $m^2 = \frac{4}{\alpha'}(0-1) = -\frac{4}{\alpha'} < 0$. States with this property are known as *tachyons* T. We interpret $m^2 = \frac{\partial^2 V(T)}{\partial^2_T}|_{T=0} < 0$ as the curvature of a scalar potential, and its negative sign indicates that T is a fluctuation for expanding around the maximum rather than the minimum of the potential. This is analogous to the Higgs H whose mexican-hat potential has both minima and maxima, and the positive mass square that has been measured at LHC stems from expansion around a minimum.

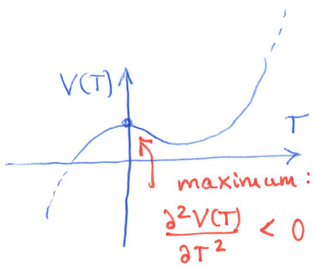


Figure 6: The negative m^2 of the tachyon is interpreted as expanding a scalar potential $V(T)$ around its maximum.

- By virtue of $a = 1$, we have $m^2 > 0$ for any higher excitation beyond $\alpha^i_{-1}\tilde{\alpha}^j_{-1}|0;p\rangle$, starting with

$$(\alpha^i_{-2} \oplus \alpha^i_{-1}\alpha^j_{-1}) \otimes (\tilde{\alpha}^k_{-2} \oplus \tilde{\alpha}^k_{-1}\tilde{\alpha}^l_{-1})|0;p\rangle \tag{2.41}$$

  The number of independent components for both $\alpha^i_{-2} \oplus \alpha^i_{-1}\alpha^j_{-1}$ and $\tilde{\alpha}^k_{-2} \oplus \tilde{\alpha}^k_{-1}\tilde{\alpha}^l_{-1}$ is easily seen to be $D-2+\frac{1}{2}(D-2)(D-1) = \frac{1}{2}(D-1)D-1$ which is the dimension of the traceless symmetric two tensor of $SO(D-1)$. Hence, the lightest string excitations with positive m^2 fall into the tensor product of two massive spin-two representations of $SO(D-1)$.

2.4.2 The mathematician's way

As a mathematician's way of fixing the normal-ordering constant a, we rearrange

$$\alpha' p^- = \frac{1}{2p^+} \sum_{i=1}^{D-2} \left(\alpha' p^i p^i + 2 \sum_{n=1}^{\infty} (\alpha^i_{-n} \alpha^i_n + \underbrace{\alpha^i_n \alpha^i_{-n}}_{\alpha^i_{-n}\alpha^i_n + n}) \right) \tag{2.42}$$

$$m^2 = 2p^+ p^- - \sum_{i=1}^{D-2} p^i p^i = \frac{4}{\alpha'} \left[\left(\sum_{n=1}^{\infty} \sum_{i=1}^{D-2} \alpha^i_{-n} \alpha^i_n \right) + \frac{D-2}{2} \sum_{n=1}^{\infty} n \right]$$

and run into the formally divergent sum $\sum_{n=1}^\infty n$. The latter can be assigned a finite value $\sum_{n=1}^\infty n = -\frac{1}{12}$, either via zeta-function regularization (to be done in the exercises) or via the following renormalization trick: The formally divergent sum in question is obtained by setting $\epsilon = 0$ on the left-hand side of

$$\sum_{n=1}^{\infty} n e^{-\epsilon n} = -\frac{\partial}{\partial \epsilon} \sum_{n=1}^{\infty} e^{-\epsilon n} = \frac{e^{-\epsilon}}{(1-e^{-\epsilon})^2} = \frac{1}{\epsilon^2} - \frac{1}{12} + \frac{\epsilon^2}{240} + O(\epsilon^4) \tag{2.43}$$

We discard the divergent term $\frac{1}{\epsilon^2}$ and take the constant piece $-\frac{1}{12}$ in the Laurent expansion as the answer. Although it might appear highly disturbing that the sum over positive integers evaluates to a negative fractional number, there will be another physics confirmation of the resulting normal-ordering constant

$$a = -\frac{D-2}{2} \sum_{n=1}^{\infty} n = \frac{D-2}{24} \tag{2.44}$$

This is consistent with $a = 1$ as required by massless states $\alpha^i_{-1} \tilde{\alpha}^j_{-1} |0; p\rangle$ if spacetime has $D = 26$ dimensions.

2.4.3 From the Lorentz algebra to $a = 1$ and $D = 26$

Another perspective on the above results $a = 1$ and $D = 26$ can be obtained from the conserved charges for the Lorentz symmetry of the Polyakov action (see exercise A.4)

$$J^{\mu\nu} = p^\mu x^\nu - p^\nu x^\mu + i \sum_{n=1}^{\infty} \frac{1}{n} (\alpha^\nu_n \alpha^\mu_{-n} - \alpha^\mu_n \alpha^\nu_{-n}) \tag{2.45}$$

They should obey the Lorentz algebra that you know well from your quantum-field-theory course

$$[J^{\mu\nu}, J^{\lambda\rho}] = \eta^{\nu\lambda} J^{\mu\rho} - \eta^{\nu\rho} J^{\mu\lambda} - \eta^{\mu\lambda} J^{\nu\rho} + \eta^{\mu\rho} J^{\nu\lambda} \tag{2.46}$$

However, we have given up manifest Lorentz invariance in lightcone quantization, so with our operator relations for $p^-, \alpha^-_n, \tilde{\alpha}^-_n$, it is non-trivial to check

$$\begin{aligned} 0 = [J^{i,-}, J^{j,-}] &= \frac{2}{(p^+)^2} \sum_{n=1}^{\infty} \left[\alpha^i_{-n} \alpha^j_n - \alpha^j_{-n} \alpha^i_n + (\alpha \leftrightarrow \tilde{\alpha}) \right] \\ &\times \left\{ \left(\frac{D-2}{24} - 1 \right) n + \left(a - \frac{D-2}{24} \right) \frac{1}{n} \right\} \end{aligned} \tag{2.47}$$

At $n = 1$, the $\{\ldots\}$ parenthesis equals $a - 1$, and we recover the conclusion $a = 1$ of our discussion of the $\alpha^i_{-1} \tilde{\alpha}^j_{-1} |0; p\rangle$ states. From additionally considering $n \neq 1$, we obtain $\frac{D-2}{24} = 1$, in lines with the above $\sum_{n=1}^\infty n = -\frac{1}{12}$. In conclusion, we have seen in multiple ways that consistency of bosonic strings in lightcone quantization requires $a = 1$ and $D = 26$.

3 Open bosonic strings

We will now adapt the discussion of closed bosonic strings to open ones. The highlight in this first look at open strings will be:

- (non)-abelian gauge bosons form $m^2 = 0$ states;
- geometric implementation of the Higgs mechanism;
- D branes as dynamical degree of freedom in string theories.

3.1 Dirichlet and Neumann boundary conditions

The worldsheets Σ of open strings have a non-periodic spatial direction whose coordinates will be normalized to live in the interval $\sigma \in (0, \pi)$. We need to revisit the variation of the Polyakov action (let's use conformal gauge to keep the discussion simple) since it was the periodicity $\sigma \cong \sigma + 2\pi$ specific to closed strings that led to the cancellation of boundary terms.

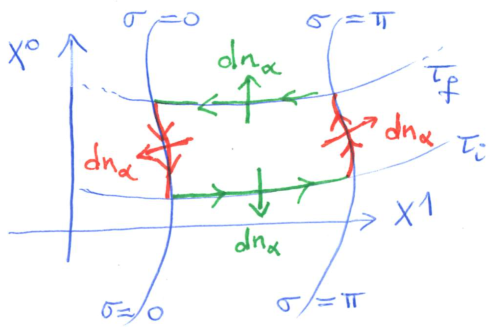


Figure 7: Examples of the outward pointing normal vector dn_α for the worldsheet of open strings.

For open strings, by contrast, there is a potential boundary term $\sim \int_{\partial\Sigma}$ when integrating by parts in the variation of the Polyakov action in conformal gauge (see the figure for the normal vector dn_α)

$$\begin{aligned}\delta S_{CG}[X] =& \frac{1}{2\pi\alpha'} \int_\Sigma d^2\sigma \, \partial_\alpha(\delta X^\mu)\partial^\alpha X_\mu \\ =& \frac{1}{2\pi\alpha'} \Bigg\{ \underbrace{\int_\Sigma d^2\sigma \, \delta X^\mu \partial_\alpha \partial^\alpha X_\mu}_{\text{see closed strings}} - \underbrace{\int_{\partial\Sigma} dn_\alpha \, \delta X^\mu \partial^\alpha X_\mu}_{\text{new for open strings}} \Bigg\}\end{aligned} \tag{3.1}$$

The first term has already been discussed for closed strings, so it remains to look at the second term.

$$\int_{\partial\Sigma} dn_\alpha \delta X^\mu \partial^\alpha X_\mu = - \underbrace{\int_0^\pi d\sigma \, \delta X^\mu \partial_\tau X_\mu \,\Big|_{\tau=\tau_i}^{\tau=\tau_f}}_{\text{vanishes by } \delta X^\mu(\tau_f) = \delta X^\mu(\tau_i) = 0} + \int_{\tau_i}^{\tau_f} d\tau \, \delta X^\mu \partial_\sigma X_\mu \,\Big|_{\sigma=0}^{\sigma=\pi} \tag{3.2}$$

Since we assume the initial and final configurations to be specified, one can set $\delta X^\mu(\tau_f) = \delta X^\mu(\tau_i) = 0$, and only the second term $\sim \int d\tau$ needs additional input to vanish: At both endpoints $\sigma = 0$ and $\sigma = \pi$, there are two choices of boundary conditions to make the product $\delta X^\mu \partial_\sigma X_\mu$ vanish – either δX^μ or $\partial_\sigma X_\mu$ has to vanish:

- Neumann conditions

$$\partial_\sigma X^\mu \big|_{\sigma=0,\pi} = 0 \tag{3.3}$$

for some $0 \leq \mu \leq D-1$, this corresponds to open-string endpoints moving freely.

- Dirichlet conditions

$$\delta X^I \big|_{\sigma=0,\sigma=\pi} = 0 \tag{3.4}$$

for some $1 \leq I \leq D-1$. This leads to a specified position c^I, d^I in space that is constant over τ,

$$X^I(\tau, \sigma=0) = c^I\,, \qquad X^I(\tau,\sigma=\pi) = d^I \tag{3.5}$$

We do not admit $I=0$ to be part of the Dirichlet directions as that would localize the open-string endpoints in time.

The mode expansions for open strings are found by solving $\partial_\alpha \partial^\alpha X^\mu = 0$ with Dirichlet / Neumann conditions instead of the periodicity $\sigma \cong \sigma + 2\pi$. We make the usual ansatz, $X^\mu(\tau,\sigma) = X^\mu_L(\sigma^+) + X^\mu_R(\sigma^-)$

$$\begin{aligned} X_L(\sigma^+) &= \frac{1}{2}x^\mu + \alpha' p^\mu_L \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{1}{n} \tilde{\alpha}^\mu_n e^{-in\sigma^+} \\ X_R(\sigma^-) &= \frac{1}{2}x^\mu + \alpha' p^\mu_R \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{1}{n} \alpha^\mu_n e^{-in\sigma^-} \end{aligned} \tag{3.6}$$

and will next see how the boundary conditions imply constraints on $x^\mu, p^\mu_L, p^\mu_R, \alpha^\mu_n, \tilde{\alpha}^\mu_n$

- Neumann conditions "NN" at both endpoints $\sigma = 0$ and $\sigma = \pi$:

$$\begin{aligned} 0 = \partial_\sigma X^\mu(\tau,\sigma)\big|_{\sigma=0,\pi} &= \partial_+ X^\mu_L(\sigma^+) - \partial_- X^\mu_R(\sigma^-)\big|_{\sigma=0,\pi} \\ &= \alpha'(p^\mu_L - p^\mu_R) + \sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} (\tilde{\alpha}^\mu_n - \alpha^\mu_n) e^{-in\tau} \times \left\{ \begin{smallmatrix} 1 & : \ \sigma=0 \\ (-1)^n & : \ \sigma=\pi \end{smallmatrix} \right. \end{aligned} \tag{3.7}$$

This is the case for any τ if we choose $p^\mu_L = p^\mu_R$ and $\tilde{\alpha}^\mu_n = \alpha^\mu_n$ for μ in Neumann directions.

$$X^\mu(\tau,\sigma)\big|_{\rm NN} = x^\mu + 2\alpha' p^\mu \tau + 2i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{1}{n} \alpha^\mu_n e^{-in\tau} \cos(n\sigma) \tag{3.8}$$

Note that the closed-string normalization of the second term $\sim p^\mu \tau$ is α' instead of $2\alpha'$. This modified normalization for open strings is to ensure that $p^\mu = \frac{1}{2\pi\alpha'} \int_0^\pi d\sigma \partial_\tau X^\mu$ can be identified with the Poincaré generator of translations studied in the exercises.

- Dirichlet conditions "DD" at both endpoints $\sigma = 0$ and $\sigma = \pi$:

$$c^I = X^I(\tau,\sigma=0)\,, \qquad d^I = X^I(\tau,\sigma=\pi) \tag{3.9}$$

This gives, in Dirichlet direction (with index I instead of μ),

$$x^I = c^I\,, \qquad p^I_L = -p^I_R = \frac{d^I - c^I}{2\pi\alpha'}\,, \qquad \tilde{\alpha}^I_n = -\alpha^I_n \tag{3.10}$$

The solution is

$$X^I(\tau,\sigma)\big|_{\rm DD} = c^I + \frac{\sigma}{\pi}(d^I - c^I) - 2\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{1}{n} \alpha^I_n e^{-in\tau} \sin(n\sigma) \tag{3.11}$$

and does not feature any momentum p^I.

The bottom line in both cases is that all the $\tilde{\alpha}_n^\mu$ are fixed by α_n^μ, i.e. open strings only have half of the oscillator degrees of freedom! As will be explored in the exercises, one could also have mixed boundary conditions, say Neumann at $\sigma = 0$ and Dirichlet at $\sigma = \pi$.

3.2 Open-string quantization and mass spectrum

Let us now apply lightcone quantization to open strings. We impose Neummann boundary conditions at $\sigma = 0, \pi$ for the range of spacetime directions with indices $0 \leq \mu \leq d_N{-}1$, and Dirichlet conditions at $\sigma = 0, \pi$ for $d_N \leq I \leq D{-}1$. Hence, there is a total of d_N Neumann directions and $D - d_N$ Dirichlet directions. This choice breaks the Lorentz group from $SO(1, D{-}1)$ to $SO(1, d_N{-}1) \times SO(D{-}d_N)$.

We pick lightcone gauge in Neumann directions

$$X^\pm = \sqrt{\frac{1}{2}}(X^0 \pm X^{d_N-1}) \tag{3.12}$$

and again use the residual gauge freedom to fix $\partial_\pm X^+ = \alpha' p^+$. Then, the classical energy-momentum constraints are solved via

$$\partial_\pm X^- = \frac{1}{2\alpha' p^+} \left(\sum_{i=1}^{d_N-2} \partial_\pm X^i \partial_\pm X^i + \sum_{I=d_N}^{D-1} \partial_\pm X^I \partial_\pm X^I \right) \tag{3.13}$$

with $d_N{-}2$ transverse Neumann directions and $D{-}d_N$ transverse Dirichlet directions. The left-hand side evaluates to $\alpha' p^-$ plus oscillators. On the right-hand side, we obtain $\alpha' p^i p^i$ plus oscillators from the first term (Neumann directions) and $\frac{1}{4\pi^2} \sum_{I=d_N}^{D-1} (c^I - d^I)^2$ plus oscillators from the second term (Dirichlet directions).

The dynamical degrees of freedom are

- x^i, p^i, α_n^i in transverse Neumann direction $1 \leq i \leq d_N - 2$;
- α_n^I in Dirichlet direction $d_N \leq I \leq D - 1$

The commutation relations are the same as for closed strings:

$$\begin{aligned} [x^i, p^j] &= i\delta^{ij} \\ [\alpha_n^i, \alpha_k^j] &= n\delta^{ij}\delta_{n+k,0} \\ [\alpha_n^I, \alpha_k^J] &= n\delta^{IJ}\delta_{n+k,0} \\ [\alpha_n^i, \alpha_k^J] &= 0 \end{aligned} \tag{3.14}$$

The Fock space of physical states is built on eigenstates $|0; p\rangle$ of $p^{\mu=0,1,\ldots,d_N-1}$ subject to $\alpha_{n>0}^{i,I}|0; p\rangle = 0$

$$\{|\text{phys}_{\text{open}}\rangle\} = \{\alpha_{-1}^{i_1}\alpha_{-1}^{i_2} \ldots \alpha_{-2}^{j_1} \ldots \alpha_{-1}^{I_1}\alpha_{-1}^{I_2} \ldots \alpha_{-2}^{J_1} \ldots |0, p\rangle\} \tag{3.15}$$

The mass spectrum is (note the rescaling $\alpha' \to 4\alpha'$ in comparison to closed strings)

$$\begin{aligned} m^2 &= 2p^+p^- - \sum_{i=1}^{d_N-2} p^i p^i \\ &= \frac{1}{(2\pi\alpha')^2} \sum_{I=d_N}^{D-1} (c^I - d^I)^2 + \frac{1}{\alpha'} \left\{ \sum_{n=1}^{\infty} \left(\sum_{i=1}^{d_N-2} \alpha_{-n}^i \alpha_n^i + \sum_{I=d_N}^{D-1} \alpha_{-n}^I \alpha_n^I \right) - a \right\} \end{aligned} \tag{3.16}$$

with the same type of normal-ordering constant a as for closed strings in both Dirichlet and Neumann directions:

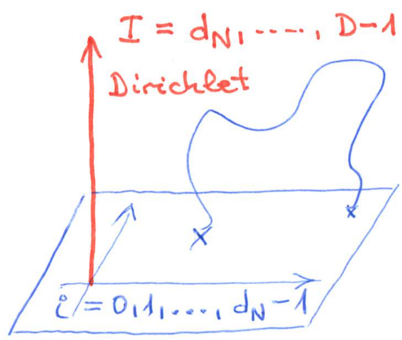
$$a = -\frac{1}{2}(D-2)\sum_{n=1}^{\infty} n = \frac{D-2}{24} \tag{3.17}$$

Again, Lorentz invariance in the Neumann directions including $X^{\pm}$ will give $a = 1$ and $D = 26$. Since these are the same values found in the discussion of closed strings, we can view open- and closed-string excitations as states in the same theory.

Let us take a brief look at the lowest mass levels (i.e. the states with the lowest oscillator numbers n of α^i_{-n} and α^I_{-n}):

- level zero: a tachyonic ground state $m^2|0;p\rangle = -\frac{1}{\alpha'}|0;p\rangle$
- level one: massless states $\alpha^i_{-1}|0;p\rangle$ and $\alpha^I_{-1}|0;p\rangle$. The former forms the vector of the little group $SO(d_N-2)$ for $m^2 = 0$ states, so the $d_N - 2$ states $\alpha^i_{-1}|0;p\rangle$ are interpreted as gauge bosons. The $D-d_N$ states $\alpha^I_{-1}|0;p\rangle$ in turn are scalars with respect to the Lorentz group in Neumann directions.
- level ≥ 2: excited states of the open string have $m^2 \in \frac{\mathbb{N}}{\alpha'}$, starting with $\alpha^i_{-2}|0;p\rangle \oplus \alpha^J_{-2}|0;p\rangle$ as well as $\alpha^i_{-1}\alpha^j_{-1}|0;p\rangle \oplus \alpha^i_{-1}\alpha^J_{-1}|0;p\rangle \oplus \alpha^I_{-1}\alpha^J_{-1}|0;p\rangle$. As for a chiral half of the analogous closed-string states, the open-string states $\alpha^i_{-2}|0;p\rangle \oplus \alpha^i_{-1}\alpha^j_{-1}|0;p\rangle$ can be recombined to a spin-two representation of the massive little group $SO(d_N-1)$. The remaining states obtained from replacing $\alpha^i_{-n} \to \alpha^I_{-n}$ are massive scalars and vectors.

3.3 D branes and non-abelian gauge fields



The open-string endpoints $X^I(\tau,\sigma{=}0) = c^I$ and $X^I(\tau,\sigma{=}\pi) = d^I$ fixed by Dirichlet boundary conditions define hypersurfaces in spacetime. However, since one may ask what is special about these choices of c^I, d^I, we shall now interpret these hypersurfaces as dynamical objects called D-branes (with D for "Dirichlet"). More precisely, the number d_N of spacetime directions with Neumann boundary conditions is usually specified, and we speak of a Dp branes with $p = d_N - 1$. In these conventions, a D0 brane corresponds to a particle, a D1 brane to a fundamental string and a D2 brane to a membrane. With $p = D-1$ or $d_N = D$ (i.e. no Dirichlet boundary conditions), we get spacetime filling branes.

Let us revisit the interpretation of massless open-string states in the light of D-branes:

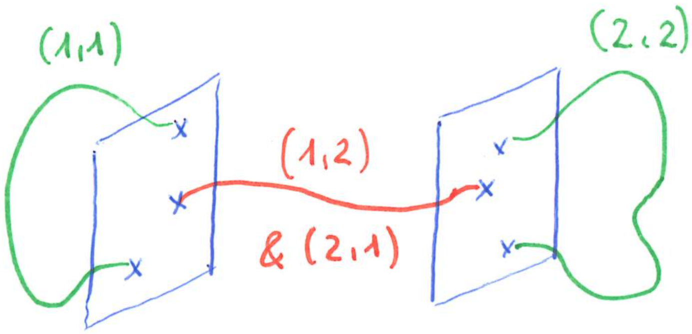
- vector $\alpha^i_{-1}|0;p\rangle \leftrightarrow$ gauge field on the brane
- scalars $\alpha^I_{-1}|0;p\rangle \leftrightarrow$ fluctuations of the brane in the transverse direction ($\rightarrow$ hint for dynamics)

However, the dynamics of D-brane is hard to see perturbatively since $M_{brane} \sim \frac{1}{g_{string}}$.

In case of two separated branes $c^I \neq d^I$, the open-string states become 2×2 matrices: Depending on the independent choices to locate the two endpoints on one of the branes, we assign labels $(1,1),(1,2),(2,1),(2,2)$ to these open-string states. The off-diagonal states having endpoints on different D-branes acquire the mass shift

$$\Delta(m^2) = \frac{1}{(2\pi\alpha')^2} \sum_{I=p+1}^{D-1} (c^I - d^I)^2 \tag{3.18}$$

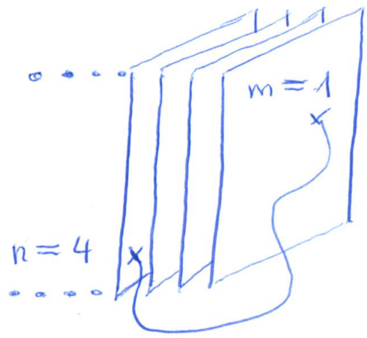
found in the previous section.



If we have a stack of N coinciding branes, both endpoints can be labelled by $m, n = 1, 2, \ldots, N$ to specify the brane they are attached to. This yields the degrees of freedom of non-abelian gauge bosons of $U(N)$:

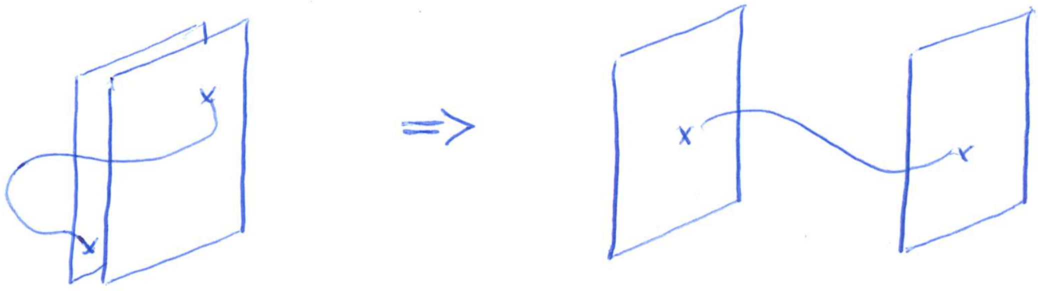
$$|\text{phys}_{\text{open}}\rangle \to |\text{phys}_{\text{open}}; (m,n)\rangle = \sum_{a=1}^{N^2} (T^a)_m{}^n |\text{phys}_{\text{open}}; a\rangle \tag{3.19}$$

where the coefficients are a spanning set of hermitian $N \times N$ matrices T^a, known as "Chan Paton factors".

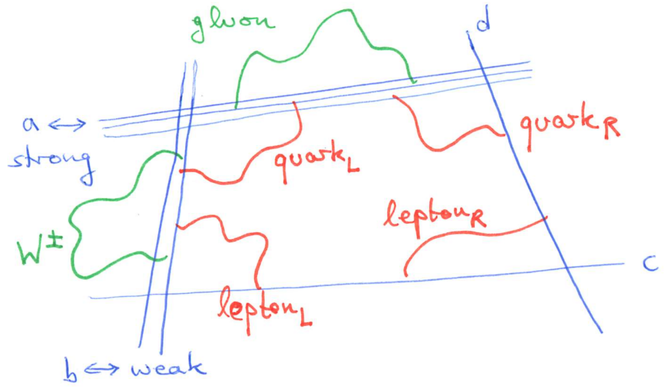


The D-brane setup admits a geometric implementation of the Higgs effect: By separating initially coinciding branes, the $U(2)$ gauge group is broken to $U(1) \times U(1)$. From the previous discussion, the off-diagonal gauge fields gain the mass $\Delta(m^2) \sim (c^I - d^I)^2$, and the diagonal scalars gain a VEV $\alpha^I_{-1}|0;p;(m,n)\rangle \leftrightarrow \left(\begin{smallmatrix} \phi_{11} & 0 \\ 0 & \phi_{22} \end{smallmatrix}\right)$. However, in contrast to the Standard Model Higgs, we are dealing with adjoint rather than fundamental scalars here.

The D-brane setup can be used to make progress towards the Standard Model gauge group $SU(3)_{\text{strong}} \times SU(2)_{\text{weak}} \times U(1)_Y$: Take 4 stacks of D-branes $U(3)_a \times U(2)_b \times U(1)_c \times U(1)_d$ (the 4th one is needed



in order to balance certain charges). Gauge bosons for one of the gauge groups stem from open strings with both endpoints on the same stack. Chiral fermions such as leptons/quarks arise from open strings located at intersecting stacks of branes, they are in bi-fundamental (rather than adjoint) representations of the two associated gauge groups.



D-branes can be wrapped around cycles in compact spacetime dimensions. The number of quark- and lepton generations is then determined by the intersection numbers of different cycles.

4 Basics of conformal field theory

This section provides an introduction to conformal field theories with the following motivation and goals:

- sidestep the complicated commutator manipulations in the quantized bosonic string such as the lengthy and subtle computation of $[L_m, L_n] \leftrightarrow [\alpha_{m-p}\alpha_p, \alpha_{n-q}\alpha_q]$; instead infer such brackets from clean complex-analysis methods, starting with Cauchy's theorem $\oint_{B_\epsilon(0)} z^n = \delta_{n,-1}$ for a small circle $B_\epsilon(0)$ around the origin
- characterize & exploit the residual Diff $\times$ Weyl invariance $\sigma^+ \to f(\sigma^+)$ and $\sigma^- \to g(\sigma^-)$ as an infinite-dimensional symmetry; use it for covariant quantization and computing string interactions.

4.1 Conformal symmetry in general dimensions d

Although we will be mainly interested in applications to the conformally invariant worldsheet theory with $d = 2$ dimensions, we will discuss some aspects of d-dimensional conformal symmetry that for instance apply to Maxwell theory and Yang Mills in $d = 4$. Conformal transformations in $\mathbb{R}^{1,d-1}$ are defined as those diffeomorphisms that preserve angles, i.e. rescale metric under $x^\mu \to y^\mu(x)$

$$\eta_{\mu\nu} \to g_{\mu\nu} = \frac{\partial x^\lambda}{\partial y^\mu}\frac{\partial x^\rho}{\partial y^\nu}\eta_{\lambda\rho} = \Omega^2(x)\eta_{\mu\nu} \tag{4.1}$$

with some possibly x-dependent scale factor (or conformal factor) $\Omega(x)$. Examples of d-dimensional conformal transformations are

Example	$y^\mu(x)$	$\Omega^2(x)$
translation	$x^\mu + c^\mu$	1
Lorentz transformation	$\Lambda^\mu{}_\nu x^\nu, \Lambda \in SO(1, d{-}1)$	1
dilatation	λx^μ	$\frac{1}{\lambda^2}$
special conformal transformation	$\frac{x^\mu + x^2 b^\mu}{1+2b\cdot x + b^2x^2}$	$(1+2b\cdot x + b^2 x^2)^2$

(4.2)

where special conformal transformations are translations by b^μ sandwiched between two inversions

$$x^\mu \to \frac{x^\mu}{x^2}\,, \qquad \Omega^2(x) = (x^2)^2 \tag{4.3}$$

The inversion is disconnected from unity and therefore does not enter the list of infinitesimal conformal transformations $y^\mu = x^\mu + \epsilon\omega^\mu(x)$. In order to determine the conformal algebra, evaluate the group action on $F(x^\mu)$ via the inverse of infinitesimal conformal transformations, $F(x^\mu) \to F(x^\mu - \epsilon\omega^\mu(x)) + O(\epsilon^2)$

infinitesimal diffeo ω^μ	action on $F(x)$	generator
c^μ	$1 - i\epsilon c^\mu P_\mu$	$P_\mu = -i\partial_\mu$
$m^{[\mu\nu]}x_\nu$	$1 - \frac{i}{2}\epsilon m^{\mu\nu}J_{\mu\nu}$	$J_{\mu\nu} = -i(x_\nu\partial_\mu - x_\mu\partial_\nu)$
λx^μ	$1 - i\epsilon\lambda D$	$D = -ix^\mu\partial_\mu$
$x^2 b^\mu - 2(x\cdot b)x^\mu$	$1 - i\epsilon b^\mu K_\mu$	$K_\mu = -ix^2\partial_\mu + 2ix_\mu x\cdot\partial$

(4.4)

The conformal group in $\mathbb{R}^{1,d-1}$ is $\frac{1}{2}(d+1)(d+2)$-dimensional, and by comparing the commutators of $\mathrm{span}\{P_\mu, J_{\mu\nu}, D, K_\mu\}$ to higher-dimensional Lorentz brackets, the conformal algebra turns out to be isomorphic to $SO(2,d)$. The conserved charges can be constructed from the energy-momentum tensor,

$$T^{\mu\nu} = \frac{4\pi}{\sqrt{-\det g}}\frac{\delta S[g,\ldots]}{\delta g_{\mu\nu}}|_{g_{\lambda\rho}=\eta_{\lambda\rho}} \tag{4.5}$$

using the fact that the infinitesimal rescaling of the metric is $\delta g_{\mu\nu} = -\epsilon(\partial_\mu\omega_\nu + \partial_\nu\omega_\mu)$,

$$\begin{aligned} \delta S[g,\ldots] &= \frac{1}{4\pi}\int d^dx\sqrt{-\det g}\,T^{\mu\nu}\delta g_{\mu\nu} \\ &= \int d^dx\,\partial_\mu J^\mu_{(\omega)}(x) \end{aligned} \tag{4.6}$$

In the last line, we have rewritten the variation of the action as a divergence of the Noether current $J^\mu_{(\omega)}$ for $y^\mu = x^\mu + \epsilon\omega^\mu$. Assuming the consequences $T^{\mu\nu} = T^{\nu\mu}$ and $\partial_\mu T^{\mu\nu} = 0$ of Poincaré symmetry, the general form of these currents is

$$J^\mu_{(\omega)} = T^{\mu\nu}\omega_\nu \tag{4.7}$$

In particular, the vanishing divergence for the dilatation current with $\omega_\nu = \lambda x_\nu$ implies that

$$\partial_\mu J^\mu_{(\lambda x)} \sim \partial_\mu(T^{\mu\nu}x_\nu) = T^{\mu\nu}\eta_{\mu\nu} = 0 \tag{4.8}$$

by $\partial_\mu T^{\mu\nu} = 0$. Hence, conformal field theories have a traceless energy-momentum tensor $T^\mu{}_\mu = 0$.

4.2 Conformal symmetry in two dimensions

The two-dimensional Minkowski metric factorizes in lightcone coordinates $\sigma^{\pm} = x^0 \pm x^1$ that have already been used to solve the free wave equation

$$ds^2 = -(dx^0)^2 + (dx^1)^2 = -d\sigma^+ d\sigma^- \tag{4.9}$$

where we identify x^0 with τ and x^1 with σ on the string worldsheet.

We shall now pass to Euclidean time

$$x^0 = \tau = -it \tag{4.10}$$

and complexify the lightcone coordinates

$$\left.\begin{aligned} z &= i\sigma^+ = t + i\sigma \\ \bar{z} &= i\sigma^- = t - i\sigma \end{aligned}\right\} \quad \Rightarrow \quad ds^2 = dz\, d\bar{z} \tag{4.11}$$

In this setting, a diffeomorphism $z \to \xi(z, \bar{z})$ and $\bar{z} \to \bar{\xi}(z, \bar{z})$ is conformal if

$$\begin{aligned} ds^2 &= \frac{\partial z}{\partial \xi}\frac{\partial \bar{z}}{\partial \xi} d\xi^2 + \frac{\partial z}{\partial \bar{\xi}}\frac{\partial \bar{z}}{\partial \bar{\xi}} d\bar{\xi}^2 + \left(\frac{\partial z}{\partial \xi}\frac{\partial \bar{z}}{\partial \bar{\xi}} + \frac{\partial z}{\partial \bar{\xi}}\frac{\partial \bar{z}}{\partial \xi}\right) d\xi d\bar{\xi} \\ &= \Omega^2(\xi, \bar{\xi}) d\xi d\bar{\xi} \end{aligned} \tag{4.12}$$

As a necessary condition, we need to enforce vanishing coefficients of $d\xi^2$ and $d\bar{\xi}^2$, i.e.

$$\frac{\partial \xi}{\partial \bar{z}} = \frac{\partial \bar{\xi}}{\partial z} = 0 \quad \Rightarrow \quad \xi(z) \text{ meromorphic and } \bar{\xi}(\bar{z}) \text{ antimeromorphic} \tag{4.13}$$

Then we can transform $z \to \xi(z)$ and $\bar{z} \to \bar{\xi}(\bar{z})$ independently, by relaxing the reality of

$$t = \frac{\xi + \bar{\xi}}{2}, \qquad \sigma = \frac{\xi - \bar{\xi}}{2i} \tag{4.14}$$

then $(z, \bar{z})$ and $(\xi, \bar{\xi})$ do not have to be complex conjugates of each other. The conformal transformations realized by meromorphic $\xi(z)$ are generated by $z \to z + \epsilon z^{n+1}$ with $n \in \mathbb{Z}$

$$\begin{aligned} F(z) &\to F(z - \epsilon z^{n+1}) + O(\epsilon^2) \\ &= F(z) - \epsilon z^{n+1} \partial_z F(z) + O(\epsilon^2) \\ &= (1 + \epsilon l_n) F(z) + O(\epsilon^2) \end{aligned} \tag{4.15}$$

so we read off generators $l_n = -z^{n+1}\partial_z$ with $n \in \mathbb{Z}$ subject to

$$[l_m, l_n] = (m - n) l_{m+n} \tag{4.16}$$

Together with $\bar{l}_n = -\bar{z}^{n+1}\partial_{\bar{z}}$, $n \in \mathbb{Z}$ this yields an infinite dimensional extension of the naively expected $\frac{1}{2}(d+1)(d+2)|_{d=2} = 6$ generators. Still, the significance of this number can be seen from the 6 generators $l_0, l_1, l_{-1}, \bar{l}_0, \bar{l}_1, \bar{l}_{-1}$:

- they form a subalgebra (i.e. they close under taking commutators)
- the remaining generators $l_{n\geq 2}, l_{n\leq -2}, \bar{l}_{n\geq 2}, \bar{l}_{n\leq -2}$ are singular as $z, \bar{z} \to 0$ or ∞; Only the $l_n, \bar{l}_n$ at $n \in \{-1, 0, 1\}$ are globally defined

- the finite form for the globally defined $l_n, \bar{l}_n$ is

$$z \to \frac{az+b}{cz+d}\,, \qquad \bar{z} \to \frac{\bar{a}\bar{z}+\bar{b}}{\bar{c}\bar{z}+\bar{d}} \tag{4.17}$$

  where one can always perform a simultaneous rescaling of numerator and denominator to enforce $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{C})$.

  - The $l_{-1}, \bar{l}_{-1}$ generates translations $\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$
  - The $l_0 + \bar{l}_0$ generates dilatations $\left(\begin{smallmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{smallmatrix}\right)$
  - The $i(l_0 - \bar{l}_0)$ generates rotations $\left(\begin{smallmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{smallmatrix}\right)$
  - The $l_1, \bar{l}_1$ generates special conformal transformation $\left(\begin{smallmatrix} 1 & 0 \\ c & 1 \end{smallmatrix}\right)$

As an example of fields transforming under conformal $z \to \xi(z)$ and $\bar{z} \to \bar{\xi}(\bar{z})$, we define conformal primaries $\phi_{h,\bar{h}}(z,\bar{z})$ of weight $(h,\bar{h})$ as follows: the conformal transformation $z \to \xi(z)$, $\bar{z} \to \bar{\xi}(\bar{z})$ sends

$$\phi_{h,\bar{h}}(z,\bar{z}) \to \phi'_{h,\bar{h}}(\xi,\bar{\xi}) = \left(\frac{\partial \xi}{\partial z}\right)^{-h} \left(\frac{\partial \bar{\xi}}{\partial \bar{z}}\right)^{-\bar{h}} \phi_{h,\bar{h}}(z,\bar{z}) \tag{4.18}$$

- This generalizes $d\xi\, d\bar{\xi} = \frac{\partial \xi}{\partial z}\frac{\partial \bar{\xi}}{\partial \bar{z}} dz\, d\bar{z}$ to values of $h, \bar{h}$ different from -1.

- $\phi_{h,\bar{h}}$ transforms like a tensor with h holomorphic indices and $\bar{h}$ antiholomorphic ones $A_{\underbrace{zz\ldots z}_{h}\underbrace{\bar{z}\bar{z}\ldots\bar{z}}_{\bar{h}}}$,

$$\text{cf.} \quad A_{\mu_1\mu_2\ldots\mu_{h+\bar{h}}} \to \frac{\partial x^{\nu_1}}{\partial y^{\mu_1}} \frac{\partial x^{\nu_2}}{\partial y^{\mu_2}} \cdots \frac{\partial x^{\nu_{h+\bar{h}}}}{\partial y^{\mu_{h+\bar{h}}}} A_{\nu_1\nu_2\ldots\nu_{h+\bar{h}}} \tag{4.19}$$

- as an example of conformal primaries,

$$\partial_z X^\mu(z,\bar{z}) \to \frac{\partial}{\partial \xi} X^\mu(\xi,\bar{\xi}) = \frac{\partial z}{\partial \xi}\frac{\partial}{\partial z} X\big(\xi(z),\bar{\xi}(\bar{z})\big) \tag{4.20}$$

  has $(h,\bar{h}) = (1,0)$, and similarly, $\partial_{\bar{z}} X^\mu(z,\bar{z})$ has $(h,\bar{h}) = (0,1)$

- infinitesimal transformations $\xi(z) = z + \epsilon\eta(z)$ and $\bar{\xi}(\bar{z}) = \bar{z}$ (working to first order in ϵ only)

$$\begin{aligned} \delta_\eta \phi_{h,\bar{h}}(z,\bar{z}) &= \phi'_{h,\bar{h}}(z) - \phi_{h,\bar{h}}(z) \\ &= -\epsilon\eta(z)\partial_z\phi_{h,\bar{h}}(z) + \left[\left(\frac{\partial \xi}{\partial z}\right)^{-h} - 1\right]\phi_{h,\bar{h}}(z) + O(\epsilon^2) \\ &= -\epsilon\big[\eta(z)\partial_z\phi_{h,\bar{h}}(z) + h\phi_{h,\bar{h}}(z)\partial_z\eta(z)\big] + O(\epsilon^2) \end{aligned} \tag{4.21}$$

  where we have used the following in passing to the second line:

$$\phi'_{h,\bar{h}}(z) = \phi'_{h,\bar{h}}\big(\xi(z) - \epsilon\eta(z)\big) = \left(\frac{\partial \xi}{\partial z}\right)^{-h} \phi_{h,\bar{h}}(z) - \epsilon\eta(z)\partial_z\phi_{h,\bar{h}}(z) + O(\epsilon^2) \tag{4.22}$$

Let us now take a closer look at the energy-momentum tensor $T_{\alpha\beta\in\{z,\bar{z}\}}$ in two-dimensional CFTs. It is traceless as in any higher-dimensional CFT which in the $z,\bar{z}$ coordinates takes the form

$$T_{z\bar{z}} = T_{\bar{z}z} = 0 \tag{4.23}$$

The conservation equations $\partial_\alpha T^{\alpha\beta}$ along with $T_{\bar{z}\bar{z}} = \frac{1}{4}T^{zz}$ give

$$\partial_z T_{\bar{z}\bar{z}} = \partial_{\bar{z}} T_{zz} = 0 \tag{4.24}$$

so we have two independent components – one meromorphic and one anti-meromorphic.

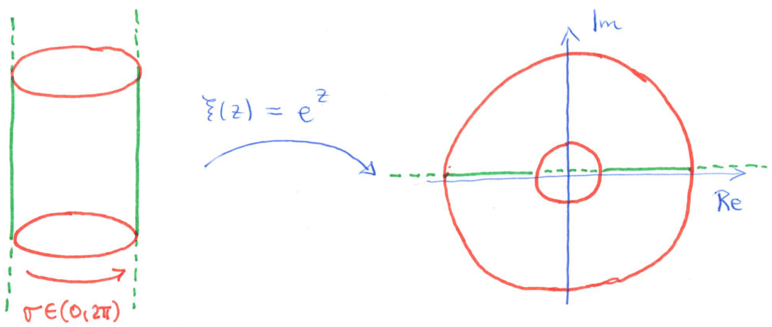
$$T(z) = T_{zz}, \qquad \bar{T}(\bar{z}) = T_{\bar{z}\bar{z}} \tag{4.25}$$

The conserved current generating the infinitesimal conformal transformation $z \to z + \epsilon\eta(z)$ is

$$J_{(\eta)}(z) = T(z)\eta(z)\,, \qquad \bar{J}_{(\bar{\eta})}(\bar{z}) = \bar{T}(\bar{z})\bar{\eta}(\bar{z}) \tag{4.26}$$

4.3 Towards quantum CFTs in two dimensions

We motivate the transition to quantum CFTs through the closed-string example. We can map the cylinder worldsheet defined by the periodic direction $\sigma \cong \sigma + 2\pi$ to the complex plane via $\xi(z) = e^z = e^{t+i\sigma}$, where periodicity in σ becomes automatic by $\xi = e^{2\pi i}\xi$. Infinite past/future w.r.t. Euclidean time $t = i\tau$ are mapped to points $\xi = 0$ and $\xi = \infty$ on the Riemann sphere, $\mathbb{C} \cup \{\infty\}$.



Let us now investigate the mode expansion on the plane, upon renaming $\alpha_k^\mu \leftrightarrow \tilde{\alpha}_k^\mu$ for convenience:

$$\frac{\partial}{\partial\sigma^+} X^\mu = \sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \alpha_n^\mu e^{-in\sigma^\pm}\,, \qquad \frac{\partial}{\partial\sigma^-} X^\mu = \sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \tilde{\alpha}_n^\mu e^{-in\sigma^\pm} \tag{4.27}$$

- the dictionary for the derivatives is $\frac{\partial}{\partial\sigma^+} = i\partial_z$ and $\frac{\partial}{\partial\sigma^-} = i\partial_{\bar{z}}$, i.e.

$$\partial_z X^\mu = -i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \alpha_n^\mu e^{-nz}\,, \qquad \partial_{\bar{z}} X^\mu = -i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \tilde{\alpha}_n^\mu e^{-n\bar{z}} \tag{4.28}$$

- after $\xi(z) = e^z$ and $\partial_z = \xi\partial_\xi$ as well as $\bar{\xi}(\bar{z}) = e^{\bar{z}}$ and $\partial_{\bar{z}} = \bar{\xi}\partial_{\bar{\xi}}$, we have

$$\partial_\xi X^\mu = -i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \alpha_n^\mu \xi^{-n-1}\,, \qquad \partial_{\bar{\xi}} X^\mu = -i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \tilde{\alpha}_n^\mu \bar{\xi}^{-n-1} \tag{4.29}$$

- henceforth rename $\xi(z) \to z$, we will work on the complex plane for the rest of the course (with infinite past at $z \to 0$, infinite future $z \to \infty$ and circular constant-τ slice $|z| = const$)

In general quantum CFTs, one promotes fields to operators

$$\underbrace{\text{fields}}_{\substack{\text{e.g. primaries } \phi_{h,\bar{h}}(z,\bar{z}) \\ \text{such as } \partial_z X^\mu \text{ at } (h,\bar{h}) = (1,0)}} \qquad \longleftrightarrow \qquad \underbrace{\text{operators}}_{\substack{\text{e.g. sums over } \alpha_n^\mu,\, \tilde{\alpha}_n^\mu \\ \text{with } [\alpha_m^\mu, \alpha_n^\nu] = m\eta^{\mu\nu}\delta_{m+n,0}}} \tag{4.30}$$

- introduce vacuum state $|0\rangle$, which is by definition annihilated by "half" the modes of the primaries,

$$\partial_z X^\mu \sim \sum_{n\in\mathbb{Z}} \alpha_n^\mu z^{-n-1} \quad \leftrightarrow \quad \alpha_{n\geq 0}^\mu |0\rangle = 0 \tag{4.31}$$

where we will clarify the relation between $|0\rangle$ and the state $|0;p\rangle$ from the construction of string spectra in section 4.5. For primaries of generic $(h,\bar{h})$, we will expand in mode operators $\phi_{m,n}$,

$$\phi_{h,\bar{h}}(z,\bar{z}) \sim \sum_{m,n\in\mathbb{Z}} \phi_{m,n} z^{-m-h}\bar{z}^{-n-\bar{h}} \quad \leftrightarrow \quad \phi_{m,n}|0\rangle = 0 \;\forall\; m > -h \text{ and } n > -\bar{h} \tag{4.32}$$

Spoiler: the conditions for $\phi_{m,n}|0\rangle = 0$ ensure that the asymptotic states $\partial_z X^\mu(z \to 0)|0\rangle$ and $\phi_{h,\bar{h}}(z \to 0)|0\rangle$ do not diverge as $z \to 0$ or $t \to -\infty$

- for energy-momentum modes in $T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}$,

$$L_n|0\rangle = 0 \;\forall\; n = -1, 0, 1, 2, \ldots \tag{4.33}$$

- the hermitian conjugates are

$$(\langle 0|\phi_{m,n})^\dagger = \phi_{-m,-n}|0\rangle \tag{4.34}$$

  such that $(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$; work with normalization of the conjugate vacuum such that $\langle 0|0\rangle = 1$.

Define n-point correlation functions as VEV's

$$\langle \phi_{h_1,\bar{h}_1}(z_1,\bar{z}_1)\phi_{h_2,\bar{h}_2}(z_2,\bar{z}_2)\ldots\phi_{h_n,\bar{h}_n}(z_n,\bar{z}_n)\rangle = \langle 0|\phi_{h_1,\bar{h}_1}(z_1,\bar{z}_1)\ldots\phi_{h_n,\bar{h}_n}(z_n,\bar{z}_n)|0\rangle \tag{4.35}$$

They must be radially ordered: the rightmost field has to be closest to origin $|z_1| > |z_2| > \ldots > |z_n|$, this is analogous to the time ordering $t_1 > t_2 > \ldots > t_n$ on the cylinder, with the latest insertion on the left. The need for radial ordering can be nicely illustrated through the following closed-string example:

$$\begin{aligned}
\langle \partial_z X^\mu(z)\partial_w X^\nu(w)\rangle &= \left(-i\sqrt{\frac{\alpha'}{2}}\right)^2 \sum_{n\in\mathbb{Z}} z^{-n-1}\sum_{p\in\mathbb{Z}} w^{-p-1}\langle 0|\alpha_n^\mu\alpha_p^\nu|0\rangle \\
&= -\frac{\alpha'}{2}\sum_{n\in\mathbb{Z}} z^{-n-1}\sum_{m\in\mathbb{Z}} w^{m-1}\langle 0|\alpha_n^\mu\alpha_{-m}^\nu|0\rangle \\
&= -\frac{\alpha'}{2}\sum_{n=1}^{\infty} z^{-n-1}\sum_{m=1}^{\infty} w^{m-1}\langle 0|n\eta^{\mu\nu}\delta_{n,m}|0\rangle \\
&= -\frac{\alpha'}{2}\eta^{\mu\nu}\frac{1}{zw}\sum_{n=1}^{\infty} n\left(\frac{w}{z}\right)^n \\
&= -\frac{\alpha'\eta^{\mu\nu}}{2zw}\sum_{n=1}^{\infty} u\partial_u u^n|_{u=\frac{w}{z}} = -\frac{\alpha'\eta^{\mu\nu}}{2zw}\frac{w}{z}\partial_u\frac{u}{1-u} \\
&= -\frac{\alpha'\eta^{\mu\nu}}{2z^2}\frac{1}{(1-u)^2} = -\frac{\alpha'\eta^{\mu\nu}}{2}\frac{1}{(z-w)^2}
\end{aligned} \tag{4.36}$$

We have used that $\sum_{n=1}^\infty \left(\frac{w}{z}\right)^n = \frac{\frac{w}{z}}{1-\frac{w}{z}}$ which only converges for $|\frac{w}{z}| < 1$. This convergence criterion imposes radial ordering $|w| < |z|$.

4.4 Comutators versus operator product expansion

Let us now explore the quantum version of the infinitesimal conformal transformation $z \to z + \epsilon\eta(z)$ on primaries $\phi_{h,\bar{h}}$ (ignoring $\bar{\eta}(\bar{z})$ and $\bar{h}$ for ease of notation).

$$\delta_\eta \phi_h(z) = -\epsilon(\eta(z)\partial_z\phi_h(z) + h\phi_h(z)\partial_z\eta_z) \tag{4.37}$$

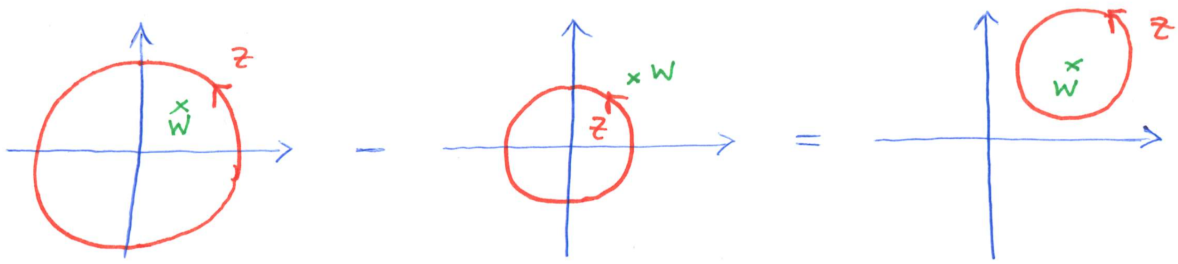
As a general rule in quantum field theory, infinitesimal transformations must stem from commutators with the conserved charge of the corresponding Noether current. In this case, this is $\delta_\eta \phi_h(z) = -\epsilon[Q(\eta), \phi_h(z)]$ with

$$Q_{(\eta)} = \int_{\substack{\text{const} \\ \text{time}}} J_{(\eta)} = \oint_{B_R(0)} \frac{dz}{2\pi i} T(z)\eta(z) \tag{4.38}$$

By the mapping $z = e^{t+i\sigma}$ to the plane, constant-time slices are circles $B_R(0)$ of radius R around the origin. By radial ordering, we have to use different R in the two terms in the commutator, depending on whether $Q_{(\eta)}$ is on the left or right of $\phi_h(w)$

$$\begin{aligned}
[Q_{(\eta)}, \phi_h(w)] &= \oint_{B_R(0)} \frac{dz}{2\pi i} [T(z)\eta(z), \phi_h(w)] \\
&= \left\{ \oint_{|z|>|w|} - \oint_{|z|<|w|} \right\} \frac{dz}{2\pi i} T(z)\eta(z)\phi_h(w) \\
&= \oint_{B_\epsilon(w)} \frac{dz}{2\pi i} T(z)\eta(z)\phi_h(w) \\
&\stackrel{!}{=} \eta(w)\partial_w \phi_h(w) + h\phi_h(w)\partial_w \eta(w)
\end{aligned} \tag{4.39}$$

As visualized in the figure below, the difference of the two circular integrals $\oint_{|z|>|w|} - \oint_{|z|<|w|}$ can be deformed to a single contour integral in z over $B_\epsilon(w)$, a circle of infinitesimal radius ϵ around w



This deformation is due to Cauchy's theorem for meromorphic integrands that allows to deform integration contours as long as no poles are crossed.

In order to explain the transition from the third line of (4.39) to the fourth one, we rely on Cauchy's formula

$$\oint_{B_\epsilon(w)} \frac{dz}{2\pi i} (z-w)^n = \delta_{n,-1} \tag{4.40}$$

in combination with the following short-distance singularity of $T(z)$ and $\phi(w)$:

$$T(z)\phi_h(w) = \frac{h\phi_h(w)}{(z-w)^2} + \frac{\partial_w \phi_h(w)}{z-w} + O((z-w)^0) \tag{4.41}$$

This is the first example of operator product expansions (OPE) that are in general valid within the scope of correlation functions $\langle \ldots \rangle$: As $z \to w$, the two operators $T(z)$ and $\phi(w)$ look like a single operator, either $\phi_h(w)$ or $\partial_w \phi_h(w)$, modular regular terms. More generally, OPEs of operators $A(z), B(w)$ take the schematic form

$$A(z)B(w) \sim \sum_{\substack{\text{operators} \\ O_I \text{ in CFT}}} \overbrace{C_I(z-w)}^{\text{singular as } z\to w} O_I(w) + \underbrace{\ldots}_{\text{regular as } z\to w} \tag{4.42}$$

where a more precise formulation will be given in a later section. As another specific example of an OPE, we can extract

$$\partial_z X^\mu(z)\partial_w X^\nu(w) \sim -\frac{\alpha'}{2}\frac{\eta^{\mu\nu}}{(z-w)^2} + \dots \tag{4.43}$$

from the exact result $-\frac{\alpha'}{2}\frac{\eta^{\mu\nu}}{(z-w)^2}$ for the corresponding two-point function.

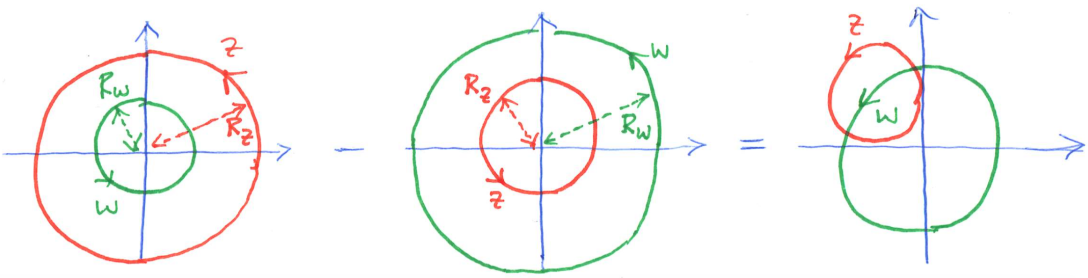
So far, we have translated a known commutation relation $[Q_{(\eta)}, \phi_h(w)]$ into an OPE. One can conversely extract commutators from OPEs, as we shall now illustrate through the example of $\partial_z X^\mu$: Its mode expansion implies the contour-integral representation of the oscillators

$$\alpha_n^\mu = i\sqrt{\frac{2}{\alpha'}}\oint_{B_R(0)} \frac{dz}{2\pi i}\, z^n \partial_z X^\mu(z) \tag{4.44}$$

which can be used to determine their commutator as follows

$$\begin{aligned}
[\alpha_m^\mu, \alpha_n^\nu] &= -\frac{2}{\alpha'}\oint_{B_{R_z}(0)}\frac{dz}{2\pi i}\oint_{B_{R_w}(0)}\frac{dw}{2\pi i}[z^m\partial_z X^\mu(z), w^n\partial_w X^\nu(w)] \\
&= -\frac{2}{\alpha'}\oint_{B_R(0)}\frac{dw}{2\pi i}w^n\oint_{B_\epsilon(w)}\frac{dz}{2\pi i}z^m\partial_z X^\mu(z)\partial_w X^\nu(w) \\
&= -\frac{2}{\alpha'}\oint_{B_R(0)}\frac{dw}{2\pi i}w^n\oint_{B_\epsilon(w)}\frac{dz}{2\pi i}(w^m + m(z-w)w^{m-1}+\dots)\left(-\frac{\alpha'\eta^{\mu\nu}}{2(z-w)^2}\right) \\
&= m\eta^{\mu\nu}\oint_{B_{R(0)}}\frac{dw}{2\pi i}w^{n+m-1} = m\eta^{\mu\nu}\delta_{m+n,0}
\end{aligned} \tag{4.45}$$

In passing to the second line, we have exploited that $R_z > R_w$ in the first "+" term of the commutator and that $R_z < R_w$ in the second term "-". The difference of the two possibilities to arrange the circular contours can be deformed via Cauchy's theorem to an infinitesimal circle for the z-integration around w, whereas w is integrated over a large circle around the origin, see the figure below



Let us use OPEs and the explicit form of the free boson's energy-momentum tensor to verify that $\partial_z X^\mu$ is a primary field of weight $(1,0)$

$$T(z) = -\frac{1}{\alpha'} : \partial_z X^\mu(z)\partial_z X_\mu(z) : \tag{4.46}$$

- we are using the following notation for normal ordering

$$:\alpha_n^\mu \alpha_m^\nu: = \begin{cases} \alpha_n^\mu\alpha_m^\nu & : m \geq n \\ \alpha_m^\nu\alpha_n^\mu & : m < n \end{cases} \tag{4.47}$$

- as in conventional quantum field theory, use Wick's theorem to convert radially ordered (i.e. time-ordered) products to normal ordered ones with vanishing VEVs by summing all (partial) pair contractions

$$\underbrace{\partial_z X^\mu(z)\partial_w X^\nu(w)} = -\frac{\alpha'}{2}\frac{\eta^{\mu\nu}}{(z-w)^2} \tag{4.48}$$

without any pair-contractions of fields within the same normal-ordering colons : . . . :.

- apply this to recover the OPE

$$\begin{aligned} T(z)\partial_w X^\nu(w) &= -\frac{1}{\alpha'} : \partial_z X^\mu(z)\partial_z X_\mu(z) : \partial_w X^\nu(w) \\ &= -\frac{1}{\alpha'}\left(-\frac{\alpha'}{2}\frac{\eta^{\mu\nu}}{(z-w)^2}\partial_z X_\mu(z) + \partial_z X^\mu(z)\left(-\frac{\alpha'}{2}\frac{\delta^\nu_\mu}{(z-w)^2}\right)\right) + \dots \\ &= \frac{1}{(z-w)^2}\Big(\partial_w X^\nu(w) + (z-w)\partial_w^2 X^\nu(w)\Big) + \dots \\ &= \frac{\partial_w X^\nu(w)}{(z-w)^2} + \frac{\partial_w(\partial_w X^\nu(w))}{z-w} + \dots \end{aligned} \tag{4.49}$$

which lines up with the general OPE $T(z)\phi_h(w)$ for a primary ϕ_h at $h=1$

By a similar sequence of Wick contractions, one can analyze the OPEs of the energy-momentum tensor and find additional features beyond conformal primary fields:

$$T(z)T(w) \sim \underbrace{\frac{D/2}{(z-w)^4}}_{\nexists \text{ for primaries}} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \dots \tag{4.50}$$

The last two terms are those of a primary with $h=2$, but the first term is incompatible with the general form of the primaries' OPE. Hence, $T(z)$ is a counterexample to a conformal primary.

- in general CFTs beyond the free boson, one refers to the numerator of $(z-w)^{-4}$ as the "central charge" c in place of the number of spacetime dimensions D
- as will be explored in the exercises, the above OPE is equivalent to the "Virasoro algebra" for

$$L_m = \oint_{B_R(0)} \frac{dz}{2\pi i} T(z) z^{m+1} \tag{4.51}$$

namely

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0} \tag{4.52}$$

- the central-charge contribution to infinitesimal conformal transformations $z \to z + \epsilon\eta(z)$ is

$$\begin{aligned} \delta_\eta T(w) &= -\epsilon \oint_{B_R(w)} \frac{dz}{2\pi i}\eta(z)T(z)T(w) \\ &= \epsilon \oint_{B_R(w)} \frac{dz}{2\pi i}\Big(\eta(w) + (z-w)\partial_w\eta(w) + \frac{1}{2}(z-w)^2\partial_w^2\eta(w) + \frac{1}{6}(z-w)^3\partial_w^3\eta(w)\Big) \\ &\quad \times \left(\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w}\right) + \dots \\ &= -\epsilon\left(\eta(w)\partial_w T(w) + 2T(w)\partial_w\eta(w) + \frac{c}{12}\partial_w^3\eta(w)\right) \end{aligned} \tag{4.53}$$

- the analogue for a finite conformal transformation $z \to \xi(z)$ is

$$T'(\xi) = \left(\frac{\partial\xi}{\partial z}\right)^{-2}\left(T(z) - \frac{c}{12}S(\xi, z)\right) \tag{4.54}$$

with the Schwarzian derivative

$$S(\xi, z) = \frac{\partial^3\xi}{\partial z^3}\left(\frac{\partial\xi}{\partial z}\right)^{-1} - \frac{3}{2}\left(\frac{\partial^2\xi}{\partial z^2}\right)^2\left(\frac{\partial\xi}{\partial z}\right)^{-2} \tag{4.55}$$

- for the special choice $\xi(z) = e^z$ that has been used to map the cylinder to the plane, we have $S(\xi = e^z, z) = -\frac{1}{2}$ and therefore get $T_{\rm cyl}(z) = \xi^2 T_{\rm plane}(\xi) - \frac{c}{24}$. At $c = 1$, this reproduces the normal-ordering offset per spacetime dimension found in the lightcone quantization

$$\frac{1}{2}\sum_{n\neq 0}\alpha^i_{-n}\alpha^i_n = \sum_{n=1}^{\infty}\alpha^i_{-n}\alpha^i_n + \frac{1}{2}\sum_{n=1}^{\infty} n = \sum_{n=1}^{\infty}\alpha^i_{-n}\alpha^i_n - \frac{1}{24} \tag{4.56}$$

- the term in the transformation law $\delta_\eta T(w) \rightarrow -\epsilon\frac{c}{12}\partial_w^3\eta(w)$ that disqualifies $T(w)$ from being a primary does not affect the globally defined generators $l_0, l_{\pm 1}$ or their finite versions $z \rightarrow \frac{az+b}{cz+d}$ with $\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in SL_2(\mathbb{C})$; conformal fields with the transformation law of $T(z)$ (i.e. primary with respect to $SL_2(\mathbb{C})$) are said to be quasi-primary of $(h, \bar{h}) = (2, 0)$.

4.5 State operator correspondence

For each primary field $\phi_{h,\bar{h}}$, we define an asymptotic in-state $|h, \bar{h}\rangle$ via evaluation at infinite past $z \rightarrow 0$

$$|h, \bar{h}\rangle = \lim_{z,\bar{z}\rightarrow 0}\phi_{h,\bar{h}}(z, \bar{z})|0\rangle \tag{4.57}$$

With the earlier mode expansion

$$\phi_{h,\bar{h}}(z, \bar{z}) = \sum_{m,n\in\mathbb{Z}}\phi_{m,n}z^{-m-h}\bar{z}^{-n-\bar{h}} \tag{4.58}$$

$$\phi_{m>-h,n}|0\rangle = |\phi_{m,n>-\bar{h}}|0\rangle = 0$$

only a single mode contributes to this limit

$$|h, \bar{h}\rangle = \lim_{z,\bar{z}\rightarrow 0}\sum_{m,n\in\mathbb{Z}}\overbrace{z^{-m-h}}^{=0 \text{ for } m<-h}\ \bar{z}^{-n-\bar{h}}\ \underbrace{\phi_{m,n}|0\rangle}_{=0 \text{ for } m>-h} = \phi_{-h,-\bar{h}}|0\rangle \tag{4.59}$$

We shall now investigate the interplay of $|h, \bar{h}\rangle$ with the Virasoro generators and ignore the antiholomorphic ($\bar{z}, \bar{h}$-dependent) part for simplicity:

$$\begin{aligned} L_m|h\rangle &= \lim_{w\rightarrow 0}\oint_{B_R(0)}\frac{dz}{2\pi i}z^{m+1}T(z)\phi_h(w)|0\rangle \\ &= \lim_{w\rightarrow 0}\oint_{B_\epsilon(w)}\frac{dz}{2\pi i}\left(w^{m+1} + (m+1)(z-w)w^m + \ldots\right)\left(\frac{h\phi_h(w)}{(z-w)^2} + \frac{\partial_w\phi_h(w)}{z-w} + \ldots\right)|0\rangle \\ &= \lim_{w\rightarrow 0}\left((m+1)hw^m\phi_h(w)|0\rangle + w^{m+1}\partial_w\phi_h(w)|0\rangle\right) \\ &= \lim_{w\rightarrow 0}\sum_{n\in\mathbb{Z}}\left((m+1)hw^{m-n-h} + w^{m-n-h}(-n-h)\right)\phi_n|0\rangle \end{aligned} \tag{4.60}$$

- For $m > 0$, all terms cancel and we have $L_{m>0}|h\rangle = 0$ since w^{m-n-h} vanishes from the limit $w \rightarrow 0$ for $n < m - h$ and $\phi_n|0\rangle$ generally vanishes for $n > -h$
- For $m = 0$, the same reasoning leaves a single contribution from $n = -h$, namely $L_0|h\rangle = h|h\rangle$
- conformal primaries generate a "highest-weight state" of the Virasoro algebra $\text{Vir} = \{L_m,\ m \in \mathbb{Z}\}$

$$L_{m>0}|h\rangle = 0\,, \qquad L_0|h\rangle = h|h\rangle \tag{4.61}$$

  which we found as the quantum version of the classical constraints $T_{\alpha\beta} = 0$ in covariant quantization

- we can build a "highest-weight representation" from $|h\rangle$ via action of $L_{m<0}$

$$\big\{L_{-n_1}L_{-n_2}\dots L_{-n_r}|h\rangle,\ n_j>0\big\} \quad\leftrightarrow\quad \text{descendant state with } L_0 \text{ eigenvalue } h+\sum_{j=1}^{r}n_j \tag{4.62}$$

Also non-primary fields $\varphi(z)$ in a CFT generate states (non-highest-weight states of Vir)

$$|\varphi\rangle = \lim_{z\to 0}\varphi(z)|0\rangle \tag{4.63}$$

- a first example is the quasi primary $T(z)$ with state correspondant

$$\lim_{z\to 0}T(z)|0\rangle = L_{-2}|0\rangle \tag{4.64}$$

  where we have used the defining property $L_{-1}|0\rangle = 0$ of the vacuum

- the state $L_{-2}|0\rangle$ does not satisfy the highest-weight condition since

$$L_2L_{-2}|0\rangle = \underbrace{[L_2,L_{-2}]}_{4L_0+c/2}|0\rangle + \underbrace{L_{-2}L_2|0\rangle}_{0} = \frac{c}{2}|0\rangle \neq 0 \tag{4.65}$$

As a string-theory relevant example of primary fields, we introduce the vertex operators

$$V_p(z,\bar z) = :e^{ip\cdot X(z,\bar z)}: \tag{4.66}$$

labelled by a Lorentz-vector p^μ.

- as will be shown in the exercises, the $V_p(z,\bar z)$ are primaries with $(h,\bar h) = \left(\frac{\alpha'}{4}p^2, \frac{\alpha'}{4}p^2\right)$.
- the vertex operator V_p generates eigenstates of the momentum operator $\sim \alpha_0^\mu$,

$$|0;p\rangle = \lim_{w\to 0}V_p(w,\bar w)|0\rangle \tag{4.67}$$

  This can be checked by evaluating the action of

$$\sqrt{\frac{2}{\alpha'}}\alpha_0^\mu = \frac{2i}{\alpha'}\oint_{B_R(0)}\frac{dz}{2\pi i}\partial_z X^\mu(z) \tag{4.68}$$

  and importing the OPE

$$\partial_z X^\mu(z)V_p(w,\bar w) \sim -\frac{i\alpha'}{2}\frac{p^\mu}{z-w}V_p(w,\bar w)+\dots \tag{4.69}$$

  As will be shown in the exercises,

$$\sqrt{\frac{2}{\alpha'}}\alpha_0^\mu|0;p\rangle = \frac{2i}{\alpha'}\oint_{B_R(0)}\frac{dz}{2\pi i}\partial_z X^\mu(z)\lim_{w\to 0}V_p(w,\bar w)|0\rangle = p^\mu|0;p\rangle \tag{4.70}$$

  which clarifies the relations between the ground states $|0;p\rangle$ w.r.t. the $\alpha^\mu_{n>0}$ oscillators and the "absolute" vacuum $|0\rangle$.

We can invert the above logic "field $\Rightarrow$ state" and define a field φ_v for each state $|v\rangle$,

$$\lim_{z\to 0}\varphi_v(z,\bar z)|0\rangle = |v\rangle \tag{4.71}$$

- The simplest descendants $L_{-1}|h\rangle$ of the highest-weight state $|h\rangle$ (defined by some primary ϕ_h) are generated by $\partial_w \phi_h(w)$

$$L_m|h\rangle\Big|_{m=-1} = \lim_{w\to 0} \big((m+1)hw^m\phi_h(w) + w^{m+1}\partial_w\phi_h(w)\big)\,|0\rangle\Big|_{m=-1} \tag{4.72}$$

$$= \lim_{w\to 0} \partial_w\phi_h(w)|0\rangle \tag{4.73}$$

- The field correspondent of $L_{-n}|h\rangle$ (with $n = 2, 3, \ldots$) can only be given as a contour integral:

$$\lim_{w\to 0} \oint_{B_\epsilon(w)} \frac{dz}{2\pi i}(z-w)^{1-n}T(z)\phi_h(w)|0\rangle = \lim_{w\to 0} L_{-n}\phi_h(w)|0\rangle = L_{-n}|h\rangle \tag{4.74}$$

  In general, one cannot give an explicit evaluation of the z-integral over $B_\epsilon(w)$ since one would need the regular terms $(z-w)^0$, $(z-w)^1$ in the OPE $T(z)\phi_h(w)$. Hence, the best we can do is the above contour-integral representation of the field correspondants to $L_{-2}|h\rangle, L_{-3}|h\rangle, etc.$

4.6 Correlation functions and Ward identities

Correlators $\langle \phi_{h_1,\bar h_1}(z_1,\bar z_1)\ldots\phi_{h_n,\bar h_n}(z_n,\bar z_n)\rangle$ must transform like the primary fields $\phi_{h_i,\bar h_i}(z_i,\bar z_i)$ themselves under conformal transformations $z_i \to \xi(z_i) = \xi_i$ and $\bar z_i \to \bar\xi(\bar z_i) = \bar\xi_i$,

$$\langle \phi'_{h_1,\bar h_1}(\xi_1,\bar\xi_1)\ldots\phi'_{h_n,\bar h_n}(\xi_n,\bar\xi_n)\rangle = \left(\prod_{j=1}^n \Big(\frac{\partial \xi_j}{\partial z_j}\Big)^{-h_j}\Big(\frac{\partial\bar\xi_j}{\partial\bar z_j}\Big)^{-\bar h_j}\right)\langle \phi_{h_1,\bar h_1}(z_1,\bar z_1)\ldots\phi_{h_n,\bar h_n}(z_n,\bar z_n)\rangle \tag{4.75}$$

These transformation properties turn out to fix the two-point functions up to a constant

- ξ of translation $\Rightarrow$ $\langle \phi_{h_1,\bar h_1}(z_1,\bar z_1)\phi_{h_2,\bar h_2}(z_2,\bar z_2)\rangle = g_{1,2}(\underbrace{z_1-z_2}_{z_{12}},\underbrace{\bar z_1-\bar z_2}_{\bar z_{12}})$
- ξ of dilatation $\Rightarrow$ $g_{1,2}(x,y) \sim x^{-h_1-h_2}y^{-\bar h_1-\bar h_2}$ since $g_{1,2}(\lambda z_{12},\lambda\bar z_{12}) = \lambda^{-h_1-h_2}\lambda^{-\bar h_1-\bar h_2}g_{1,2}(z_{12},\bar z_{12})$
- ξ of rotation / special conformal transformation $\Rightarrow$ $g_{1,2}(x,y) = 0$ unless $h_1 = h_2$ and $\bar h_1 = \bar h_2$
- putting everything together (with undetermined normalization d_{12})

$$\langle \phi_{h_1,\bar h_1}(z_1,\bar z_1)\phi_{h_2,\bar h_2}(z_2,\bar z_2)\rangle = \frac{d_{12}\delta_{h_1,h_2}\delta_{\bar h_1,\bar h_2}}{z_{12}^{2h_1}\,\bar z_{12}^{2\bar h_1}} \tag{4.76}$$

- consistent with earlier two-point function of $\partial_z X^\mu$ with $(h,\bar h) = (1,0)$, can read off $d_{12} \to -\frac{\alpha'}{2}\eta^{\mu\nu}$

By similar arguments, three-point functions of conformal primaries are fixed up to constants C_{123} (later examples include Dirac gamma matrices $\gamma^\mu_{\alpha\beta}$ with $SO(1,d-1)$ spinor indices $\alpha\beta$ of certain worldsheet fields of the RNS superstring),

$$\begin{aligned}&\langle \phi_{h_1,\bar h_1}(z_1,\bar z_1)\phi_{h_2,\bar h_2}(z_2,\bar z_2)\phi_{h_3,\bar h_3}(z_3,\bar z_3)\rangle \\ &= \frac{C_{123}}{z_{12}^{h_1+h_2-h_3}\,z_{13}^{h_1+h_3-h_2}\,z_{23}^{h_2+h_3-h_1}\times\bar z_{12}^{\bar h_1+\bar h_2-\bar h_3}\,\bar z_{13}^{\bar h_1+\bar h_3-\bar h_2}\,\bar z_{23}^{\bar h_2+\bar h_3-\bar h_1}}\end{aligned} \tag{4.77}$$

($n \geq 4$)-point functions may depend on conformally invariant cross-ratios $\frac{z_{ij}z_{kl}}{z_{ik}z_{jl}}$, and they can depend on a total of $n-3$ such variables (for both z_j and $\bar z_j$) by relations like

$$\frac{z_{12}z_{34}}{z_{13}z_{24}} + \frac{z_{14}z_{23}}{z_{13}z_{24}} = 1 \tag{4.78}$$

Still, higher point functions are severely constrained by

- asymptotic falloff with $z_j^{-2h_j}$ as $z_j \to \infty$
- associativity of the OPE

In the rest of this section, we will discuss how correlators involving descendant fields are determined from the correlators of the corresponding primaries. We introduce the notation

$$\phi_h^{(-\ell)}(w) = \oint_{B_\epsilon(w)} \frac{dz}{2\pi i}(z-w)^{1-\ell}T(z)\phi_h(w) \tag{4.79}$$

for descendant fields corresponding to the state $L_{-\ell}|h\rangle$. From this representation of $\phi^{(-\ell)}(w)$, one can anticipate that correlators of the schematic form $\langle T(z)\{\text{primaries}\}\rangle$ are needed as an intermediate result. We proceed in two steps (while abbreviating $\phi_{h_j,\bar{h}_j}(w_j,\bar{w}_j) \to \phi_{h_j}(w_j)$ here and below)

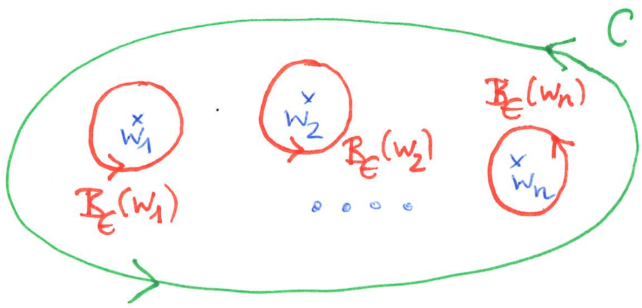


Figure 8: Deformation of the contour $\mathcal{C}$ (green) in (4.80) to a combination of infinitesimal circles (red) around the insertion points $w_1, w_2, \ldots, w_n$ of the primary fields $\phi_{h_j}(w_j)$.

- by the contour deformations in the above figure 8

$$\begin{aligned}
&\Big\langle \oint_{\mathcal{C}} \frac{dz}{2\pi i}\eta(z)T(z)\phi_{h_1}(w_1)\phi_{h_2}(w_2)\ldots\phi_{h_n}(w_n)\Big\rangle \\
&= \sum_{j=1}^{n}\Big\langle \phi_{h_1}(w_1)\ldots\phi_{h_{j-1}}(w_{j-1})\Big(\oint_{B_\epsilon(w_j)}\frac{dz}{2\pi i}\eta(z)T(z)\phi_{h_j}(w_j)\Big)\phi_{h_{j+1}}(w_{j+1})\ldots\phi_{h_n}(w_n)\Big\rangle \\
&= \sum_{j=1}^{n}\Big\langle \phi_{h_1}(w_1)\ldots\phi_{h_{j-1}}(w_{j-1})\Big(\eta(w_j)\frac{\partial\phi_{h_j}(w_j)}{\partial w_j} + h_j\phi_{h_j}(w_j)\frac{\partial\eta(w_j)}{\partial w_j}\Big)\phi_{h_{j+1}}(w_{j+1})\ldots\phi_{h_n}(w_n)\Big\rangle
\end{aligned} \tag{4.80}$$

  where we have used the OPE $T(z)\phi_{h_j}(w_j)$ in order to find the singularities $(z-w_j)^{-2}$ and $(z-w_j)^{-1}$ and to perform the z-integrals over the contour $B_\epsilon(w_j)$

- imposing this identity to hold for any conformal transformation $\eta(z)$ and for any contour $\mathcal{C}$, we obtain the following conformal Ward identity

$$\langle T(z)\phi_{h_1}(w_1)\phi_{h_2}(w_2)\ldots\phi_{h_n}(w_n)\rangle = \sum_{j=1}^{n}\Big(\frac{h_j}{(z-w_j)^2} + \frac{1}{z-w_j}\frac{\partial}{\partial w_j}\Big)\langle\phi_{h_1}(w_1)\phi_{h_2}(w_2)\ldots\phi_{h_n}(w_n)\rangle \tag{4.81}$$

From this, we can infer the desired correlator $\langle \phi_h^{(-\ell)}(w)\{\text{primaries}\}\rangle$

$$\langle \phi_h^{(-\ell)}(w)\phi_{h_1}(w_1)\phi_{h_2}(w_2)\dots\phi_{h_n}(w_n)\rangle = \underbrace{\oint_{B_\epsilon(w)}\frac{dz}{2\pi i}(z-w)^{1-\ell}}_{\text{contour deform to } -\sum_{j=1}^n \oint_{B_\epsilon(w_j)}} \underbrace{\langle T(z)\phi_h(w)\phi_{h_1}(w_1)\dots\phi_{h_n}(w_n)\rangle}_{\text{see above Ward identity}}$$

$$= -\sum_{j=1}^n \oint_{B_\epsilon(w_j)}\frac{dz}{2\pi i} \underbrace{(z-w)^{1-\ell}}_{(w_j-w)^{1-\ell}+(1-\ell)(z-w_j)(w_j-w)^{-\ell}+\dots} \Big(\frac{h_j}{(z-w_j)^2}+\frac{1}{z-w_j}\frac{\partial}{\partial w_j}\Big)\langle\phi_h(w)\phi_{h_1}(w_1)\phi_{h_2}(w_2)\dots\phi_{h_n}(w_n)\rangle$$

$$= \underbrace{\sum_{j=1}^n\Big\{\frac{(\ell-1)h_j}{(w_j-w)^\ell}-\frac{1}{(w_j-w)^{\ell-1}}\frac{\partial}{\partial w_j}\Big\}}_{\mathcal{L}_{-\ell}}\langle\phi_h(w)\phi_{h_1}(w_1)\phi_{h_2}(w_2)\dots\phi_{h_n}(w_n)\rangle \tag{4.82}$$

The result is a differential operator $\mathcal{L}_{-\ell}$ corresponding to $L_{-\ell}$ acting on a correlation function of primaries. We have reduced the correlator involving one descendant field $\phi_h^{(-\ell)}(w)$ to the simpler correlator of primaries only.

- for the simplest descendants at $\ell=1$, the expression for

$$\mathcal{L}_{-1} = -\sum_{j=1}^n\frac{\partial}{\partial w_j} = \frac{\partial}{\partial w} \tag{4.83}$$

  is consistent with the earlier result $\phi_h^{(-1)}(w)=\partial_w\phi_h(w)$, where we used translation invariance $\sum_{j=1}^n\frac{\partial}{\partial w_j}+\frac{\partial}{\partial w}=0$ of the correlator

- the same strategy applies to correlators with several descendant insertions, but the amount of algebra will grow, and the expressions start to depend on the central charge c since one encounters double-insertions of $T(z)$, e.g.

$$\langle\phi_{h_1}^{(-\ell_1)}(w_1)\phi_{h_2}^{(-\ell_2)}(w_2)\{\text{primaries}\}\rangle = \oint_{B_\epsilon(w_1)}\frac{dz_1}{2\pi i}(z_1-w_1)^{1-\ell_1}\oint_{B_\epsilon(w_2)}\frac{dz_2}{2\pi i}(z_2-w_2)^{1-\ell_2}$$
$$\times\langle T(z_1)\phi_{h_1}(w_1)T(z_2)\phi_{h_2}(w_2)\{\text{primaries}\}\rangle \tag{4.84}$$

  There are similar Ward identities to express the correlator $\langle T(z_1)T(z_2)\{\text{primaries}\}\rangle$ in terms of primary correlators without energy-momentum insertions, where $T(z_1)T(z_2)$ are replaced by any term generated from OPEs including $\frac{c}{2z_{12}^4}$.

4.7 Correlation functions versus OPEs

The goal of this section is to refine the general expression for OPEs of primary fields

$$\phi_{h_i}(z)\phi_{h_j}(w) \sim \sum_{\substack{\text{operators}\\ O_k \text{ in CFT}}} \overbrace{C_k(z-w)}^{\text{singular as } z\to w} O_k(w) + \underbrace{\dots}_{\text{regular as } z\to w} \tag{4.85}$$

The key tool will be to impose consistency of two- and three-point functions. More specifically, we start from an ansatz for the primaries on the right-hand side

$$\phi_{h_i}(z)\phi_{h_j}(w) \sim \sum_{\text{primary } k}(z-w)^{N_{ij}{}^k}C_{ij}{}^k\big[\phi_{h_k}(w)+O(z-w)\big]+\text{descendants} \tag{4.86}$$

- Laurent expand the three-point function around $z_1=z_2$,

$$\langle\phi_{h_i}(z_1)\phi_{h_j}(z_2)\phi_{h_k}(z_3)\rangle\Big|_{z_1\to z_2} = \frac{C_{ijk}}{z_{12}^{h_i+h_j-h_k}z_{23}^{h_j+h_k-h_i}}\Big\{\frac{1}{z_{23}^{h_i+h_k-h_j}}-\frac{z_{12}(h_i+h_k-h_j)}{z_{23}^{h_i+h_k-h_j+1}}+O(z_{12}^2)\Big\}$$
$$= \frac{C_{ijk}}{z_{12}^{h_i+h_j-h_k}z_{23}^{2h_k}}\Big\{1-(h_i+h_k-h_j)\frac{z_{12}}{z_{23}}+O(z_{12}^2)\Big\} \tag{4.87}$$

- match with the OPE ansatz and two-point function

$$\begin{aligned}\langle \phi_{h_i}(z_1)\phi_{h_j}(z_2)\phi_{h_k}(z_3)\rangle &\sim \sum_{\text{primary } \ell} C_{ij}{}^{\ell} z_{12}^{N_{ij}{}^{\ell}} \langle \phi_{h_\ell}(z_2)\phi_{h_k}(z_3)\rangle \\ &= \sum_{\substack{\text{primary } \ell \\ \text{at } h_k=h_\ell}} C_{ij}{}^{\ell} d_{\ell,k} \frac{z_{12}^{N_{ij}{}^{\ell}}}{z_{23}^{2h_k}} \end{aligned} \tag{4.88}$$

- this is compatible with the Laurent expansion if

$$N_{ij}{}^{\ell} = h_\ell - h_i - h_j \, , \qquad C_{ijk} = \sum_{\substack{\text{primary } \ell \\ \text{at } h_k=h_\ell}} C_{ij}{}^{\ell} d_{\ell,k} \tag{4.89}$$

  i.e. the three-point couplings C_{ijk} are related to the OPE data $C_{ij}{}^{\ell}$ and the normalization factors $d_{\ell,k}$ of the two-point functions.

As we will see next, the contributions of descendant fields to OPEs are interlocked with those of the primaries via conformal invariance. Refine the above ansatz to also incorporate the descendants of each primary field ϕ_{h_k}

$$\begin{aligned}\phi_{h_i}(z)\phi_{h_j}(w) \sim &\sum_{\text{primaries } k} \frac{C_{ij}{}^{k}}{z_{12}^{h_i+h_j-h_k}} \\ &\times \Big\{ \phi_{h_k}(z_2) + z_{12}\beta^{(-1)}_{ijk}\partial_{z_2}\phi_{h_k}(z_2) + z_{12}^2\beta^{(-1,-1)}_{ijk}\partial^2_{z_2}\phi_{h_k}(z_2) + z_{12}^2\beta^{(-2)}_{ijk}\phi^{(-2)}_{h_k}(z_2) + O(z_{12}^3)\Big\}\end{aligned} \tag{4.90}$$

with some coefficients $\beta^{(-\ell_1,-\ell_2,\ldots)}_{ijk} \leftrightarrow L_{-\ell_1}L_{-\ell_2}|h\rangle$ that can be determined from conformal invariance

- the terms $+O(z_{12}^3)$ refer to all descendant fields corresponding to $L_{-\ell_1}L_{-\ell_2}\ldots L_{-\ell_m}|h_k\rangle$ at the order of $z_{12}^{h_k-h_i-h_j+\ell_1+\ell_2+\ldots+\ell_m}$ of the OPE with $\ell_1+\ell_2+\ldots \geq 3$
- their coefficients $\beta^{(-\ell_1,\ldots,-\ell_m)}_{ijk}$ only depend on h_i, h_j, h_k and c; as will be worked out in the exercises, the subleading order of (4.87) implies

$$\beta^{(-1)}_{ijk} = \frac{h_i+h_k-h_j}{2h_k} \tag{4.91}$$

4.8 Boundary conformal field theory

We shall now embed open strings into the CFT framework. The central new features of open strings as compared to closed strings are their endpoints (at $\sigma = 0, \pi$ in the parametrization of earlier sections). These endpoints introduce worldsheet boundaries at $z \in \mathbb{R}$ after mapping the cylinder to the plane via $z = e^{t+i\sigma}$, see figure 9. The bulk of the worldsheet is then mapped to the upper half plane.

The boundary reduces the conformal symmetry to those maps $z \to \xi(z)$ that preserve $\operatorname{Im} z \geq 0$ and in particular the set of $z \in \mathbb{R}$.

- Such boundary preserving diffeomorphism are generated by $T(z)$ and $\bar{T}(\bar{z})$ restricted to the upper half plane and subject to the gluing condition

$$T(z) = \bar{T}(\bar{z}) \quad \forall \; z, \bar{z} \in \mathbb{R} \tag{4.92}$$

  In case of the CFT of the free boson, the gluing conditions encode the fact that Neumann or Dirichlet boundary conditions determine the $\tilde{\alpha}^\mu_n$ oscillators in terms of the α^ν_m.

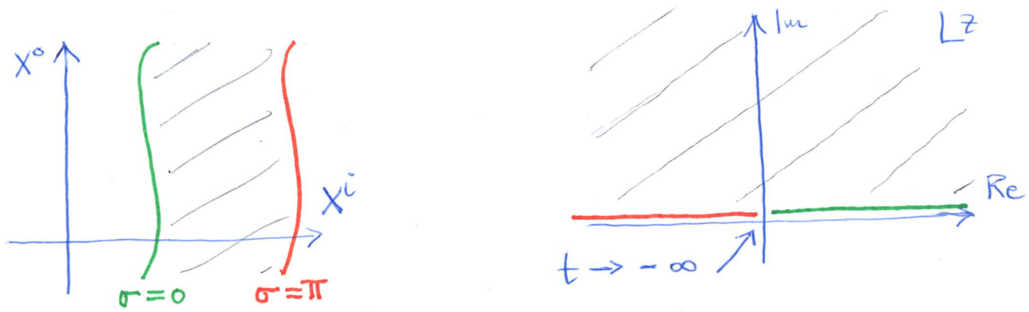


Figure 9: The contribution of the endpoint at $\sigma = 0$ (at $\sigma = \pi$) to the open-string worldsheet is mapped to $\mathbb{R}^+$ (to $\mathbb{R}^-$) on the plane under $z = e^{t+i\sigma}$.

- The gluing conditions can be satisfied by combining $T(z)$ and $\bar{T}(\bar{z})$ to a single field $T^{\rm op}(z)$ defined on $z \in \mathbb{C}$ which is continuous at $z \in \mathbb{R}$

$$T^{\rm op}(z) = \begin{cases} T(z) & : \ \mathrm{Im}\, z \geq 0 \\ \bar{T}(\bar{z}) & : \ \mathrm{Im}\, z < 0 \end{cases} \tag{4.93}$$

- The combined field $T^{\rm op}(z)$ generates a single copy of the Virasoro algebra via

$$L_n^{\rm op} = \oint_{B_\epsilon(0)} \frac{dz}{2\pi i} T^{\rm op}(z) z^{n+1} \tag{4.94}$$

In case of the free boson, the gluing conditions for $T(z) \sim \big(\partial_z X(z)\big)^2$ and $\bar{T}(\bar{z}) \sim \big(\partial_{\bar{z}} X(\bar{z})\big)^2$ can be enforced in two different ways: We attain $T(z) = \bar{T}(\bar{z})$ for $z \in \mathbb{R}$ if

$$\partial_z X^\mu(z \in \mathbb{R}) = \begin{cases} +\partial_{\bar{z}} X^\mu(\bar{z} \in \mathbb{R}) & : \ \text{Neumann}; \ \partial_\sigma \sim \partial_z - \partial_{\bar{z}} = 0 \\ -\partial_{\bar{z}} X^\mu(\bar{z} \in \mathbb{R}) & : \ \text{Dirichlet}; \ \partial_\tau \sim \partial_z + \partial_{\bar{z}} = 0 \end{cases} \tag{4.95}$$

- We can again combine $\partial_z X_\mu$ & $\partial_{\bar{z}} X_\mu$ to a single field $\partial_z X_\mu^{\rm op}$ on $\mathbb{C}$ with one copy of oscillators $\alpha_\mu^{\rm op}$.

- Recall the closed-string two-point function (or Green function)

$$\langle X^\mu(z) X^\nu(w) \rangle = \eta^{\mu\nu} G_{\rm cl}(z,w) \, , \qquad G_{\rm cl}(z,w) = -\frac{\alpha'}{2} \log |z-w|^2 \tag{4.96}$$

(we have derived $\langle \partial_z X^\mu(z) \partial_w X^\nu(w) \rangle$ earlier on via operator methods). Neither $G_{\rm cl}(z,w)$ nor any of its derivatives vanish for $z, w \in \mathbb{R}$.

- Define open-string Green functions $G^N_{\rm op}, G^D_{\rm op}$ adapted to Dirichlet and Neumann boundary conditions by the boundary-value problems

$$\partial_z \partial_{\bar{z}} G^{N,D}_{\rm op}(z,w) = \partial_z \partial_{\bar{z}} G_{\rm cl}(z,w) = -\frac{\alpha'}{2} \pi \delta^2(z-w) \, , \qquad \mathrm{Im}\, z, \mathrm{Im}\, w > 0 \tag{4.97}$$

$$(\partial_z - \partial_{\bar{z}}) G^N_{\rm op}(z,w) = 0 = (\partial_z + \partial_{\bar{z}}) G^D_{\rm op}(z,w) \, , \qquad z \in \mathbb{R} \tag{4.98}$$

- Solve via method of image charges that you know from your electrodynamics course, e.g.

$$\begin{aligned} G^N_{\rm op}(z,w) &= G_{\rm cl}(z,w) + G_{\rm cl}(\bar{z},w) \\ &= -\frac{\alpha'}{2} \big(\log |z-w|^2 + \log |\bar{z}-w|^2 \big) \end{aligned} \tag{4.99}$$

The contribution $G_{\rm cl}(\bar{z}, w)$ from the image charge $\bar{z}$ of z does not pose any obstructions to having $\partial_z \partial_{\bar{z}} G^N_{\rm op}(z, w) = -\frac{\alpha'}{2} \pi \delta^2(z - w)$ in the upper half plane since $\partial_z \partial_{\bar{z}} G_{\rm cl}(\bar{z}, w)$ has delta-function support on the lower half plane.

- With both of z and w on the boundary $z, w \in \mathbb{R}$, the image charge yields a factor of two between the open- and closed-string Green functions $G^N_{\rm op}|_{\mathbb{R}} = 2G_{\rm cl}|_{\mathbb{R}}$. As we will see later on, this reflects the rescaling of the open- and closed-string mass spectra $m^2_{\rm cl} = 4m^2_{\rm op}$ observed in the discussion of lightcone quantization.

5 Path integrals and ghosts

We have seen powerful techniques to constrain and determine correlation functions in two-dimensional conformal field theories which do not rely on the availability of an action functional. The worldsheet theory of bosonic strings in a rather "atypical" CFT in the sense that it admits a description through the Polyakov action in conformal gauge

$$S[X] = \frac{1}{\pi \alpha'} \int d^2 z \ \partial_z X_\mu(z) \partial_{\bar{z}} X^\mu(z) \tag{5.1}$$

for free bosons X^μ with $\mu = 0, 1, \ldots, D{-}1$ and total central charge $c = D$. We will now discuss path-integral methods to both perform CFT computations and to introduce a b, c ghost system that drives covariant quantization.

Warning: Our conventions will differ by factors of 2 from those of the Blumenhagen / Lüst / Theisen textbook, David Tong's lecture notes and the Polchinski textbooks. More specifically, for $z = x + iy$,

$$d^2 z \,\big|_{\rm here} = dx \wedge dy = \frac{1}{2} d^2 z \,\big|_{\rm there} \tag{5.2}$$

$$\delta^2(z) \,\big|_{\rm here} = \delta(x)\delta(y) = 2\delta^2(z) \,\big|_{\rm there} \tag{5.3}$$

leading to the following normalized integral in all cases:

$$\int_{B_\epsilon(0)} d^2 z \, \delta^2(z) = 1 \tag{5.4}$$

5.1 Path-integral methods for correlators of the free boson

Starting from the above Polyakov action $S[X]$ in conformal gauge, we will now introduce path-integral methods to compute correlation functions. These methods will provide an alternative to the operator formalism that could in principle be used to determine correlation functions as VEVS $\langle 0| \ldots |0\rangle$.

- We initiate the definition of path integrals by introducing the partition function Z as a normalization for "zero-point functions"

$$Z = \int \mathcal{D}[X] \, e^{-S[X]} \tag{5.5}$$

- Let $\phi_i(z_i)$ denote any field built from $X^\mu(z, \bar{z})$ and its derivatives (not necessarily a conformal primary). Then, we give a formal definition of n-point functions through the path integral

$$\langle \phi_1(z_1) \phi_2(z_2) \ldots \phi_n(z_n) \rangle = \frac{1}{Z} \int \mathcal{D}[X] \, \phi_1(z_1) \phi_2(z_2) \ldots \phi_n(z_n) \, e^{-S[X]} \tag{5.6}$$

where the normalization by Z^{-1} ensures that $\langle 1 \rangle = 1$.

- We have identified $\partial_z X^\mu$ and $\partial_{\bar z} X^\mu$ to be conformal primaries of weight $(h, \bar h) = (1,0)$ and $(0,1)$. The path integral $\int \mathcal{D}[X]$ should be thought of as integrating over all field configurations for $X^\mu(z,\bar z)$ but not as independently integrating over all the descendant fields.

In order to define the path integral in the numerator of $\langle \phi_1(z_1) \ldots \phi_n(z_n) \rangle$, we impose two basic rules: First, partial derivatives ∂_{z_j} in the insertion points commute with the path integral. Second, functional derivatives integrate to zero

$$0 = \int \mathcal{D}[X] \frac{\delta}{\delta X_\mu(z)} \big(\ldots e^{-S[X]} \big) \tag{5.7}$$

We will now derive some corollaries by simple choices of the insertions $\ldots$ in the functional derivatives:

- Ehrenfest theorem: VEVs obey the classical equations of motion

$$\begin{aligned} 0 = \int \mathcal{D}[X] \frac{\delta}{\delta X_\mu(z)} e^{-S[X]} &= \int \mathcal{D}[X] \underbrace{\Big(- \frac{\delta S[X]}{\delta X_\mu(z)} \Big)}_{\frac{2}{\pi \alpha'} \partial_z \partial_{\bar z} X^\mu(z)} e^{-S[X]} \\ \implies \quad \langle \partial_z \partial_{\bar z} X^\mu(z) \rangle = 0 \end{aligned} \tag{5.8}$$

- Reproducing the Laplace equation of the two-point function known from the operator formalism:

$$\begin{aligned} 0 &= \int \mathcal{D}[X] \frac{\delta}{\delta X_\mu(z)} \big(X^\nu(w) e^{-S[X]} \big) \\ &= \int \mathcal{D}[X] \Big(\eta^{\mu\nu} \delta^2(z-w) + \frac{2}{\pi \alpha'} \partial_z \partial_{\bar z} X^\mu(z) X^\nu(w) \Big) e^{-S[X]} \\ &\implies \quad \partial_z \partial_{\bar z} \langle X^\mu(z) X^\nu(w) \rangle = - \frac{\alpha'}{2} \pi \delta^2(z-w) \end{aligned} \tag{5.9}$$

As a next step, we will use path integrals for an efficient evaluation of n-point functions of vertex operators $V_p(z,\bar z) =: e^{ip\cdot X(z,\bar z)}:$ that we identified as conformal primaries of weight $(h,\bar h) = \big(\frac{\alpha'}{4} p^2, \frac{\alpha'}{4} p^2 \big)$ in earlier sections

- Given that the Polyakov action is quadratic in X^μ, the finite-dimensional prototype for correlators involving $e^{ip\cdot X(z,\bar z)}$ is the Gaussian integral

$$\int_{\mathbb{R}^n} d^n x \, e^{\vec x \cdot A \vec x + i \vec b \cdot \vec x} = e^{\frac14 \vec b \cdot A^{-1} \vec b} \int_{\mathbb{R}^n} d^n x \, e^{\vec x \cdot A \vec x} \tag{5.10}$$

  with some invertible $n \times n$ matrix A (whose eigenvalues need to have a negative definite real part to obtain a convergent integral).

- Since the path-integral analogue of the matrix A is $\partial_z \partial_{\bar z}$, the analogue of A^{-1} is the Green function, i.e. the solution to $\partial_z \partial_{\bar z} g(z-w) = \delta^2(z-w)$ (the closed-string Green function $g(z-w) \sim G_{\rm cl}(z,w)$ in unit normalization). One can think of

$$g(z-w) = \frac{1}{\pi} \log |z-w|^2 \tag{5.11}$$

  as the matrix element of $(\partial_z \partial_{\bar z})^{-1}$.

- We shall now use this to compute the n-point function of $V_{p_j}(z_j,\bar z_j)$:

$$\begin{aligned}\left\langle \prod_{j=1}^n : e^{ip_j\cdot X(z_j)} : \right\rangle &= \frac{1}{Z}\int \mathcal{D}[X]\,\exp\left(i\sum_{j=1}^n p_j\cdot X(z_j) + \int \frac{d^2 z}{\pi\alpha'} X_\mu(z)\partial_z\partial_{\bar z}X^\mu(z)\right)\\ &= \frac{1}{Z}\int \mathcal{D}[X]\,\exp\left(\int d^2 z \left[X_\mu(z)\frac{\partial_z\partial_{\bar z}}{\pi\alpha'}X^\mu(z) + iX_\mu(z)J^\mu(z)\right]\right)\\ &= \delta^D\left(\sum_{j=1}^n p_j\right)\exp\left(\frac{\pi\alpha'}{4}\int d^2 z\int d^2 w\, J^\mu(z)g(z-w)J_\mu(w)\right)\end{aligned} \tag{5.12}$$

In passing to the second line, we have introduced the shorthand for the source term

$$J^\mu(z) = \sum_{j=1}^n p_j^\mu \delta^2(z-z_j) \tag{5.13}$$

and the delta function in the third line is due to the finite-dimensional integral over zero modes x^μ that is part of $\mathcal{D}[X]$ over infinitely many modes. The double-integral over z and w in the exponent of the third line can be thought of as the continuum version of $\vec J\cdot(\partial_z\partial_{\bar z})^{-1}\vec J$, i.e. each vector-index contraction generalizing to one of the d^2z, d^2w integrations. By inserting the delta-function representation of the source term into the double integral, we find

$$\begin{aligned}\left\langle \prod_{j=1}^n : e^{ip_j\cdot X(z_j)} : \right\rangle &= \delta^D\left(\sum_{j=1}^n p_j\right)\exp\left(\frac{\alpha'}{4}\sum_{\substack{i,j=1\\ i\neq j}}^n p_i\cdot p_j \log|z_i-z_j|^2\right)\\ &= \delta^D\left(\sum_{j=1}^n p_j\right)\prod_{1\leq i<j}^n |z_i-z_j|^{\alpha' p_i\cdot p_j}\end{aligned} \tag{5.14}$$

after discarding the terms from $i=j$. This is the famous "Koba-Nielsen factor" on the sphere which will be shown to crucially shape the properties of string tree-level amplitudes.

- The Koba-Nielsen factor at $n=3$ is consistent with the general result on the three-point function of conformal primaries of weights $(h_j,\bar h_j) = \left(\frac{\alpha'}{4}p_j^2, \frac{\alpha'}{4}p_j^2\right)$ with $j=1,2,3$: We can match

$$\left\langle \prod_{j=1}^3 : e^{ip_j\cdot X(z_j)} : \right\rangle = C_{123}\left(z_{12}^{\frac{\alpha'}{4}(p_3^2-p_1^2-p_2^2)}\bar z_{12}^{\frac{\alpha'}{4}(p_3^2-p_1^2-p_2^2)}\times \text{cyclic}(1,2,3)\right) \tag{5.15}$$

by identifying $C_{123}=\delta^D(p_1{+}p_2{+}p_3)$ and noting that

$$p_3^2-p_1^2-p_2^2 = (p_1+p_2)^2-p_1^2-p_2^2 = 2p_1\cdot p_2 \tag{5.16}$$

on the support of three-point momentum conservation.

5.2 Polyakov path integral and b,c ghost system

We shall now revisit the role of the diffeomorphism and Weyl invariance in quantizing the Polyakov action

$$S_P[X,h] = -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}\partial_\alpha X^\mu\partial_\beta X_\mu \tag{5.17}$$

in presence of a general worldsheet metric $h_{\alpha\beta}$. Following the path-integral approach to gauge theories, the path integral for $S_P[X,h]$ should only involve diffeomorphism $\times$ Weyl inequivalent configurations of the fields $X(\sigma)$ and $h(\sigma)$. At the level of the partition function, this means that we need the inverse volume factor in

$$Z_P = \frac{1}{\text{vol}}\int \mathcal{D}[X]\mathcal{D}[h]\,e^{-S_P[X,h]} \tag{5.18}$$

which instructs to formally divide out the infinite overcount by diffeomorphism $\times$ Weyl (though there is no way of assigning a finite volume to the group of diffeomorphisms and Weyl transformations). We proceed on the basis of the following key ideas

- as visualized in figure 10, change field variables from (X, h) to

$$\{\text{physically distinct configurations}\} \times \{\text{gauge orbits}\} \tag{5.19}$$

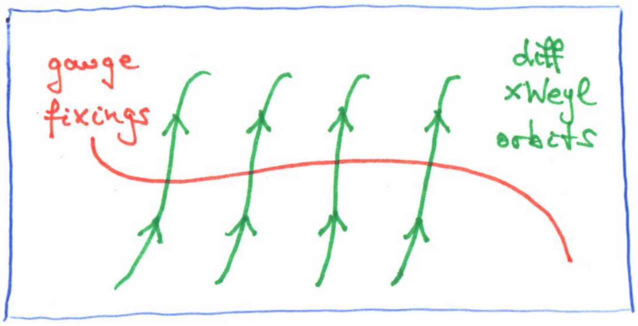


Figure 10: Space of (X, h) field configurations.

- this change of field variables yields a Jacobian, the so-called "Faddeev-Popov determinant", which may be viewed as the infinite-dimensional analogue of the following Jacobian in n dimensions

$$\int d^n x\, \phi(\vec{x}) = \int d^n y \left| \det \frac{\partial x^\mu}{\partial y^\nu} \right| \phi\big(\vec{x}(\vec{y})\big) \tag{5.20}$$

In order to make these ideas more explicit, we introduce the collective notation

$$\zeta = \{\tilde{\sigma}, \phi\} = \{\text{diffeomorphism } \sigma^\alpha \to \tilde{\sigma}^\alpha(\sigma),\ \text{Weyl transformation } h_{\alpha\beta} \to e^{2\phi(\sigma)} h_{\alpha\beta}\} \tag{5.21}$$

for diffeomorphism $\times$ Weyl transformations which act on the worldsheet metric as follows

$$h_{\alpha\beta}(\sigma) \to e^{2\phi(\sigma)} \frac{\partial \sigma^\gamma}{\partial \tilde{\sigma}^\alpha} \frac{\partial \sigma^\delta}{\partial \tilde{\sigma}^\beta} h_{\gamma\delta}(\tilde{\sigma}) = h^\zeta_{\alpha\beta}(\tilde{\sigma}) \tag{5.22}$$

- we can locally attain any metric $\hat{h}_{\alpha\beta}$ of our choice (e.g. $\hat{h}_{\alpha\beta} = \eta_{\alpha\beta}$) by suitably dialing $\zeta = \{\tilde{\sigma}, \phi\}$
- conversely, any $h_{\alpha\beta}$ can be realized as $\hat{h}^\zeta_{\alpha\beta}$, again for a suitable choice of ζ
- implementing this via functional delta distribution

$$\int \mathcal{D}[\zeta]\, \delta[h - \hat{h}^\zeta] = \Delta^{-1}_{FP}[h] \tag{5.23}$$

we get the inverse Faddeev-Popov determinant, the infinite-dimensional version of

$$\int d^n x\ \delta^n\big(\vec{y}(\vec{x})\big) = \sum_{\vec{y}(\vec{x})=\vec{0}} \frac{1}{|\det \frac{\partial y^\mu}{\partial x^\nu}|} \tag{5.24}$$

- the Faddeev-Popov determinant is diffeomorphism $\times$ Weyl invariant:

$$\begin{aligned}\Delta_{FP}^{-1}[h^\zeta] &= \int \mathcal{D}[\zeta']\,\delta[h^\zeta - \hat{h}^{\zeta'}] \\ &= \int \mathcal{D}[\zeta']\,\delta[h - \hat{h}^{\zeta^{-1}\zeta'}] \\ &= \int \mathcal{D}[\zeta'']\,\delta[h - \hat{h}^{\zeta''}] = \Delta_{FP}^{-1}[h]\end{aligned} \tag{5.25}$$

  In passing to the second line, we have used that $h^\zeta = \hat{h}^{\zeta'}$ is equivalent to $h = \hat{h}^{\zeta^{-1}\zeta'}$, and the last step is based on the property $\mathcal{D}[\zeta^{-1}\zeta'] = \mathcal{D}[\zeta']$ of the measure.

- insert $1 = \Delta_{FP}[h] \int \mathcal{D}[\zeta]\delta[h - \hat{h}^\zeta]$ into the path integral,

$$\begin{aligned}Z_P &= \frac{1}{\text{vol}} \int \mathcal{D}[X]\mathcal{D}[h]\mathcal{D}[\zeta]\Delta_{FP}[h]\delta[h - \hat{h}^\zeta]e^{-S_P[X,h]} \\ &= \frac{1}{\text{vol}} \int \mathcal{D}[X]\mathcal{D}[\zeta]\Delta_{FP}[\hat{h}^\zeta]e^{-S_P[X,\hat{h}^\zeta]} \\ &= \underbrace{\frac{1}{\text{vol}} \int \mathcal{D}[\zeta]}_{=1} \int \mathcal{D}[X]\Delta_{FP}[\hat{h}]e^{-S_P[X,\hat{h}]}\end{aligned} \tag{5.26}$$

  In the last step, we have replaced $\hat{h}^\zeta \to \hat{h}$ since both Δ_{FP} and S_P are diffeomorphism $\times$ Weyl invariant.

The next step is to compute $\Delta_{FP}[\hat{h}]$:

- work infinitesimally, with small $\phi(\sigma)$ and $\eta^\alpha(\sigma)$ in $\tilde{\sigma}^\alpha = \sigma^\alpha - \eta^\alpha(\sigma)$

$$\delta\hat{h}_{\alpha\beta} = 2\phi\hat{h}_{\alpha\beta} + \nabla_\alpha\eta_\beta + \nabla_\beta\eta_\alpha \tag{5.27}$$

- with $\mathcal{D}[\zeta] = \mathcal{D}[\phi]\mathcal{D}[\eta]$,

$$\Delta_{FP}^{-1}[\hat{h}] = \int \mathcal{D}[\phi]\mathcal{D}[\eta]\,\delta[2\phi\hat{h}_{\alpha\beta} + \nabla_\alpha\eta_\beta + \nabla_\beta\eta_\alpha] \tag{5.28}$$

- insert the Fourier representation of the functional delta distribution analogous to $\delta^n(\vec{x}) = \int d^n\beta e^{2\pi i\vec{\beta}\cdot\vec{x}}$,

$$\Delta_{FP}^{-1}[\hat{h}] = \int \mathcal{D}[\phi]\mathcal{D}[\eta]\mathcal{D}[\beta]\,\exp\Big(2\pi i \int d^2\sigma\sqrt{-\hat{h}}\,\beta^{\alpha\beta}[2\phi\hat{h}_{\alpha\beta} + \nabla_\alpha\eta_\beta + \nabla_\beta\eta_\alpha]\Big) \tag{5.29}$$

- integrate over ϕ which takes the role of a Lagrange multiplier enforcing $\beta^{\alpha\beta}\hat{h}_{\alpha\beta} = 0$

$$\Delta_{FP}^{-1}[\hat{h}] = \int \mathcal{D}[\eta]\mathcal{D}[\beta]\,\exp\Big(4\pi i \int d^2\sigma\sqrt{-\hat{h}}\,\beta^{\alpha\beta}\nabla_\alpha\eta_\beta\Big) \tag{5.30}$$

However, we need $\Delta_{FP}^{+1}[\hat{h}]$ rather than $\Delta_{FP}^{-1}[\hat{h}]$ for Z_P. This can be adapted by changing the statistics of both β and η:

- think of $\Delta_{FP}^{-1}[\hat{h}]$ as the infinite-dimensional version of the complex integral

$$\int d^{2n}z\, e^{-\vec{z}\cdot B\vec{z}^*} = \frac{\pi^n}{\det B} \tag{5.31}$$

- get $\det B$ into the numerator by passing from z to Grassmann-odd variables

$$\int d^n\theta\, d^n\theta^*\, e^{\vec{\theta}\cdot B\vec{\theta}^*} = \det B \tag{5.32}$$

- convert $(\beta^{\alpha\beta}, \eta_\alpha)$ to Grassmann-odd fields $(b_{\alpha\beta}, c^\alpha)$

$$\Delta_{FP}[\hat{h}] = \int \mathcal{D}[b]\mathcal{D}[c]\, e^{-S_{gh}[b,c,\hat{h}]} \tag{5.33}$$

$$S_{gh}[b,c,\hat{h}] = \frac{1}{2\pi} \int d^2\sigma \sqrt{-\hat{h}}\, b_{\alpha\beta} \nabla^\alpha c^\beta \tag{5.34}$$

The subscript of the action indicates that $(b_{\alpha\beta}, c^\alpha)$ are ghost variables (as usual for Grassmann-odd variables in vector and tensor representations).

- in summary, the partition function for the Polyakov action with gauge-fixed metric $\hat{h}$ reads

$$Z_P = \int \mathcal{D}[X]\mathcal{D}[b]\mathcal{D}[c]\, e^{-S_P[X,\hat{h}]-S_{gh}[b,c,\hat{h}]} \tag{5.35}$$

While above discussion holds for an arbitrary choice of the gauge-fixed metric $\hat{h}_{\alpha\beta}$, we will now study the additional simplifications in conformal gauge where

$$\hat{h}_{\alpha\beta} = e^{2\phi}\eta_{\alpha\beta} \tag{5.36}$$

- use $\sqrt{-\hat{h}} = e^{2\phi}$ and complex coordinates $(z, \bar{z})$: in this case

$$b_{z\bar{z}} = 0\,, \qquad \nabla^z = \hat{h}^{z\bar{z}}\nabla_{\bar{z}} = 2e^{-2\phi}\nabla_{\bar{z}}\,, \qquad \nabla^{\bar{z}} = 2e^{-2\phi}\nabla_z \tag{5.37}$$

and therefore

$$S_{gh}[b,c,\hat{h}_{\alpha\beta} = e^{2\phi}\eta_{\alpha\beta}] = \frac{1}{\pi}\int d^2z\, (b_{zz}\nabla_{\bar{z}}c^z + b_{\bar{z}\bar{z}}\nabla_z c^{\bar{z}}) \tag{5.38}$$

- since the mixed Christoffel symbols $\Gamma^z_{\bar{z}\alpha}$ vanish by exercise A.2, one can replace $\nabla \to \partial$,

$$\nabla_{\bar{z}}c^z = \partial_{\bar{z}}c^z + \Gamma^z_{\bar{z}\alpha}c^\alpha = \partial_{\bar{z}}c^z \tag{5.39}$$

- with the shorthands $(b, \bar{b}) = (b_{zz}, b_{\bar{z}\bar{z}})$ and $(c, \bar{c}) = (c^z, c^{\bar{z}})$,

$$S_{gh}[b,c,\hat{h}_{\alpha\beta} = e^{2\phi}\eta_{\alpha\beta}] = \frac{1}{\pi}\int d^2z\, (b\partial_{\bar{z}}c + \bar{b}\partial_z\bar{c}) \tag{5.40}$$

which yields the following classical equations of motion

$$\partial_{\bar{z}}c = \partial_{\bar{z}}b = 0\,, \qquad \partial_z\bar{c} = \partial_z\bar{b} = 0 \tag{5.41}$$

5.3 The ghost CFT

The dynamics of the (b, c)-ghost system is governed by a CFT. To begin with, the computation of the energy-momentum tensor from $\frac{\delta S_{gh}[b,c,\hat{h}]}{\delta \hat{h}_{\alpha\beta}}$ is subtle due to the constraint $\hat{h}^{\alpha\beta}b_{\alpha\beta} = 0$. We simply state the result for the non-vanishing components $T = T_{zz}$ and $\bar{T} = T_{\bar{z}\bar{z}}$

$$T = 2(\partial_z c)b + c\partial_z b\,, \qquad \bar{T} = 2(\partial_{\bar{z}}\bar{c})\bar{b} + \bar{c}\partial_{\bar{z}}\bar{b} \tag{5.42}$$

We can extract a naive expression for the two-point function from the path integral

$$\begin{aligned} 0 &= \int \mathcal{D}[b]\mathcal{D}[c]\, \frac{\delta}{\delta b(z)}\big(b(w)e^{-S_{gh}[b,c]}\big) \qquad (5.43) \\ &= \int \mathcal{D}[b]\mathcal{D}[c]\, \Big(-\frac{1}{\pi}\partial_{\bar{z}}c(z)b(w) + \delta^2(z-w)\Big)e^{-S_{gh}[b,c]} \end{aligned}$$

$$\Longrightarrow \quad \partial_{\bar{z}}\langle c(z)b(w)\rangle_{\text{naive}} = \pi\delta^2(z-w) \tag{5.44}$$

$$\Longrightarrow \quad \langle c(z)b(w)\rangle_{\text{naive}} = \frac{1}{z-w} \tag{5.45}$$

However, there are subtleties about the zero-mode integrals of b, c that we have not yet taken into account. As we will see later on, the above expression for the two-point function only holds in presence of suitable zero-mode insertions. Still, the naive expression for the two-point function is reliable to extract the OPE

$$c(z)b(w) \sim \frac{1}{z-w} + \ldots\,, \qquad b(z)c(w) \sim \frac{1}{z-w} + \ldots \tag{5.46}$$

which in turn can be used to determine the conformal weights and the central charge:

- normal order the energy-momentum tensor

$$T(z) = 2 : \big(\partial_z c(z)\big) b(z) : + : c(z) \partial_z b(z) : \tag{5.47}$$

- read off $h_b = +2$ and $h_c = -1$ from

$$T(z)c(w) \sim \frac{-c(w)}{(z-w)^2} + \frac{\partial_w c(w)}{z-w} + \ldots \tag{5.48}$$

$$T(z)b(w) \sim \frac{2b(w)}{(z-w)^2} + \frac{\partial_w b(w)}{z-w} + \ldots \tag{5.49}$$

- extract the central charge $c_{gh} = -26$ from

$$T(z)T(w) \sim \frac{-13}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \ldots \tag{5.50}$$

This result for the central charge yields a new perspective on the critical dimension $D = 26$ of the bosonic string which we derived from Lorentz invariance in lightcone quantization earlier on:

- combine the CFT of X^μ in D spacetime dimensions with the (b, c) system

$$\Rightarrow \quad c_{\rm total} = c_X + c_{gh} = D - 26 \tag{5.51}$$

- in order to avoid Weyl anomalies, we have to impose $c_{\rm total}$ to vanish: this can be seen from the following one-point function when the CFT is put into a curved background with Ricci curvature R,

$$\langle T^\alpha{}_\alpha \rangle = -\frac{c_{\rm total}}{12} R \tag{5.52}$$

  which is in conflict with the necessary condition $T^\alpha{}_\alpha = 0$ for conformal symmetry

- the previous argument seems to fix the number of spacetime dimensions to $D = 26$, but there is still the possibility to adjoint additional CFT sectors to the X^μ and (b, c): in presence of some "internal" CFT with central charge $c_{\rm int}$, the condition for the cancellation of Weyl anomalies is $c_{\rm int} + D = 26$, which leaves the possibility of $D < 26$; the internal CFT can but does not need to have a geometrical interpretation as describing compactified (say circular / toroidal) extra dimensions

- in the analogous central-charge bookkeeping for the superstring, the bosons X^μ for each spacetime direction are accompanied by fermions ψ^μ which contribute $\frac{D}{2}$ to the central charge; on the other hand, there is an additional system of (β, γ) ghosts (the "superpartners" of the (b, c) system) with central-charge contribution $+11$; hence, the total central charge of the superstring is

$$c^{\rm super}_{\rm total} = D\big(1 + \tfrac{1}{2}\big) - 26 + 11 = \frac{3D}{2} - 15 \tag{5.53}$$

  which vanishes in the critical dimension $D = 10$; again, central-charge cancellation in lower dimensions is still possible by adjoining internal CFT sectors

5.4 Background charge in the (b, c) system

In this section, we resolve the following two worrisome observations on the (b, c) system

(i) Since the c-ghost has negative conformal weight $h_c = -1$, the general mode expansion for conformal primaries takes the form

$$c(z) = \sum_{n \in \mathbb{Z}} c_n z^{-n+1}, \qquad c_{n \geq 2}|0\rangle = 0 \tag{5.54}$$

Hence, c_1 is the first mode that does not annihilate $|0\rangle$. However, it still lowers the conformal weight since $[L_0, c_1] = -c_1$. Consequently, $|0\rangle$ is not the ground state of the (b, c)-oscillator algebra

$$\{b_m, c_n\} = \delta_{m+n,0} \tag{5.55}$$

(though it is of course a highest-weight state of Vir).

(ii) The naive two-point function $\langle c(z)b(w)\rangle_{\text{naive}} = (z - w)^{-1}$ discussed above is in conflict with the orthogonality property $\langle \phi_{h_1}(z)\phi_{h_2}(w)\rangle \sim \delta_{h_1,h_2}$ of weight-h_i primaries: with $(h_1, h_2) = (-1, 2)$ of (c, b), the two-point function should vanish with δ_{h_1,h_2}.

The resolutions to (i) and (ii) are in fact closely related:

- first note that one can construct two possible L_0 ground states from the ghost oscillators

$$|c\rangle = c_1|0\rangle, \qquad |(\partial c)c\rangle = c_0 c_1|0\rangle \tag{5.56}$$

subject to the highest-weight condition w.r.t. the ghost oscillators (using $c_1^2 = 0$)

$$c_{n\geq 1}|c\rangle = c_{n\geq 1}|(\partial c)c\rangle = 0 \tag{5.57}$$

- impose unusual hermiticity conditions (such that exchanging the role of $|c\rangle \leftrightarrow |(\partial c)c\rangle$ does not have any impact on observables, i.e. inner products)

$$\underbrace{(c_1|0\rangle)^\dagger = \langle 0|c_{-1}c_0}_{\substack{\text{“democratic” between the} \\ \text{2 choices } |c\rangle \text{ and } |(\partial c)c\rangle}} \quad \Leftrightarrow \quad \underbrace{\langle 0|c_{-1}c_0c_1|0\rangle = 1}_{\substack{\text{SL}_2\text{-invariant vacuum carries} \\ \text{3 units of “ghost charge”}}} \tag{5.58}$$

- yields expected three-point functions of $h_c = -1$ primaries

$$\begin{aligned}
\langle c(z_1)c(z_2)c(z_3)\rangle &= \sum_{n_1,n_2,n_3 \in \mathbb{Z}} \langle 0| \prod_{j=1}^{3} c_{n_j} z_j^{-n_j+1} |0\rangle \\
&= \langle 0|c_{-1}c_0c_1|0\rangle \big(z_1^2 z_2 - z_1 z_2^2 + \text{cyc}(z_1, z_2, z_3)\big) \\
&= z_{12}z_{13}z_{23} = z_{12}^{-h_1-h_2+h_3} \times \text{cyc}(z_1, z_2, z_3)
\end{aligned} \tag{5.59}$$

- non-zero $\langle 0|0\rangle$ would give inconsistent two-point function

$$\begin{aligned}
\langle b(z)c(w)\rangle &= \sum_{\substack{m\leq 1 \\ n \geq 2}} \langle 0|b_n z^{-n-2} c_m w^{-m+1}|0\rangle \\
&= \langle 0|0\rangle \frac{w}{z^2} \sum_{n\geq 2} \left(\frac{w}{z}\right)^n = \frac{w^3}{z^3} \frac{\langle 0|0\rangle}{z - w}
\end{aligned} \tag{5.60}$$

$\Rightarrow$ need to set $\langle 0|0\rangle = 0$.

- actual two-point function requires extra c_n-insertion:

$$\langle 0|c_{-1}c_0c_1 b(z)c(w)|0\rangle = \frac{1}{z-w} = \langle 0|c_{-1}c_0c_1 c(z)b(w)|0\rangle \tag{5.61}$$

  where anticommuting the b_m-modes to the left-vacuum (with $\langle 0|b_m = 0 \ \forall \ m \leq 1$) generates the desired three extra terms relative to the previous computation

$$\frac{w^3}{z^3}\frac{1}{z-w} + \frac{w^2}{z^3} + \frac{w}{z^2} + \frac{1}{z} = \frac{1}{z-w} \tag{5.62}$$

- in path-integral language, $\int \mathcal{D}[c]$ contains three fermionic zero-mode integrals

$$\int dc_{-1}\, dc_0\, dc_1\, c_{-1}^x c_0^y c_1^z = \delta_{x,1}\delta_{y,1}\delta_{z,1} \tag{5.63}$$

  which have been neglected in $\langle b(z)c(w)\rangle_{\text{naive}} = (z-w)^{-1}$.

Let us elaborate on the formal origin of the properties $\langle 0||0\rangle = 0$ and $\langle 0|c_{-1}c_0c_1|0\rangle = 1$, starting from the anomalous ghost current

$$j(z) = -:b(z)c(z): \tag{5.64}$$

- $j(z)$ is *not* a conformal primary

$$T(z)j(w) \sim \frac{-3}{(z-w)^3} + \frac{j(w)}{(z-w)^2} + \frac{\partial_w j(w)}{z-w} + \ldots \tag{5.65}$$

- the resulting anomalous transformation $j(z) \to j(\frac{1}{z})$ under exchange $z \to \frac{1}{z}$ of infinite past and future affects the hermiticity properties of the modes:

$$j_n = \oint_{B_\epsilon(0)} \frac{dz}{2\pi i}\, z^n j(z)\,, \qquad j_0^\dagger = -j_0 + 3 \tag{5.66}$$

- recall the state-operator correspondence: given some operator $\mathcal{O}_q$ subject to

$$j(z)\mathcal{O}_q(w) \sim \frac{q\mathcal{O}_q(w)}{z-w} + \ldots\,, \qquad \text{e.g.} \ \begin{cases} \mathcal{O}_{q=1}(w) = c(w) \\ \mathcal{O}_{q=-1}(w) = b(w) \end{cases} \tag{5.67}$$

  can generate a j_0-eigenstate via

$$|q\rangle = \lim_{z\to 0} \mathcal{O}_q(z)|0\rangle \quad \Rightarrow \quad j_0|q\rangle = q|q\rangle\,. \tag{5.68}$$

- from the anomalous $j_0^\dagger$, inner products $\langle q'|q\rangle$ vanish unless $q + q' = 3$,

$$\begin{aligned} q\langle q'|q\rangle &= \langle q'|j_0|q\rangle = \left(j_0^\dagger|q'\rangle\right)^\dagger |q\rangle \\ &= \left((3-j_0)|q'\rangle\right)^\dagger |q\rangle = (3-q')\langle q'|q\rangle \end{aligned} \tag{5.69}$$

- as a result, the hermitian-conjugate pairs are

$$\begin{aligned} q = 0 \text{ state } |0\rangle \ &\leftrightarrow \ q = 3 \text{ state } c_{-1}c_0c_1|0\rangle \\ q = 1 \text{ state } c_1|0\rangle \ &\leftrightarrow \ q = 2 \text{ state } c_0c_1|0\rangle \end{aligned} \tag{5.70}$$

- on a curved worldsheet with Ricci scalar R, the ghost current obeys an anomalous conservation law

$$\partial_{\bar{z}} j(z) = -\frac{3}{4}\sqrt{-h}R \tag{5.71}$$

The right-hand side is a total derivative in two dimensions and induces a topological term upon integration

$$\chi = \frac{1}{4\pi}\int_\Sigma d^2\sigma\,\sqrt{-h}R = 2(1-g) \tag{5.72}$$

where χ is known as the "Euler characteristics" of the surface Σ and g is its "genus" (informally speaking, the number of holes in Σ) which we will shortly see to count the loop order in string perturbation theory

- the number of zero modes N_c, N_b of c, b depends on g

$$N_c - N_b = 3 - 3g \tag{5.73}$$

which is related to the Riemann–Roch theorem, see e.g. section 6.2 in Blumenhagen-Lüst-Theisen textbook. In this way, the zero-mode integrals $\int dc_{-1}dc_0dc_1$ within $\int \mathcal{D}[c]$ arise as the special case at $g = 0$

5.5 Vertex operators

With the tools from the previous sections on conformal field theory and the ghost system, we can now give a covariant description of the bosonic-string spectrum without any recourse to the lightcone.

disclaimer: This section will take a shortcut around the BRST quantization which will be introduced in the tutorial on Nov 13 2020. I.e. the ghosts will play a rather minor role in the following, and a more elaborate treatment can for instance be found in section 5 of the Blumenhagen-Lüst-Theisen textbook.

Recall the physical-state condition of section 2.2

$$\begin{aligned} L_{n>0}|\text{phys}\rangle = \tilde{L}_{n>0}|\text{phys}\rangle = 0 \\ (L_0 - a)|\text{phys}\rangle = (\tilde{L}_0 - \tilde{a})|\text{phys}\rangle = 0 \end{aligned} \tag{5.74}$$

- identifies field correspondant

$$|\text{phys}\rangle = \lim_{z\to 0} V_{\text{phys}}(z)|0\rangle \tag{5.75}$$

to be a conformal primary of weight $(a, \tilde{a})$

- while reference to some insertion point z conflicts with diffeomorphism $\times$ Weyl invariance, can generate an invariant by integration over z:

$$\int d^2z\, V_{\text{phys}}(z) \quad \text{diffeomorphism} \times \text{Weyl invariant} \quad \Leftrightarrow \quad (a, \tilde{a}) = (1, 1) \tag{5.76}$$

the values $(a, \tilde{a}) = (1, 1)$ of the normal-ordering constants compensate the transformation of d^2z with weight $(-1, -1)$

By exercise C.4, plane waves $:e^{ip\cdot X(z)}:$ generate eigenstates of the momentum operator

$$\sqrt{\frac{2}{\alpha'}}\alpha_0^\mu = \frac{2}{\alpha'}\oint_{B_\epsilon(0)}\frac{dz}{2\pi i}\, i\partial_z X^\mu(z) \tag{5.77}$$

For the vertex operator of a physical state, make the ansatz

$$V_{\rm phys}(z) =: W_{\rm phys}(z)e^{ip\cdot X(z)}: \tag{5.78}$$

with $W_{\rm phys}(z)$ built from the field correspondents $\partial_z^n X^\mu(z)$ of α_{-n}^μ and $\partial_{\bar z}^n X^\mu(z)$ of $\tilde\alpha_{-n}^\mu$ with $n\geq 1$

- plane waves contribute conformal weights $(\frac{\alpha' p^2}{4},\frac{\alpha' p^2}{4})$ with $p^2=-m^2$, so $W_{\rm phys}$ must carry

  $$(h_W,\bar h_W) = \Big(1{+}\frac{\alpha' m^2}{4},\, 1{+}\frac{\alpha' m^2}{4}\Big) \tag{5.79}$$

  In this way, the level-matching condition (same oscillator number for the α_{-n}^μ and $\tilde\alpha_{-n}^\mu$) is automatically satisfied.

- without ghosts, we clearly have $h_W,\bar h_W\geq 0$, so m^2 is bounded from below by $-\frac{4}{\alpha'}$ which is the closed-string tachyon identified before

  $$V_{\rm tachyon}(z) = :e^{ip\cdot X(z)}:\,,\qquad p^2=\frac{4}{\alpha'} \tag{5.80}$$

- can add integer units of h_W via $\partial_z^{n\geq 1}X^\mu$

  $$m^2=\frac{4}{\alpha'}(N-1)\quad\longleftrightarrow\quad W_{\rm phys}\sim:\prod_{j=1}^r\partial_z^{n_j}X^{\mu_j}:\quad\text{where}\quad\sum_{j=1}^r n_j=N \tag{5.81}$$

  but one still has to impose $(h,\bar h)=(1,1)$ primary condition on $V_{\rm phys}(w)$ via OPE with $T(z)$.

For massless vertex operators, the above counting leads to the ansatz

$$V_{m^2=0}(z)=\zeta_{\mu\nu}:\partial_z X^\mu\partial_{\bar z}X^\nu e^{ip\cdot X(z)}:\,,\qquad p^2=0 \tag{5.82}$$

for $W_{\rm phys}$ of weight $(1,1)$ with some polarization tensor $\zeta_{\mu\nu}$. As you have worked out in exercise C.4, the normal-ordered combination with the plane wave is a primary of weight $(h,\bar h)=(1,1)$ if

$$p^\mu\zeta_{\mu\nu}=\zeta_{\mu\nu}p^\nu=0 \tag{5.83}$$

since these contractions appear as the residues of $(z{-}w)^{-3}$ in the OPE $T(z)V_{m^2=0}(w)$.

- In general, vertex operators that can be written as total derivatives are spurious states (we will discuss in the next chapter why surface terms in presence of the Koba-Nielsen factor do not contribute to amplitudes of both open and closed strings):

  $$\int d^2z\, V_{\rm sp}(z)=\int d^2z\,\frac{d}{dz}D_{\rm sp}(z)=\int d^2z\,\frac{d}{d\bar z}\bar D_{\rm sp}(z) \tag{5.84}$$

- For massless states, the relevant total-derivative generator is

  $$D_{m^2=0}(z)=-i\xi_\mu:\partial_{\bar z}X^\mu e^{ip\cdot X(z)}:\,,\qquad \xi\cdot p=0 \tag{5.85}$$

  which leads to primary fields

  $$\int d^2z\, V_{{\rm sp},m^2=0}(z)=\int d^2z\,:p_\mu\xi_\nu\partial_z X^\mu\partial_{\bar z}X^\nu e^{ip\cdot X(z)}: \tag{5.86}$$

  Together with the complex conjugates $\bar D_{m^2=0}(z)=-i\bar\xi_\mu:\partial_z X^\mu e^{ip\cdot X(z)}:$, the spurious choices of the polarization tensor for massless states are

  $$\zeta_{\mu\nu}\to p_\mu\xi_\nu\,,\qquad \zeta_{\mu\nu}\to\bar\xi_\mu p_\nu \tag{5.87}$$

- When restricting to the symmetric-traceless component of $\zeta_{\mu\nu}$ that describes spin-two particles, the spurious states with $\bar{\xi}_\mu = \xi_\mu$ correspond to the shift

$$\delta\zeta_{\mu\nu} = p_\mu \xi_\nu + \xi_\mu p_\nu \ , \qquad p \cdot \xi = 0 \tag{5.88}$$

This is nothing but linearized diffeomorphism symmetry – it arises from vertex operators of total-derivative form and can ultimately be traced back to BRST-exact states (see exercise D.3). Massless spin-two states with diffeomorphism symmetry are necessarily gravitons.

- Let us now do the state counting of (not necessarily symmetric-traceless) massless states, i.e. non-spurious $\zeta_{\mu\nu}$

  (i) the primary-field condition $p^\mu \zeta_{\mu\nu} = \zeta_{\mu\nu} p^\nu = 0$ leaves $(D{-}1)^2$ states

  (ii) subtract $2(D{-}1)$ states from ξ_μ and $\bar{\xi}_\mu$ associated with total z- and $\bar{z}$-derivatives

  (iii) add back 1 since $p_\mu p_\nu : \partial_z X^\mu \partial_{\bar{z}} X^\nu e^{ip\cdot X(z)}:$ was subtracted twice in the previous step

  In total, we arrive at $(D{-}2)^2$ massless states, i.e. we have reproduced the conclusion from lightcone quantization in a covariant way.

- The $\mathrm{SL}_2(\mathbb{C})$-weight $(-1,-1)$ of $d^2 z$ can be alternatively realized via insertion of ghosts $c(z)\bar{c}(\bar{z})$. This may seem to contradict the earlier argument that specification of unintegrated punctures is at odds with diffeomorphism invariance on the worldsheet. However, there is a loophole for punctures that the correlation functions do not depend on: As we discussed in the context of correlation functions, n-point functions of conformal primaries on the sphere only depend on $n{-}3$ cross-ratios. Hence, we are free to insert a total of three "unintegrated vertex operators" at genus zero

$$\int d^2 z\, V_{\mathrm{phys}}(z) \quad \to \quad c(z)\bar{c}(\bar{z}) V_{\mathrm{phys}}(z) \tag{5.89}$$

  At the same time, the three insertions of $c(z)\bar{c}(z)$ saturate the background charge and lead to non-vanishing correlators of the ghost system.

For open strings, vertex operators are conformal primaries in the boundary CFT that are confined on $\mathbb{R}$ and only integrated over the worldsheet boundary

$$\int_{\mathbb{C}} d^2 z\, V^{\mathrm{cl}}_{\mathrm{phys}}(z) \quad \leftrightarrow \quad \int_{\mathbb{R}} dz\, V^{\mathrm{op}}_{\mathrm{phys}}(z) \tag{5.90}$$

- constituents of $V^{\mathrm{op}}_{\mathrm{phys}}$ obey gluing conditions

$$\partial_z X_\mu(z \in \mathbb{R}) = \begin{cases} +\partial_{\bar{z}} X_\mu(z \in \mathbb{R}) & : \text{ Neumann, } \partial_\sigma \to 0 \\ -\partial_{\bar{z}} X_\mu(z \in \mathbb{R}) & : \text{ Dirichlet, } \partial_\tau \to 0 \end{cases} \tag{5.91}$$

  which were explained to be realized by a single field $\partial_z X^{\mathrm{op}}_\mu(z)$ on $\mathbb{C}$

- by the mirror image in the open-string Green function

$$G^N_{\mathrm{op}}(z,w) = G_{\mathrm{cl}}(z,w) + G_{\mathrm{cl}}(z,\bar{w}) \tag{5.92}$$

  the OPE of each $\partial_z X^{\mathrm{op}}$ picks up a factor of two,

$$i\partial_z X^{\mathrm{op}}_\mu(z) : e^{ip\cdot X^{\mathrm{op}}(w)} : \sim \frac{\alpha' p_\mu}{2} \Big(\frac{1}{z{-}w} + \frac{1}{z{-}\bar{w}} \Big) : e^{ip\cdot X^{\mathrm{op}}(w)} : + \ldots \tag{5.93}$$

$$T^{\mathrm{op}}(z) : e^{ip\cdot X^{\mathrm{op}}(w)} : \sim \frac{\alpha' p^2}{4} \Big(\frac{1}{z{-}w} + \frac{1}{z{-}\bar{w}} \Big)^2 : e^{ip\cdot X^{\mathrm{op}}(w)} : + \ldots \tag{5.94}$$

This introduces an extra factor of four into the conformal weights for $w \in \mathbb{R}$,

$$T^{\rm op}_\mu(z) : e^{ip\cdot X^{\rm op}(w\in\mathbb{R})} :\sim \frac{\alpha' p^2}{(z-w)^2} : e^{ip\cdot X^{\rm op}(w)} : + \ldots \tag{5.95}$$

- resulting open-string vertex operators

$$V^{\rm op}_{\rm phys}(z) = : W^{\rm op}_{\rm phys}(z) e^{ip\cdot X^{\rm op}(w)} : \tag{5.96}$$

where $W^{\rm op}_{\rm phys}$ is now built from $\partial^n_\tau X^{\rm op}_\mu$ with the derivative $\partial_\tau = \partial_z + \partial_{\bar z}$ in the boundary direction (effectively $\partial_\tau \to 2\partial_z$ if $z \in \mathbb{R}$). For states of mass m, the conformal weight of $W^{\rm op}_{\rm phys}$ has to be $1+\alpha' m^2$ in order to attain a conformal primary of weight one, i.e. a conformally invariant $\int_{\mathbb{R}} V^{\rm op}_{\rm phys}$

$$m^2_{\rm op} = \frac{N-1}{\alpha'} \, , \qquad N \in \mathbb{N}_0 \tag{5.97}$$

starting with the open-string tachyon at $m^2_{\rm op} \to -\frac{1}{\alpha'}$

- construct massless open-string states from an ansatz

$$V^{\rm op}_{m^2=0}(z) = \zeta^\mu : \partial_z X^{\rm op}_\mu e^{ip\cdot X^{\rm op}(z)} : \tag{5.98}$$

which leads to primary fields if $\zeta \cdot p = 0$. As before, polarizations $\zeta^\mu \to p^\mu$ descending from total derivatives (here $\partial_\tau : e^{ip\cdot X^{\rm op}(z)} :$) are spurious. Hence, the number of physical degrees of freedom (after subtracting one component from transversality $\zeta \cdot p = 0$ and one component from the gauge freedom $\delta\zeta^\mu = p^\mu$) is $D{-}1{-}1$, corresponding to an abelian gauge boson.

- just as for closed strings, one can convert effective conformal weight -1 of dz into $c(z)$,

$$\int V^{\rm op}_{\rm phys}(z) \quad \to \quad c(z) V^{\rm op}_{\rm phys}(z) \tag{5.99}$$

which needs to be done for 3 vertex operators per genus-zero correlation function

As mentioned earlier on, one can engineer $D < 26$ spacetime dimensions by adjoining an internal CFT at central charge $c_{\rm int} = 26 - D$.

- the conformal primaries $\phi^{\rm int}_h$ of the internal CFT serve as building blocks for vertex operators, e.g.

$$V^{\rm op}_{\rm phys}(z) \to \left\{ \begin{array}{r} : \phi^{\rm int}_h(z) e^{ip\cdot X^{\rm op}(z)} : \ \ \Rightarrow \ m^2_{\rm op} = \frac{h-1}{\alpha'} \\ \zeta^\mu : \partial_z X^{\rm op}_\mu \phi^{\rm int}_h(z) e^{ip\cdot X^{\rm op}(z)} : \ \ \Rightarrow \ m^2_{\rm op} = \frac{h}{\alpha'} \end{array} \right. \tag{5.100}$$

- more generally, each internal primary $\phi^{\rm int}_h$ modifies the open- and closed-string spectra as follows

$$m^2_{\rm op} = \frac{1}{\alpha'}(h{-}1{+}N) \, , \qquad m^2_{\rm cl} = \frac{4}{\alpha'}(h{-}1{+}N) \, , \qquad N \in \mathbb{N}_0 \tag{5.101}$$

in particular: we get an extra massless states for each $h = 1$ primary $\phi^{\rm int}_{h=1}(z)$, i.e. the spectrum of the internal CFT has profund impact on the low-energy dynamics of the resulting string compactification.

6 String amplitudes

The rough idea in string perturbation theory is similar to the Feynman-diagram organization of point-particle amplitudes in field theory: Path integrate over all histories for a scattering process that are compatible with a given configuration of external states, see figure 11. The goal of this section is to make the form of this "path integral" more precise in the context of scattering amplitudes of open and closed bosonic strings. Cross sections and probabilities are then obtained from the absolute-value square of amplitudes.

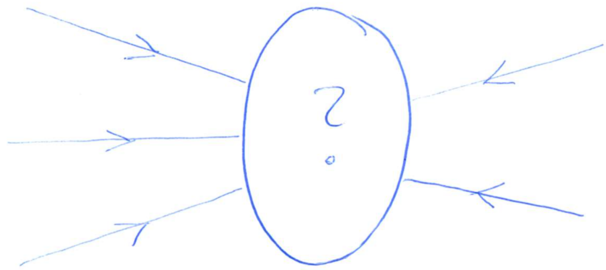


Figure 11: Cartoon of a scattering process described by string amplitudes.

There are several lines of motivation to the study of string amplitudes:

- Comparison of open- and closed-string amplitudes identifies gravitational amplitudes to arise from squares of suitably chosen gauge-theory building blocks. Hence, string amplitudes point towards a double-copy structure of perturbative gravity. This notion of double copy grew into a vibrant research field with various kinds of applications and extensions to non-gravitational theories, see 1909.01358 for a review.
- The α'-expansion of string amplitudes determines the low-energy effective action of different string theories encoding the dynamics of the massless states after integrating out the infinite tower of massive ones. Low-energy effective actions are key quantities in testing or exploiting string dualities and also offer a window into non-perturbative effects.
- The coefficients and building blocks in the α'-expansion of string amplitudes triggered fruitful interplay with number theory: The low-energy expansion of string amplitudes became a rewarding laboratory to explore and apply special functions in a simple context, e.g. multiple zeta values, polylogarithms, elliptic generalizations of both and modular forms. These systems of functions appear in various other areas of high-energy physics, e.g. elliptic polylogarithms are a driving force in the state-of-the-art evaluation of Feynman integrals in quantum field theory and therefore precision calculations for LHC processes.

6.1 Basic ideas in string perturbation theory

The opening line for string amplitudes is to path-integrate over all (diffeomorphism $\times$ Weyl)-inequivalent field configurations $(X, h, \ldots)$ for worldsheets that connect given external states. The analogue of the expansion of field-theory amplitudes in the loop order of Feynman graphs is a topological expansion of string amplitudes in the genus of the worldsheet, see figure 12

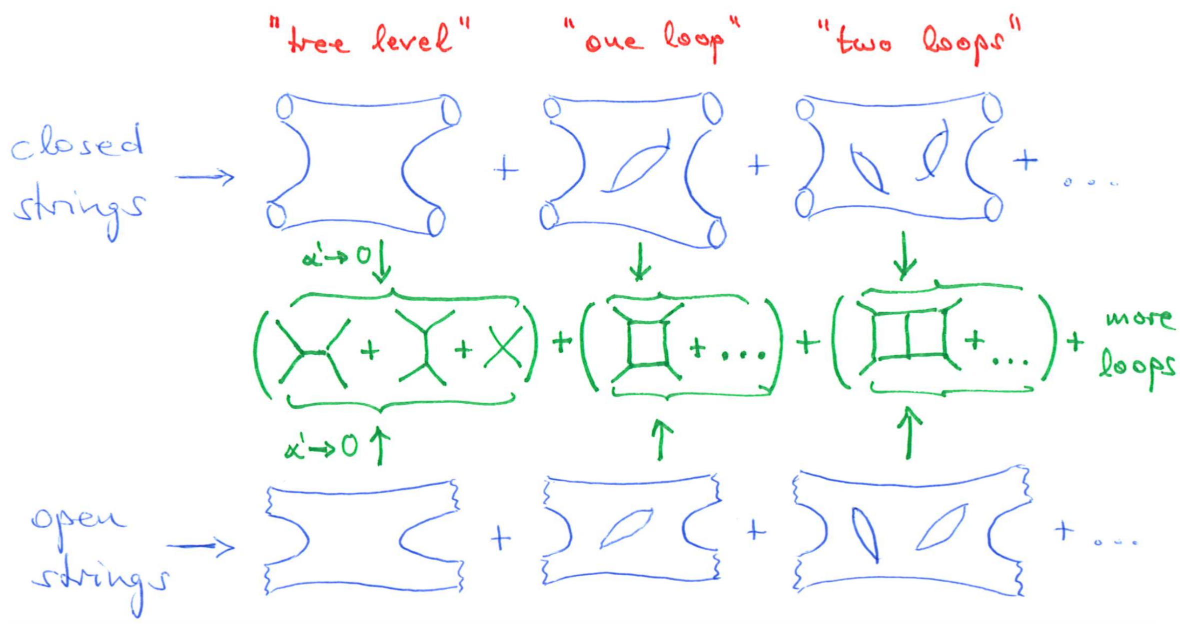


Figure 12: The topological expansion of string amplitudes corresponds to the expansion of field-theory amplitudes according to the number of loops in the Feynman diagrams.

For massless external string states, a particularly important question concerns the point-particle- or low-energy limit $\alpha' \to 0$:

- The gravity states and gauge bosons at $m^2 = 0$ dominate in this limit while the tower of massive states at $m^2 = \frac{4(N-1)}{\alpha'}$ (closed strings) or $m^2 = \frac{N-1}{\alpha'}$ (open strings) with $N \neq 1$ becomes infinitely heavy and is effectively integrated out. This is similar to the effective four-fermion interaction vertices that emerge when integrating out the W or Z bosons that couple to the Standard-Model fermions.

- In the low-energy expansion of string amplitudes (i.e. their Taylor-expansion in α'), we get field-theory amplitudes in non-abelian gauge theories and perturbative gravity at the leading order. Subleading orders in α' (in fact, an infinite series of α'-corrections) signal higher-derivative operators in an effective Lagrangian, e.g. $\alpha'^2 \mathrm{Tr}(F^4)$ as a first correction to the Yang–Mills Lagrangian $\mathrm{Tr}(F^2)$ for gauge bosons and $\alpha'^3 R^4$ as a first correction to the Einstein–Hilbert Lagrangian R for gravitons.

Let us compare the worldsheet diagrams in the topological expansion of string amplitudes with "fattened Feynman diagrams":

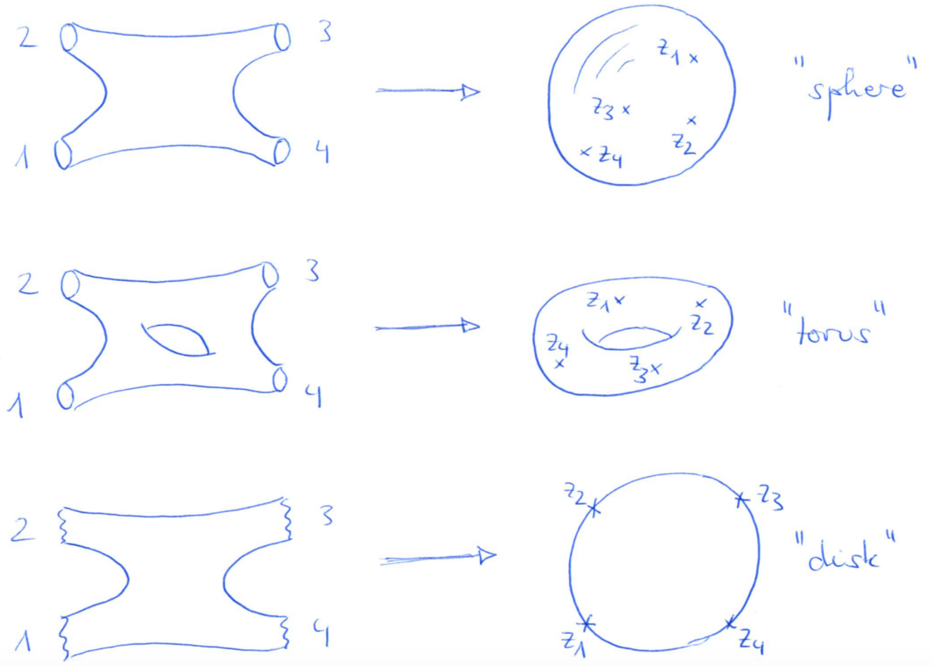
- A single worldsheet can capture all the Feynman diagrams at the relevant loop order, see figure 12; for instance, all the $(2n-5)!!$ cubic-vertex diagrams in an n-point tree-level amplitude arise from various degeneration limits of a single sphere worldsheet (closed strings) or disk worldsheet (open strings).

- The interaction zone of a string worldsheet is smeared out and does not have any analogue of the point-like collision of particles in Feynman diagrams, see figure 13. This aims to provide intuition for the exponentially soft high-energy behaviour of string amplitudes: There are no ultraviolet divergences in loop amplitudes of the closed superstring (in contrast to the divergences expected for supergravity amplitudes from a naive powercounting on the basis of their Feynman rules).

Figure 13: String worldsheets give rise to smeared out interaction zones in comparison with the pointlike intersections of worldlines in Feynman diagrams.

We need to make the program "integrate over diffeomorphism $\times$ Weyl-inequivalent worldsheets" more precise:

- By conformal transformations, the infinitely long tubes or strips for external closed- or open-string states connecting to an interacting worldsheet are mapped to punctures, i.e. point-like defects. These are the insertion points of vertex operators



- Conformal gauge for the worldsheet metric $h_{\alpha\beta} \to e^{2\phi}\eta_{\alpha\beta}$ can only be attained locally (in some open neighborhood of any given point on the surface). Globally, this is only possible at genus zero.

- At $g > 0$ loops, gauge fixing the path integral $\int \mathcal{D}[h]$ over metrics leaves a finite-dimensional integral over "complex-structure moduli" τ. At $g = 1$, for instance, there is a single modulus $\tau \in \mathbb{C}$ with $\operatorname{Im}\tau > 0$ that describes the shape of a torus. At genus $g \geq 2$, there is a total of $3g-3$ complex-structure moduli.

How does one need to weight the contributions from different worldsheet topologies to a string amplitude?

- extend the Polyakov action by a topological term

$$S_{\text{bos}}[X,h] = S_P[X,h] + \lambda\chi \tag{6.1}$$

with Euler characteristics χ and genus g

$$\chi = \frac{1}{4\pi}\int d^2\sigma\sqrt{-h}\,R = 2-2g \tag{6.2}$$

The parameter λ which sets the string coupling turns out to be the VEV of the dilaton (the trace part of the polarization tensor $\zeta_{\mu\nu}$ of massless closed-string states). Hence, the string coupling is not a parameter or "by-hand" input of string theories but a dynamically determined quantity.

- the first approximation for a closed-string amplitude reads

$$\begin{aligned}\sum_{\substack{\text{topologies}\\ \text{and metrics}}} \int \mathcal{D}[X]\, e^{-S_{\text{bos}}[X,h]} &= e^{-2\lambda}\int \mathcal{D}[X]\,\Delta_{FP}[\hat{h}]\,e^{-S_P[X,\hat{h}]} \\ &+ \int \mathcal{D}[X]\int d^2\tau\,\Delta_{FP}[\hat{h}(\tau)]\,e^{-S_P[X,\hat{h}(\tau)]} \\ &+ \sum_{g=2}^{\infty} e^{2\lambda(g-1)}\int \mathcal{D}[X]\int d^{2(3g-3)}\tau\,\Delta_{FP}[\hat{h}(\tau)]\,e^{-S_P[X,\hat{h}(\tau)]}\end{aligned} \tag{6.3}$$

where τ is a collective notation for the complex-structure moduli that couple the ghosts (entering through the Faddeev-Popov determinant) to the matter variable X

6.2 Closed-string tree-level amplitudes

The opening line for the computation of tree-level amplitudes among n closed-string states is pictorially shown in figure 14: Instead of integrating over all insertion points $z_1, z_2, \ldots, z_n$ of the vertex operators, conformal transformations can be used to fix three of them to any desired positions on the Riemann sphere, for instance to the convenient values $(0,1,\infty)$. This can be understood from the properties of the generators $l_n = -z^{n+1}\partial_z$ and $\bar{l}_n = -\bar{z}^{n+1}\partial_{\bar{z}}$ of the conformal algebra with $n \in \mathbb{Z}$.

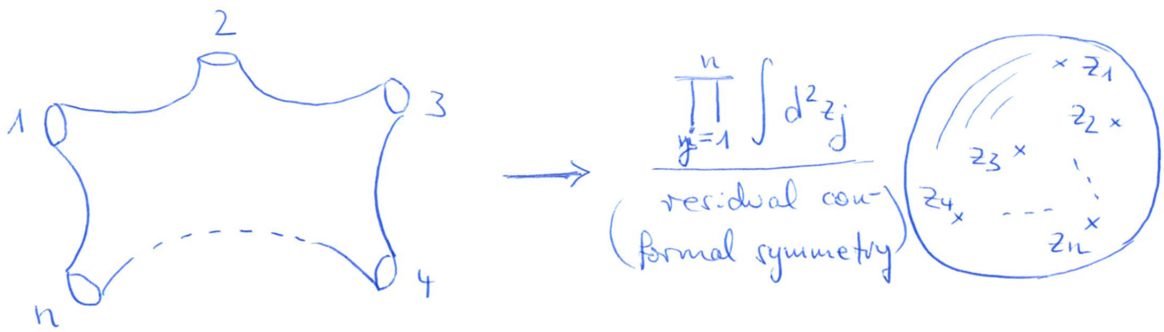


Figure 14: The schematic idea for n-point closed-string tree-level amplitudes. The formal division through residual conformal symmetry amounts to fixing any triplet (z_i, z_j, z_k) of punctures to $(0,1,\infty)$.

- Only the generators $\ell_1, \ell_0, \ell_{-1}$ and $\bar{\ell}_1, \bar{\ell}_0, \bar{\ell}_{-1}$ are globally defined on the Riemann sphere $\mathbb{C}\cup\{\infty\}$, and the finite version of the associated conformal transformations is parametrized by $\mathrm{SL}_2(\mathbb{C})$,

$$z \to \frac{az+b}{cz+d}\,, \qquad \begin{pmatrix} a & b \\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathbb{C}) \tag{6.4}$$

- The three independent entries of the $\mathrm{SL}_2(\mathbb{C})$ matrix can be dialed to map any three z_i, z_j, z_k to arbitrary positions, say $(0, 1, \infty)$. These will be the insertion points of the three unintegrated vertex operators,

$$\int d^2 z_a \, V_{\rm phys}(z_a) \rightarrow c(z_a)\bar{c}(\bar{z}_a) V_{\rm phys}(z_a) \, , \qquad a = i, j, k \tag{6.5}$$

  and we will denote the prescription to perform this replacement for any $i, j, k \in \{1, 2, \ldots, n\}$ by $(\mathrm{vol}\,\mathrm{SL}_2(\mathbb{C}))^{-1}$ in later equations.

- The associated c-ghost correlator $\langle c(z_i)c(z_j)c(z_k)\rangle = z_{ij}z_{ik}z_{jk}$ can be viewed as a Jacobian relating the coefficients $\eta_{-1}, \eta_0, \eta_1$ of the globally defined generators to the punctures z_i, z_j, z_k to be fixed,

$$\delta z_a = -\eta_{-1} - \eta_0 z_a - \eta_1 z_a^2 \quad \Rightarrow \quad \det\left(\frac{\partial(\delta z_i, \delta z_j, \delta z_k)}{\partial(\eta_{-1}, \eta_0, \eta_1)} \right) = z_{ij}z_{ik}z_{jk} \tag{6.6}$$

- In the $\mathrm{SL}_2(\mathbb{C})$-frame with $(z_i, z_j, z_k) = (0, 1, \infty)$, one can identify the remaining variables z_a with $a \neq i, j, k$ as cross ratios,

$$z_a = \frac{z_{ai} z_{jk}}{z_{ji} z_{ak}} \, , \qquad (z_i, z_j, z_k) = (0, 1, \infty) \tag{6.7}$$

With these preliminary considerations, the tree-level amplitude of closed-bosonic-string states is given by

$$\mathcal{M}^{\rm tree}_{\rm bos}(\{\Phi_i\}; \alpha') \sim \int \frac{d^2 z_1 \, d^2 z_2 \ldots d^2 z_n}{\mathrm{vol}\,\mathrm{SL}_2(\mathbb{C})} \Big\langle \prod_{a=1}^n V_{\Phi_a}(z_a) \Big\rangle \tag{6.8}$$

where the notation $\Phi_1, \Phi_2, \ldots, \Phi_n$ for the external states collectively refers to their momenta p_j and their polarization tensors such as $\zeta_{\mu\nu}$ in case of the graviton.

- The proportionality sign comprises $e^{-2\lambda}$ and normalization factors of the vertex operators; a procedure to determine such normalizations by matching with field-theory amplitudes and unitarity will be introduced later on.

- The bracket $\langle \ldots \rangle$ enclosing the vertex operators refers to the path integral $\int \mathcal{D}[X]$ relevant to $e^{ip\cdot X}$ and $\partial_z^n X^\mu, \partial_{\bar{z}}^m X^\nu$ entering V_{Φ_a} as well as $\int \mathcal{D}[c]$ relevant to the $c\bar{c}$ introduced by the $\mathrm{SL}_2(\mathbb{C})$-fixing.

- The closed-string amplitude is independent on the choice of the triplet i, j, k which can be understood from the c-ghost correlators taking the role of a Jacobian, and the conformal weights $(h, \bar{h}) = (1, 1)$ of each V_{Φ_a}.

- The correlator of n vertex operators V_{Φ_a} and three insertions of $c\bar{c}$ is finite at $z_k \to \infty$: The growth of the c- and $\bar{c}$-correlators as $|z_k|^4$ is compensated by the falloff $\langle \ldots V_{\Phi_k}(z_k \to \infty)\rangle \sim |z_k|^{-4}$ following from V_{Φ_k} being a conformal primary with $(h, \bar{h}) = (1, 1)$.

- Since $V_{\Phi_a}(z_a)$ is built from plane waves $e^{ip\cdot X}$ accompanied by polynomials in $\partial_z^n X^\mu, \partial_{\bar{z}}^m X^\nu$, all the correlation functions involve the Koba–Nielsen factor $\prod_{1\leq i<j}^n |z_{ij}|^{\alpha' p_i \cdot p_j}$ and a momentum-conserving delta function enforcing $\sum_{j=1}^n p_j = 0$. These two universal building blocks are multiplied by rational expression in the $z_1, z_2, \ldots, z_n$ and $\bar{z}_1, \bar{z}_2, \ldots, \bar{z}_n$ which is determined by the $\partial_z^n X^\mu, \partial_{\bar{z}}^m X^\nu$ in the vertex operators.

6.2.1 Three tachyons

As a first example of the above prescription for closed-string tree-level amplitudes, let us consider three-tachyon scattering. By momentum conservation $p_1 + p_2 + p_3 = 0$ and the mass-shell condition $p_j^2 = -m_j^2 = \frac{4}{\alpha'}$, the exponents in the Koba-Nielsen factor are

$$\alpha' p_1 \cdot p_2 = \frac{\alpha'}{2}\big[(p_1{+}p_2)^2 - p_1^2 - p_2^2\big] = \frac{\alpha'}{2}(m_1^2 + m_2^2 - m_3^2) = -2 \tag{6.9}$$

Hence, the correlator of the vertex operators $V_{T_a} \sim : e^{ip_a\cdot X(z_a)} :$ in the integrand is

$$\Big\langle \prod_{a=1}^{3} : e^{ip_a\cdot X(z_a)} : \Big\rangle = \prod_{1\leq i<j}^{3} |z_{ij}|^{\alpha' p_i\cdot p_j} = \frac{1}{|z_{12}z_{13}z_{23}|^2} \tag{6.10}$$

Together with the c-ghost correlators $\langle c\bar{c}(z_1)c\bar{c}(z_2)c\bar{c}(z_3)\rangle = |z_{12}z_{13}z_{23}|^2$ introduced by the $\mathrm{SL}_2(\mathbb{C})$ fixing, this cancels any dependence on the punctures,

$$\begin{aligned}\mathcal{M}^{\rm tree}_{\rm bos}(\{T_1,T_2,T_3\};\alpha') &\sim \int \frac{d^2z_1\, d^2z_2 d^2z_3}{\mathrm{vol}\,\mathrm{SL}_2(\mathbb{C})} \Big\langle \prod_{a=1}^{3} V_{T_a}(z_a)\Big\rangle \\ &\sim \Big\langle \prod_{a=1}^{3} c\bar{c}(z_a) : e^{ip_a\cdot X(z_a)} : \Big\rangle\Big|_{(z_1,z_2,z_3)\to(0,1,\infty)} \\ &= 1\end{aligned} \tag{6.11}$$

in lines with the absence of integration variables at three points (the only three punctures in the problems can fixed to any three positions in $\mathbb{C}$). The cancellation of the z_j-dependence is built in since $c\bar{c}V_{T_a}$ are conformal primaries of weight $(h,\bar{h}) = (0,0)$.

6.2.2 Two tachyons, one graviton

With two tachyons $m_2^2 = m_3^2 = -\frac{4}{\alpha'}$ and one external graviton with vertex operator

$$V_{\zeta_1}(z_1) \sim \zeta_1^{\mu\nu} : i\partial_z X_\mu i\partial_{\bar{z}} X_\nu(z_1) e^{ip_1\cdot X(z_1)} : , \qquad p_1^2 = 0 \tag{6.12}$$

the $\mathrm{SL}_2(\mathbb{C})$-fixed amplitude representation is

$$\begin{aligned}\mathcal{M}^{\rm tree}_{\rm bos}(\{\zeta_1,T_2,T_3\};\alpha') &\sim \zeta_1^{\mu\nu}\Big\langle c\bar{c}(z_1){:}i\partial_z X_\mu i\partial_{\bar{z}} X_\nu(z_1) e^{ip_1\cdot X(z_1)}{:} \prod_{a=2}^{3} c\bar{c}(z_a){:}e^{ip_a\cdot X(z_a)}{:}\Big\rangle\Big|_{(z_1,z_2,z_3)\to(0,1,\infty)} \\ &= |z_{12}z_{13}z_{23}|^2 \zeta_1^{\mu\nu}\Big\langle {:}i\partial_z X_\mu i\partial_{\bar{z}} X_\nu(z_1) e^{ip_1\cdot X(z_1)}{:} \prod_{a=2}^{3} {:}e^{ip_a\cdot X(z_a)}{:}\Big\rangle\Big|_{(z_1,z_2,z_3)\to(0,1,\infty)}\end{aligned} \tag{6.13}$$

By adapting $\alpha' p_1\cdot p_2 = \frac{\alpha'}{2}(m_1^2{+}m_2^2{-}m_3^2)$ and permutations to the mass configuration of two tachyons and one graviton, the Koba-Nielsen factor evaluates to $\prod_{1\leq i<j}^{3} |z_{ij}|^{\alpha' p_i\cdot p_j} = |z_{23}|^{-4}$. The extra contributions from $i\partial_z X_\mu i\partial_{\bar{z}} X_\nu(z_1)$ have been determined in exercise D.4: Ignoring the chiral half with $i\partial_{\bar{z}} X_\nu(z_1)$ for the moment, the answer can be assembled from the double copy of

$$\begin{aligned}\zeta_1^{\mu\nu}\Big\langle {:}i\partial_z X_\mu e^{ip_1\cdot X(z_1)}{:} \prod_{a=2}^{3} {:}e^{ip_a\cdot X(z_a)}{:}\Big\rangle &= \frac{\alpha'}{2}\Big(\frac{p^2_\mu}{z_{12}} + \frac{p^3_\mu}{z_{13}}\Big)\zeta_1^{\mu\nu}|z_{23}|^{-4} \\ &= \frac{\alpha'}{2}\frac{z_{23}}{z_{12}z_{13}} p^2_\mu \zeta_1^{\mu\nu}|z_{23}|^{-4}\end{aligned} \tag{6.14}$$

after stripping off the Koba-Nielsen contribution $|z_{23}|^{-4}$. We have used that $p_3 = -p_1 - p_2$ and $p^1_\mu \zeta_1^{\mu\nu} = 0$ in passing to the second line, and the identical calculation for $i\partial_{\bar z} X_\nu(z_1)$ with $\mu \leftrightarrow \nu$ and $z_j \leftrightarrow \bar z_j$ interchanged introduces an extra factor of $\frac{\alpha'}{2} \frac{\bar z_{23}}{\bar z_{12} \bar z_{13}} p^2_\nu$ into the closed-string amplitude

$$
\begin{aligned}
\mathcal{M}^{\rm tree}_{\rm bos}(\{\zeta_1, T_2, T_3\}; \alpha') &\sim \Big(\frac{\alpha'}{2}\Big)^2 |z_{12}z_{13}z_{23}|^2 \underbrace{\frac{z_{23}}{z_{12}z_{13}} p^2_\mu}_{\text{from } i\partial_z X_\mu(z_1)} \zeta_1^{\mu\nu} \underbrace{\frac{\bar z_{23}}{\bar z_{12}\bar z_{13}} p^2_\nu}_{\text{from } i\partial_{\bar z} X_\nu(z_1)} |z_{23}|^{-4} \Big|_{(z_1,z_2,z_3)\to(0,1,\infty)} \\
&= \Big(\frac{\alpha'}{2}\Big)^2 p^2_\mu \zeta_1^{\mu\nu} p^2_\nu
\end{aligned} \tag{6.15}
$$

We have made essential use of the primary-field condition $p^1_\mu \zeta_1^{\mu\nu} = 0$ in demonstrating that the dependence on the z_j drops out as expected.

6.2.3 Four tachyons

As a first look at four-point amplitudes, we write the Koba–Nielsen factor in terms of the dimensionless Mandelstam invariants

$$
s = \alpha'(p_1 + p_2)^2\,, \qquad t = \alpha'(p_2 + p_3)^2\,, \qquad u = \alpha'(p_1 + p_3)^2 \tag{6.16}
$$

where the fourth momentum can be reinstated via $\sum_{j=1}^4 p_j = 0$, for instance $s = \alpha'(p_3 + p_4)^2$. In case of four external tachyons, the exponents are determined by permutations of $\alpha' p_1 \cdot p_2 = \frac{s}{2} - 4$ such that

$$
\prod_{1\le i<j}^4 |z_{ij}|^{\alpha' p_i \cdot p_j} = \frac{|z_{12}z_{34}|^{s/2} |z_{13}z_{24}|^{u/2} |z_{14}z_{23}|^{t/2}}{|z_{12}z_{13}z_{14}z_{23}z_{24}z_{34}|^4} \tag{6.17}
$$

In order to evaluate the amplitude in an $\mathrm{SL}_2(\mathbb{C})$ frame with $(z_1, z_3, z_4) \to (0, 1, \infty)$, it is instructive to see why the $z_4 \to \infty$ limit is finite: The contributions from $z_{*4} \in \{z_{14}, z_{24}, z_{34}\}$ to the Koba–Nielsen factor (6.17) are equally dominant, so we do not need to distinguish them in the limit $z_4 \to \infty$,

$$
\frac{|z_{34}|^{s/2}|z_{24}|^{u/2}|z_{14}|^{t/2}}{|z_{14}z_{24}z_{34}|^4} \to \frac{|z_{*4}|^{\frac{1}{2}(s+t+u)}}{|z_{*4}|^{12}} = \frac{1}{|z_{*4}|^4} \tag{6.18}
$$

In the second step, we have used that $s + t + u = 16$ in case of four tachyons as a consequence of $p_j^2 = \frac{4}{\alpha'}$ and $\sum_{j=1}^4 p_j = 0$. This leading contribution of $z_4 \to \infty$ is cancelled by that of the c-ghost correlator $\langle c\bar c(z_1) c\bar c(z_3) c\bar c(z_4)\rangle \to |z_{13}|^2 |z_{*4}|^4$, with $z_{13} = -1$ in the SL_2-frame of interest. Hence, the limit $z_4 \to \infty$ is finite, and we can effectively ignore all the contributions of z_{14}, z_{24}, z_{34} to (6.17) since they cancel against the c-ghost correlator in our SL_2-frame. As a result, the four-tachyon amplitude is given by

$$
\begin{aligned}
\mathcal{M}^{\rm tree}_{\rm bos}(\{T_1, T_2, T_3, T_4\}; \alpha') &\sim \int_{\mathbb{C}} d^2 z_2 \, \langle V_{T_2}(z_2) c\bar c V_{T_1}(z_1) c\bar c V_{T_3}(z_3) c\bar c V_{T_4}(z_4)\rangle \Big|_{(z_1,z_3,z_4)\to(0,1,\infty)} \\
&\sim \int_{\mathbb{C}} d^2 z_2 \, |z_2|^{s/2-4} |1 - z_2|^{t/2-4}
\end{aligned} \tag{6.19}
$$

This integral may not be in your standard toolkit, but we will see in the next lectures that it can be evaluated in terms of Gamma functions subject to $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$. With the analytic continuation determined by $x\Gamma(x) = \Gamma(x + 1)$ and the convergent integral representation for $\mathrm{Re}(x) > 0$,

$$
\Gamma(x) = \int_0^\infty dt \, t^{x-1} e^{-t} \tag{6.20}
$$

we can evaluate the four-tachyon amplitude for various complex s, t, u subject to $s + t + u = 16$:

$$
\int_{\mathbb{C}} d^2 z_2 \, |z_2|^{s/2-4} |1 - z_2|^{t/2-4} = \frac{\pi \Gamma(\frac{s}{4} - 1)\Gamma(\frac{t}{4} - 1)\Gamma(\frac{u}{4} - 1)}{\Gamma(2 - \frac{s}{4})\Gamma(2 - \frac{t}{4})\Gamma(2 - \frac{u}{4})} \tag{6.21}
$$

This is the famous *Virasoro–Shapiro amplitude* which will be derived in detail during the first lectures of string theory II. In fact, it will be simpler to first discuss the analogous open-string amplitudes since

- the correlator contributions from $\langle \ldots \partial_z^N X \ldots \partial_{\bar{z}}^N X e^{ip\cdot X}\rangle$ do not appear "squared"
- the four-point amplitude involves a line integral "dz_2" rather than a surface integral "$d^2 z_2$" which is easier to evaluate and exhibits fewer pole channels in s, t and u

6.3 Open-string tree-level amplitudes

For open strings, the perturbative expansion is based on worldsheet topologies with boundaries, and their tree-level amplitudes are computed from punctured disks. As visualized in figure 15, the placement of open-string vertex operators on the disk boundary leads to a notion of cyclic ordering, where for instance leg 2 is neighboring to legs 1 and 3 but not to $4, 5, \ldots, n$.

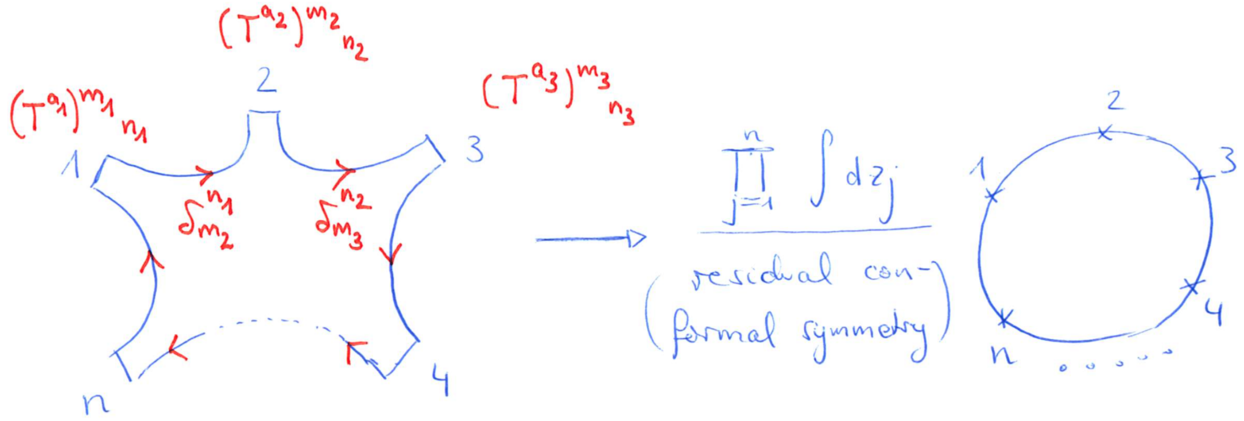


Figure 15: The schematic idea for n-point open-string tree-level amplitudes. The formal division through residual conformal symmetry amounts to fixing any triplet (z_i, z_j, z_k) of punctures to $(0, 1, \infty)$.

The choice of cyclic ordering is also crucial for the dependence of open-string amplitudes on the Chan-Paton factors T^a, i.e. the non-abelian gauge degrees of freedom: As one can see from the left panel of figure 15, the endpoints of two neighboring open-string states are connected by segments of the disk boundary which amounts to contracting the (anti-)fundamental indices m_j, n_j of the respective gauge-group generators $(T^{a_j})^{m_j}{}_{n_j}$. The contractions from the boundary segments in the depicted cyclic ordering give rise to the trace $\text{Tr}(T^{a_1}T^{a_2}\ldots T^{a_n})$ of gauge group generators, which we need to correlate with the integration domain for the $z_j \in \mathbb{R}$ in the right panel of figure 15.

For n external open-string states, there is a total of $(n-1)!$ cyclically inequivalent orderings that can be labelled by permutations $\rho \in S_{n-1}$ of legs $1, 2, \ldots, n-1$. All of them will contribute to the open-string tree amplitude according to our program of path-integrating over all conformally inequivalent histories. With the traces of T^{a_j} reflecting the cyclic orderings, this leads to the *color decomposition*

$$\mathcal{M}^{\text{tree}}_{\substack{\text{open}\\\text{bos}}}(\{\varphi_i, a_i\}; \alpha') = \sum_{\rho \in S_{n-1}} \text{Tr}(T^{a_{\rho(1)}}T^{a_{\rho(2)}}\ldots T^{a_{\rho(n-1)}}T^{a_n})\mathcal{A}^{\text{tree}}_{\text{bos}}(\varphi_{\rho(1)}, \varphi_{\rho(2)}, \ldots, \varphi_{\rho(n-1)}, \varphi_n; \alpha') \tag{6.22}$$

where we have without loss of generality kept leg n in the last position of the cyclically invariant traces. The full open-string tree amplitude $\mathcal{M}^{\text{tree}}_{\substack{\text{open}\\\text{bos}}}$ depends on both the adjoint indices a_j and the kinematic degrees of freedom φ_j, i.e. polarizations and momenta (which we have represented through the collective notation Φ_j for closed strings). The color-ordered amplitudes $\mathcal{A}^{\rm tree}_{\rm bos}$ in turn are only functions of φ_j that depend on their cyclic ordering in the brackets.

Each cyclic ordering gives is associated with its own integration domain for the z_j – with the parametrization of the disk boundary via $\mathbb{R}$, the ordering in figure 15 translates into inequalities

$$\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_2,\ldots,\varphi_n) \ \leftrightarrow\ -\infty < z_1 < z_2 < \ldots < z_n < \infty \tag{6.23}$$

Finally, the restriction of residual conformal symmetry to $z_j \in \mathbb{R}$ again allows to fix any three punctures to $(0,1,\infty)$. This can always be done by the globally defined conformal transformations $z \rightarrow \frac{az+b}{cz+d}$ with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in {\rm SL}_2(\mathbb{R})$ that preserve the boundary $z \in \mathbb{R}$. By picking an ${\rm SL}_2$-frame compatible with the inequalities of a given cyclic ordering, the prescription for color-ordered disk amplitudes is

$$\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_2,\ldots,\varphi_n;\alpha') \sim \int\limits_{-\infty<z_1<z_2<\ldots<z_n<\infty} \frac{dz_1\, dz_2 \ldots dz_n}{{\rm vol}\ {\rm SL}_2(\mathbb{R})} \left\langle \prod_{a=1}^n V^{\rm op}_{\varphi_a}(z_a) \right\rangle \tag{6.24}$$

where the inverse vol ${\rm SL}_2(\mathbb{R})$ instructs to insert a c-ghost at the three punctures that are fixed to $(0,1,\infty)$. The open-string vertex operators $V^{\rm op}_{\varphi_a}$ are constructed from $e^{ip\cdot X^{\rm op}}$ and $\partial^n_z X^{\rm op}_\mu$ as explained in section 5.5. The correlators are computed from the same Wick rules as in the closed-string case, and the net effect of having $X^{\rm op}_\mu$ in the place of X_μ is a rescaling of α' by factors of two and four. Accordingly, we will no longer display the $^{\rm op}$ superscripts for ease of notation. The normalization factors suppressed by the proportionality sign $\sim$ will again be fixed later on.

6.3.1 Three gluons

We recall the form of the massless open-string vertex operators (with $p_j^2 = 0$ and polarization vectors ϵ_j^μ subject to $\epsilon_j \cdot p_j = 0$)

$$V_{\epsilon_j}(z_j) \sim \epsilon_j^\mu : i\partial_z X_\mu e^{ip_j\cdot X(z_j)} : \tag{6.25}$$

The color-ordered amplitude associated with ${\rm Tr}(T^{a_1}T^{a_2}T^{a_3})$ is

$$\begin{aligned}
\mathcal{A}^{\rm tree}_{\rm bos}(\epsilon_1,\epsilon_2,\epsilon_3;\alpha') &\sim \langle cV_{\epsilon_1}(z_1) cV_{\epsilon_2}(z_2) cV_{\epsilon_3}(z_3)\rangle \\
&= |z_{12}z_{13}z_{23}| \epsilon_1^{\mu_1}\epsilon_2^{\mu_2}\epsilon_3^{\mu_3} \left\langle \prod_{j=1}^3 : i\partial X_{\mu_j} e^{ip_j\cdot X(z_j)} : \right\rangle \\
&= |z_{12}z_{13}z_{23}| \prod_{1\leq i\leq j}^3 |z_{ij}|^{2\alpha' p_i\cdot p_j} \epsilon_1^{\mu_1}\epsilon_2^{\mu_2}\epsilon_3^{\mu_3} \bigg\{ (2\alpha')^3 \frac{p^2_{\mu_1} z_{23}}{z_{12}z_{13}} \frac{p^3_{\mu_2} z_{31}}{z_{23}z_{21}} \frac{p^1_{\mu_3} z_{12}}{z_{31}z_{32}} \\
&\qquad\qquad + (2\alpha')^2 \Big[\frac{\eta_{\mu_1\mu_2}}{z_{12}^2} \frac{p^1_{\mu_3} z_{12}}{z_{31}z_{32}} + {\rm cyc}(1,2,3) \Big] \bigg\} \\
&\sim \big[(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot p_1) + {\rm cyc}(1,2,3) \big] + 2\alpha'(\epsilon_1\cdot p_2)(\epsilon_2\cdot p_3)(\epsilon_3\cdot p_1)
\end{aligned} \tag{6.26}$$

In passing to the last line, we have used that all the Koba–Nielsen exponents $\sim p_i\cdot p_j$ vanish in the momentum phase space of three massless particles subject to momentum conservation. In an open-string context, the c-ghost correlator is given by $|z_{12}z_{13}z_{23}|$ with absolute values. This can for instance be understood from its role as a Jacobian, and it leads to a negative relative sign between the two color-orderings,

$$\mathcal{A}^{\rm tree}_{\rm bos}(\epsilon_1,\epsilon_2,\epsilon_3;\alpha') = -\mathcal{A}^{\rm tree}_{\rm bos}(\epsilon_3,\epsilon_2,\epsilon_1;\alpha') \tag{6.27}$$

As a consequence, the accompanying color traces enter $\mathcal{M}^{\text{tree}}_{\substack{\text{open}\\\text{bos}}}$ with a relative sign $\text{Tr}([T^{a_1}, T^{a_2}]T^{a_3}) \sim f^{a_1 a_2 a_3}$ involving the structure constants of the gauge group, $[T^{a_1}, T^{a_2}] = i f^{a_1 a_2 a_3} T^{a_3}$.

- At the leading order in α', the three-gluon amplitude (6.26) reproduces the result of the Feynman rule in figure 16 that follows from the Lagrangian of Yang–Mills field theory

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4}\text{Tr}(F^{\mu\nu}F_{\mu\nu}) \sim \text{Tr}(A\partial^2 A + A^2 \partial A + A^4) \tag{6.28}$$

  In the second step, we have inserted the expression $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - g_{\text{YM}}[A_\mu, A_\nu]$ for non-linear field strength without tracking the structure of Lorentz indices.

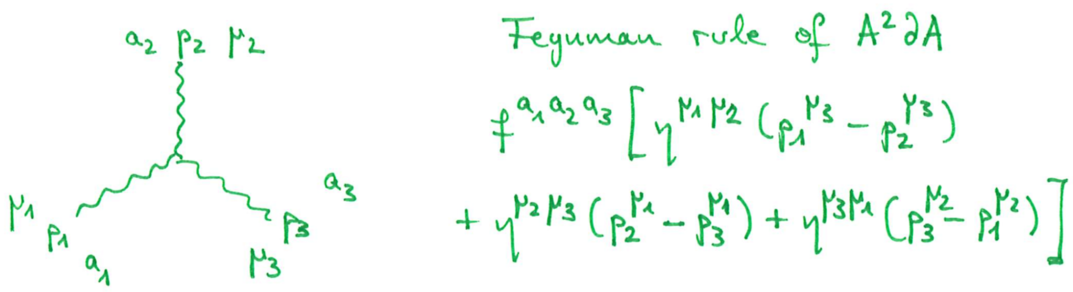


Figure 16: Feynman rule for the cubic vertex in Yang–Mills field theory from the term of schematic form $\text{Tr}(A^2\partial A)$ in the Lagrangian.

- The subleading order of (6.26) in α' can be traced back to the Feynman rules of a higher-mass-dimension operator $\alpha'\text{Tr}(F^3)$, see figure 17. The non-linear parts $ig_{\text{YM}}[A_\mu, A_\nu]$ of the gluon field strength introduce additional four-, five- and six-point vertices into the Feynman rules that do not contribute to the three-point amplitude under discussion.

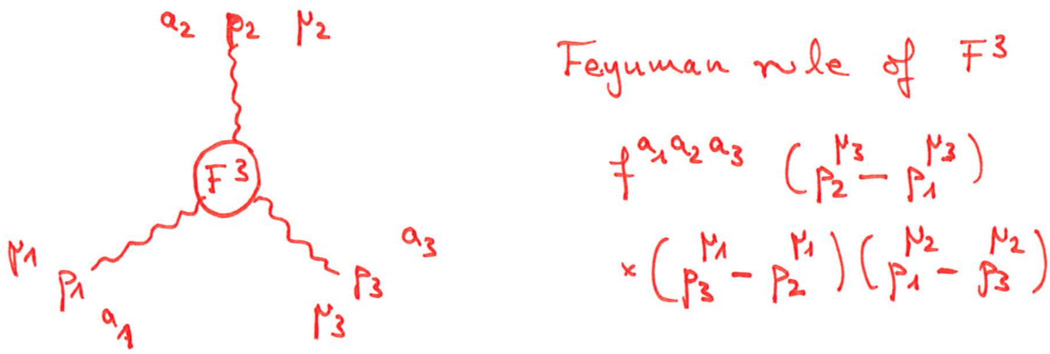


Figure 17: Feynman rule for the cubic vertex from the operator $\text{Tr}(F^3)$.

- Hence, the α'-expansion of the three-point amplitude is captured by the low-energy effective Lagrangian

$$\mathcal{L}_{\substack{\text{open}\\\text{bos}}} = \mathcal{L}_{\text{YM}} + \alpha'\text{Tr}(F^\mu{}_\nu F^\nu{}_\lambda F^\lambda{}_\mu) + \mathcal{O}(\alpha'^2) \tag{6.29}$$

  and we will see from the four-point amplitude that additional operators starting from $\alpha'^2\text{Tr}(F^4)$ are needed to reproduce higher-order terms in the α'-expansion.

6.3.2 Four-point generalities

Before considering the specific example of four external tachyons, we note some general properties of four-point open-string amplitudes. First, the Koba–Nielsen exponents $|z_{ij}|^{2\alpha' p_i\cdot p_j}$ have integer offsets (depending on the external masses) in comparison to the dimensionless Mandelstam invariants

$$s_{ij} = \alpha'(p_i+p_j)^2 = 2\alpha' p_i\cdot p_j - \alpha'(m_i^2+m_j^2) \tag{6.30}$$

In case of four tachyons or four massless states, we rewrite the Koba–Nielsen factor as follows

$$\prod_{1\leq i<j}^4 |z_{ij}|^{2\alpha' p_i\cdot p_j} = \begin{cases} \prod_{1\leq i<j}^4 |z_{ij}|^{s_{ij}} & : \ \text{four external gluons, } m_j^2=0 \\ \prod_{1\leq i<j}^4 |z_{ij}|^{s_{ij}-2} & : \ \text{four external tachyons, } m_j^2=-1/\alpha' \end{cases} \tag{6.31}$$

which is motivated by the fact that scattering amplitudes have poles in s_{ij} and not in $2\alpha' p_i\cdot p_j$. In an $\mathrm{SL}_2(\mathbb{R})$-frame with $(z_1,z_3,z_4)\to(0,1,\infty)$, color-ordered four-point amplitudes involving generic open-string states φ_j read

$$\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_2,\varphi_3,\varphi_4;\alpha') \sim \int_0^1 dz_2 \lim_{z_4\to\infty}\Big(z_4^2 \langle V_{\varphi_1}(z_1)V_{\varphi_2}(z_2)V_{\varphi_3}(z_3)V_{\varphi_4}(z_4)\rangle\Big) \tag{6.32}$$

according to the general prescription in (6.24). The integration limits $z_2\in(0,1)$ implement the constraints $z_1<z_2<z_3$ inherited from the color ordering. The limit $z_4\to\infty$ is finite since V_{φ_j} are conformal primaries of weight $h_j=1$, which can be verified at the level of the Koba–Nielsen factors by relations such as $s+t+u\big|_{4 \text{ tachyons}}=4$ and $s+t+u\big|_{4\text{ gluons}}=0$.

6.3.3 Four tachyons

When specializing φ_i to four external open-string tachyons, the above color-ordered four-point disk amplitudes becomes

$$\begin{aligned} \mathcal{A}^{\rm tree}_{\rm bos}(T_1,T_2,T_3,T_4;\alpha') &\sim \int_0^1 dz_2\, |z_2|^{s-2}|1-z_2|^{t-2} \\ &= \frac{\Gamma(s-1)\Gamma(t-1)}{\Gamma(s+t-2)} \\ &= \frac{\Gamma(s-1)\Gamma(t-1)}{\Gamma(2-u)} \end{aligned} \tag{6.33}$$

The dz_2 integral over $(0,1)$ is simpler to evaluate than the earlier d^2z_2 integral over $\mathbb{C}$ seen in the closed-string four-point amplitude. This integral can be identified as the Euler beta function

$$\int_0^1 dz_2\, |z_2|^{a-1}|1-z_2|^{b-1} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \tag{6.34}$$

which in turn is well-known to yield a ratio of Gamma functions

$$\Gamma(a) = \int_0^\infty dx\, x^{a-1}e^{-x} \tag{6.35}$$

subject to

$$\Gamma(a{+}1) = a\Gamma(a)\,, \qquad \Gamma(n) = (n{-}1)!\ \forall\, n\in\mathbb{N} \tag{6.36}$$

The functional relation $\Gamma(a{+}1)=a\Gamma(a)$ yields an analytic continuation to complex values of a outside the region $\mathrm{Re}(a)>0$ of where the integral converges. This analytic continuation has simple poles at non-positive integers $a=0,-1,-2,-3,\ldots$ which have important physical meaning in four-point string amplitudes.

- The numerators $\Gamma(s-1)$ and $\Gamma(t-1)$ in the four-tachyon amplitudes reveal poles when either $s = s_{12} = s_{34}$ or $t = s_{23} = s_{14}$ take values $1, 0, -1, -2, \ldots$. These poles signal the propagators $\frac{1}{(p_1+p_2)^2+m^2_{\rm int}} = \frac{\alpha'}{s+\alpha' m^2_{\rm int}}$ of the state-exchange diagram in figure 18, where the mass-squares of the internal states can be read off as $m^2_{\rm int} = -\frac{1}{\alpha'}, 0, \frac{1}{\alpha'}, \frac{2}{\alpha'}, \ldots$. This selection of internal masses precisely matches the open-string spectrum (starting with the tachyon mass-square $-\frac{1}{\alpha'}$ and covering the entire massive tower at mass-squares $\frac{\mathbb{N}_0}{\alpha'}$). It is an important cross-check that no states at unphysical masses appear in internal propagators.

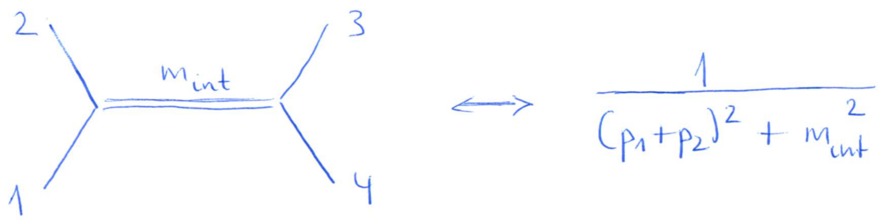


Figure 18: Interpretation of the s-channel poles of the four-tachyon amplitude as an exchange diagram with internal states of mass $m_{\rm int}$.

- There are no poles in the u-channel, the Gamma function $\Gamma(2-u)$ in the denominator of the four-tachyon amplitude only yields zeros at $u = \alpha'(p_1+p_3)^2 = 2, 3, 4, \ldots$. This is consistent with the accompanying Chan-Paton factor ${\rm Tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$ having T^{a_1} and T^{a_3} at non-adjacent positions.

- Since all negative integers occur among the s- and t-channel poles, there must be at least one state at each mass level that couples to two tachyons. As a next step, we infer more information about these states by computing the residues of the poles: From the binomial expansion $(1-z_2)^{t-2} = \sum_{n=0}^\infty (-z_2)^n \binom{t-2}{n}$ with $\binom{t-2}{n} = \frac{\Gamma(t-1)}{n!\Gamma(t-n-1)}$, we can rewrite the four-tachyon amplitude as

$$\begin{aligned} \int_0^1 dz_2\, z_2^{s-2}(1-z_2)^{t-2} &= \int_0^1 dz_2\, z_2^{s-2} \sum_{n=0}^\infty (-z_2)^n \binom{t-2}{n} = \sum_{n=0}^\infty \frac{(-1)^n \binom{t-2}{n}}{s+n-1} \\ &= \frac{1}{s-1} - \frac{t-2}{s} + \frac{(t-2)(t-3)}{2(s+1)} + \ldots \end{aligned} \tag{6.37}$$

The Mandelstam t-variable in the numerators signals contractions $p_2 \cdot p_3$ or $p_1 \cdot p_4$ across the propagator in figure 18. Such contractions are only possible if the propagators contain $\eta^{\mu\nu}$, i.e. if the exchanged particles have spin, and the degree of the polynomial in t at the residue of the pole in $s+n$ imposes a lower bound of the spin. Given that $(s+N)^{-1}$ is accompanied by degree-$(N+1)$ polynomials, there must be spins $\geq N+1$ at mass level N, e.g. spin 0,1 and 2 at mass-level $-1, 0$ and 1. Indeed, these bounds are saturated by the bosonic-string spectrum since the construction of vertex operators from multiple insertions of $:\partial_z X^\mu \partial_z X^\nu \ldots :$ yields spins up to $N+1$ at level N.

- From a quantum-field-theory perspective, it may appear surprising that the infinite sum of s-channel poles is an exact expression for the color-ordered amplitude: The latter is symmetric under $s \leftrightarrow t$ and also has an infinite tower of t-channel poles. By the infinitely many terms in the pole expansion in the s-channel, the t-channel poles are automatically accounted for – the polynomials in t of unbounded degree can be viewed as a smoking gun for the singular behaviour as $t \to 1, 0, -1, -2, \ldots$.

The astonishing property of the four-tachyon amplitude that an infinite sum over s-channel pole incorporates the analogous poles in the (dual) t-channel is known as *duality*. The expression $\frac{\Gamma(s-1)\Gamma(t-1)}{\Gamma(2-u)}$ for the four-tachyon amplitude is the possibly first formula in the history of string theory – it was Veneziano who recognized its remarkable duality property in 1968. The Veneziano amplitude was initially proposed to describe hadron scattering, and its interpretation in terms of string scattering was found later on.

6.4 Revisiting closed strings as a double copy of open strings

We shall now revisit closed-string tree-level amplitudes and identify their building blocks as double copies of open-string quantities. The simpler part is to recognize the closed-string integrands as double copies of open-string correlators at rescaled value of α'. Moreover, we will also see that the double-copy structure of tree amplitudes even persists at the integrated level: The so-called *Kawai–Lewellen–Tye* relations allow to unwind the integrals $\int d^2 z_j$ over the sphere and to replace them by independent integrations $\int dz_2$ and $\int d\bar{z}_2$ along segments of the real line, i.e. the disk boundary.

6.4.1 Integrand level

We start by demonstrating the factorization of closed-string correlation functions into holomorphic and antiholomorphic parts, where the vertex operators take the schematic form $:(\partial_z X_\mu)^n(\partial_{\bar{z}} X_\nu)^n e^{ip\cdot X}:$. The comparison with the analogous open-string correlators of $:(\partial_z X^{\rm op}_\mu)^n e^{ip\cdot X^{\rm op}}:$ is most conveniently done at the level of their generating functions: According to exercise D.4, the open-string correlators are obtained from expanding

$$\left\langle \prod_{j=1}^n :e^{ip\cdot X^{\rm op}+i\xi_j\cdot(\partial_z+\partial_{\bar{z}})X^{\rm op}}: \right\rangle = \prod_{1\leq i<j}^n |z_{ij}|^{2\alpha' p_i\cdot p_j} \exp\left(2\alpha' \sum_{1\leq i<j}^n \left[\frac{\xi_i\cdot\xi_j}{z_{ij}^2}+\frac{\xi_i\cdot p_j}{z_{ij}}+\frac{\xi_j\cdot p_i}{z_{ji}}\right]\right) \tag{6.38}$$

in the formal bookkeeping vectors ξ_j^μ, where $(\partial_z+\partial_{\bar{z}})$ is the derivative along the boundary. The generating functions for closed strings involve separate bookkeeping variables $\bar{\xi}_j$ to distinguish $\partial_z X^\mu$ for left movers from $\partial_{\bar{z}} X^\mu$ for right movers.

$$\begin{aligned}\left\langle \prod_{j=1}^n :e^{ip\cdot X+i\xi_j\cdot\partial_z X+i\bar{\xi}_j\cdot\partial_{\bar{z}}X}: \right\rangle &= \prod_{1\leq i<j}^n |z_{ij}|^{\alpha' p_i\cdot p_j} \exp\left(\frac{\alpha'}{2} \sum_{1\leq i<j}^n \left[\frac{\xi_i\cdot\xi_j}{z_{ij}^2}+\frac{\xi_i\cdot p_j}{z_{ij}}+\frac{\xi_j\cdot p_i}{z_{ji}}\right]\right)\\ &\quad\times \exp\left(\frac{\alpha'}{2} \sum_{1\leq i<j}^n \left[\frac{\bar{\xi}_i\cdot\bar{\xi}_j}{\bar{z}_{ij}^2}+\frac{\bar{\xi}_i\cdot p_j}{\bar{z}_{ij}}+\frac{\bar{\xi}_j\cdot p_i}{\bar{z}_{ji}}\right]\right)\end{aligned} \tag{6.39}$$

By comparing the two types of generating functions, we observe that the correlators of the first derivatives of X_μ and $X^{\rm op}_\mu$ are consistent with

$$\left\langle \prod_{j=1}^n V_{\Phi_j}(z_j)\right\rangle = \left|\left\langle \prod_{j=1}^n V^{\rm op}_{\varphi_j}(z_j)\right\rangle_{\alpha'\to\alpha'/4}\right|^2 \tag{6.40}$$

The need for the rescaling of α' is obvious for the exponentials and also plays out with the Koba–Nielsen factor since $|z_{ij}|^{\alpha' p_i\cdot p_j} = z_{ij}^{\alpha' p_i\cdot p_j/2}\bar{z}_{ij}^{\alpha' p_i\cdot p_j/2}$. The absolute-value square is understood to act on functions of z_j and ξ_j via $|f(z,\xi)|^2 = f(z,\xi)f(\bar{z},\bar{\xi})$, and the closed-string polarizations are taken to factorize into open-string ones according to $\Phi_j = \varphi_j\otimes\bar{\varphi}_j$. By the same type of arguments, the double-copy formula for correlators also applies to higher derivatives $\partial_z^{n\geq 2}X$, with separate bookkeeping variables $\xi^{(2)},\xi^{(3)},\ldots$ and $\bar{\xi}^{(2)},\bar{\xi}^{(3)},\ldots$ for each derivative order.

6.4.2 Application to three massless closed-string states

At the massless level, the closed-string polarization tensors

$$\zeta_j^{\mu\nu} = \epsilon_j^\mu \otimes \bar{\epsilon}_j^\nu \tag{6.41}$$

obtained from double copies of gluon polarizations describe gravitons, B-fields and dilatons through their Lorentz irreducibles. In the field-theory limit, B-fields and dilatons only couple in pairs to gravitons, i.e. the tree-level scattering amplitudes corresponding to the perturbative expansion of general relativity are unaffected by the extra states.

By the double-copy formula for closed-string correlators, we obtain the following factorized form of three-point amplitudes

$$\begin{aligned}
\mathcal{M}_{\rm bos}^{\rm tree}(\{\zeta_1,\zeta_2,\zeta_3\};\alpha') &\sim \Big\langle \prod_{j=1}^{3} c\bar{c}V_{\zeta_j}(z_j)\Big\rangle \\
&= \Big| z_{12}z_{13}z_{23}\epsilon_1^\mu\epsilon_2^\nu\epsilon_3^\lambda \Big\langle \prod_{j=1}^{3} :i\partial_z X^{\rm op}_{\mu_j} e^{ip_j\cdot X}(z_j): \Big\rangle_{\alpha'\to\alpha'/4}\Big|^2 \\
&= \Big|(2\alpha')^2\big[(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot p_1) + {\rm cyc}(1,2,3)\big] + (2\alpha')^3(\epsilon_1\cdot p_2)(\epsilon_2\cdot p_3)(\epsilon_3\cdot p_1)\Big|^2 \\
&\sim \mathcal{A}_{\rm bos}^{\rm tree}(\epsilon_1,\epsilon_2,\epsilon_3;\alpha')\mathcal{A}_{\rm bos}^{\rm tree}(\bar{\epsilon}_1,\bar{\epsilon}_2,\bar{\epsilon}_3;\alpha')
\end{aligned} \tag{6.42}$$

At the leading order in α', the spin-two components in this tensor product correspond to perturbative gravity, i.e. the Feynman rules coming from the Einstein-Hilbert Lagrangian upon expansion of the spacetime metric

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}\,, \qquad \kappa = \sqrt{32\pi^2 G_{\rm Newton}} \tag{6.43}$$

around Minkowski spacetime. With this form of the metic, the Einstein-Hilbert Lagrangian (with D-dimensional Ricci scalar $\mathcal{R}$) takes the schematic form

$$\begin{aligned}
\mathcal{L}_{\rm EH} &\sim \sqrt{|\det g|}\,\mathcal{R} \\
&= h\partial^2 h + \kappa h^2\partial^2 h + \kappa^2 h^3\partial^2 h + \begin{pmatrix} \infty \text{ more vertices} \\ \text{of order } \mathcal{O}(\kappa^3)\end{pmatrix}
\end{aligned} \tag{6.44}$$

When the Lorentz indices are spelt out, the Feynman rules for the cubic and quartic vertex due to $\kappa h^2\partial^2 h$ and $\kappa^2 h^3\partial^2 h$ have 171 and 2850 terms, respectively. By the double-copy form of the closed-string amplitude, the three-graviton amplitude due to the 171-term cubic vertex is contained in the considerably simpler expression

$$\mathcal{M}_{\rm EH}^{\rm tree}(\{\zeta_1,\zeta_2,\zeta_3\}) \sim \big[(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot p_1) + {\rm cyc}(1,2,3)\big]\big[(\bar{\epsilon}_1\cdot\bar{\epsilon}_2)(\bar{\epsilon}_3\cdot p_1) + {\rm cyc}(1,2,3)\big]\Big|_{\zeta_j^{\mu\nu}=\epsilon_j^\mu\otimes\bar{\epsilon}_j^\nu} \tag{6.45}$$

The graviton components of the α'-corrections can be identified with higher-derivative operators of schematic form $\mathcal{R}^2$ and $\mathcal{R}^3$ at the two- and four-derivative order beyond the Einstein–Hilbert term

$$\mathcal{L}_{\substack{\rm closed \\ \rm bos}} = \mathcal{L}_{\rm EH} + \sqrt{|\det g|}\Big(\alpha'\mathcal{R}^2 + \alpha'^2\mathcal{R}^3 + \mathcal{O}(\alpha'^3)\Big) \tag{6.46}$$

In fact, the squaring relations of three-point amplitudes are universal to all closed-string states $\Phi_j = \varphi_j \otimes \bar{\varphi}_j$ since they solely rely on the factorization property of the correlators:

$$\mathcal{M}_{\rm bos}^{\rm tree}(\{\Phi_1,\Phi_2,\Phi_3\};\alpha') \sim \mathcal{A}_{\rm bos}^{\rm tree}(\varphi_1,\varphi_2,\varphi_3;\tfrac{\alpha'}{4})\mathcal{A}_{\rm bos}^{\rm tree}(\bar{\varphi}_1,\bar{\varphi}_2,\bar{\varphi}_3;\tfrac{\alpha'}{4}) \tag{6.47}$$

6.4.3 Kawai–Lewellen–Tye relations

Given the factorized form of closed-string three-point amplitudes, one may wonder if the same is possible in presence of non-trivial sphere and disk integrals at four points and beyond. As we will see, it is indeed possible to also "factorize the integrals", and the Kawai–Lewellen–Tye (KLT) relations to be spelled out below schematically relate

$$\int_{\text{sphere}} d^2 z_j \leftrightarrow \int_{\text{disk bdy}} dz_j \int_{\text{disk bdy}} d\bar{z}_j \tag{6.48}$$

modulo phase factors $\exp(\frac{i\pi\alpha'}{2} p_i \cdot p_j)$ from the Koba–Nielsen factor.

At four points, the KLT relations reduce arbitrary closed-string tree-level amplitudes to products of color-ordered open-string ones

$$\mathcal{M}^{\text{tree}}_{\text{bos}}(\{\Phi_1, \Phi_2, \Phi_3, \Phi_4\}; \alpha') \sim -\sin\Big(\frac{\pi\alpha'}{2} p_1 \cdot p_2\Big) \mathcal{A}^{\text{tree}}_{\text{bos}}(\varphi_1, \varphi_2, \varphi_3, \varphi_4; \tfrac{\alpha'}{4}) \mathcal{A}^{\text{tree}}_{\text{bos}}(\bar{\varphi}_1, \bar{\varphi}_2, \bar{\varphi}_4, \bar{\varphi}_3; \tfrac{\alpha'}{4}) \tag{6.49}$$

where we note the swap of the 3rd and 4th leg in the color-orderings of the left movers $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ and the right movers $\bar{\varphi}_1, \bar{\varphi}_2, \bar{\varphi}_4, \bar{\varphi}_3$. The proof of the four-point KLT relations is based on a contour deformation for the sphere integral over the unfixed puncture. A detailed derivation can be found in the you-tube video of the lecture "string amplitudes 3" of the Sagex summer school 2019 (DESY, Hamburg), starting from minute 38:

https://indico.desy.de/event/22450/contributions/46535/

Alternatively, you can consult the original KLT paper: "A Relation Between Tree Amplitudes of Closed and Open Strings," Nucl. Phys. B **269** (1986), 1-23.

For four open- and closed-string tachyons ($m_j^2 = -\frac{4}{\alpha'}$ on the closed-string side), the KLT relations involve double copies of the rescaled Veneziano amplitude on the open-string side

$$\begin{aligned} \mathcal{A}^{\text{tree}}_{\text{bos}}(T_1, T_2, T_3, T_4; \tfrac{\alpha'}{4}) &\sim \int_0^1 dz_2 \, |z_2|^{\frac{s}{4}-2} |1-z_2|^{\frac{t}{4}-2} \\ &= \frac{\Gamma(\frac{s}{4}-1)\Gamma(\frac{t}{4}-1)}{\Gamma(2-\frac{u}{4})} \end{aligned} \tag{6.50}$$

where $\mathcal{A}^{\text{tree}}_{\text{bos}}(T_1, T_2, T_4, T_3; \frac{\alpha'}{4})$ can be obtained by swapping $t \leftrightarrow u$ in the arguments of the Gamma functions. The sine function in the KLT relation can also be expressed in terms of Gamma functions

$$\sin\Big(\frac{\pi\alpha'}{2} p_1 \cdot p_2\Big) = \sin\Big(\pi\Big[\frac{s}{4} - 2\Big]\Big) = \frac{-\pi}{\Gamma(\frac{s}{4}-1)\Gamma(2-\frac{s}{4})} \tag{6.51}$$

using $p_j^2 = \frac{4}{\alpha'}$ in the four-tachyon case. Upon insertion into the KLT formula, two of the Gamma functions cancel, and one is left with the permutation-invariant expression

$$\begin{aligned} \mathcal{M}^{\text{tree}}_{\text{bos}}(\{T_1, T_2, T_3, T_4\}; \alpha') &\sim \frac{\pi}{\Gamma(\frac{s}{4}-1)\Gamma(2-\frac{s}{4})} \times \frac{\Gamma(\frac{s}{4}-1)\Gamma(\frac{t}{4}-1)}{\Gamma(2-\frac{u}{4})} \times \frac{\Gamma(\frac{s}{4}-1)\Gamma(\frac{u}{4}-1)}{\Gamma(2-\frac{t}{4})} \\ &= \frac{\pi\Gamma(\frac{s}{4}-1)\Gamma(\frac{t}{4}-1)\Gamma(\frac{u}{4}-1)}{\Gamma(2-\frac{s}{4})\Gamma(2-\frac{t}{4})\Gamma(2-\frac{u}{4})} \end{aligned} \tag{6.52}$$

By the discussion in earlier lectures, the last line is the outcome of the integral

$$\int_{\mathbb{C}} |z_2|^{\frac{s}{2}-4} |1-z_2|^{\frac{t}{2}-4} \tag{6.53}$$

i.e. from a mathematical viewpoint, KLT relations are an efficient way of evaluating sphere integrals. In physical terms, the role of the sine function in the numerator can be understood in terms of the pole structure: The bilinear in α'-rescaled open-string amplitudes at color orderings $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ and $\bar{\varphi}_1, \bar{\varphi}_2, \bar{\varphi}_4, \bar{\varphi}_3$ has infinite towers of s-channel poles at $s = 4, 0, -4, -8, \ldots$ from both factors. Hence, the net effect of the numerator factor $\sin(\frac{\pi\alpha'}{2} p_1 \cdot p_2)$ with zeros at $s \in 4\mathbb{Z}$ is to convert the double poles of $|\mathcal{A}^{\rm tree}_{\rm bos}|^2$ into simple poles as expected for $\mathcal{M}^{\rm tree}_{\rm bos}$ from state-exchange diagrams as in figure 18. The Gamma-function representation of $\mathcal{M}^{\rm tree}_{\rm bos}$ manifests simple poles at $s, t, u = 4, 0, -4, -8, \ldots$ which corresponds to the internal masses $m^2_{\rm int} = -\frac{4}{\alpha'}, 0$ and $m^2_{\rm int} \in \frac{\mathbb{N}}{\alpha'}$ of the closed-string spectrum. In contrast to the Veneziano amplitude where the color ordering only admits s- and t-channel poles, the result for $\mathcal{M}^{\rm tree}_{\rm bos}$ is permutation symmetric in p_1, p_2, p_3, p_4 and has the same collection of poles in the s-, t- and u channel.

The general form of four-point integrals computed by the KLT relations involves rational meromorphic and antimeromorphic functions $f(z)$ and $g(\bar{z})$, respectively: The complex-analysis methods in the proof of the KLT relations are universal in showing that

$$\int_{\mathbb{C}} d^2z\, |z|^{2s} |1-z|^{2t} f(z) g(\bar{z}) = -\sin(\pi s) \bigg(\int_0^1 dz\, |z|^s |1-z|^t f(z) \bigg) \bigg(\int_{-\infty}^0 d\bar{z}\, |\bar{z}|^s |1-\bar{z}|^t g(\bar{z}) \bigg) \tag{6.54}$$

6.4.4 KLT relations in field theory

A particularly important corollary of the KLT relations can be found in the field-theory limit limit of massless amplitudes. We have seen in a three-point context that the $\alpha' \to 0$ limit of graviton amplitudes $\mathcal{M}^{\rm tree}_{\rm bos}(\{\zeta_1, \ldots, \zeta_n\}; \alpha')$ and gluon amplitudes $\mathcal{A}^{\rm tree}_{\rm bos}(\epsilon_1, \ldots, \epsilon_n; \alpha')$ in string theory reproduces those of general relativity and Yang-Mills theory, respectively. The B-field and dilaton states in the tensor product $\epsilon^\mu_j \otimes \bar{\epsilon}^\nu_j$ of massless states only couple to gravitons in pairs as mentioned before, so they do not appear in the propagators of tree-level diagrams of perturbative gravity.

Hence, the $\alpha' \to 0$ limit of the string-theory relation yields

$$\mathcal{M}^{\rm tree}_{\rm EH}(\{\zeta_1, \zeta_2, \zeta_3, \zeta_4\}) \sim -(p_1+p_2)^2 \mathcal{A}^{\rm tree}_{\rm YM}(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) \mathcal{A}^{\rm tree}_{\rm YM}(\bar{\epsilon}_1, \bar{\epsilon}_2, \bar{\epsilon}_4, \bar{\epsilon}_3) \tag{6.55}$$

where the sine function has been replaced by its leading order in α', i.e. $\sin(\frac{\pi\alpha'}{2} p_1 \cdot p_2) = \frac{\pi\alpha'}{2} p_1 \cdot p_2 + \mathcal{O}(\alpha'^3)$. The resulting inverse propagator $(p_1 + p_2)^2$ in the field-theory KLT relation can again be understood as a necessity to cancel the massless double pole in the s-channel of $|\mathcal{A}^{\rm tree}_{\rm YM}|^2$.

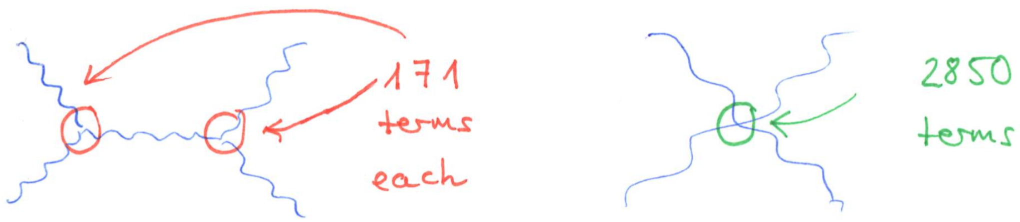


Figure 19: Number of terms in Feynman diagrams that contribute to the four-graviton tree amplitude

As a key virtue of the field-theory KLT relations, they capture the net result of the gravitational diagrams in figure 19 in an extremely compact way! Already the s-channel diagram would have 171^2 terms in a naive Feynman-diagram computation, while the KLT formula for the four-graviton amplitude is a bilinear in color-ordered Yang–Mills trees with around 15 terms each.

Together with their n-point generalizations to be discussed below, the KLT relations for the tree-level amplitudes of general relativity are the oldest incarnation of the gravitational double copy. There are similar double-copy prescriptions at loop level to assemble gravitational loop integrands from squares of gauge-theory quantities, and a comprehensive review can be found in 1909.01358.

6.4.5 Higher-point KLT relations

One can similarly unwind the integration domain for $d^2z_j = dz_j\, d\bar{z}_j$ at higher multiplicity up to phases $\exp(\frac{i\pi\alpha'}{2}p_i \cdot p_j)$, e.g. at five points

$$\begin{aligned}\mathcal{M}^{\rm tree}_{\rm bos}(\{\Phi_1,\Phi_2,\Phi_3,\Phi_4,\Phi_5\};\alpha') &\sim \mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_2,\varphi_3,\varphi_4,\varphi_5;\tfrac{\alpha'}{4})\sin\Big(\frac{\pi\alpha'}{2}p_1\cdot p_2\Big)\\ &\times\Big[\sin\Big(\frac{\pi\alpha'}{2}(p_1{+}p_2)\cdot p_3\Big)\mathcal{A}^{\rm tree}_{\rm bos}(\bar\varphi_1,\bar\varphi_2,\bar\varphi_3,\bar\varphi_5,\bar\varphi_4;\tfrac{\alpha'}{4})\\ &\quad+\sin\Big(\frac{\pi\alpha'}{2}p_1\cdot p_3\Big)\mathcal{A}^{\rm tree}_{\rm bos}(\bar\varphi_1,\bar\varphi_3,\bar\varphi_2,\bar\varphi_5,\bar\varphi_4;\tfrac{\alpha'}{4})\Big]+(2\leftrightarrow 3)\end{aligned}\tag{6.56}$$

where $+(2 \leftrightarrow 3)$ instructs to add the image of all the three lines under simultaneous exchange of $(p_2,\varphi_2,\bar\varphi_2) \leftrightarrow (p_3,\varphi_3,\bar\varphi_3)$. Upon comparison with the three- and four-point KLT formulae, the number of sine functions is seen to grow linearly with multiplicity. Accordingly, each term in the n-point KLT formula involves factors of the schematic form $\big[\sin(\frac{\pi\alpha'}{2}p_i\cdot p_j)\big]^{n-3}$

$$\begin{aligned}\mathcal{M}^{\rm tree}_{\rm bos}(\{\Phi_1,\Phi_2,\ldots,\Phi_n\};\alpha') \sim \sum_{\rho,\tau\in S_{n-3}} &\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\rho(\varphi_2,\varphi_3,\ldots,\varphi_{n-2}),\varphi_{n-1},\varphi_n;\tfrac{\alpha'}{4})\\ &\times S_{\alpha'}(\rho|\tau)\mathcal{A}^{\rm tree}_{\rm bos}(\bar\varphi_1,\tau(\bar\varphi_2,\bar\varphi_3,\ldots,\bar\varphi_{n-2}),\bar\varphi_n,\bar\varphi_{n-1};\tfrac{\alpha'}{4})\end{aligned}\tag{6.57}$$

which are organized as the entries of an $(n{-}3)! \times (n{-}3)!$ matrix indexed by permutations ρ,τ of external legs $2,3,\ldots,n{-}2$. In the $\alpha' \to 0$ limit, the structure propagates to gravitational amplitudes $\mathcal{M}^{\rm tree}_{\rm bos} \to \mathcal{M}^{\rm tree}_{\rm EH}$ with $\mathcal{A}^{\rm tree}_{\rm bos} \to \mathcal{A}^{\rm tree}_{\rm YM}$ and all factors of $\sin(\frac{\pi\alpha'}{2}p_i\cdot p_j)$ replaced by $p_i \cdot p_j$. This is again a dramatic improvement in comparison to the Feynman diagrammatics of perturbative gravity, where an infinite number of vertices arises from the expansion of the Einstein–Hilbert action around a Minkowski background.

6.4.6 Permutation invariance and monodromy relations

Closed-string amplitudes of identical bosonic external states have to be permutation invariant functions of $(p_i,\varphi_i,\bar\varphi_i)$ with $i = 1,2,\ldots,n$. However, the above KLT formulae only manifest invariance under permutations of a subset $i = 2,3,\ldots,n{-}2$ of the external legs. Permutation invariance involving the remaining legs $i = 1,n{-}1,n$ is based on so-called *monodromy relations* among color-ordered open-string amplitudes: At four points, for instance, symmetry under $(p_2,\varphi_2,\bar\varphi_2) \leftrightarrow (p_4,\varphi_4,\bar\varphi_4)$ can be checked by the monodromy relations

$$\sin(2\pi\alpha' p_1\cdot p_2)\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_2,\varphi_4,\varphi_3;\alpha') = \sin(2\pi\alpha' p_2\cdot p_3)\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\varphi_3,\varphi_2,\varphi_4;\alpha')\tag{6.58}$$

to be derived in exercise E.4 which imply that all color-ordered four-point amplitudes are proportional to each other and related by a ratio of sine-functions. More generally, monodromy relations among color-ordered n-point amplitudes leave a $(n{-}3)!$-dimensional basis that can be chosen as

$$\big\{\mathcal{A}^{\rm tree}_{\rm bos}(\varphi_1,\rho(\varphi_2,\varphi_3,\ldots,\varphi_{n-2}),\varphi_{n-1},\varphi_n;\alpha'),\ \rho\in S_{n-3}\big\}\tag{6.59}$$

in lines with the number of terms in each chiral half of the n-point KLT formula. Note that both KLT and monodromy relations hold universally for any combination of external states $\Phi_i = \varphi_i \otimes \bar\varphi_i$ since their derivation only uses the analytic properties of the ubiquitous Koba–Nielsen factor.

6.5 Low-energy physics from string amplitudes

The goals of this section are to

- recover Yang–Mills and gravity from the $\alpha' \to 0$ limit of string amplitudes and to infer the systematics of higher-derivative operators $\mathrm{Tr}(D^{2m}F^n), D^{2m}\mathcal{R}^n$ in the effective Lagrangians via α'-expansion (with D_μ denoting the covariant derivatives w.r.t. non-linear gauge transformations or diffeomorphisms)
- settle the normalization of vertex operators by matching with the effective field theory and unitarity

As already illustrated in the three-point examples of earlier sections, the α'-expansion of massless string amplitudes can be used to reverse-engineer Feynman rules and thereby the low-energy effective Lagrangian that reproduces their expressions. When restoring the couplings $g_{\rm YM}$ and $\kappa^2 = 32\pi^2 G_{\rm Newton}$ of Yang–Mills theory and general relativity in D spacetime dimensions, the contributions to the low-energy effective Lagrangian visible from three-point string amplitudes are

$$\mathcal{L}_{\substack{\rm open\\ \rm bos}} = -\frac{1}{4}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{2\alpha'}{3} g_{\rm YM}\mathrm{Tr}(F^\mu{}_\nu F^\nu{}_\lambda F^\lambda{}_\mu) + \mathcal{O}(\alpha'^2) \tag{6.60}$$

$$\begin{aligned}\mathcal{L}_{\substack{\rm closed\\ \rm bos}} &= \frac{2}{\kappa^2}\sqrt{\det|g|}\Big\{\mathcal{R} + \frac{\alpha'}{4}(\mathcal{R}_{\mu\nu\lambda\rho}\mathcal{R}^{\mu\nu\lambda\rho} - 4\mathcal{R}_{\mu\nu}\mathcal{R}^{\mu\nu} + \mathcal{R}^2) \\ &\quad + \Big(\frac{\alpha'}{4}\Big)^2(\mathcal{R}^{\mu\nu}{}_{\alpha\beta}\mathcal{R}^{\alpha\beta}{}_{\lambda\rho}\mathcal{R}^{\lambda\rho}{}_{\mu\nu} - \tfrac{4}{3}\mathcal{R}^{\mu\nu\alpha\beta}\mathcal{R}_{\nu\lambda\beta\rho}\mathcal{R}^\lambda{}_\mu{}^\rho{}_\alpha) + (B\text{-field and dilaton})\Big\} + \mathcal{O}(\alpha'^3)\end{aligned} \tag{6.61}$$

where the non-linear field strength and the expansion of the gravitational field around a Minkowski background are

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - g_{\rm YM}[A_\mu, A_\nu]\,, \qquad g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu} \tag{6.62}$$

The operators $\alpha'\mathrm{Tr}(F^3), \alpha'\mathcal{R}^2, etc.$ of higher mass dimension are the joint effort of integrating out the tachyon and the infinite tower of massive states at $m^2_{\rm open} \in \frac{\mathbb{N}}{\alpha'}$ and $m^2_{\rm closed} \in \frac{4\mathbb{N}}{\alpha'}$. From the viewpoint of a Taylor expansion in the dimensionless quantities $\alpha' p_i \cdot p_j \ll 1$, all non-zero masses are very large. The low-energy effective action corresponding to the α'-expansion of string amplitudes follows from a path integral over all the massive degrees of freedom denoted by $\hat\varphi$, i.e. schematically

$$\int \mathcal{D}[\hat\varphi_{m^2\neq 0}]\mathcal{D}[\varphi_{m^2=0}]e^{-S^{\rm exact}_{\rm bos}[\varphi,\hat\varphi]} = \int \mathcal{D}[\varphi_{m^2=0}]e^{-S^{\rm eff}_{\rm bos}[\varphi]} \tag{6.63}$$

The general idea in *effective field theory* is to consider the exact Lagrangian ($S^{\rm exact}_{\rm bos}[\varphi,\hat\varphi]$ in this case) as currently out of reach and to set the more modest goal of determining the effective Lagrangian $S^{\rm eff}_{\rm bos}[\varphi]$ that is accurate up to a fixed order in the energy scales. In a variety of examples, the effective-field-theory approach allows to make predictions at the LHC scale based on ordinary quantum-field-theory methods while acknowledging the ignorance of Planck-scale physics. Conversely, one cannot probe Planck-scale physics based on LHC experiments which can only be used to fix the parameters in the effective Lagrangian order by order. As an example of this limitation within string amplitudes, the α'-expansion of the factor $\Gamma(1+s)$ in the Veneziano amplitude cannot detect its poles at $s \in -\mathbb{N}$.

6.5.1 First look at four-point α'-expansions

Apart from the above three-point examples, the effective interactions of open and closed bosonic strings can be conveniently organized according to the accompanying Riemann zeta values

$$\zeta_n = \sum_{k=1}^{\infty}\frac{1}{k^n}\,, \qquad n \geq 2 \tag{6.64}$$

which are said to carry "weight" n. The massless four-gluon amplitude is a rational function of the Lorentz invariants $(\epsilon_i \cdot \epsilon_j), (\epsilon_i \cdot p_j)$ and $s_{ij} = 2\alpha' p_i \cdot p_j$, and the exact polarization dependence can be worked out just as in the three-point case by going through the possible Wick contractions of the vertex operators $\sim: \partial_z X_\mu e^{ip\cdot X}$: (with zero, one or two pair contractions of $\partial_z X_\mu$). Each term contains an integral that evaluates to an Euler beta function and is proportional to

$$\frac{\Gamma(1{+}s)\Gamma(1{+}t)}{\Gamma(1{+}s{+}t)} = 1 - \zeta_2 st - \zeta_3 stu - \zeta_4 st(s^2 + \tfrac{1}{4}st + t^2) + \zeta_2\zeta_3 s^2t^2u - \zeta_5 stu(s^2+st+t^2) + \mathcal{O}(\alpha'^6) \tag{6.65}$$

Regardless on the power of z_2 and $(1{-}z_2)$ in the integrand, one can always attain arguments $1{+}s, 1{+}t$ and $1{+}s{+}t$ in the Gamma functions by repeatedly using their functional identity $\Gamma(x{+}1) = x\Gamma(x)$. These manipulations introduce rational functions of s and t. The α'-expansion of the beta integral is said to be uniformly transcendental since the powers of α' always match the overall weight of the accompanying zeta values. The analogous expansion for the sphere integral in four-graviton amplitudes is

$$\frac{\Gamma(1{+}\frac{s}{4})\Gamma(1{+}\frac{t}{4})\Gamma(1{+}\frac{u}{4})}{\Gamma(1{-}\frac{s}{4})\Gamma(1{-}\frac{t}{4})\Gamma(1{-}\frac{u}{4})} = 1 - 2\zeta_3 stu - 2\zeta_5 stu(s^2+st+t^2) + \mathcal{O}(\alpha'^6) \tag{6.66}$$

which shares the uniform transcendentality of the sphere integral but does not exhibit any instance of ζ_2 and ζ_4.

When factoring out the beta integral (6.65) from the four-gluon amplitude, the remaining terms will by themselves be series in α'. The latter are determined from the correlator of four gluon vertex operators and the rational functions in s and t that result from $\Gamma(x{+}1) = x\Gamma(x)$ and may include geometric series $(1{-}s)^{-1} = \sum_{k=0}^\infty s^k$. This separate α'-expansion of the coefficient of (6.65) does not involve any zeta values since the underlying Wick contractions can only produce rational numbers and is collected in the curly bracket of

$$\mathcal{A}^{\rm tree}_{\rm bos}(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4;\alpha') = \Big\{ \underbrace{\mathcal{A}^{\rm tree}_{\rm YM}(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4) + \alpha' \mathcal{A}^{\rm tree}_{{\rm YM}+F^3}(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4) + \mathcal{O}(\alpha'^2)}_{\text{no zeta values}} \Big\} \frac{\Gamma(1{+}s)\Gamma(1{+}t)}{\Gamma(1{+}s{+}t)} \tag{6.67}$$

The leading order in α' must reproduce the color-ordered four-point tree amplitude $\mathcal{A}^{\rm tree}_{\rm YM}$ of YM field theory – the vertex $\sim g^2_{\rm YM}{\rm Tr}(A^4)$ from the Yang–Mills Lagrangian will inevitably contribute in view of gauge invariance. The subleading order $\alpha' \mathcal{A}^{\rm tree}_{{\rm YM}+F^3}$ arises from the Feynman diagrams including a single insertion of $\alpha' g_{\rm YM}{\rm Tr}(F^3)$ (either in a cubic-vertex graph with a Yang–Mills vertex $A^2\partial A$ and a $\alpha'\partial^3 A^3$ vertex each, or from the four-point vertex in the non-linear extension of ${\rm Tr}(F^3)$).

The $+\mathcal{O}(\alpha'^2)$ in the curly bracket of (6.67) refer to a variety of further operators starting from $\alpha'^2{\rm Tr}(F^4)$ none of which have a zeta value in their coefficient. The crossterm of $\mathcal{A}^{\rm tree}_{\rm YM}$ with the subleading term $-\zeta_2 st$ in the Gamma functions signals another operator $\alpha'^2{\rm Tr}(F^4)$, but the latter has a ζ_2-coefficient and therefore does not mix with the $\alpha'^2{\rm Tr}(F^4)$ from the curly bracket. Given that $\zeta_2 = \frac{\pi^2}{6}$, the tensor structure of the accompanying effective operator in the low-energy effective action is

$$\mathcal{L}_{\substack{\rm open\\ \rm bos}} = \ldots + \frac{\pi^2\alpha'^2}{2}{\rm STr}\Big(F_\mu{}^\nu F_\nu{}^\lambda F_\lambda{}^\rho F_\rho{}^\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}F_{\lambda\rho}F^{\lambda\rho}\Big) + \ldots \tag{6.68}$$

where STr instructs to symmetrize over the orderings of the four enclosed Chan-Paton matrices $T^a{\in}U(N)$. Also note that there are diagrams with double insertion of $\alpha'{\rm Tr}(F^3)$ contributing at the order of α'^2.

6.5.2 Higher orders and higher multiplicity

Higher orders in the α'-expansion of the Gamma functions are captured by the all-order formula

$$\frac{\Gamma(1+s)\Gamma(1+t)}{\Gamma(1+s+t)} = \exp\bigg(\sum_{n=2}^{\infty} \frac{\zeta_n}{n} (-1)^n \big[s^n + t^n - (s+t)^n \big] \bigg) \tag{6.69}$$

which reproduces (6.65) upon Taylor expansion, manifests uniform transcendentality at all orders in α' and signals an infinite tower of effective interactions of the form

$$(\alpha')^w \zeta_{w_1}\zeta_{w_2}\ldots\zeta_{w_m} \mathrm{Tr}(D^{2(w-2)}F^4)\,, \qquad \sum_{j=1}^m w_j \leq w \tag{6.70}$$

with gauge-covariant derivatives D_μ. Terms with weight $\sum_{j=1}^m w_j = w$ take all their powers of α' from the Gamma functions and are accompanied by $\mathcal{A}^{\rm tree}_{\rm YM}$ in the curly bracket in (6.67), so the α'-order matches the weight of the zeta values. Higher orders in the curly brackets lead to operators with $\sum_{j=1}^m w_j < w$.

The α'-expansion of the four-point function is insensitive to operators involving $F^{n\geq 5}$. Hence, the latter can only be determined from higher-point amplitudes and have a schematic structure of $(\alpha')^w \mathrm{Tr}(D^{2(w-3)}F^5)$ and $(\alpha')^w \mathrm{Tr}(D^{2(w-4)}F^6)$ *etc.*. For instance, the complete coefficient of $\zeta_3\alpha'^3$ and $\zeta_4\alpha'^4$ in the effective action is a combination of operators

$$\zeta_3\alpha'^3 \mathrm{Tr}(D^2F^4 + F^5)\,, \qquad \zeta_4\alpha'^4 \mathrm{Tr}(D^4F^4 + D^2F^5 + F^6) \tag{6.71}$$

The same kind of analysis applies to Riemann-curvature operators in the gravitational sector of closed-string effective actions: The subleading terms in the expansion of the Gamma functions belong to effective operators of the form

$$\zeta_3\alpha'^3\mathcal{R}^4\,, \qquad \zeta_5\alpha'^5(D^4\mathcal{R}^4 + D^2\mathcal{R}^5 + \mathcal{R}^6) \tag{6.72}$$

In general, the α'-expansion of $(n\geq 5)$-point amplitudes contains so-called multiple zeta values

$$\zeta_{n_1,n_2,\ldots,n_r} = \sum_{0<k_1<k_2<\ldots<k_r} k_1^{-n_1} k_2^{-n_2}\ldots k_r^{-n_r}\,, \qquad n_r \geq 2 \tag{6.73}$$

which generalize the Riemann zeta values to multiple arguments $n_j \in \mathbb{N}$ and are said to have weight $\sum_{j=1}^r n_j$ and depth r.

6.5.3 Normalization of vertex operators and amplitudes

The undetermined normalization factors in the previous sections boil down to

- one state-independent normalization constant per worldsheet topology (e.g. $\mathcal{N}_{D^2}$ and $\mathcal{N}_{S^2}$ for the disk and sphere)
- additionally one normalization g_Φ for each species of vertex operators $V_\Phi(z)$

These normalization factors enter for instance the tree amplitudes of m tachyons and $n-m$ gravitons via

$$\begin{aligned}\mathcal{M}^{\rm tree}_{\rm bos}(\{T_1,\ldots,T_m,\zeta_{m+1},\ldots,\zeta_n\};\alpha') &= \mathcal{N}_{S^2}(g_T)^m(g_\zeta)^{n-m} \\ &\times \int \frac{d^2z_1\ldots d^2z_n}{\mathrm{vol}\,\mathrm{SL}_2(\mathbb{C})} \bigg\langle \prod_{j=1}^m : e^{ip_j\cdot X(z_j)} : \prod_{k=m+1}^n \zeta_k^{\mu_k\nu_k} : i\partial_z X_{\mu_k} i\partial_{\bar z} X_{\nu_k} e^{ip_k\cdot X(z_k)} : \bigg\rangle\end{aligned} \tag{6.74}$$

In order to determine all the normalization factors, a first step is to infer $\mathcal{N}_{D^2}, \mathcal{N}_{S^2}$ and g_ϵ, g_ζ for massless vertex operators from gluon and graviton amplitudes: The $\alpha' \to 0$ limits of correctly normalized three- and four-point amplitudes have to match those of Yang-Mills and general relativity, e.g. for open strings

$$\lim_{\alpha' \to 0} \mathcal{N}_{D^2} (g_\epsilon)^3 (2\alpha')^2 \big[(\epsilon_1 \cdot \epsilon_2)(\epsilon_3 \cdot p_1) + \text{cyc}(1,2,3) \big] = A^{\text{tree}}_{\text{YM}}(\epsilon_1, \epsilon_2, \epsilon_3) \tag{6.75}$$

Together with a similar matching at four points, we obtain two equations for two unknowns that are solved by

$$g_\epsilon = g_{\text{YM}} \, , \qquad \mathcal{N}_{D^2} = \frac{2}{(2 g_{\text{YM}} \alpha')^2} \tag{6.76}$$

and the same logic can be used to fix g_ζ and $\mathcal{N}_{S^2}$ in terms of α' and the gravitational coupling κ. For massive vertex operators V_Φ at $m^2 \neq 0$, one cannot perform a comparable field-theory match but instead determine g_Φ from unitarity: In figure 20 below, the massless four-point amplitude on the left-hand side is taken to have a known normalization. Its residues on the kinematic poles $s{+}N$, $N \geq -1$ at all mass levels must be products of three-point trees, summed over the polarizations of the massive states Φ_N at the given level as visualized on the right-hand side. Specifically, one can immediately solve this constraint on the $N=-1$ pole to determine the normalization $(g_T)^2$ of the closed-string tachyon.

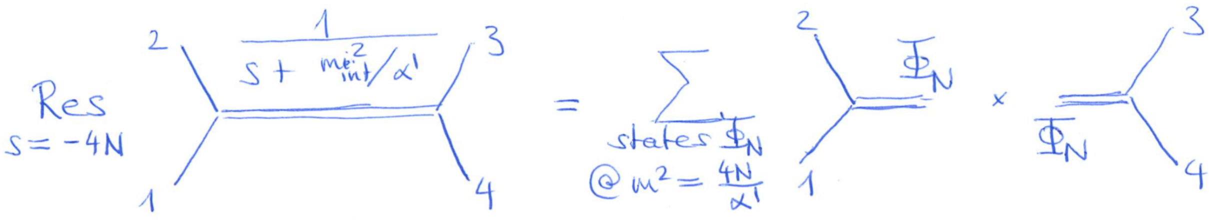


Figure 20: The factorization of massless string amplitudes on massive poles can be used to constrain or determine normalization factors of vertex operators.

6.6 High-energy scattering

While a lot of emphasis in the previous subsections has been placed on the low-energy expansion of string amplitudes at small dimensionless Mandelstam variables s, t, we shall now consider the high-energy limit. More specifically, we shall discuss the hard-scattering limit $s, t \to \infty$ at fixed angle θ or s/t. The key result is that closed- and open-string amplitudes are exponentially suppressed at large center-of-mass energy $E = \sqrt{-\frac{s}{\alpha'}}$ which is very different from the behaviour of gravitational tree amplitudes: Both the cubic and quartic Feynman vertex from the Einstein-Hilbert action lead to a growth of the four-graviton tree amplitude as E^2 for large energies, i.e. a quadratic divergence as $E \to \infty$.

The main mathematical tool in exploring the high-energy regime of string tree amplitudes is the approximation

$$\Gamma(1{+}x) \longrightarrow \exp(x \log x) \, , \qquad x \to \infty \tag{6.77}$$

which can be inserted into the universal Virasoro-Shapiro factor at $s+t+u=0$,

$$\frac{\Gamma(1{+}\frac{s}{4})\Gamma(1{+}\frac{t}{4})\Gamma(1{+}\frac{u}{4})}{\Gamma(1{-}\frac{s}{4})\Gamma(1{-}\frac{t}{4})\Gamma(1{-}\frac{u}{4})} \to \exp\left(\frac{s}{2} \log s + \frac{t}{2} \log t + \frac{u}{2} \log u \right) \tag{6.78}$$

The same conclusion for the high-energy limit can be obtained by performing the sphere integral in the saddle-point approximation, i.e. by localizing

$$\int_{\mathbb{C}} d^2 z_2 \, |z_2|^{s/2} |1-z_2|^{t/2} \to |z_2|^{s/2} |1-z_2|^{t/2} \Big|_{z_2=z_*} \tag{6.79}$$

at the extremal point $z_2 = z_*$ of the integrand:

$$\frac{\partial}{\partial z_2} |z_2|^{s/2} |1-z_2|^{t/2} \Big|_{z_2=z_*} = 0 \quad \Rightarrow \quad z_* = -\frac{s}{u} \tag{6.80}$$

Upon insertion into the Koba–Nielsen factor, we arrive at

$$|z_*|^{s/2} |1-z_*|^{t/2} = |s|^{s/2} |t|^{t/2} |u|^{-s/2-t/2} = \exp\left(\frac{s}{2} \log|s| + \frac{t}{2} \log|t| + \frac{u}{2} \log|u| \right) \tag{6.81}$$

6.6.1 Angular dependence of massless amplitudes

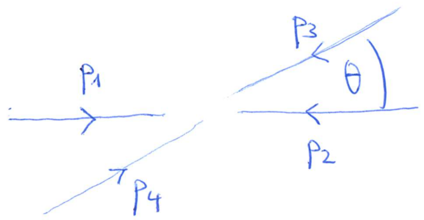


Figure 21: Scattering angle in massless four-point amplitudes

We shall now pick the Lorentz frame depicted in figure 21 to resolve the dependence of massless high-energy amplitudes on the scattering angle θ. The vanishing Minkowski norm of the momenta

$$\begin{aligned} p_1 &= \frac{1}{2}\sqrt{-\frac{s}{\alpha'}}(1,\, 1\,, 0,\, 0,\, \ldots) & p_3 &= -\frac{1}{2}\sqrt{-\frac{s}{\alpha'}}(1,\, \cos\theta\,, \sin\theta,\, 0,\, \ldots) \\ p_2 &= \frac{1}{2}\sqrt{-\frac{s}{\alpha'}}(1,\, -1\,, 0,\, 0,\, \ldots) & p_4 &= -\frac{1}{2}\sqrt{-\frac{s}{\alpha'}}(1,\, -\cos\theta\,, -\sin\theta,\, 0,\, \ldots) \end{aligned} \tag{6.82}$$

reflects our choice of massless external states. The Mandelstam invariants t, u are expressible in terms of s and θ:

$$\begin{aligned} t &= \alpha'(p_1+p_4)^2 = \alpha'\Big(-\frac{s}{\alpha'}\Big)\Big(0,\, \frac{1+\cos\theta}{2},\, \frac{\sin\theta}{2},\, 0,\, \ldots\Big)^2 = -s\cos^2\Big(\frac{\theta}{2}\Big) \\ u &= -s-t = -s\sin^2\Big(\frac{\theta}{2}\Big) \end{aligned} \tag{6.83}$$

Upon plugging back into the high-energy approximation of closed-string amplitudes,

$$\begin{aligned} \mathcal{M}^{\rm tree}(\{\Phi_1,\Phi_2,\Phi_3,\Phi_4\};\alpha') &\to \exp\left(\frac{s}{2}\log|s| - \frac{s}{2}\cos^2\Big(\frac{\theta}{2}\Big)\log\Big|s\cos^2\Big(\frac{\theta}{2}\Big)\Big| - \frac{s}{2}\sin^2\Big(\frac{\theta}{2}\Big)\log\Big|s\sin^2\Big(\frac{\theta}{2}\Big)\Big|\right) \\ &= \exp\left(-\frac{s}{2}\Big[\cos^2\Big(\frac{\theta}{2}\Big)\log\Big|\cos^2\Big(\frac{\theta}{2}\Big)\Big| + \sin^2\Big(\frac{\theta}{2}\Big)\log\Big|\sin^2\Big(\frac{\theta}{2}\Big)\Big|\Big]\right) \end{aligned} \tag{6.84}$$

we can read off the exponential suppression with $s = -\alpha' E^2$ as $E \to \infty$.

The high-energy regime of tree amplitudes is important in view of the ultraviolet properties of loop amplitudes: Similar to Feynman integrals in field theory, genus-one amplitudes in string theory can be obtained from integrating over one D-dimensional loop momentum ℓ^μ that correspond to a joint zero

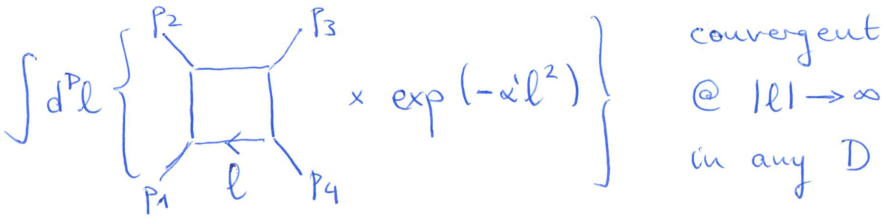


Figure 22: Schematic form of loop integrands in genus-one closed-string amplitudes

mode of $\partial_z X^\mu, \partial_{\bar z} X^\mu$ and can be associated with one of the edges in the Feynman graphs of the $\alpha' \to 0$ limit, see figure 22.

Roughly speaking, the ultraviolet behaviour of string tree amplitudes introduces an exponential damping $\exp(-\alpha'\ell^2)$ into the loop integrand of genus-one amplitudes (a refined argument based on modular invariance will be mentioned at the end of this course). In this way, the ultraviolet divergences that plague loop amplitudes of general relativity and various extensions are clearly cancelled in massless closed-string amplitudes at genus one.

Together with modular invariance, this argument to rule out ultraviolet divergences generalizes to arbitrary genus in closed-string perturbation theory. However, it does not yet tell you about other kinds of divergences, say from the infrared regime of massless external particles or subtle technicalities coming from the ghost systems of various formulations of the superstring.

The pole expansion of the Veneziano amplitude has the following closed-string counterpart:

$$\frac{\Gamma(1+\frac{s}{4})\Gamma(1+\frac{t}{4})\Gamma(1+\frac{u}{4})}{\Gamma(1-\frac{s}{4})\Gamma(1-\frac{t}{4})\Gamma(1-\frac{u}{4})} \sim \sum_{n=1}^{\infty} \frac{(t^{2n} + \text{lower degree in } t)}{s/4 + n} \tag{6.85}$$

Term by term, the residues $\sim t^{2n}$ of the poles at different mass levels exhibit horrible ultraviolet divergences as $E \to \infty$. However, the joint effort of all the terms in the infinite series over all mass levels n gives rise to the above exponential softness. This illustrates the benefit of the infinite spectrum in string theories which yields much milder ultraviolet properties as any finite subset of its particles.

High-energy scattering at fixed angle θ probes the internal structure of the target. While power-law growth with s would indicate point particles, the falloff $\sim \exp(-\alpha' k_i \cdot k_j)$ is a smoking gun for extended objects of length scale $\sqrt{\alpha'}$ (similar to scattering off a Gaussian potential with fluctuation at spatial scales of $\sqrt{\alpha'}$). Hence, the high-energy regime of string tree-level amplitudes reveals that strings are fuzzy at length scale $\leq \sqrt{\alpha'}$.

7 Bosonic strings in background fields

So far, we have taken flat Minkowski $\mathbb{R}^{1,D-1}$ to be the background spacetime. But string theory is meant to incorporate general relativity, so there must be a way of addressing a background metric $G_{\mu\nu}(X) \neq \eta_{\mu\nu}$ (as well as background electromagnetic fields $A_a(X)$ on D branes).

In more general terms, the goal of this section on strings in background fields is to connect the worldsheet and target-space point of view and to describe the treatment of non-trivial backgrounds at the level of the worldsheet CFT.

7.1 Strings in curved spacetime

The flat-space Polyakov action has a natural uplift to curved spacetime via $\eta_{\mu\nu} \rightarrow G_{\mu\nu}(X)$:

$$S_P[G,X,g] = -\frac{1}{4\pi\alpha'}\int d^2\sigma\, \sqrt{-g}g^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu G_{\mu\nu}(X) \tag{7.1}$$

In contrast to earlier sections, we switch notation for the 2×2 worldsheet metric from $h_{\alpha\beta}$ to $g_{\alpha\beta}$ since the letter h will be used for the spin-two fluctuation of the spacetime metric in the expansion

$$G_{\mu\nu}(X) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(X) \tag{7.2}$$

around the Minkowski case. These spin-two fluctuation connect with the worldsheet techniques in earlier sections as follows:

- rewriting the partition function in terms of $h_{\mu\nu}(X)$,

$$\begin{aligned} Z_P[h] &\sim \int \mathcal{D}[X]\mathcal{D}[g]\, \exp\big(-S_P[\eta+\kappa h, X, g]\big) \\ &= \int \mathcal{D}[X]\mathcal{D}[g]\, \underbrace{\exp\big(-S_P[\eta, X, g]\big)}_{\text{action in flat spacetime}} \exp\big(V[h]\big) \end{aligned} \tag{7.3}$$

  one is led to an exponentiated vertex operator

$$V[h] = \frac{\kappa}{4\pi\alpha'}\int d^2\sigma\, \sqrt{-g}g^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu h_{\mu\nu}(X) \tag{7.4}$$

- more precisely, we recover the integrated graviton vertex operator $\int d^2z V_\zeta(z)$ upon

$$g_{\alpha\beta} \rightarrow \eta_{\alpha\beta}\,, \qquad h_{\mu\nu}(X) \rightarrow \zeta_{\mu\nu}e^{ip\cdot X} \tag{7.5}$$

- while a single copy of $V[h]$ in the path integral corresponds to a single-graviton state, the insertion of $\exp(V[h])$ yields a coherent state of gravitons whose joint effort changes the spacetime metric from $\eta_{\mu\nu}$ to $\eta_{\mu\nu} + \kappa h_{\mu\nu}$

On way of dealing with the X-dependent profile of the spacetime metric $G_{\mu\nu}(X)$ is based on worldsheet perturbation theory in α':

- in conformal gauge $g_{\alpha\beta} \rightarrow \eta_{\alpha\beta}$, we have

$$S_P[G,X,\eta] = -\frac{1}{4\pi\alpha'}\int d^2\sigma\, \partial_\alpha X^\mu \partial^\alpha X^\nu G_{\mu\nu}(X) \tag{7.6}$$

  For general $G_{\mu\nu}(X)$, this action describes an interacting two-dimensional QFT on the worldsheet. Actions of this type are referred to as "non-linear σ-model (NLSM)", where the historic origin of the name stems from the study of pions.

- expand the NLSM action around the situation where the string is localized at spacetime point $\bar{x}^\mu$,

$$X^\mu(\sigma) = \bar{x}^\mu + \sqrt{\alpha'}Y^\mu(\sigma) \tag{7.7}$$

  where $\sqrt{\alpha'}$ (with units of length) is introduced to render the fluctuation $Y^\mu(\sigma)$ dimensionless. Expanding to second order in Y^μ, we have

$$\begin{aligned} S_P[G,\bar{x}+\sqrt{\alpha'}Y,\eta] &= -\frac{1}{4\pi}\int d^2\sigma\, \partial_\alpha Y^\mu \partial^\alpha Y^\nu \\ &\times \Big\{ G_{\mu\nu}(\bar{x}) + \underbrace{\sqrt{\alpha'}Y^\lambda \partial_\lambda G_{\mu\nu}(\bar{x})}_{\text{cubic vertex}} + \frac{\alpha'}{2}\underbrace{Y^\lambda Y^\rho \partial_\lambda\partial_\rho G_{\mu\nu}(\bar{x})}_{\text{quartic vertex}} + \mathcal{O}(\sqrt{\alpha'}^3) \Big\} \end{aligned} \tag{7.8}$$

We emphasize that this expansion in $\sqrt{\alpha'}Y$ amounts to perturbation theory from the two-dimensional worldsheet point of view, not in D-dimensional spacetime as studied in the previous section!

- One can read off an infinite number of couplings $G_{\mu\nu}(\bar{x}), \partial_\lambda G_{\mu\nu}(\bar{x}), \dots$ from $S_P[G, \bar{x}+\sqrt{\alpha'}Y, \eta]$ with generating function $G_{\mu\nu}(X)$
- Suppose the target-space metric fluctuates at length scales of $\partial_\lambda G_{\mu\nu} \sim \frac{1}{r}$, then the perturbative expansion is in the dimensionless "effective coupling" $\frac{\sqrt{\alpha'}}{r}$.
  - $\Rightarrow$ NLSM perturbation theory is reliable if $r \gg \sqrt{\alpha'}$, i.e. if the metric only fluctuates on sufficiently large length scales
  - $\Rightarrow$ once more, strings are seen to be fuzzy at length scales $\ll \sqrt{\alpha'}$ since this is where NLSM perturbation theory breaks down

The setup of worldsheet perturbation theory admits an alternative way of recovering the Einstein–Hilbert action of general relativity as $\alpha' \to 0$:

- the classical conformal symmetry of $S_P[G, X, \eta]$ might get broken at the quantum level since the interactions starting from $\partial_\lambda G_{\mu\nu}(\bar{x})$ yield divergences in loop diagrams which in turn introduce a scale μ upon regularization
- in order to compute the one-loop divergence of the two-point function $\langle YY \rangle$ or propagator, pick Riemann normal coordinates (that were probably ubiquitous in your general-relativity course)

$$G_{\mu\nu}(X) = \eta_{\mu\nu} - \frac{\alpha'}{3}\mathcal{R}_{\mu\lambda\nu\rho}(\bar{x})Y^\lambda Y^\rho + \mathcal{O}(\sqrt{\alpha'}^3) \tag{7.9}$$

  which identifies a four-point vertex $\mathcal{R}_{\mu\lambda\nu\rho}\partial_\alpha Y^\mu \partial^\alpha Y^\nu Y^\lambda Y^\rho$; the resulting one-loop correction is computed from the diagrams in figure 23 with dimensional regularization (i.e. taking a $(2+\epsilon)$-dimensional worldsheet) and evaluates as follows to leading orders in α':

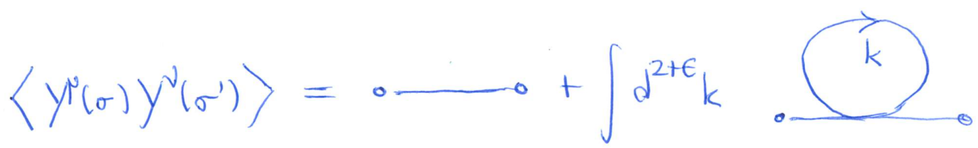


Figure 23: one-loop correction to the two-point function of the fluctuation $Y^\mu(\sigma)$ in Riemann normal coordinates

$$\langle Y^\mu(\sigma)Y^\nu(\sigma')\rangle = -\frac{\eta^{\mu\nu}}{2}\log|\sigma-\sigma'|^2 + \alpha'\frac{\eta^{\mu\nu}}{\epsilon} + \mathcal{O}(\sqrt{\alpha'}^3) \tag{7.10}$$

- the $\frac{1}{\epsilon}$ pole in the propagator can be cancelled by the following counterterm in the interacting NLSM:

$$\begin{aligned} S_P[G,X,\eta] &\to S_P[\eta,X,\eta] \\ &+ \frac{\alpha'}{12\pi}\int d^2\sigma \left(\mathcal{R}_{\mu\lambda\nu\rho}\partial_\alpha Y^\mu \partial^\alpha Y^\nu Y^\lambda Y^\rho - \frac{1}{\epsilon}\mathcal{R}_{\mu\nu}\partial_\alpha Y^\mu \partial^\alpha Y^\nu\right) + \mathcal{O}(\sqrt{\alpha'}^3) \end{aligned} \tag{7.11}$$

- regularization always introduces some scale μ

⇒ quantum conformal invariance requires $\mathcal{R}_{\mu\nu} = 0$

⇒ in this case, the target-space metric solves the vacuum Einstein equations

⇒ this is yet another derivation of general relativity coming from the $\alpha' \to 0$ limit of string theory (complementary to the study of graviton amplitudes)

7.2 B-field and dilaton backgrounds

Apart from the graviton, we also found the B-field and dilaton state Φ among the massless closed-string excitations. Extend the worldsheet action obtained from exponentiating graviton vertex operators by

$$\begin{aligned} S_\sigma[G, B, \Phi, X, g] = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-g} \Big\{ g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu G_{\mu\nu}(X) \\ + \varepsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu B_{\mu\nu}(X) + \alpha' \Phi(X) R^{(2)} \Big\} \end{aligned} \tag{7.12}$$

with the *two-dimensional* Ricci scalar $R^{(2)}$ as opposed to the one $\mathcal{R}$ in target space. While the B-field terms in the σ-model action can again be recognized as an exponentiated vertex operator, the dilaton term $\alpha'\Phi(X)R^{(2)}$ is more tricky to justify. As will be seen later on, its tree-level terms in worldsheet perturbation theory will cancel one-loop violations of conformal invariance coming from $G_{\mu\nu}(X), B_{\mu\nu}(X)$.

The B-field coupling generalizes the electromagnetic worldline action $\int d\tau \, A_\mu(X)\dot{X}^\mu = \int A_\mu \, dX^\mu$ with the pullback of the one-form $A = A_\mu \, dX^\mu$ in the integrand.

- Analogously, the pullback of the two-form $B = B_{\mu\nu} dX^\mu \wedge dX^\nu$ from spacetime to the worldsheet reproduces the B-field term of the above S_σ

$$\int B = \int B_{\mu\nu} \, dX^\mu \wedge dX^\nu = \int d^2\sigma \, B_{\mu\nu} \, \varepsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \tag{7.13}$$

- Gauge variations $\delta A_\mu = \partial_\mu \alpha$ of the components of an abelian gauge-field one-form A translate into $\delta A = d\alpha$; therefore, gauge invariance of the worldline action follows from the Stokes theorem

$$\delta \int_{\mathcal{C}} A = \int_{\mathcal{C}} d\alpha = \alpha \Big|_{\partial\mathcal{C}} = 0 \tag{7.14}$$

  As a generalization for the B-field, gauge variations $\delta B = dC$ become $\delta B_{\mu\nu} = \partial_\mu C_\nu - \partial_\nu C_\mu$ at the level of two-form components; as a consequence, the above B-field coupling in S_σ enjoys a gauge invariance

$$\delta \int_\Sigma B = \int_\Sigma dC = \int_{\partial\Sigma} C = 0 \tag{7.15}$$

  which corresponds to the double copy of $\delta\epsilon^\mu = p^\mu$ at the level of the B-field polarizations $\epsilon^{[\mu} \otimes \bar{\epsilon}^{\nu]}$.

- the gauge-invariant field strength $F = dA$ with components $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ generalizes as follows to a three-form field strength H of the B-field

$$H = dB \, , \qquad H_{\mu\nu\lambda} = \partial_\mu B_{\nu\lambda} + \partial_\nu B_{\lambda\mu} + \partial_\lambda B_{\mu\nu} \tag{7.16}$$

Let us now find the necessary conditions for conformal invariance of the σ-model action S_σ in (7.12): Instead of investigating specific one-loop divergences, we shall demand the energy-momentum tensor $T_{\alpha\beta}$ to be traceless (as implied by conformal invariance):

- each spacetime field brings its own *beta function*, i.e. tentative departure from conformal invariance

$$\langle T^\alpha{}_\alpha \rangle = -\frac{1}{2\alpha'}\beta_{\mu\nu}(G) g^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu - \frac{1}{2\alpha'}\beta_{\mu\nu}(B)\varepsilon^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu - \frac{1}{2}\beta(\Phi) R^{(2)} \tag{7.17}$$

- to one-loop order in NLSM perturbation theory (i.e. first order in α'),

$$\begin{aligned} \beta_{\mu\nu}(G) &= \alpha' \mathcal{R}_{\mu\nu} + 2\alpha' \nabla_\mu \nabla_\nu \Phi - \frac{\alpha'}{4} H_{(\mu}{}^{\lambda\rho} H_{\nu)\lambda\rho} + \mathcal{O}(\alpha'^2) \\ \beta_{\mu\nu}(B) &= -\frac{\alpha'}{2}\nabla^\lambda H_{\lambda\mu\nu} + \alpha'(\nabla^\lambda \Phi) H_{\lambda\mu\nu} + \mathcal{O}(\alpha'^2) \\ \beta(\Phi) &= \frac{D-26}{6} - \frac{\alpha'}{2}\nabla^2\Phi + \alpha' \nabla_\mu \Phi \nabla^\mu \Phi - \frac{\alpha'}{24} H_{\mu\nu\rho} H^{\mu\nu\rho} + \mathcal{O}(\alpha'^2) \end{aligned} \tag{7.18}$$

- consistent backgrounds must preserve Weyl invariance

$$\beta_{\mu\nu}(G) = \beta_{\mu\nu}(B) = \beta(\Phi) = 0 \tag{7.19}$$

  which, by the first term of $\beta_{\mu\nu}(G) = \alpha' \mathcal{R}_{\mu\nu} + \ldots$, extends Einstein's equations to B-field and dilaton

- a particularly simple class of solutions is given by $D = 26$ Minkowski background with $B = 0$ and constant (i.e. X^μ-indepdendent) dilaton profile $\Phi(X) \to \Phi_0$

$$S_\sigma[\eta, 0, \Phi_0, X, g] = S_P[\eta, X, g] - \frac{1}{4\pi}\int d^2\sigma \sqrt{-g}\, \Phi_0 R^{(2)} \tag{7.20}$$

  The second term is topological and gives the Euler characteristics

$$\chi = \frac{1}{4\pi}\int d^2\sigma \sqrt{-g} R^{(2)} = 2 - 2g \tag{7.21}$$

  where the genus g is the order in string perturbation theory. This backs up the earlier claim that the dilaton VEV sets the string coupling.

Given that each beta function induces a field equation, the σ-model action on the worldsheet offers a new approach to the low-energy effective action: Which spacetime action ($\int d^D X$ of some Lagrangian) yields $\beta(*) = 0$ as its equations of motion?

- alternative to reconstructing spacetime effective actions from amplitudes

- α'-expansion $\leftrightarrow$ loop order in the σ model, e.g. general relativity represented by the Einstein-Hilbert action $\mathcal{R}$ stems from the one-loop order $\sim \alpha'$ while the $\zeta_3 \alpha'^4 \mathcal{R}^4$ interaction stems from four loops in the worldsheet σ model

- we obtain $\beta(*) = 0$ to linear order in α' from

$$S_{\text{eff}}[G, B, \Phi] = \frac{1}{2\kappa_0^2}\int d^D X \sqrt{-\det G}\, e^{-2\Phi}\Big\{\mathcal{R} - \frac{1}{12} H_{\mu\nu\lambda} H^{\mu\nu\lambda} + 4\partial_\mu \Phi \partial^\mu \Phi - \frac{2(26-D)}{3\alpha'} + \mathcal{O}(\alpha'^2)\Big\} \tag{7.22}$$

However, the kinetic terms in S_{eff} do not have the canonical form and sign which can be fixed through the following change of frame

- redefine $\Phi = \Phi_0 + \tilde{\Phi}$ and

$$\tilde{G}_{\mu\nu} = \exp\Big(-\frac{4\tilde{\Phi}}{D-2}\Big) G_{\mu\nu} \tag{7.23}$$

- results in

$$S_{\text{eff}}[\tilde{G}, B, \tilde{\Phi}] = \frac{1}{2\kappa^2} \int \mathrm{d}^D X \sqrt{-\det \tilde{G}} \Big\{ \tilde{R} - \frac{1}{12} e^{-\frac{8\tilde{\Phi}}{D-2}} H_{\mu\nu\lambda} H^{\mu\nu\lambda} \\ - \frac{4}{D-2} \partial_\mu \tilde{\Phi} \partial^\mu \tilde{\Phi} + \frac{4(D-1)}{D-2} \nabla^2 \tilde{\Phi} - e^{\frac{4\tilde{\Phi}}{D-2}} \frac{2(D-26)}{3\alpha'} \Big\} \tag{7.24}$$

  with gravitational coupling $\kappa^2 = \kappa_0^2 e^{2\Phi_0} = 32\pi^2 G_{\text{Newton}}$

- original fields $\{G, B, \Phi\}$ are said to be in "string frame", the redefined ones $\{\tilde{G}, B, \tilde{\Phi}\}$ in "Einstein frame"

7.3 Background gauge fields and Born-Infeld

Similar to closed-string backgrounds, can exponentiate photon vertex operators on Dp branes

$$\Rightarrow \quad S_{\text{endpt}}[A, X] = \int_{\partial\Sigma} d\tau \, A_a(X) \dot{X}^a \,, \qquad a = 0, 1, \ldots, p \tag{7.25}$$

$\Rightarrow$ open-string endpoints (with worldlines $\partial\Sigma$) charged under A_a

Such electromagnetic backgrounds lead to deformed Neumann boundary conditions (with $F_{ab} = \partial_a A_b - \partial_b A_a$ and $\partial_a = \frac{\partial}{\partial X^a}$) from imposing the boundary terms to cancel

$$\begin{aligned} &\delta\Big(-\frac{1}{4\pi\alpha'} \int_\Sigma d^2\sigma \, \partial_\alpha X_\mu \partial^\alpha X^\mu + \int_{\partial\Sigma} d\tau \, A_a(X) \dot{X}^a \Big) \\ &= \int_{\partial\Sigma} d\tau \left\{ -\frac{1}{2\pi\alpha'} \delta X_\mu \partial_\sigma X^\mu + \partial_b A_a \delta X^b \dot{X}^a - \delta X^a \frac{dA_a}{d\tau} \right\} \\ &= -\frac{1}{2\pi\alpha'} \int_{\partial\Sigma} d\tau \, \delta X_a \Big(\partial_\sigma X^a - 2\pi\alpha' F^{ab} \partial_\tau X_b \Big) \end{aligned} \tag{7.26}$$

In passing to the third line, we have used that δX_μ vanishes for Dirichlet directions, that is why we are left with the a-index of $\delta X_a \partial_\sigma X^a$. These boundary terms necessitate a modification of Neumann boundary conditions to

$$\partial_\sigma X^a = 2\pi\alpha' F^{ab} \partial_\tau X_b \tag{7.27}$$

in the place of $\partial_\sigma X^a = 0$ in absence of A_a backgrounds.

Let us study the impact of the F_{ab}-dependent Neumann boundary conditions on the propagator in complex variables $z = i(\tau + \sigma)$ and $\bar{z} = i(\tau - \sigma)$:

- recall the open-string Green function at $A_a = 0$ subject to $(\partial_z - \partial_{\bar{z}}) G^N_{\text{op}}(z, w)\big|_{\sigma=0} = 0$:

$$G^N_{\text{op}}(z, w) = -\frac{\alpha'}{2} \Big(\log|z-w|^2 + \log|z-\bar{w}|^2 \Big) \tag{7.28}$$

- in presence of F^{ab}, need a matrix-valued Green function that vanishes at the boundary $z = \bar{z}$

$$\Big(\delta_{ab}(\partial_z - \partial_{\bar{z}}) - 2\pi\alpha' F_{ab}(\partial_z + \partial_{\bar{z}}) \Big) G^{bc}_{\text{op}}(z, w)\big|_{\sigma=0} = 0 \tag{7.29}$$

- this is accomplished by the following expression

$$G^{ab}_{\text{op}}(z, w) = -\frac{\alpha'}{2} \left\{ \delta^{ab} \log|z-w|^2 + \left(\frac{1 - 2\pi\alpha' F}{1 + 2\pi\alpha' F} \right)^{ab} \log(\bar{z}-w) + \left(\frac{1 + 2\pi\alpha' F}{1 - 2\pi\alpha' F} \right)^{ab} \log(z-\bar{w}) \right\} \tag{7.30}$$

  where the appearance of F_{ab} in the denominator is understood via power-series expansion $\frac{1}{1-2\pi\alpha' F} = \sum_{n=0}^\infty (2\pi\alpha')^n F^n$ and matrix multiplication, e.g. $(F^2)_{ab} = F_{ac} F^c{}_b$.

We shall now explore the leading-order contribution of F^{ab} to the beta function:

- expand $S_{\rm endpt}[A,X]$ around a background configuration $\bar{x}$ in $X^a = \bar{x}^a + \sqrt{\alpha'}Y^a$ which obeys $\partial_\alpha \partial^\alpha \bar{x}^a = A_a(\bar{x}) = 0$; then, at first order in α',

$$S_{\rm endpt}[A,\bar{x}+\sqrt{\alpha'}Y]\,\big|_{\alpha'} = \int_{\partial\Sigma} d\tau\,\Big(\alpha' \partial_a A_b Y^a \dot{Y}^b + \frac{\alpha'}{2}\partial_a \partial_b A_c Y^a Y^b \dot{\bar{x}}^c\Big) \tag{7.31}$$

- integrate by parts for one of the terms in $\partial_a A_b Y^a \dot{Y}^b = \frac{1}{2}\partial_a A_b Y^a \dot{Y}^b + \frac{1}{2}\partial_a A_b Y^a \dot{Y}^b$

$$\int_{\partial\Sigma} d\tau\,\partial_a A_b Y^a \dot{Y}^b = \frac{1}{2}\int_{\partial\Sigma} d\tau\,\Big\{\partial_a A_b Y^a \dot{Y}^b - \partial_a A_b \dot{Y}^a Y^b - \partial_a(\partial_c A_b \dot{\bar{x}}^c) Y^a Y^b\Big\} \tag{7.32}$$

- manifest gauge invariance

$$S_{\rm endpt}[A,\bar{x}+\sqrt{\alpha'}Y]\,\big|_{\alpha'} = \frac{\alpha'}{2}\int_{\partial\Sigma}\Big(F_{ab} Y^a \dot{Y}^b + \partial_a F_{bc} Y^a Y^b \dot{\bar{x}}^c\Big) \tag{7.33}$$

- the one-loop divergence in σ-model perturbation theory can be obtained from replacing

$$\begin{aligned} Y^a(z)Y^b(w) \rightarrow G^{ab}_{\rm op}(z\in\mathbb{R},w\in\mathbb{R}) &= -\alpha'\log(z{-}w)\Big\{\delta^{ab} + \frac{1}{2}\Big(\frac{1-2\pi\alpha' F}{1+2\pi\alpha' F}\Big)^{ab} + \frac{1}{2}\Big(\frac{1+2\pi\alpha' F}{1-2\pi\alpha' F}\Big)^{ab}\Big\} \\ &= -2\alpha'\log(z{-}w)\Big(\frac{1}{1-4\pi^2\alpha'^2 F^2}\Big)^{ab} \end{aligned} \tag{7.34}$$

- get counterterm and beta function from insertion into the term $\partial_a F_{bc} Y^a Y^b \dot{x}^c$ of $S_{\rm endpt}[A,\bar{x}{+}\sqrt{\alpha'}Y]$,

$$\beta_c(A)\,\big|_{\alpha'} \sim \alpha' \partial_a F_{bc}\Big(\frac{1}{1-4\pi^2\alpha'^2F^2}\Big)^{ab} = 0 \tag{7.35}$$

The effective action defined by its equation of motion $\beta_c(A)=0$ is known as the *Born-Infeld action*, see exercise F.2

$$S_{\rm BI}[A] = -T_p \int d^{p+1}\xi\,\sqrt{-\det(\eta_{ab}+2\pi\alpha' F_{ab})} \tag{7.36}$$

with tension T_p and coordinates ξ^a on the $(p{+}1)$-dimensional worldvolume of the brane, which can for instance be chosen as $\xi^a = X^a\ \forall\ a\leq p$.

- as explored in exercise F.2, the low-energy expansion

$$\begin{aligned} S_{\rm BI}[A] = -T_p \int d^{p+1}\xi\,\Big\{&1 + (\pi\alpha')^2 F_{ab}F^{ab} \\ &+ (\pi\alpha')^4\Big(\frac{1}{2}F_{ab}F^{ab}F_{cd}F^{cd} - 2F^{ab}F_{bc}F^{cd}F_{da}\Big) + \mathcal{O}(\alpha'^6)\Big\} \end{aligned} \tag{7.37}$$

  augments the Maxwell action by an infinite series of corrections starting with $\frac{1}{2}({\rm tr}(F^2))^2 - 2{\rm tr}(F^4)$, where ${\rm tr}(\ldots)$ denotes a trace over Lorentz indices $a,b,\ldots$

- in the above discussion, $S_{\rm BI}[A]$ arises as the one-loop beta function in the σ-model but is in fact an exact result in case of constant F_{ab} (i.e. for $\partial_a F_{bc}=0$)

We shall now incorporate transverse fluctuations on the brane which were related in earlier lectures to the oscillators of

$$\phi^I = \frac{X^I}{2\pi\alpha'}\,,\qquad I = p{+}1,\ldots,D{-}1 \tag{7.38}$$

- generalizes $S_{\rm BI}[A]$ to the *Dirac-Born-Infeld* action

$$S_{\rm DBI}[A,\phi] = -T_p \int d^{p+1}\xi \sqrt{-\det\left(\eta_{ab} + 2\pi\alpha' F_{ab} + (2\pi\alpha')^2 \frac{\partial\phi^I}{\partial\xi^a}\frac{\partial\phi_I}{\partial\xi^b}\right)} \tag{7.39}$$

  where $\eta_{ab} + (2\pi\alpha')^2 \frac{\partial\phi^I}{\partial\xi^a}\frac{\partial\phi_I}{\partial\xi^b}$ may be interpreted as the induced metric on the brane

- in the low-energy limit reduces to the action of Maxwell plus free scalars

$$S_{\rm DBI}[A,\phi] = -T_p \int d^{p+1}\xi \left\{1 + (2\pi\alpha')^2\left(\frac{1}{4}F_{ab}F^{ab} + \frac{1}{2}\partial_a\phi^I\partial^a\phi_I\right) + \mathcal{O}(\alpha'^4)\right\} \tag{7.40}$$

Another simple extension of Born-Infeld concerns the following coupling to massless closed-string modes

$$S_{\rm DBI}[A,\phi] = -T_p \int d^{p+1}\xi \, e^{-\tilde\Phi}\sqrt{-\det\left(\frac{\partial X^\mu}{\partial\xi^a}\frac{\partial X^\nu}{\partial\xi^b}G_{\mu\nu}(X) + 2\pi\alpha' F_{ab} + B_{ab}\right)} \tag{7.41}$$

- dilaton dependence $T_p e^{-\tilde\Phi} \sim e^{-\Phi_0}$ lines up with tree level of open-string perturbation theory

- B-field coupling dictated by gauge invariance of

$$\underbrace{-\frac{1}{4\pi\alpha'}\int_\Sigma d^2\sigma\,\varepsilon^{\alpha\beta}\partial_\alpha X^\mu\partial_\beta X^\nu B_{\mu\nu}(X)}_{\substack{\text{under } \delta B_{\mu\nu} = \partial_\mu C_\nu - \partial_\nu C_\mu \\ \text{varies by } \frac{1}{2\pi\alpha'}\int_{\partial\Sigma} d\tau\,\dot X^a C_a}} + \underbrace{\int_{\partial\Sigma} d\tau\, A_a(X)\dot X^a}_{\substack{\text{get compensating} \\ \text{trf. if } \delta A_a = -\frac{C_a}{2\pi\alpha'}}} \tag{7.42}$$

  since the action is exclusively written in terms of the gauge invariant combination $B_{ab} + 2\pi\alpha' F_{ab}$.

7.4 Point particles compactified on a circle

How does point-particle physics in some D-dimensional spacetime M_D change upon compactification of one spatial direction on a circle of radius r?

$$M_D \to M_{D-1}\times S^1\,, \qquad X_{D-1} \cong X_{D-1} + 2\pi r \tag{7.43}$$

Consider general relativity with all fields independent on X_{D-1}:

$$D\text{-dim metric } G \quad \longrightarrow \quad \begin{cases} (D{-}1)\text{-dim metric } \tilde G_{\mu\nu} \\ \text{gauge field } A_\mu \text{ in } D{-}1 \text{ dim} \\ \text{scalar } \sigma \end{cases} \tag{7.44}$$

The line element in terms of the $(D{-}1)$-dim fields is given as follows:

$$ds^2 = \tilde G_{\mu\nu}\,dX^\mu\,dX^\nu + e^{2\sigma}(dX_{D-1} + A_\mu\,dX^\mu)^2 \tag{7.45}$$

with $\mu,\nu = 0,1,2,\ldots,D{-}2$ and $\tilde G_{\mu\nu}$ in Einstein frame.

- under diffeomorphisms $\delta X^\mu = -\Lambda^\mu(X^\nu)$ and $\delta X_{D-1} = -\Lambda(X^\nu)$,

$$\delta\tilde G_{\mu\nu} = \nabla_\mu\Lambda_\nu + \nabla_\nu\Lambda_\mu\,, \qquad \delta A_\mu = \partial_\mu\Lambda \tag{7.46}$$

  i.e. the $(D{-}1)$-cpt. of the diffeomorphism parameter Λ_μ takes the role of the gauge scalar for A_μ

- interactions of $\tilde{G}_{\mu\nu}, A_\mu, \sigma$ are determined by dimensional reduction of $\mathcal{L}_{\text{EH}}$, in particular by the following behaviour of the Ricci scalar

$$\mathcal{R}^{(D)} = \mathcal{R}^{(D-1)} - 2e^{-\sigma}\nabla^2 e^\sigma - \frac{1}{4}e^{2\sigma}F_{\mu\nu}F^{\mu\nu} \tag{7.47}$$

such that the Einstein-Hilbert action becomes (recall that the fields are taken to be X_{D-1}-independent)

$$\begin{aligned} S_{\text{EH}} &= \frac{1}{2\kappa^2}\int d^DX\,\sqrt{-\det G}\,\mathcal{R}^{(D)} \\ &= \frac{2\pi r}{2\kappa^2}\int d^{D-1}X\,\sqrt{-\det\tilde{G}}\,e^\sigma\left\{\mathcal{R}^{(D-1)} - \frac{1}{4}e^{2\sigma}F_{\mu\nu}F^{\mu\nu} + \partial_\mu\sigma\partial^\mu\sigma\right\} \end{aligned} \tag{7.48}$$

This is the action of $(D-1)$-dimensional Einstein gravity coupled to a $U(1)$ gauge field and a scalar σ. The factor of $2\pi r$ coming from the X_{D-1} integral effectively rescales the gravitational coupling κ, i.e. the normalization factor takes the standard form $\frac{1}{2\kappa_{D-1}^2}$ if we set $\kappa_{D-1} = \frac{\kappa}{\sqrt{2\pi r}}$.

- The VEV $\langle\sigma\rangle$ controls the radius r of the circular dimension since σ sets the diagonal entry of the metric in the $(D-1)$-direction. There is no potential in the reduced S_{EH} that could select a preferred VEV. This leaves the freedom to change r, i.e. there is no mechanism for *moduli stabilization* (where moduli in general refer to massless scalars like the dilaton Φ or here the radion σ).

- A reduction similar to that of G can be performed for the B-field (we have seen a typical field-theory Lagrangian at the first order of the earlier S_{eff} in α')

  $\Rightarrow$ another $U(1)$ gauge field $\tilde{A}_\mu$ from $B_{\mu,D-1}$

  $\Rightarrow$ massless d.o.f. of closed bosonic strings on a circle $= \{\tilde{G}_{\mu\nu}, B_{\mu\nu}, A_\mu, \tilde{A}_\mu, \Phi, \sigma\}$

Let us now consider X_{D-1}-dependent fields, whose main features on a circle are already illustrated by a scalar example. Periodicity in $X_{D-1} \cong X_{D-1} + 2\pi r$ is implemented via Fourier expansion

$$\varphi(X^\mu, X_{D-1}) = \sum_{n=-\infty}^{\infty} \varphi_n(X^\mu)\exp\left(\frac{inX_{D-1}}{r}\right) \tag{7.49}$$

where each Fourier mode $\varphi_n(X^\mu)$ is considered as a separate scalar field from the $(D-1)$-dimensional point of view.

- reduction of the kinetic term

$$\begin{aligned} -\int d^DX\,\varphi\Box_D\varphi &= \int d^DX\left\{\partial_\mu\varphi\partial^\mu\varphi + \partial_{D-1}\varphi\partial_{D-1}\varphi\right\} \\ &= 2\pi r\int d^{D-1}X\sum_{n=-\infty}^{\infty}\left(\partial_\mu\varphi_n\partial^\mu\varphi_{-n} + \frac{n^2}{r^2}\varphi_n\varphi_{-n}\right) \end{aligned} \tag{7.50}$$

- from $(D-1)$-dimensional point of view: ∞ tower of massive scalars with $m^2 = \frac{n^2}{r^2},\ n \in \mathbb{Z}$

  $\Rightarrow$ different scaling of the masses as for string spectra where M^2 grow with $n \in \mathbb{N}_0$ rather than n^2

- massive X_{D-1}-dependent states are called "Kaluza-Klein" modes – they are negligible at energies $\ll \frac{1}{r}$ or distances $\gg r$, that is why signals of small circles are very hard to detect

- the n'th mode carries charge $\frac{n}{r}$ under $A_\mu = G_{\mu,D-1}$ since

$$\delta X_{D-1} = -\Lambda \quad \Rightarrow \quad \begin{cases} A_\mu \to A_\mu + \partial_\mu\Lambda \\ \varphi_n \to \exp(-\frac{in\Lambda}{r})\varphi_n \end{cases} \tag{7.51}$$

7.5 Strings compactified on a circle and T duality

As the main result of this section, we will see that strings have a much richer interplay with circular dimensions than point particles and cannot distinguish radius r from $\frac{\alpha'}{r}$.

The starting point is again a D-dimensional spacetime with a circular direction $X_{D-1} \cong X_{D-1} + 2\pi r$.

- just like for point particles, need to quantize momentum $p_{D-1} = \frac{n}{r}$, $n \in \mathbb{Z}$, e.g. to get single-valued $e^{ip_{D-1}X_{D-1}}$ in the plane-wave part of vertex operators
- closed strings may wind $m \in \mathbb{Z}$ times around a circle, i.e. the admissible boundary conditions are

$$X_{D-1}(\sigma + 2\pi) = X_{D-1}(\sigma) + 2\pi r m \tag{7.52}$$

- combining both motion and winding in the $(D-1)$-direction

$$X_{D-1}(\tau, \sigma) = x_{D-1} + \frac{\alpha' n}{r}\tau + m r \sigma + \text{oscillators} \tag{7.53}$$

- splitting into left- and right-movers $X_{D-1}(\tau,\sigma) = X^L_{D-1}(\sigma^+) + X^R_{D-1}(\sigma^-)$ with $\sigma^\pm = \tau \pm \sigma$

$$X^L_{D-1}(\sigma^+) = \frac{x_{D-1}}{2} + \frac{\alpha'}{2} p_L \sigma^+ + \text{oscillators} \ \sim \ e^{\pm in\sigma^+} \tag{7.54}$$

$$X^R_{D-1}(\sigma^-) = \frac{x_{D-1}}{2} + \frac{\alpha'}{2} p_R \sigma^- + \text{oscillators} \ \sim \ e^{\pm in\sigma^-} \tag{7.55}$$

- momentum- and winding-numbers $n, m \in \mathbb{Z}^2$ yield the following lattice of admissible momenta

$$p_L = \frac{n}{r} + \frac{mr}{\alpha'}\,, \qquad p_R = \frac{n}{r} - \frac{mr}{\alpha'} \tag{7.56}$$

From the $(D-1)$-dimensional perspective, masses only involve p_μ-components $\mu = 0, 1, 2, \ldots, D-2$

- Virasoro conditions $L_0 = \bar{L}_0 = 1$ yield modified mass formulae

$$M^2 = -p_\mu p^\mu = p_L^2 + \frac{4}{\alpha'}(\tilde{N}-1) = p_R^2 + \frac{4}{\alpha'}(N-1) \tag{7.57}$$

  with $\tilde{N}, N$ the left- and right-moving oscillator numbers

- with simultaneous winding and momentum, can violate level matching

$$N - \tilde{N} = \frac{\alpha'}{4}(p_L^2 - p_R^2) = mn \tag{7.58}$$

- eliminating the mn term, we can rewrite the mass formula as

$$M^2 = \frac{n^2}{r^2} + \frac{m^2 r^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2) \tag{7.59}$$

  and interpret the first two terms as follows: The mass^2 contribution of $\frac{n^2}{r^2}$ is identical to the Kaluza-Klein spectrum of point particles $m^2 = \frac{n^2}{r^2}$. The mass-contribution of winding strings with tension T is $2\pi r|m|T$; by matching with the second term in M^2, the tension can be read off to be $T = \frac{1}{2\pi\alpha'}$.

- at $p_L = p_R = 0$, recover the massless states from KK reduction of the D-dimensional matrix $G_{\mu\nu}$ and B-field,

$$\begin{aligned}
&V_\zeta \sim \zeta_{\mu\nu} : \partial_z X^\mu \partial_{\bar{z}} X^\nu e^{ip\cdot X} :\,, \qquad \mu,\nu \le D-2 \ \leftrightarrow \ (D-1)\text{-dim } G \ \& \ B \\
&V_\pm \sim \epsilon_\mu : (\partial_z X_{D-1} \partial_{\bar{z}} X^\mu \pm \partial_z X^\mu \partial_{\bar{z}} X_{D-1}) e^{ip\cdot X} : \ \leftrightarrow \ \text{gauge fields } A_\mu, \tilde{A}_\mu \\
&V_\sigma \sim : \partial_z X_{D-1} \partial_{\bar{z}} X_{D-1} e^{ip\cdot X} : \ \leftrightarrow \ \text{radion scalar}
\end{aligned} \tag{7.60}$$

Relate p_L, p_R to charges w.r.t. $A_\mu, \tilde{A}_\mu$

- consider X_{D-1}-dependent tachyon modes

$$V_{m,n} =: e^{ip_L X^L_{D-1} + ip_R X^R_{D-1}} e^{ip\cdot X} : , \qquad (p_L, p_R) = \left(\frac{n}{r} + \frac{mr}{\alpha'} , \; \frac{n}{r} - \frac{mr}{\alpha'} \right) \tag{7.61}$$

- can use three-point couplings

$$\langle c\tilde{c} V_\pm(z_1) c\tilde{c} V_{m,n}(z_2) c\tilde{c} V_{-m,-n}(z_3) \rangle \sim \epsilon \cdot (p_2 - p_3)(p_L \pm p_R) \tag{7.62}$$

  to read off the charges of $V_{m,n}$ w.r.t. the two gauge fields:

$$\begin{cases} p_L + p_R = \frac{2n}{r} \;\; \text{under} \;\; V_+ \;\; \leftrightarrow \;\; \text{gauge field from } G \\ p_L - p_R = \frac{2mr}{\alpha'} \;\; \text{under} \;\; V_- \;\; \leftrightarrow \;\; \text{gauge field from } B \end{cases} \tag{7.63}$$

  While the charge of Kaluza-Klein modes w.r.t. the gauge field from the metric was already observed for point particles, the charge of winding modes w.r.t. the gauge field from the B-field is a new feature of strings.

We find enhanced massless spectra at special values or r, e.g.

$$n = 0 = N = \tilde{N} \quad \Rightarrow \quad M^2 = \left(\frac{mr}{\alpha'} \right)^2 - \frac{4}{\alpha'} = 0 \quad \text{if} \quad r^2 = \frac{4\alpha'}{m^2} \tag{7.64}$$

$$m = 0 = N = \tilde{N} \quad \Rightarrow \quad M^2 = \frac{n^2}{r^2} - \frac{4}{\alpha'} \quad = \quad 0 \quad \text{if} \quad r^2 = \frac{\alpha' n^2}{4} \tag{7.65}$$

The richest collection of massless states occurs at the *self-dual radius* $r = \sqrt{\alpha'}$. As will be explained below, the upshot is that

$$\Rightarrow \quad U(1) \times U(1) \text{ gauge symmetry of } V_\pm \text{ enhanced to } SU(2) \times SU(2)$$

- at $r = \sqrt{\alpha'}$, the momenta simplify to $p_L \to \frac{m+n}{\sqrt{\alpha'}}$ and $p_R \to \frac{n-m}{\sqrt{\alpha'}}$, this can be used to get integer conformal weights for

$$h(e^{ip_L X^L_{D-1}}) = \frac{(m+n)^2}{4} , \qquad \bar{h}(e^{ip_R X^R_{D-1}}) = \frac{(m-n)^2}{4} \tag{7.66}$$

- setting $m = n = \pm 1$, obtain gauge fields with (V_+, V_-)-charges $(\pm 1, \mp 1)$ and vertex operators

$$V^L_\pm \sim \epsilon_\mu : \partial_{\bar{z}} X^\mu e^{\pm \frac{2i}{\sqrt{\alpha'}} X^L_{D-1}} e^{ip\cdot X} : \tag{7.67}$$

- setting $n = -m = \pm 1$, obtain gauge fields with (V_+, V_-)-charges $(\pm 1, \pm 1)$ and vertex operators

$$V^R_\pm \sim \epsilon_\mu : \partial_z X^\mu e^{\pm \frac{2i}{\sqrt{\alpha'}} X^R_{D-1}} e^{ip\cdot X} : \tag{7.68}$$

- charged massless spin-one states only make sense within non-abelian gauge symmetry: The four extra gauge fields at the self-dual radius enhance

$$r = \sqrt{\alpha'} \quad \Rightarrow \quad U(1) \times U(1) \to SU(2) \times SU(2) \tag{7.69}$$

As will be argued below, the exchange of "big and small circles" $r \to \frac{\alpha'}{r}$ known as *T duality* is a symmetry of both string spectra and string interactions.

- M^2 at $r \to \frac{\alpha'}{r}$ simply has $m \leftrightarrow n$ interchanged. Since both of $m, n \in \mathbb{Z}$ run over the same range, T duality keeps the spectrum invariant and simply changes the interpretation of states as winding or moving in the circular $(D-1)$-direction.

- In the large-radius limit $r \to \infty$, momentum modes $M^2 = \frac{n^2}{r^2}$ form mass continuum as usual for large dimensions. In the opposite limit $r \to 0$, however, it is the winding modes which form a mass continuum as if a large extra dimension opened up.

- T duality maps $(p_L, p_R) \to (p_L, -p_R)$ and in fact

$$X_{D-1} = X^L_{D-1} + X^R_{D-1} \to Y_{D-1} = X^L_{D-1} - X^R_{D-1} \tag{7.70}$$

  Since the OPEs for X_{D-1} and Y_{D-1} are the same, both the correlation functions of vertex operators and the resulting amplitudes are unchanged by T duality, that is why it is also a symmetry of string interactions.

- From the compactified Einstein-frame action

$$S_{\text{eff}}[G, B, \Phi] \sim \int d^{D-1}X \, 2\pi i \, e^{-2\Phi} \left\{ \frac{\sqrt{-\det G}}{2\kappa_0^2} \mathcal{R} + \ldots \right\} \tag{7.71}$$

  the response of the dilaton to T duality must leave $re^{-2\Phi}$ invariant; hence, the dilaton transforms by a shift

$$\Phi \to \Phi + \frac{1}{2} \log \frac{\alpha'}{r^2} \tag{7.72}$$

- Given that $\partial_\alpha X_{D-1} = \varepsilon_{\alpha\beta} \partial^\beta Y_{D-1}$ and $X_{D-1} \leftrightarrow Y_{D-1}$ under T duality, it also interchanges $\partial_\tau \leftrightarrow \partial_\sigma$ and therefore Dirichlet and Neumann boundary conditions:

$$\text{Dp brane} \to \begin{cases} \text{D(p+1) brane} & : \ \text{T duality} \perp \text{brane} \\ \text{D(p−1) brane} & : \ \text{T duality} \parallel \text{brane} \end{cases} \tag{7.73}$$

- For superstrings, T duality relates two different theories (so-called type IIA $\leftrightarrow$ type IIB, see later) and different target-space geometries (mirror symmetry among Calabi-Yau manifolds with big impact on mathematics).

8 Basics of the RNS superstring

8.1 Overview

Bosonic strings suffer from tachyons and cannot accommodate spacetime fermions (quarks, leptons, etc.), and both of these shortcomings can be cured by superstrings.

A key ingredient of superstrings is supersymmetry (SUSY), an extension of the Poincaré symmetry (Lorentz group and translations) that relates bosons and fermions. As will be detailed in the rest of this course, one will have to distinguish two incarnations of SUSY

- worldsheet SUSY (in 2 dimensions), e.g. $X^\mu(\tau, \sigma) \leftrightarrow \Psi^\mu(\tau, \sigma)$, with a worldsheet spinor Ψ^μ as a SUSY partner

- spacetime SUSY (for the purpose of this course, in 4 or 10 dimensions), relating for instance the photon (spin 1) to a fermionic spin-1/2 state dubbed *photino*

There are roughly speaking three formulations of superstrings in Minkowski background

- Ramond-Neveu-Schwarz (RNS): manifest worldsheet SUSY, hidden spacetime SUSY
- Green-Schwarz (GS): manifest spacetime SUSY, no worldsheet SUSY, usually break $SO(1, D-1)$ covariance upon quantization ($\rightarrow$ lightcone approach similar to our discussion of lightcone quantization of the bosonic string)
- pure-spinor formalism (Berkovits 2000): manifest spacetime SUSY and $SO(1,9)$ covariance, no manifest worldsheet SUSY; the pure-spinor formulation is currently leading in the evaluation of multi-loop and -leg string amplitudes

There are five superstring theories; their rough characterization involves the usual decomposition of the closed-string spectrum into

$$|\text{closed}\rangle \in \left\{ \begin{smallmatrix} L\text{: left-} \\ \text{movers} \end{smallmatrix} \right\} \otimes \left\{ \begin{smallmatrix} R\text{: right-} \\ \text{movers} \end{smallmatrix} \right\} \tag{8.1}$$

- both L and R contain Weyl-spinors of $SO(1, D-1)$, must decide for one chirality each[4]; depending on the relative chirality of L vs. R, get
  - same chirality of L and R: type-IIB superstrings
  - opposite chirality of L and R: type-IIA superstrings

  Dp branes in type IIB / IIA have odd / even p, respectively!
- only one chirality choice for open superstrings: "type I" = type IIB with spacetime filling D9 brane + "orientifold plane" + gauge group $SO(32)$ (the latter arising as a consistency condition from cancellations of gauge anomalies and UV divergences of one-loop open-string amplitudes)
- can combine left-movers of bosonic strings with right-movers of superstrings $\Rightarrow$ "heterotic strings" (het), two possible gauge groups $SO(32)$ or $E_8 \times E_8$

There is a web of dualities among superstring theories and beyond:

- T duality $r \rightarrow \frac{\alpha'}{r}$ exchanges type IIA $\leftrightarrow$ type IIB (cf. their respective Dp brane content) as well as $\text{het}_{SO(32)} \leftrightarrow \text{het}_{E_8 \times E_8}$
- S duality among other things acts on the string coupling $g_s \rightarrow \frac{1}{g_s}$ and relates

  $$\text{type I at } \left\{ \begin{smallmatrix} \text{weak} \\ \text{strong} \end{smallmatrix} \right\} \text{ coupling} \quad \stackrel{\text{S-duality}}{\longleftrightarrow} \quad \text{het}_{SO(32)} \text{ at } \left\{ \begin{smallmatrix} \text{strong} \\ \text{weak} \end{smallmatrix} \right\} \text{ coupling} \tag{8.2}$$

  S duality maps the type IIB theory to itself and exchanges weak $\leftrightarrow$ strong coupling
- Since both r and g_s are dynamically determined by scalar VEVs, T- and S-duality relate different phases of the same theory.
- For type IIA and $\text{het}_{E_8 \times E_8}$ at strong coupling g_s, an extra dimension of size $\ell_s g_s$ emerges (with string length-scale ℓ_s); the overarching 11-dimensional quantum field theory is referred to as *M theory*.

[4]In even dimensions, a Dirac spinor is the direct sum of a left- and right-handed Weyl spinor which in turn furnish irreducible representations of the even-dimensional Lorentz groups. The chirality or "handedness" is the eigenvalue of the Dirac matrix γ_{D+1} in even D, namely ± 1 for left- and right-handed Weyl spinors.

- The earlier theories are reached via dualities and different limits of M theory as visualized in figure 24 below; also: 11-dimensional supergravity (SUSY version of gravity) arises from the $\alpha' \to 0$ limit of M theory. Note that $D = 11$ is the highest dimension where gravitons can be embedded into supersymmetry multiplets without running into spin $\geq \frac{5}{2}$ states.

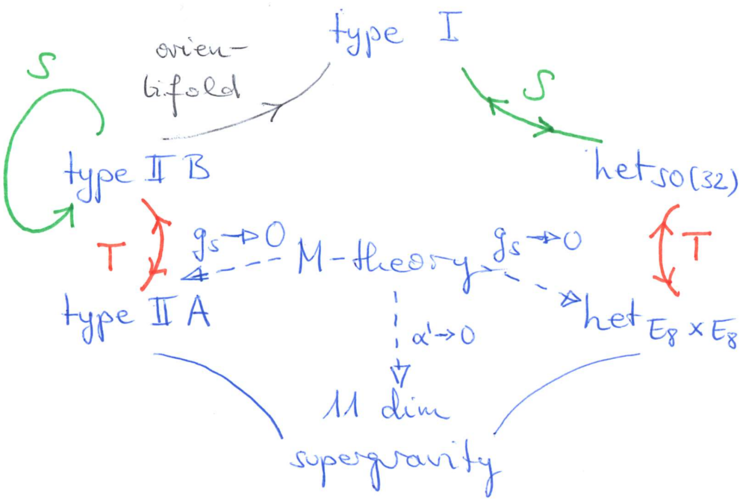


Figure 24: The web of dualities relating different superstring theories

8.2 The RNS worldsheet action

Before gauge fixing, the Polyakov action for bosonic strings could be thought of as general relativity coupled to bosonic matter X^μ on a two-dimensional worldsheet.

We shall now introduce the RNS superstring as two-dimensional supergravity coupled to SUSY matter (X^μ, Ψ^μ) on a worldsheet. More specifically, worldsheet SUSY relates

$$\begin{aligned} \text{metric } g_{\alpha\beta} \quad &\leftrightarrow \quad \text{“gravitino” } \chi_\alpha \\ \text{bos. matter } X^\mu \quad &\leftrightarrow \quad \text{fermionic matter } \Psi^\mu \end{aligned} \tag{8.3}$$

The RNS generalization of the Polyakov action in flat spacetime reads as follows:

$$\begin{aligned} S_{\text{RNS}}[X, \Phi, g, \chi] = \frac{1}{4\pi} \int d^2\sigma \, \sqrt{-\det g} \, \Big\{ &-\frac{1}{\alpha'} g^{\alpha\beta} \partial_\alpha X_\mu \partial_\beta X^\mu + \overline{\Psi}^\mu \rho^\alpha \nabla_\alpha \Psi_\mu \\ &+ (\overline{\chi}_\alpha \rho^\beta \rho^\alpha \Psi^\mu) \Big(\frac{\partial_\beta X_\mu}{\sqrt{2\alpha'}} + \frac{1}{8} \overline{\chi}_\beta \Psi_\mu \Big) \Big\} \end{aligned} \tag{8.4}$$

- Ψ^μ and χ_α are two-component Dirac spinors on a two-dimensional worldsheet (Dirac spinors have $2^{D/2}$ components in $D \in 2\mathbb{N}$ dimensions) for each choice of $\mu = 0, 1, \ldots, D-1$ and $\alpha = 0, 1$

- two-dimensional Dirac "gamma"-matrices subject to the Clifford algebra

$$\{\rho^\alpha, \rho^\beta\} = \rho^\alpha \rho^\beta + \rho^\beta \rho^\alpha = 2 g^{\alpha\beta} \mathbb{1}_{2\times 2} \tag{8.5}$$

where a possible choice of explicit representations is $\rho^0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix} \right)$ and $\rho^1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix} \right)$

- the above $S_{\rm RNS}$ is again invariant under σ^α-diffeomorphisms, but also under local SUSY (with spinorial parameter $\varepsilon(\sigma)$)

$$\begin{aligned} \delta_\varepsilon g_{\alpha\beta} &\sim \overline{\varepsilon}\rho_{(\alpha}\chi_{\beta)}\,, & \delta_\varepsilon X^\mu &\sim \overline{\varepsilon}\Psi^\mu \\ \delta_\varepsilon \chi_\alpha &\sim \nabla_\alpha \varepsilon\,, & \delta_\varepsilon \Psi^\mu &\sim \rho^\alpha\Big(\frac{\partial_\alpha X^\mu}{\sqrt{2\alpha'}} + \frac{1}{4}\overline{\chi}_\alpha \Psi^\mu\Big)\varepsilon \end{aligned} \tag{8.6}$$

- $S_{\rm RNS}$ is Weyl invariant if the transformations are extended as follows to fermions (while $\delta_\phi X^\mu = 0$):

$$\delta_\phi g_{\alpha\beta} = 2\phi g_{\alpha\beta}\,, \qquad \delta_\phi \Psi^\mu = -\frac{\phi}{2}\Psi^\mu\,, \qquad \delta_\phi \chi_\alpha = \frac{\phi}{2}\chi_\alpha \tag{8.7}$$

- finally, Weyl invariance in fact extends to *super-Weyl* invariance with spinorial transformation parameter ζ

$$\delta_\zeta \chi_\alpha = \rho_\alpha \zeta\,, \qquad \delta_\zeta(\text{other fields}) = 0 \tag{8.8}$$

The above symmetries can be used to pick *superconformal gauge*, i.e. the SUSY analogue of conformal gauge,

$$(g_{\alpha\beta}, \chi_\gamma) \to (e^{2\phi}\eta_{\alpha\beta},\, \rho_\gamma \zeta) \tag{8.9}$$

where both of ϕ and the worldsheet spinor ζ can be removed via super-Weyl transformations.

- gauge-fixed action in terms of $z = i(\tau+\sigma)$ and $\bar{z} = i(\tau-\sigma)$

$$S_{\rm RNS}[X,\psi,\bar{\psi}] = \frac{1}{2\pi}\int d^2z \left\{\frac{2}{\alpha'}\partial_z X_\mu \partial_{\bar{z}} X^\mu + \psi_\mu \partial_{\bar{z}}\psi^\mu + \bar{\psi}_\mu \partial_z \bar{\psi}^\mu\right\} \tag{8.10}$$

  with Majorana-Weyl basis $\Psi^\mu = \big(\begin{smallmatrix}\psi^\mu \\ \bar{\psi}^\mu\end{smallmatrix}\big)$

- the equations of motion imply ψ^μ and $\bar{\psi}^\mu$ to be meromorphic and antimeromorphic, respectively

$$\partial_{\bar{z}}\psi^\mu = 0 = \partial_z \bar{\psi}^\mu \tag{8.11}$$

- superconformal gauge can only be attained locally, there are subtleties for higher-genus super-Riemann surfaces that cause challenges in the evaluation of higher-loop amplitudes

- the residual gauge symmetry is formed by diffeomorphisms and SUSY transformations that can be undone via (super-)Weyl transformations which amount to *superconformal symmetry*

8.3 RNS superstrings and superconformal field theory

Recall that conformal transformations are generated by a traceless energy-momentum tensor $T_{\alpha\beta} = \frac{4\pi}{\sqrt{-\det g}}\frac{\delta S}{\delta g^{\alpha\beta}}$. In two dimensions, the latter is characterized by one meromorphic and one antimeromorphic component each

$$Q_{(\eta)} = \oint_{B_R(0)} \frac{dz}{2\pi i}\, T(z)\eta(z)\,, \qquad \bar{Q}_{(\bar{\eta})} = \oint_{B_R(0)} \frac{d\bar{z}}{2\pi i}\, \bar{T}(\bar{z})\bar{\eta}(\bar{z}) \tag{8.12}$$

In superconformal theories, additionally define a *supercurrent* G_α via gravitino variation

$$G_\alpha \sim \frac{1}{\sqrt{-\det g}}\frac{\delta S}{\delta \chi^\alpha} \tag{8.13}$$

In applications to the RNS worldsheet action, χ^α and G_α are two-component Dirac spinors on the worldsheet for both values $\alpha = 0,1$ of their vector index.

- the e.o.m. of $X^\mu, \psi^\mu, \bar{\psi}^\mu$ imply conservation laws

$$\partial^\alpha T_{\alpha\beta} = 0 = \partial^\alpha G_\alpha \tag{8.14}$$

- in the same way as conformal symmetry implies $g^{\alpha\beta} T_{\alpha\beta} = 0$, superconformal symmetry implies "γ-tracelessness" $\rho^\alpha G_\alpha = 0$
- left with 2 instead of 4 independent supercurrent components, one meromorphic $G(z)$ and one antimeromorphic $\bar{G}(\bar{z})$ by the conservation laws
- accordingly, the conserved charges are characterized by one meromorphic and one antimeromorphic function $\varepsilon(z)$ and $\bar{\varepsilon}(\bar{z})$

$$Q_{(\varepsilon)} = \oint_{B_R(0)} \frac{dz}{2\pi i} \, \varepsilon(z) G(z) \, , \qquad \bar{Q}_{(\bar{\varepsilon})} = \oint_{B_R(0)} \frac{d\bar{z}}{2\pi i} \, \bar{\varepsilon}(\bar{z}) \bar{G}(\bar{z}) \tag{8.15}$$

From S_{RNS} before fixing superconformal gauge, one can derive the explicit form

$$\begin{aligned} T(z) &= -\frac{1}{\alpha'} : \partial_z X_\mu \partial_z X^\mu : + \frac{1}{2} : (\partial_z \psi_\mu) \psi^\mu : \\ G(z) &= \frac{1}{\sqrt{2\alpha'}} : i (\partial_z X_\mu) \psi^\mu : \end{aligned} \tag{8.16}$$

- in order to determine SUSY transformations of X^μ and ψ^μ, first get relevant OPEs
- usual path-integral methods give rise to the two-point function

$$\begin{aligned} \langle \psi^\mu(z) \psi^\nu(w) \rangle &= \frac{\eta^{\mu\nu}}{z-w} \\ \langle \bar{\psi}^\mu(\bar{z}) \bar{\psi}^\nu(\bar{w}) \rangle &= \frac{\eta^{\mu\nu}}{\bar{z}-\bar{w}} \\ \langle \psi^\mu(z) \bar{\psi}^\nu(\bar{w}) \rangle &= 0 \end{aligned} \tag{8.17}$$

  that reflects the anticommuting nature of ψ^μ
- resulting OPEs are (with $\psi^\mu \leftrightarrow \bar{\psi}^\nu$ non-singular)

$$\psi^\mu(z) \psi^\nu(w) \sim \frac{\eta^{\mu\nu}}{z-w} + \ldots \, , \qquad \bar{\psi}^\mu(\bar{z}) \bar{\psi}^\nu(\bar{w}) \sim \frac{\eta^{\mu\nu}}{\bar{z}-\bar{w}} + \ldots \tag{8.18}$$

- by Wick rules, can reduce the OPEs of T and G to those of X^μ and ψ^μ:

$$\begin{aligned} T(z) i\partial_w X^\mu(w) &\sim \frac{i\partial_w X^\mu(w)}{(z-w)^2} + \frac{i\partial_w^2 X^\mu(w)}{z-w} + \ldots \\ T(z) \psi^\mu(w) &\sim \frac{\psi^\mu(w)}{2(z-w)^2} + \frac{\partial_w \psi^\mu(w)}{z-w} + \ldots \\ G(z) \sqrt{\frac{2}{\alpha'}} i\partial_w X^\mu(w) &\sim \frac{\psi^\mu(w)}{2(z-w)^2} + \frac{\partial_w \psi^\mu(w)}{2(z-w)} + \ldots \\ G(z) \psi^\mu(w) &\sim \frac{i\partial_w X^\mu(w)}{2(z-w)} \sqrt{\frac{2}{\alpha'}} + \ldots \end{aligned} \tag{8.19}$$

- can read off $(h, \bar{h}) = (\frac{1}{2}, 0)$ for ψ^μ from $T(z)\psi^\mu(w)$
- OPE $G(z) \leftrightarrow \partial_w X^\mu(w)$ integrates to

$$G(z) \sqrt{\frac{2}{\alpha'}} i X^\mu(w) \sim \frac{\psi^\mu(w)}{2(z-w)} + \ldots \tag{8.20}$$

- resulting superconformal transformations

$$Q_{(\varepsilon)} i\sqrt{\frac{2}{\alpha'}} X^\mu(w) = \frac{1}{2}\varepsilon(w)\psi^\mu(w)\,, \qquad Q_{(\varepsilon)}\psi^\mu(w) = \frac{1}{2}\sqrt{\frac{2}{\alpha'}} i\partial_w X^\mu(w)\varepsilon(w) \tag{8.21}$$

- by iterating the above Wick contractions, infer a representation of the *super-Virasoro algebra*

$$\begin{aligned} T(z)T(w) &\sim \frac{3D/4}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \ldots \\ T(z)G(w) &\sim \frac{3G(w)}{2(z-w)^2} + \frac{\partial_w G(w)}{z-w} + \ldots \\ G(z)G(w) &\sim \frac{D/4}{(z-w)^3} + \frac{T(w)}{2(z-w)} + \ldots \end{aligned} \tag{8.22}$$

  with central charge $c = \frac{3D}{2}$; in other words, each spacetime dimension contributes $\frac{3}{2}$ units of central charge (1 from X^μ and $\frac{1}{2}$ from ψ^μ), and the factor of $D/4$ along with the triple pole in the OPE of $G(z)G(w)$ generalizes to $c/6$

The above OPEs line up with those in a general two-dimensional superconformal field theory (SCFT) which are often defined without any reference to an action functional.

- superconformal primaries at weight h are defined as a pair of conformal primaries ϕ_h and $\phi_{h+1/2}$ of weights h and $h + \frac{1}{2}$, respectively

- the defining properties of superconformal primaries $(\phi_h, \phi_{h+1/2})$ include the OPEs

$$\begin{aligned} T(z)\phi_h(w) &\sim \frac{h\phi_h(w)}{(z-w)^2} + \frac{\partial_w \phi_h(w)}{z-w} + \ldots \\ T(z)\phi_{h+\frac{1}{2}}(w) &\sim \frac{(h+\frac{1}{2})\phi_{h+\frac{1}{2}}(w)}{(z-w)^2} + \frac{\partial_w \phi_{h+\frac{1}{2}}(w)}{z-w} + \ldots \\ G(z)\phi_h(w) &\sim \frac{\phi_{h+\frac{1}{2}}(w)}{2(z-w)} + \ldots \\ G(z)\phi_{h+\frac{1}{2}}(w) &\sim \frac{h\phi_h(w)}{(z-w)^2} + \frac{\partial_w \phi_h(w)}{2(z-w)} + \ldots \end{aligned} \tag{8.23}$$

- by comparing with the OPEs of X^μ, ψ^ν, the pair of $(\psi^\mu, i\sqrt{\frac{\alpha'}{2}}\partial_z X^\mu)$ is a superconformal primary of weight $h = \frac{1}{2}$

- the general form of the SUSY transformation is

$$\begin{aligned} Q_{(\varepsilon)}\phi_h(z) &= \frac{1}{2}\varepsilon(z)\phi_{h+\frac{1}{2}}(z) \\ Q_{(\varepsilon)}\phi_{h+\frac{1}{2}}(z) &= \frac{1}{2}\varepsilon(z)\partial_z\phi_h(z) + h(\partial_z\varepsilon(z))\phi_h(z) \end{aligned} \tag{8.24}$$

- $T(z)$ and $G(z)$ are generators of conformal transformations and SUSY transformations, respectively; given that $G(z)G(w)$ contains $\frac{T(w)}{2(z-w)}$ among its singular terms, one can think of $Q_{(\varepsilon)}$ at globally defined $\varepsilon(z)$ as the square root of a conformal transformation

8.4 (Anti-)periodic boundary conditions

Since the spatial coordinate σ is taken to range from 0 to 2π for closed strings, we need to find a fermionic analogue of the boundary condition $X^\mu(\sigma + 2\pi) = X^\mu(\sigma)$. The guiding principle is to avoid boundary terms in varying S_{RNS} and deriving the e.o.m. $\partial_{\bar z}\psi^\mu = \partial_z \bar\psi^\mu = 0$,

$$\delta S_{\text{RNS}}[X,\psi,\bar\psi] \quad \Rightarrow \quad \int d\tau \, (\psi^\mu \delta\psi_\mu - \bar\psi^\mu \delta\bar\psi_\mu)\Big|_{\sigma=0}^{\sigma=2\pi} \stackrel{!}{=} 0 \tag{8.25}$$

For this purpose, it is sufficient to impose periodic boundary conditions up to a sign

$$\psi^\mu(\sigma+2\pi) = \pm\psi^\mu(\sigma) \ , \qquad \bar\psi^\mu(\sigma+2\pi) = \pm\bar\psi^\mu(\sigma) \tag{8.26}$$

since physical quantities are always bilinear in fermionic variables. In fact, the two signs for ψ^μ and $\bar\psi^\mu$ can be chosen independently.

- different sign choices in the fermionic boundary conditions are referred to as follows

$$\begin{aligned} \psi^\mu \text{ antiperiodic in } \sigma \quad &\longleftrightarrow \quad \text{Neveu-Schwarz sector (NS)} \\ \psi^\mu \text{ periodic in } \sigma \quad &\longleftrightarrow \quad \text{Ramond sector (R)} \end{aligned} \tag{8.27}$$

- together with the analogous choices for $\bar\psi^\mu$, we arrive at a total of 4 closed-string sectors (NS,NS), (NS,R), (R,NS), (R,R)
- supercurrents $G \sim \partial_z X_\mu \psi^\mu$ and $\bar G \sim \partial_{\bar z} X_\mu \bar\psi^\mu$ inherit their boundary conditions from $\psi^\mu, \bar\psi^\mu$

You already got a first glimpse at fermionic mode expansions in exercise C.2:

- on the cylinder with $z = i(\tau+\sigma)$ and $\bar z = i(\tau-\sigma)$,

$$\psi^\mu(\tau,\sigma) = \begin{cases} \displaystyle\sum_{r\in\mathbb{Z}-1/2} \psi^\mu_r e^{-rz} & : \text{ NS, antiperiodic on cylinder} \\ \displaystyle\sum_{r\in\mathbb{Z}} \psi^\mu_r e^{-rz} & : \text{ R, periodic on cylinder} \end{cases} \tag{8.28}$$

- OPEs can be equivalently formulated in terms of anticommutators, namely $\{\psi^\mu_r, \bar\psi^\nu_s\} = 0$ and

$$\{\psi^\mu_r, \psi^\nu_s\} = \eta^{\mu\nu}\delta_{r+s,0} = \{\bar\psi^\mu_r, \bar\psi^\nu_s\} \tag{8.29}$$

- in mapping to the complex plane $\xi = e^z$ with $\frac{\partial\xi}{\partial z} = \xi$, the primaries ψ^μ and $\bar\psi^\mu$ transform with weight $(h,\bar h) = (\frac{1}{2},0)$ and $(0,\frac{1}{2})$, respectively

$$\psi^\mu(\xi) = \left(\frac{\partial\xi}{\partial z}\right)^{-1/2} \psi^\mu(\tau,\sigma) = \begin{cases} \displaystyle\sum_{r\in\mathbb{Z}-1/2} \psi^\mu_r \xi^{-r-1/2} & : \text{ NS, periodic on plane} \\ \displaystyle\sum_{r\in\mathbb{Z}} \psi^\mu_r \xi^{-r-1/2} & : \text{ R, antiperiodic on plane} \end{cases} \tag{8.30}$$

- transformation factor $(\frac{\partial\xi}{\partial z})^{-1/2}$ reverses periodicity properties under $\sigma \to \sigma+2\pi$ or $\xi \to e^{2\pi i}\xi$, i.e.

$$\begin{aligned} \text{NS} \quad &\longleftrightarrow \quad \text{antiperiodic on the cylinder but periodic on the plane} \\ \text{R} \quad &\longleftrightarrow \quad \text{periodic on the cylinder but antiperiodic on the plane} \end{aligned} \tag{8.31}$$

- henceforth work on the plane and rename $\xi \to z$

As will be explained below, the Ramond-sector states in D dimensions exhibit a "vacuum energy" or an offset in conformal weights by $h_0 = \frac{D}{16}$

- mode expansion of the energy-momentum tensor

$$L_m = \frac{1}{2}\sum_{n\in\mathbb{Z}} :\alpha_{m-n}\cdot\alpha_n: + \begin{cases} \frac{1}{2}\sum_{r\in\mathbb{Z}-1/2} r :\psi_{m-r}\cdot\psi_r: & \text{(NS)} \\ \frac{1}{2}\sum_{r\in\mathbb{Z}} r :\psi_{m-r}\cdot\psi_r: +\delta_{m,0}h_0 & \text{(R)} \end{cases} \tag{8.32}$$

- need to dial the offset h_0 in L_0 of the Ramond sector to get the same Virasoro algebra at $c=\frac{3D}{2}$ as in the NS sector

$$[L_m, L_n] \stackrel{!}{=} (m-n)L_{m+n} + \frac{D}{8}m(m^2-1)\delta_{m+n,0} \tag{8.33}$$

which uniquely fixes $h_0 = \frac{D}{16}$ as the R-sector vacuum energy

- take NS- and R-sector ground states $|0\rangle_{\rm NS}$ and $|A\rangle_{\rm R}$ to be annihilated by $\psi_{r>0}$ and hence by $L_{m>0}$

- by the state-operator map in CFTs, $|A\rangle_{\rm R}$ must stem from some conformal primary at infinite past $z\to 0$ acting on the vacuum; the primaries relevant to $|A\rangle_{\rm R}$ are so-called *spin fields* $S_A(z)$ of weight $h=\frac{D}{16}$,

$$|A\rangle_{\rm R} = \lim_{z\to 0} S_A(z)|0\rangle_{\rm NS}\,, \qquad {}_{\rm R}\langle B| = \lim_{z\to\infty} {}_{\rm NS}\langle 0|S_B(z)z^{D/8} \tag{8.34}$$

Let us elaborate on the basic properties of spin fields:

- R ground states are degenerate since the D zero modes ψ_0^μ do not change the energy, $[L_0,\psi_0^\mu]=0$

- zero modes form a Clifford algebra like the Dirac Γ-matrices in D dimensions

$$\{\psi_0^\mu,\psi_0^\nu\} = \eta^{\mu\nu} \quad \leftrightarrow \quad \{\Gamma^\mu,\Gamma^\nu\} = 2\eta^{\mu\nu} \tag{8.35}$$

Hence, $|A\rangle_{\rm R}$ and therefore $S_A(z)$ must be spinors of $SO(1,D{-}1)$, and all the R-sector states (obtained from $\psi^\mu_{r<0}$ action on $|A\rangle_{\rm R}$) are spacetime fermions!

- in even dimensions $D=2n$, Dirac spinors have $2^{D/2}=2^n$ components and decompose into two Lorentz-irreducibles, so-called Weyl spinors with $2^{D/2-1}$ components each; at the level of the spinor indices, decompose $A=(\alpha,\dot\alpha)$, e.g. in ten dimensions

$$D=10 \quad\Rightarrow\quad \underbrace{A=1,2,\ldots,32}_{\text{Dirac spinor}} \;=\; \underbrace{\alpha=1,2,\ldots,16}_{\text{left-handed Weyl}} \oplus \underbrace{\dot\alpha=\dot 1,\dot 2,\ldots,\dot{16}}_{\text{right-handed Weyl}} \tag{8.36}$$

We next infer the OPEs of spin fields

- their two-point functions must be $\sim (z-w)^{-D/8}$ since S_A are primaries of $h=\frac{D}{16}$, and the normalization factor must be $\sim C_{AB}$ (the so-called charge conjugation matrix) by Lorentz-covariance

$$\langle S_A(z)S_B(w)\rangle = \frac{C_{AB}}{(z-w)^{D/8}} \quad\Rightarrow\quad S_A(z)S_B(w) \sim \frac{C_{AB}}{(z-w)^{D/8}} + \ldots \tag{8.37}$$

- use the corresponendce $\psi_0^\mu \leftrightarrow \frac{1}{\sqrt2}\Gamma^\mu$ at the level of matrix elements:

$${}_{\rm R}\langle A|\psi_0^\mu|B\rangle_{\rm R} = \frac{1}{\sqrt2}(\Gamma^\mu C)_{AB} = \frac{1}{\sqrt2}(\Gamma^\mu)_A{}^D C_{DB} \tag{8.38}$$

this must be reproduced from the following limit of a three-point function

$${}_{\rm R}\langle A|\psi_0^\mu|B\rangle_{\rm R} = \underbrace{\lim_{z_1\to\infty} {}_{\rm NS}\langle 0|z_1^{D/8}S_A(z_1)}_{{}_{\rm R}\langle A|}\,\underbrace{\lim_{z_2\to 0} z_2^{1/2}\psi^\mu(z_2)}_{\psi_0^\mu}\,\underbrace{S_B(0)|0\rangle_{\rm NS}}_{|B\rangle_{\rm R}} \tag{8.39}$$

as a three-point function of conformal primaries, it is determined by the h_j up to normalization; the only normalization choice compatible with ${}_{\mathrm{R}}\langle A|\psi_0^\mu|B\rangle_{\mathrm{R}} = \frac{1}{\sqrt{2}}(\Gamma^\mu C)_{AB}$ is

$$\langle S_A(z_1)\psi^\mu(z_2)S_B(z_3)\rangle = \frac{1}{\sqrt{2}}\frac{(\Gamma^\mu C)_{AB}}{(z_{12}z_{23})^{1/2}z_{13}^{D/8-1/2}} \tag{8.40}$$

- the $z_1 \to z_2$ limit of the relabelled three-point function

$$\langle \psi^\mu(z_1)S_A(z_2)S_B(z_3)\rangle = \frac{1}{\sqrt{2}}\frac{(\Gamma^\mu C)_{AB}}{(z_{12}z_{13})^{1/2}z_{23}^{D/8-1/2}} \tag{8.41}$$

  together with the above expression for $\langle S_A(z_2)S_B(z_3)\rangle$ leads to the OPE

$$\psi^\mu(z)S_A(w) \sim \frac{(\Gamma^\mu)_A{}^B S_B(w)}{\sqrt{2}(z-w)^{1/2}} + \ldots \tag{8.42}$$

  the square-root dependence $\sim (z-w)^{-1/2}$ is expected: $S_A(0)$ opens up a branch cut for ψ^μ to enforce the R-sector condition $\psi^\mu(e^{2\pi i}z) = -\psi^\mu(z)$ on the plane; the (ψ^μ, S_A)-system turns out to be interacting, i.e. correlators with insertions of both ψ^μ and S_A cannot be computed by simply adding up Wick contractions (at least, not in a manifestly Lorentz-covariant way)

- the $z_2 \to z_3$ limit of the above three-point function together with $\langle \psi^\mu(z_1)\psi^\nu(z_2)\rangle = \eta^{\mu\nu}/z_{12}$ determine the subleading term in

$$\langle S_A(z)S_B(w)\rangle = \frac{C_{AB}}{(z-w)^{D/8}} \quad \Rightarrow \quad S_A(z)S_B(w) \sim \frac{C_{AB}}{(z-w)^{D/8}} + \frac{(\Gamma^\mu C)_{AB}\psi_\mu(w)}{\sqrt{2}(z-w)^{D/8-1/2}} + \ldots \tag{8.43}$$

- we will later on impose vertex operators to have no mutual branch cuts after combining matter and ghost sectors; this will lead to the *GSO projection* that removes the tachyon from the superstring spectrum and eliminates one of the Weyl components from the spin fields, i.e. all spacetime fermions have definite chirality

8.5 The β, γ superghost system

We have seen for bosonic strings that the anticommuting (b,c)-ghost system removes the gauge redundancy of the Polyakov action. In conformal gauge, the ghost contribution to the worldsheet action was found to be

$$S_{gh}[b,c] = \frac{1}{\pi}\int d^2z\,\big(b\partial_{\bar z}c + \bar b\partial_z \bar c\big) \tag{8.44}$$

We shall now adjoin commuting β,γ ghosts to fix the fermionic symmetries (i.e. worldsheet supersymmetry and super-Weyl transformations) of the RNS action; in superconformal gauge, this results in the following extra term in the action

$$S_{sgh}[\beta,\gamma] = \frac{1}{\pi}\int d^2z\,\big(\beta\partial_{\bar z}\gamma + \bar\beta\partial_z\bar\gamma\big) \tag{8.45}$$

where the choice of NS / R boundary conditions must be the same for $\beta,\gamma \leftrightarrow \psi^\mu$ and separately for $\bar\beta,\bar\gamma \leftrightarrow \bar\psi^\mu$.

The (β,γ)-system is yet another example of a CFT with the following data

- the above $S_{sgh}[\beta,\gamma]$ implies non-singular OPEs $\beta(z)\beta(w)$ and $\gamma(z)\gamma(w)$ as well as

$$\gamma(z)\beta(w) \sim \frac{1}{z-w} + \ldots \tag{8.46}$$

- the meromorphic component of the energy-momentum tensor of the superghost system is

$$T_{\beta,\gamma} = -\frac{3}{2} : (\partial_z \gamma)\beta : -\frac{1}{2} : \gamma \partial_z \beta : \tag{8.47}$$

which identifies β and γ to be conformal primaries of weights $h_\beta = \frac{3}{2}$ and $h_\gamma = -\frac{1}{2}$, respectively (cf. $h_b = 2$ and $h_c = -1$ for the (b,c)-system); moreover, the central charge of the (β,γ)-system is found to be $c_{\beta,\gamma} = 11$ (whereas $c_{b,c} = -26$)

- the critical dimension of the RNS superstring is determined by the vanishing of

$$c_{\text{total}} = c_X + c_\psi + c_{b,c} + c_{\beta,\gamma} = \frac{3D}{2} - 15 \tag{8.48}$$

this is the case in $D = 10$ dimensions, though one can again adjoin internal or non-geometric CFT sectors with $c_{\text{int}} > 0$ which allow for $D < 10$ large spacetime dimensions

- the pairs (β, b) and (c, γ) form superconformal primaries of $h = \frac{3}{2}$ and $h = -1$ w.r.t. the supercurrent

$$G_{gh} = - : (\partial_z \beta) c : -\frac{3}{2} : \beta \partial_z c : +\frac{1}{2} : b\gamma : \tag{8.49}$$

Similar to the (b,c)-system, the counting of zero modes and the notion of hermitian conjugates is not completely intuitive for the (β,γ)-system:

- just like $j_{b,c} = - : bc :$, there is an anomalous superghost current

$$j_{\beta,\gamma}(z) = - : \beta(z)\gamma(z) : \tag{8.50}$$

- the departure from being a conformal primary

$$T_{\beta,\gamma}(z) j_{\beta,\gamma}(w) \sim \frac{2}{(z-w)^3} + \frac{j_{\beta,\gamma}(w)}{(z-w)^2} + \frac{\partial_w j_{\beta,\gamma}(w)}{z-w} + \ldots \tag{8.51}$$

identifies the background charge $Q_{\beta,\gamma} = +2$ (in the place of $Q_{b,c} = -3$ of the (b,c)-system)

- again, the role of the $j_{\beta,\gamma}$-charge in the hermiticity properties is unusual

$$|\beta{-}\gamma \text{ charge } q\rangle^\dagger = \langle \beta{-}\gamma \text{ charge } (-q{-}2)| \tag{8.52}$$

- the numbers of zero modes N_γ, N_β depend on the genus g

$$N_\gamma - N_\beta = -Q_{\beta,\gamma}(g-1) = 2 - 2g \tag{8.53}$$

generalizing $N_c - N_b = 3 - 3g$

Let us now construct the superghost ground states in the NS and R sectors

- in the NS mode expansion $\gamma(z) = \sum_{r \in \mathbb{Z}-1/2} \gamma_r z^{-r+1/2}$ only $\gamma_{r \geq 3/2}$ annihilate the ground state $|0\rangle_{\beta,\gamma}$ but not $\gamma_{r \leq 1/2}$

- by $[L_0, (\gamma_{1/2})^n] = -\frac{n}{2}(\gamma_{1/2})^n$, the fact that $\gamma_{1/2}|0\rangle_{\beta,\gamma} \neq 0$ admits states $(\gamma_{1/2})^n|0\rangle_{\beta,\gamma}$ of arbitrarily negative L_0-eigenvalues, there is no analogue of the bound from $(c_1)^2 = 0$ from the (b,c)-ghost analogue $c_1|0\rangle_{b,c} \neq 0$

- cure the unbounded spectrum by engineering an alternative ground state via change of variables $(\beta,\gamma)\to$ fermions $(\eta,\xi)+$ chiral boson ϕ

$$\beta(z) =: e^{-\phi(z)} : \partial_z\xi(z)\,, \qquad \gamma(z) =: e^{\phi(z)} : \eta(z) \tag{8.54}$$

this reparametrization preserves the OPEs of β,γ as one can check from the non-singular OPEs $\xi(z)\xi(w)$ and $\eta(z)\eta(w)$ as well as

$$\eta(z)\xi(w) \sim \frac{1}{z-w} + \ldots\,, \qquad \phi(z)\phi(w) \sim -\log(z-w) + \ldots \tag{8.55}$$

- the energy-momentum tensor of the new variables reads

$$T_{\eta,\xi} = - : \eta\partial_z\xi :\,, \qquad T_\phi = -\frac{1}{2} : (\partial_z\phi)\partial_z\phi : -\partial_z^2\phi \tag{8.56}$$

(with the last term $\sim \partial_z^2\phi$ reflecting the background charge) and assigns

$$h(\eta) = 1\,, \qquad h(\xi) = 0\,, \qquad h(e^{q\phi}) = -\frac{q^2}{2} - q \tag{8.57}$$

- by the same techniques that determine the Koba-Nielsen factor $\langle \prod_{j=1}^n : e^{ip_j\cdot X(z_j)} :\rangle$,

$$: e^{q_1\phi(z)} :: e^{q_2\phi(w)} :\sim (z-w)^{-q_1q_2} : e^{(q_1+q_2)\phi(w)} : + \ldots \tag{8.58}$$

- use this to construct the following state which is annihilated by $\gamma_{1/2} = \oint \frac{dz}{2\pi i}\frac{\gamma(z)}{z}$

$$|q{=}-1\rangle_{\beta,\gamma} =: e^{-\phi(0)} : |0\rangle_{\beta,\gamma}\,, \qquad h(e^{-\phi}) = \frac{1}{2} \tag{8.59}$$

this qualifies for a NS-sector ground state since $\gamma_{1/2}|q{=}-1\rangle_{\beta,\gamma} = 0$

- similarly, the R-sector ground state must be annihilated by the γ_1 mode and open a branch cut for β,γ in the same way as $S_A(z)$ does for $\psi^\mu(z)$

$$|q{=}-\tfrac{1}{2}\rangle_{\beta,\gamma} =: e^{-\phi(0)/2} : |0\rangle_{\beta,\gamma}\,, \qquad h(e^{-\phi/2}) = \frac{3}{8} \tag{8.60}$$

i.e. one can think of $e^{-\phi(0)/2}$ as a "spin field for superghosts"

8.6 The superstring spectrum and GSO projection

Recall that the physical states of the bosonic string are generated by vertex operators $V_{\rm phys}$, conformal primaries of $h=1$,

$$|{\rm phys}\rangle = \lim_{z\to 0} V_{\rm phys}(z)|0\rangle \tag{8.61}$$

We will now discuss the rules for finding similar $h=1$ primaries for physical states of the RNS superstring. For simplicity, we only consider the left movers (with insertions of $\partial_z X^\mu, \psi^\mu$ and their derivatives in the matter sector) while taking the right-moving contributions such as $\partial_{\bar z}X^\mu, \bar\psi^\mu$ to follow from double copy.

From the discussion of the (β,γ)-system, we build the NS- and R-sector spectra on the respective ground states $|q{=}-1,-\frac{1}{2}\rangle_{\beta,\gamma}$:

- NS sector $\longleftrightarrow \alpha^\mu_{-\mathbb{N}}, \psi^\mu_{1/2-\mathbb{N}}$ acting on $|q{=}-1\rangle_{\beta,\gamma} =: e^{-\phi(0)} : |0\rangle_{\beta,\gamma}$; the oscillators only give rise to vector indices of $SO(1,D{-}1)$, i.e. no spinor indices; hence, the resulting physical states are spacetime bosons with vertex operators

$$V^{\rm NS}_{\rm phys}(z) = \Phi^{\rm NS}_{\rm phys}(z) : e^{-\phi(z)} :\,, \qquad \text{need } h(\Phi^{\rm NS}_{\rm phys}) = \frac{1}{2} \tag{8.62}$$

where we separate off the matter contribution $\Phi^{\rm NS}_{\rm phys}$ to not have the ubiquitous $:e^{-\phi(z)}:$ in all the subsequent formulae

- R sector $\longleftrightarrow$ $\alpha^\mu_{-\mathbb{N}}, \psi^\mu_{-\mathbb{N}}$ acting on $S_A(0)|q{=}-\frac{1}{2}\rangle_{\beta,\gamma} = S_A(0):e^{-\phi(0)/2}:|0\rangle_{\beta,\gamma}$; the spinor index of $S_A(0)$ implies that all the resulting physical states are spacetime fermions with vertex operators

$$V^{\rm R}_{\rm phys}(z) = \Phi^{\rm R}_{\rm phys}(z):e^{-\phi(z)/2}:, \qquad \text{need } h(\Phi^{\rm R}_{\rm phys}) = \frac{5}{8} \tag{8.63}$$

  which $\Phi^{\rm R}_{\rm phys}(z)$ depending only on matter variables; the fractional conformal weight can be easily attained since S_A in $D=10$ dimension has $h(S_A)=\frac{5}{8}$.

- the closed-superstring spectrum is built from the following combinations of left- and right-movers

$$\begin{aligned}\text{(NS,R) and (R,NS)} &\longleftrightarrow \text{spacetime fermions}\\ \text{(NS,NS) and (R,R)} &\longleftrightarrow \text{spacetime bosons}\end{aligned}$$

  where the bispinors in the (R,R) sector will be later on explained to yield antisymmetric tensors

As for bosonic strings, each $\Phi_{\rm phys}$ will have a plane-wave factor $:e^{ip\cdot X}:$ of conformal weight $(h,\bar h) = (\frac{\alpha' p^2}{4}, \frac{\alpha' p^2}{4})$. Since $\alpha^\mu_{<0}$ and $\psi^\mu_{<0}$ can only increase h of $:e^{q\phi}:$ and $:e^{ip\cdot X}:$, we have the following bounds on $m^2=-p^2$

$$m^2 \geq \begin{cases} -\frac{2}{\alpha'} & : \text{ NS sector}\\ 0 & : \text{ R sector}\end{cases} \tag{8.64}$$

- there is a potential bosonic tachyon at mass level $-1/2$,

$$\Phi^{\rm NS}_{\rm tachyon}(z) =: e^{ip\cdot X(z)}:, \qquad p^2 = \frac{2}{\alpha'} = -m^2 \tag{8.65}$$

- massless bosons from a single $\psi^\mu_{-1/2}$ oscillator

$$\Phi^{\rm NS}_{m^2=0}(z) = \epsilon_\mu \psi^\mu(z):e^{ip\cdot X(z)}:, \qquad p^2=0 \tag{8.66}$$

- massless fermions characterized by spinor wavefunction χ^A

$$\Phi^{\rm R}_{m^2=0}(z) = \chi^A S_A(z):e^{ip\cdot X(z)}:, \qquad p^2=0 \tag{8.67}$$

- the next task is to find consistency conditions to eliminate the tachyon and to constrain the polarizations ϵ_μ, χ^A of massless states; in contrast to the bosonic string, transversality $p\cdot\epsilon=0$ does *not* follow from imposing a fixed conformal weight on $\epsilon_\mu\psi^\mu:e^{ip\cdot X(z)}:$

A central constraint on the physical spectrum of the superstring is known as *GSO projection* referring to Gliozzi, Scherk and Olive. Its selection rules can be summarized as "OPEs among physical vertex operators must not have any branch cuts".

- plane-wave correlators are single-valued for any collection of momenta p_i by the absolute values in the Koba–Nielsen factor $\langle\prod_{j=1}^n :e^{ip_j\cdot X(z_j)}:\rangle \sim \prod_{i<j}^n |z_{ij}|^{\alpha' p_i\cdot p_j}$

- OPEs among $V^{\rm NS}_{\rm phys}(z)$ and $V^{\rm R}_{m^2=0}(w)$ are potentially multivalued

$$\begin{aligned}&\sim (z-w)^{-1/2} \text{ from } e^{-\phi(z)} \text{ and } e^{-\phi(w)/2}\\ &\sim (z-w)^{-\#(\psi)/2} \text{ from } \partial_z^{n\geq 0}\psi^\mu(z) \text{ and } S_A(w)\end{aligned}$$

  where all of ψ^μ and its derivatives $\partial_z^{n\geq1}\psi^\mu$ are assigned 1 unit of worldsheet fermion number $\#(\psi)$

- branch cuts of the superghost sector and (ψ^μ, S_A) cancel only for odd numbers of ψ^μ within NS vertex operators; this selection rule removes the tachyon with $\#(\psi)=0$ and keeps the $k^2=0$ vector

- next investigate the OPE of two massless fermions $V^{\rm R}_{m^2=0}(z)$ and $V^{\rm R}_{m^2=0}(w)$ in $D=10$ where

$$:e^{-\phi/2}S_A(z)::e^{-\phi/2}S_B(w):\sim \frac{C_{AB}:e^{-\phi(w)}:}{(z-w)^{3/2}}+\frac{(\Gamma^\mu C)_{AB}\psi_\mu(w)}{\sqrt{2}(z-w)}:e^{-\phi(w)}:+\dots \tag{8.68}$$

- get a branch cut from the OPE of massless fermions unless all pairs of physical wavefunctions obey $\chi_1^A\chi_2^B C_{AB}=0$; this can be accomplished by truncating the 32-component Dirac spinors $\chi^A S_A(z)$ to 16-component Weyl spinors $\chi^\alpha S_\alpha(z)$ since the charge-conjugation matrix is block-off-diagonal in a Weyl basis with $C_{\alpha\beta}=C^{\dot\alpha\dot\beta}=0$ and therefore $C_{AB}=\left(\begin{smallmatrix} 0 & C_\alpha{}^{\dot\beta} \\ C^{\dot\alpha}{}_\beta & 0\end{smallmatrix}\right)$

- also the Dirac matrices are block off-diagonal in a Weyl basis, $(\Gamma^\mu)_A{}^B=\left(\begin{smallmatrix} 0 & \gamma^\mu_{\alpha\dot\beta} \\ \gamma^{\mu\dot\alpha\beta} & 0\end{smallmatrix}\right)$ with 16×16 Clifford algebra $\gamma^\mu_{\alpha\dot\beta}\gamma^{\nu\dot\beta\omega}+\gamma^\nu_{\alpha\dot\beta}\gamma^{\mu\dot\beta\omega}=2\eta^{\mu\nu}\delta^\omega_\alpha$, therefore get nonzero $\chi_1^A(\Gamma^\mu C)_{AB}\chi_1^B\to\chi_1^\alpha(\gamma^\mu C)_{\alpha\beta}\chi_1^\beta$

- for R-sector states with $m^2>0$, the integer moding $\psi^\mu_{r\in\mathbb{Z}}$ only admits integer jumps in conformal weights via insertions of $\partial_z^{n\geq 0}\psi^\mu$, which automatically gives an even number of ψ^μ in massive vertex operators (e.g. states $\psi^\mu_{-1}|A\rangle_{\rm R}$ are generated by fields of the form $\psi^\nu\psi^\lambda S_B|0\rangle_{\rm NS}$)

- the above selection rules define GSO projection for admissible physical vertex operators:

$$\begin{array}{lcl} V^{\rm NS}_{\rm phys} & \longleftrightarrow & \text{odd } \#(\psi) \;\;\Rightarrow\;\; \text{no tachyon} \\ V^{\rm R}_{\rm phys} & \longleftrightarrow & \text{even } \#(\psi) \;\&\; S_A\to S_\alpha \text{ Weyl spinor}\end{array}$$

  Moreover, the requirement of odd $\#(\psi)$ removes all half-odd integer mass levels from the NS-sector spectrum, i.e. states $\psi^\mu\psi^\nu:e^{-\phi}e^{ip\cdot X}:$ and $\partial_z X^\mu:e^{-\phi}e^{ip\cdot X}:$ with would-be mass level $\frac{1}{2}$. Hence, both the NS- and the R-sector states exclusively populate integer mass levels with $m^2\in\frac{4\mathbb{N}}{\alpha'}$ which is a necessary condition for spacetime supersymmetry.

Still, there are two loose ends about GSO projected states

- polarizations ϵ^μ,χ^α of the massless states are unconstrained by $h(\Phi^{\rm NS}_{\rm phys})=\frac{1}{2}$ and $h(\Phi^{\rm R}_{\rm phys})=\frac{5}{8}$ – this is different from the bosonic string where $\epsilon_\mu:i\partial_z X^\mu e^{ip\cdot X}:$ only transforms as a conformal primary of weight one if $\epsilon\cdot p=0$

- by the background charge of the (β,γ)-system, correlators $\langle\prod_{j=1}^n:e^{q_j\phi(z_j)}:\rangle$ at genus zero vanish unless $\sum_{j=1}^n q_j=-2$ (or $=2g-2$ at higher genus); however, correlation functions with more than two $V^{\rm NS}_{\rm phys}$ or more than four $V^{\rm R}_{\rm phys}$ necessarily have a sum of charges <-2 even though the associated tree-level amplitudes are expected to be nonzero

- the resolution of both is based on worldsheet supersymmetry

In order to obtain non-vanishing n-point tree-level amplitudes, we need representatives of physical vertex operators $V_{\rm phys}(z)\sim:e^{q\phi(z)}:$ with "ghost picture" $q\neq 1,-\frac{1}{2}$, e.g.

- for bosons $\widehat{V}^{\rm NS}_{\rm phys}(z)\sim\widehat{\Phi}^{\rm NS}_{\rm phys}(z):e^{0\phi(z)}:$ in the 0 picture

- for fermions $\widehat{V}^{\rm R}_{\rm phys}(z)\sim\widehat{\Phi}^{\rm R}_{\rm phys}(z):e^{\phi(z)/2}:$ in the $\frac{1}{2}$ picture

- recalling that $h(e^{q\phi}) = -\frac{q^2}{2} - q$, we attain $h(\widehat{V}^{\rm NS}_{\rm phys}) = h(\widehat{V}^{\rm R}_{\rm phys}) = 1$ for the new representatives if the contributions $\widehat{\Phi}^{\rm NS}_{\rm phys}, \widehat{\Phi}^{\rm R}_{\rm phys}$ from the matter variables obey

$$h(\widehat{\Phi}^{\rm NS}_{\rm phys}) = 1 \, , \qquad h(\widehat{\Phi}^{\rm R}_{\rm phys}) = \frac{13}{8} \tag{8.69}$$

- want to represent the same $|{\rm phys}\rangle$ via $(V^{\rm NS}_{\rm phys}, \widehat{V}^{\rm NS}_{\rm phys})$ and $(V^{\rm R}_{\rm phys}, \widehat{V}^{\rm R}_{\rm phys})$, i.e. we have to make sure that the respective $\Phi_{\rm phys}$ and $\widehat{\Phi}_{\rm phys}$ are related by a symmetry of the RNS action, more specifically by the worldsheet supercurrent $G(z) \sim i\partial_z X_\mu(z) \psi^\mu(z)$

For spacetime bosons, $(\Phi^{\rm NS}_{\rm phys}, \widehat{\Phi}^{\rm NS}_{\rm phys})$ must form a superconformal primary of $h = \frac{1}{2}$ (which yields the desired $h(\widehat{\Phi}^{\rm NS}_{\rm phys}) = 1$). By the general OPEs of superconformal primaries, this requires

$$G(z) \Phi^{\rm NS}_{\rm phys}(w) \sim \frac{\widehat{\Phi}^{\rm NS}_{\rm phys}}{2(z-w)} + \ldots \tag{8.70}$$

- absence of double poles in the OPE $G(z)\Phi^{\rm NS}_{\rm phys}(w)$ imposes constraints on polarization tensors, and we will in particular find $\epsilon \cdot p = 0$ in the massless case
- investigate the massless example in detail:

$$\begin{aligned} G(z)\Phi^{\rm NS}_{m^2=0}(w) &\sim : i\partial_z X_\mu(z)\psi^\mu(z) : \epsilon_\lambda : \psi^\lambda(w) e^{ip\cdot X(w)} : \\ &\sim \epsilon_\lambda : \left(\frac{2\alpha' p_\mu}{z-w} + i\partial_w X_\mu(w) + \ldots \right)\left(\frac{\eta^{\mu\lambda}}{z-w} + \psi^\mu \psi^\lambda(w) + \ldots \right) e^{ip\cdot X(w)} : \\ &\sim: \left(\frac{2\alpha' \epsilon\cdot p}{(z-w)^2} + \frac{\epsilon_\mu \big[i\partial_w X^\mu(w) + 2\alpha'(p\cdot\psi)\psi^\mu(w) \big]}{z-w} + \ldots \right) e^{ip\cdot X(w)} : \end{aligned} \tag{8.71}$$

- need $\epsilon \cdot p = 0$ for absence of double poles, and from the simple pole, one can read off the zero-picture vertex operator

$$\widehat{\Phi}^{\rm NS}_{m^2=0}(z) \sim \epsilon_\mu : \big[i\partial_z X^\mu(z) + 2\alpha'(p\cdot\psi)\psi^\mu(z) \big] e^{ip\cdot X(z)} : \tag{8.72}$$

- from the zero-picture representative, one can see that $\epsilon_\mu \to p_\mu$ yields a total derivative $\widehat{\Phi}^{\rm NS}_{m^2=0} \to \partial_z : e^{ip\cdot X(z)} :$ as in the bosonic string (while $p_\mu p_\nu : \psi^\mu \psi^\nu := 0$ by the anticommuting nature of ψ^μ)

For spacetime fermions, we shall now generalize the supersymmetry transformation $G_{-1/2} : \Phi^{\rm NS}_{\rm phys} \to \widehat{\Phi}^{\rm NS}_{\rm phys}$ to $G_{-1} : \Phi^{\rm R}_{\rm phys} \to \widehat{\Phi}^{\rm R}_{\rm phys}$:

- supercurrent modes can be picked up from the contour integral

$$G_r = \oint_{B_\epsilon(w)} \frac{dz}{2\pi i} G(z)(z-w)^{r+1/2} \tag{8.73}$$

- at $r = -1$, we find

$$\widehat{\Phi}^{\rm R}_{\rm phys}(w) = \big(G_{-1}\Phi^{\rm R}_{\rm phys} \big)(w) = \oint_{B_\epsilon(w)} \frac{dz}{2\pi i} G(z)(z-w)^{-1/2} \Phi^{\rm R}_{\rm phys}(w) \tag{8.74}$$

- have to make sure that $\widehat{\Phi}^{\rm R}_{\rm phys}$ is a conformal primary and annihilated by L_1; however, by $[L_1, G_{-1}] = \frac{3}{2}G_0$, only get $h = 1$ primary if the vertex operator in the $-1/2$ picture obeys

$$0 = \big(G_0 \Phi^{\rm R}_{\rm phys} \big)(w) = \oint_{B_\epsilon(w)} \frac{dz}{2\pi i} G(z)(z-w)^{1/2} \Phi^{\rm R}_{\rm phys}(w) \tag{8.75}$$

which requires the pole $\sim (z-w)^{-3/2}$ of the OPE $G(z)\Phi^{\rm R}_{\rm phys}(w)$ to vanish

- in the massless example,

$$G(z)\Phi^{\rm R}_{\rm phys}(w) \sim: i\partial_z X_\mu(z)\psi^\mu(z) : \chi^\alpha S_\alpha(w) : e^{ip\cdot X(w)} :$$
$$\sim \left(\frac{2\alpha' p_\mu}{z-w} + \dots\right)\left(\frac{\chi^\alpha \gamma^\mu_{\alpha\dot\beta} S^{\dot\beta}(w)}{\sqrt{2}(z-w)^{1/2}} + \dots\right) : e^{ip\cdot X(w)} : + \dots \tag{8.76}$$

- again, the G_0-action vanishes as desired if the coefficient of $(z-w)^{-3/2}$ does, i.e. need the fermion polarization χ^α to satisfy the massless Dirac equation

$$\chi^\alpha \gamma^\mu_{\alpha\dot\beta} p_\mu = 0 \tag{8.77}$$

  which eliminates 8 out of 16 components of $\chi^{\alpha=1,2,\dots,16}$; this is similar to the massless Dirac equation in four dimensions with $\gamma^\mu_{\alpha\dot\beta} p_\mu \to \left(\begin{smallmatrix} 2E & 0 \\ 0 & 0 \end{smallmatrix}\right)$ in a suitable $SO(1,3)$ frame, which clearly eliminates one out of two Weyl-spinor components

- after imposing the above requirements on $G(z)\Phi^{\rm NS}_{\rm phys}(w)$ and $G(z)\Phi^{\rm R}_{\rm phys}(w)$, the resulting 8 physical massless fermion polarizations χ^α match the 8 bosonic degrees of freedom in ϵ^μ, which is again an important hallmark of spacetime supersymmetry!

We conclude this section with the following comments:

- instead of $V^{\rm NS}_{\rm phys}, \widehat V^{\rm NS}_{\rm phys}, V^{\rm R}_{\rm phys}, \widehat V^{\rm R}_{\rm phys}$, denote vertex operators for open-string state $|\varphi\rangle$ by $V^{(q)}_\varphi(z) \sim: e^{q\phi(z)} :$, i.e. with the ghost picture q in the superscript; for massless states, for instance

$$\begin{aligned} V^{(-1)}_\epsilon(z) &\sim \epsilon_\mu \psi^\mu(z) : e^{-\phi(z)} e^{ip\cdot X(z)} : \\ V^{(0)}_\epsilon(z) &\sim \epsilon_\mu : \big[i\partial_z X^\mu(z) + 2\alpha'(p\cdot\psi)\psi^\mu(z)\big] e^{ip\cdot X(z)} : \\ V^{(-1/2)}_\chi(z) &\sim \chi^\alpha S_\alpha(z) : e^{-\phi(z)/2} e^{ip\cdot X(z)} : \end{aligned} \tag{8.78}$$

- decoupling of the longitudinal polarization $\epsilon_\mu \to p_\mu$ propagates from $V^{(0)}_\epsilon(z)$ to $V^{(-1)}_\epsilon(z)$ by ghost-picture equivalence, but it is harder to see that $p_\mu \psi^\mu : e^{-\phi + ip\cdot X} :$ is spurious

- for closed strings, $\phi(z)$ and $\bar\phi(\bar z)$ need to independently saturate the background charge; one can independently pick left- and right-moving ghost pictures, e.g. the graviton vertex operator

$$V^{(-1,-1)}_\zeta(z,\bar z) \sim \zeta_{\mu\nu} \psi^\mu(z) \bar\psi^\nu(\bar z) : e^{-\phi(z) - \bar\phi(\bar z) + ip\cdot X(z,\bar z)} : \tag{8.79}$$

  exists in the alternative ghost pictures $V^{(-1,0)}_\zeta, V^{(0,-1)}_\zeta$ and $V^{(0,0)}_\zeta$.

9 String theory and spacetime supersymmetry

Parts of the material in this section can be found in the lecture notes arXiv:1011.1491 on supersymmetry and extra dimensions which are recommended for further reading.

9.1 Supersymmetry in four spacetime dimensions

To avoid collisions with the ten-dimensional notation (with vector indices $\mu, \nu = 0, 1, \dots, 9$ and Weyl-spinor indices $\alpha = 1, 2, \dots, 16$ & $\dot\beta = \dot 1, \dot 2, \dots, \dot{16}$) our conventions for tensors of the four-dimensional Lorentz group $SO(1,3)$ will be as follows

- four-dimensional vector indices $m, n, p, \ldots = 0, 1, 2, 3$
- left- and right-handed Weyl-spinor indices $a, b, \ldots = 1, 2$ and $\dot{a}, \dot{b}, \ldots = \dot{1}, \dot{2}$
- four-dimensional Minkowski metric in mostly-minus signature $\eta^{mn} = \text{diag}(1, -1, -1, -1)$
- four-dimensional Weyl-spinor indices are raised / lowered using the antisymmetric ε-tensor

$$\varepsilon_{ab} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon_{\dot{a}\dot{b}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\psi_a = \varepsilon_{ab}\psi^b, \quad \psi^b = \varepsilon^{bc}\psi_c \quad \text{with } \varepsilon^{bc} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{9.1}$$

$$\bar{\chi}_{\dot{a}} = \varepsilon_{\dot{a}\dot{b}}\bar{\chi}^{\dot{b}}, \quad \bar{\chi}^{\dot{b}} = \varepsilon^{\dot{b}\dot{c}}\bar{\chi}_{\dot{c}} \quad \text{with } \varepsilon^{\dot{b}\dot{c}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

i.e. the $SO(1,3)$ charge-conjugation matrix acting on 4-component Dirac spinors $(\psi_a, \bar{\chi}^{\dot{a}})$ is block diagonal (in contrast to the block-off-diagonal charge-conjugation matrix in $D = 10$)

From a representation-theoretic viewpoint, supersymmetry (SUSY) is an extension of the Poincaré algebra $\{P^m, J^{mn}\}$ by spinor generators $\{Q_a, \bar{Q}_{\dot{b}}\}$ to the so-called *super Poincaré algebra*

- recall the commutation relations among translations P^m and Lorentz transformations J^{mn}

$$[J_{mn}, P_q] = i(\eta_{nq}P_m - \eta_{mq}P_n) \tag{9.2}$$

$$[J_{mn}, J_{pq}] = i\big(\eta_{np}J_{mq} - \eta_{nq}J_{mp} - (m \leftrightarrow n)\big)$$

- (anti-)commutators of Q_a, $\bar{Q}_{\dot{b}}$ involve Pauli matrices

$$\sigma^m_{a\dot{b}} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\} = (\mathbb{1}_{2\times 2}, \vec{\sigma}) \tag{9.3}$$

as well as

$$\bar{\sigma}^{m\dot{a}b} = \varepsilon^{\dot{a}\dot{c}}\varepsilon^{bd}\sigma^m_{d\dot{c}} = (\mathbb{1}_{2\times 2}, -\vec{\sigma}) \tag{9.4}$$

subject to the Clifford algebra $\sigma^m\bar{\sigma}^n + \sigma^n\bar{\sigma}^m = 2\eta^{mn}$

- the commutators of the spinor generators with those of the Poincaré group are

$$[Q_a, J^{mn}] = \frac{i}{4}(\sigma^m\bar{\sigma}^n - \sigma^n\bar{\sigma}^m)_a{}^b Q_b$$

$$[Q_a, P^m] = [\bar{Q}_{\dot{b}}, P^m] = 0 \tag{9.5}$$

- anticommutators among the spinor generators introduce translations

$$\{Q_a, Q_b\} = \{\bar{Q}_{\dot{a}}, \bar{Q}_{\dot{b}}\} = 0$$

$$\{Q_a, \bar{Q}_{\dot{b}}\} = 2\sigma^m_{a\dot{b}}P_m \tag{9.6}$$

In order to study massless representations of the super-Poincaré algebra, we pick a Lorentz frame where the momentum eigenvalues are aligned to $P_m = (E, 0, 0, E)$

$$\Rightarrow\ \{Q_a, \bar{Q}_{\dot{b}}\} = 2E(\sigma^0 + \sigma^3)_{a\dot{b}} = 4E\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow\ \text{vanishing matrix elements for } Q_2 \text{ and } \bar{Q}_{\dot{2}} \tag{9.7}$$

- $Q_1, \bar{Q}_{\dot{1}}$ as raising and lowering operators for helicity λ – the eigenvalue of $J^3 = J^{12}$ adjusted to the spatial components of the P^m eigenvalues

$$[Q_a, J^{12}] = \frac{1}{2} \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)_a{}^b Q_b \tag{9.8}$$
$$\Rightarrow \quad [Q_1, J^3] = \frac{1}{2} Q_1 \quad \text{and} \quad [\bar{Q}_{\dot{1}}, J^3] = -\frac{1}{2} \bar{Q}_{\dot{1}}$$

- can map the action of $Q_1, \bar{Q}_{\dot{1}}$ to the fermionic oscillator by normalizing as follows

$$a = \frac{Q_1}{2\sqrt{E}}\,, \qquad a^\dagger = \frac{\bar{Q}_{\dot{1}}}{2\sqrt{E}} \tag{9.9}$$
$$\Rightarrow \quad a^2 = (a^\dagger)^2 = 0\,, \qquad \{a, a^\dagger\} = 1$$

- build super-Poincaré representations on lowest-helicity state $|\lambda\rangle$ subject to

$$J^3|\lambda\rangle = \lambda|\lambda\rangle\,, \qquad a|\lambda\rangle = 0 \tag{9.10}$$

- SUSY adjoins state $a^\dagger|\lambda\rangle$ of helicity $(\lambda + \frac{1}{2})$ and therefore opposite statistics:

$$\begin{aligned} J^3\big(a^\dagger|\lambda\rangle\big) &= \big([J^3, a^\dagger] + a^\dagger J^3\big)|\lambda\rangle \\ &= \big(\lambda + \tfrac{1}{2}\big)\big(a^\dagger|\lambda\rangle\big) \end{aligned} \tag{9.11}$$

- simple examples of massless SUSY multiplets:

$$\begin{aligned} \underbrace{(\lambda{=}0)\text{ scalar}}_{\substack{\text{e.g. Higgs} \\ \text{or “squark”}}} \; &\underset{a^\dagger}{\overset{a}{\leftrightarrows}} \; \underbrace{(\lambda{=}\tfrac{1}{2})\text{ fermion}}_{\substack{\text{e.g. “Higgsino”} \\ \text{or quark}}} \\ (\lambda{=}\tfrac{1}{2})\text{ “gaugino”} \; &\underset{a^\dagger}{\overset{a}{\leftrightarrows}} \; (\lambda{=}1)\text{ gauge boson} \\ (\lambda{=}\tfrac{3}{2})\text{ “gravitino”} \; &\underset{a^\dagger}{\overset{a}{\leftrightarrows}} \; (\lambda{=}2)\text{ graviton} \end{aligned} \tag{9.12}$$

- gravitinos necessarily introduce local supersymmetry $\sim e^{i\varepsilon^a(x)Q_a}$ (with spacetime dependent spinor parameter $\varepsilon^a(x)$) as a gauge symmetry to avoid negative-norm states; by $\{Q_a, \bar{Q}_{\dot{b}}\} \sim P_m$, this leads to local translation invariance and therefore diffeomorphism symmetry, which introduces gravity and thereby massless spin-two states

Extended SUSY admits $\mathcal{N} \geq 1$ pairs $(Q_a, \bar{Q}_{\dot{b}})$, with an additional index $I, J = 1, 2, \ldots, \mathcal{N}$ in

$$\{Q_a^I, \bar{Q}_{\dot{b}J}\} = 2\delta^I_J \sigma^m_{a\dot{b}} P_m \tag{9.13}$$

- with massless $P_m = (E, 0, 0, E)$, the $\mathcal{N}$-extended super-Poincaré algebra is equivalent to $\mathcal{N}$ copies of the fermionic oscillator

$$\begin{aligned} \{a^I, a^J\} &= \{a^\dagger_I, a^\dagger_J\} = 0 \\ \{a^I, a^\dagger_J\} &= \delta^I_J \end{aligned} \tag{9.14}$$

- representations of $\mathcal{N}$-extended super-Poincaré again built on lowest-helicity state $a^{I=1,2,\ldots,\mathcal{N}}|\lambda\rangle = 0$

$$
\begin{array}{c|c|cc}
\text{state} & \text{helicity} & \#(\text{states}) & \\
\hline
|\lambda\rangle & \lambda & 1 & = \binom{\mathcal{N}}{0} \\
a_I^\dagger|\lambda\rangle & \lambda+\frac{1}{2} & \mathcal{N} & = \binom{\mathcal{N}}{1} \\
a_I^\dagger a_J^\dagger|\lambda\rangle & \lambda+1 & \frac{1}{2}\mathcal{N}(\mathcal{N}-1) & = \binom{\mathcal{N}}{2} \\
\vdots & \vdots & \ldots & \\
a_1^\dagger\ldots\widehat{a_J^\dagger}\ldots a_{\mathcal{N}}^\dagger|\lambda\rangle & \lambda+\frac{\mathcal{N}}{2}-\frac{1}{2} & \mathcal{N} & = \binom{\mathcal{N}}{\mathcal{N}-1} \\
a_1^\dagger a_2^\dagger\ldots a_{\mathcal{N}}^\dagger|\lambda\rangle & \lambda+\frac{\mathcal{N}}{2} & 1 & = \binom{\mathcal{N}}{\mathcal{N}}
\end{array}
\tag{9.15}
$$

for a total of $\sum_{k=0}^{\mathcal{N}}\binom{\mathcal{N}}{k} = 2^{\mathcal{N}}$ states ($2^{\mathcal{N}-1}$ bosonic and $2^{\mathcal{N}-1}$ fermionic ones each)

- $\mathcal{N}=2$ example: vector multiplet

$$
\begin{tikzcd}
 & |\lambda=\tfrac{1}{2};1\rangle \arrow[r, "a_1^\dagger"] \arrow[l, bend left, "a_1"] \arrow[drr, "a_2^\dagger"] & 0 & \\
|\lambda=0\rangle \arrow[ur, bend left, "a_1^\dagger"] \arrow[dr, bend right, "a_2^\dagger"'] & & & |\lambda=1;12\rangle \\
 & |\lambda=\tfrac{1}{2};2\rangle \arrow[r, "a_2^\dagger"'] \arrow[ul, bend right, "a_2"'] \arrow[urr, "a_1^\dagger"] & 0 &
\end{tikzcd}
$$

- SUSY multiplets have to be invariant under CPT (combination of charge conjugation, parity and time reversal) which maps $|\lambda\rangle \to |-\lambda\rangle$; in the present case, this amounts to adding additional states $|\overline{\lambda=0}\rangle, |\lambda=-\frac{1}{2};1\rangle, |\lambda=-\frac{1}{2};2\rangle$ and $|\lambda=-1;12\rangle$ such that for instance $|\lambda=\pm 1;12\rangle$ yields the two physical polarizations of a gauge boson

The maximal SUSY which is possible for a gauge theory (i.e. in absence of gravitational states of helicities $|\lambda| \geq \frac{3}{2}$) is $\mathcal{N}=4$. The states of the so-called $\mathcal{N}=4$ super-Yang–Mills (SYM) theory are built on the lowest-helicity state with $a^{I=1,2,3,4}|\lambda=-1\rangle = 0$:

$$
\begin{tikzcd}
1\times & & 4\times & & 6\times & & 4\times & & 1\times \\
|-1\rangle \arrow[rr, bend left, "a_I^\dagger"] & & |-\tfrac{1}{2};I\rangle \arrow[ll, bend left, "a_I"] \arrow[rr, bend left, "a_J^\dagger"] & & |0;[IJ]\rangle \arrow[ll, bend left, "a_J"] \arrow[rr, bend left, "a_K^\dagger"] & & |\tfrac{1}{2};[IJK]\rangle \arrow[ll, bend left, "a_K"] \arrow[rr, bend left, "a_L^\dagger"] & & |1;[IJKL]\rangle \arrow[ll, bend left, "a_L"]
\end{tikzcd}
$$

6 scalars $\phi^{[IJ]}$

4 "gluinos" $\lambda_a^I, \bar{\lambda}_J^{\dot{b}}$

2 phys. polarizations of the gluon

- $Q_a^I, \bar{Q}_J^{\dot{b}}$ do not act on gauge-group generators, the full multiplet $\{A_m, \chi_a^I, \bar{\chi}_J^{\dot{b}}, \phi^{IJ} = -\phi^{JI}\}$ transforms in the adjoint representation (in contrast to the Standard-Model fermions in (anti-)fundamental representations)

- follows from dimensional reduction of $\mathcal{N}=1$ SYM in $D=10$, which in turn comes from the massless states of the open superstring

- the above states of the $\mathcal{N}=4$ SYM multiplet are CPT self-conjugate, i.e. there is no need or room to add states for CPT invariantization

As our last example of a massless multiplet of extended SUSY, we shall discuss $\mathcal{N}=8$ supergravity with lowest-helicity state $a^{I=1,2,\ldots,8}|\lambda=-2\rangle=0$

"gravitini"

"dilatini"

# states	1	8	28	56	70	56	28	8	1
$\|\lambda\rangle$	$\|-2\rangle$	$\|-\frac{3}{2}\rangle$	$\|-1\rangle$	$\|-\frac{1}{2}\rangle$	$\|0\rangle$	$\|+\frac{1}{2}\rangle$	$\|+1\rangle$	$\|+\frac{3}{2}\rangle$	$\|+2\rangle$

scalars

"graviphotons"

2 phys. polarizations of graviton

- higher SUSY $\mathcal{N}\geq 9$ would necessarily involve helicities $|\lambda|\geq\frac{5}{2}$ and thereby exceed spin two

$$\Longrightarrow \ \mathcal{N}=8 \text{ is the max. supergravity excluding higher spin}$$

- $\mathcal{N}=8$ supergravity can be obtained from dimensional reduction of the unique $\mathcal{N}=1$ supergravity in $D=11$ or from the type IIA/IIB $\mathcal{N}=2$ supergravities in $D=10$ (the massless states of type IIA/IIB superstrings)

- alternatively, the $\mathcal{N}=8$ supergravity multiplet can be constructed from a double copy of $\mathcal{N}=4$ SYM, and the KLT formula uplifts this relation to tree amplitudes of arbitrary $\mathcal{N}=8$ states

- again, this construction of the $\mathcal{N}=8$ supergravity multiplet is automatically CPT self-conjugate

9.2 Open superstrings and higher-dimensional supersymmetry

Recall the worldsheet realization of Poincaré generators of open bosonic string

$$\begin{aligned} P^{\rm bos}_\mu &= \frac{1}{2\alpha'}\oint \frac{dz}{2\pi i} i\partial_z X_\mu(z) \\ J^{\rm bos}_{\mu\nu} &= \frac{1}{2\alpha'}\oint \frac{dz}{2\pi i}(iX_\mu i\partial_z X_\nu - iX_\nu i\partial_z X_\mu) \end{aligned} \tag{9.16}$$

One can similarly construct super-Poincaré generators from ingredients of RNS superstrings:

- observe that $P^{\rm bos}_\mu$ matches the gluon vertex operator at zero momentum; this turns out to generalize to the open superstring, where the two ghost pictures $V^{(q)}_\epsilon$ at $q=-1,0$ yield two incarnations of the translation operator

$$\begin{aligned} P^{(-1)}_\mu &= \frac{1}{\sqrt{2\alpha'}}\oint \frac{dz}{2\pi i}\psi_\mu(z) : e^{-\phi(z)} : \\ P^{(0)}_\mu &= \frac{1}{2\alpha'}\oint \frac{dz}{2\pi i} i\partial_z X_\mu(z) \end{aligned} \tag{9.17}$$

- the Lorentz generators augment $J^{\rm bos}_{\mu\nu}$ by a bi-spinor contribution acting on the spacetime vector ψ^μ

$$J^{(0)}_{\mu\nu} = \oint \frac{dz}{2\pi i}\left[\frac{1}{2\alpha'}(iX_\mu i\partial_z X_\nu - iX_\nu i\partial_z X_\mu) + :\psi_\mu\psi_\nu:\right] \tag{9.18}$$

- by analogy with $\lim_{p\to 0} V_\epsilon^{(q)} \sim P_\mu^{(q)}$, obtain SUSY generators $Q_\alpha^{(q)}$ in ghost pictures $q = \pm\frac{1}{2}$ from the fermion vertex $V_\chi^{(q)}$ at $p \to 0$

$$Q_\alpha^{(-1/2)} = \sqrt{2}(\alpha')^{-1/4} \oint \frac{dz}{2\pi i} S_\alpha : e^{-\phi/2} : \tag{9.19}$$

$$Q_\alpha^{(+1/2)} = \frac{1}{\sqrt{2}(\alpha')^{3/4}} \oint \frac{dz}{2\pi i} i\partial_z X_\mu \gamma^\mu_{\alpha\dot\beta} S^{\dot\beta} : e^{\phi/2} :$$

- in both pictures $q = \pm\frac{1}{2}$, the spin-field OPE in $D = 10$ implies

$$\{Q_\alpha^{(-1/2)}, Q_\beta^{(q)}\} = 2(\gamma^\mu C)_{\alpha\beta} P_\mu^{(q-1/2)} \tag{9.20}$$

which resembles the anticommutator of the four-dimensional SUSY algebra

$$\{Q_a, \bar{Q}_{\dot b}\} = 2(\sigma^m)_{a\dot b} P_m \tag{9.21}$$

note, however, that the supercharges in the $D = 10$ anticommutator have alike chiralities, whereas non-vanishing anticommutators in $D = 4$ involve opposite chiralities

- will now reproduce the extended $\mathcal{N} = 4, 8$ SUSY multiplets in four dimensions from dimensional reduction of $D = 10$ SUSY

The open superstring enjoys $\mathcal{N} = 1$ SUSY in $D = 10$ dimensions, i.e. its supercharges form a single left-handed Weyl spinor $Q_{\alpha=1,2,\ldots,16}$ of $SO(1,9)$

$\Rightarrow$ 16 components or "supercharges" just like $\mathcal{N} = 4$ in four spacetime dimensions

- dimensional reduction of Q_α to $D = 4$ is based on the decomposition

$$SO(1,9) \to SO(1,3) \times SO(6) \tag{9.22}$$

$SO(1,3)$ is interpreted as the four-dimensional Lorentz group and acts on the first four components of $P_\mu = (E, 0, 0, E, \vec{0})$, where Q_α shifts the eigenvalue of J^{12} by $\pm\frac{1}{2}$ just as in four dimensions; the second factor $SO(6)$ has Cartan generators J^{45}, J^{67}, J^{89}, and Q_α shifts their eigenvalues as well

- more specifically, the decomposition of the ten-dimensional SUSY generators (left-handed Weyl spinor of $SO(1,9)$) is

$$Q_\alpha = Q_{a=1,2}^{I=1,2,3,4} \oplus \bar{Q}^{\dot b=1,2}_{J=1,2,3,4} \tag{9.23}$$

where $Q_{a=1,2}^{I=1,2,3,4}$ carries a left-handed Weyl-spinor index a of $SO(1,3)$ and a left-handed Weyl-spinor index I of $SO(6)$; similarly, $\bar{Q}^{\dot b=1,2}_{J=1,2,3,4}$ features right-handed Weyl-spinor indices $\dot b$ and J of $SO(1,3)$ and $SO(6)$, respectively

$\Longrightarrow$ the four-dimensional label $I = 1, 2, \ldots, \mathcal{N}$ of extended SUSY
refers to spinors of $SO(6)$ when dimensionally reduced from $D = 10$

- in this setting, we obtain the $\mathcal{N} = 4$ SYM multiplet in $D = 4$ by dimensionally reducing the $\mathcal{N} = 1$ SYM multiplet in $D = 10$ dimensions: the massless states of the ten-dimensional open superstring were found to arise from vertex operators

$$\text{gluon } A_\mu \leftrightarrow V_\epsilon^{(q=-1,0)} \tag{9.24}$$

$$\text{gluino } \chi_\mu \leftrightarrow V_\chi^{(q=-1/2,1/2)}$$

- decompose both fields under $SO(1,9) \to SO(1,3) \times SO(6)$

$$A_{\mu=0,1,\ldots,9} \to \begin{cases} A_{m=0,1,2,3} & : \ \text{vector in } D=4 \\ A_{i=4,5,\ldots,9} & : \ \text{scalar in } D=4 \end{cases} \tag{9.25}$$

$$\chi^{\alpha=1,2,\ldots,16} \to \begin{cases} \chi^{a=1,2}_{I=1,2,3,4} & : \ \text{left-handed spinor in } D=4 \\ \bar{\chi}^{J=1,2,3,4}_{\dot{b}=\dot{1},\dot{2}} & : \ \text{right-handed spinor in } D=4 \end{cases} \tag{9.26}$$

The explicit SUSY transformations of vertex operators and hence of physical states can be computed from CFT methods, by inserting OPEs into the contour-integral representations of the supercharges $Q^{(\mp 1/2)}_\alpha$. We will focus on the massless states and for bookkeeping purposes contract the spinor index α of the supercharges with a reference Weyl-spinor η^α

- $Q^{(\mp 1/2)}_\alpha$: boson $\to$ fermion (with normalization g_ϵ of the gluon vertex operator, see section 6.5.3)

$$\begin{aligned}
[\eta^\alpha Q^{(-1/2)}_\alpha, V^{(0)}_\epsilon(w)] &= \frac{\sqrt{2}}{\alpha'^{1/4}} \oint \frac{dz}{2\pi i} \eta^\alpha S_\alpha(z) :e^{-\phi(z)/2}: \frac{g_\epsilon}{\sqrt{2\alpha'}} \epsilon_\mu : \big[i\partial_w X^\mu(w) + 2\alpha'(p\cdot\psi)\psi^\mu(w)\big] e^{ip\cdot X(w)}: \\
&= g_\epsilon \alpha'^{1/4} \epsilon_\mu p_\nu (\eta\gamma^\mu\gamma^\nu)^\alpha S_\alpha : e^{-\phi(w)/2} e^{ip\cdot X(w)} :
\end{aligned} \tag{9.27}$$

where the right-hand side can be identified as a gluino vertex operator $V^{(-1/2)}_\chi = g_\epsilon \alpha'^{1/4} \chi^\alpha S_\alpha : e^{-\phi(w)/2} e^{ip\cdot X(w)} :$ with wavefunction $\chi^\alpha = \epsilon_\mu p_\nu (\eta\gamma^\mu\gamma^\nu)^\alpha$. The latter obeys the massless Dirac equation $\chi^\alpha \gamma^\lambda_{\alpha\dot{\beta}} p_\lambda$ since $p_\nu p_\lambda \gamma^\nu \gamma^\lambda = p_\nu p_\lambda \eta^{\nu\lambda} = p^2 = 0$ in the massless case.

- $Q^{(\mp 1/2)}_\alpha$: fermion $\to$ boson (by repeating the above kind of OPE computation)

$$[\eta^\alpha Q^{(-1/2)}_\alpha, V^{(-1/2)}_\chi(w)] = \ldots = g_\epsilon (\eta\gamma_\mu C\chi)\psi^\mu(w) : e^{-\phi(w)} e^{ip\cdot X(w)} : \tag{9.28}$$

this results in a gluon vertex operator with polarization vector $\epsilon_\mu = (\eta\gamma_\mu C\chi)$ which is transversal since χ obeys the massless Dirac equation

- with the shorthand notation to summarize the above transformations

$$\delta_\eta \epsilon^\mu = (\eta\gamma^\mu C\chi)\,, \qquad \delta_\eta \chi^\beta = \eta^\alpha (\gamma^{\mu\nu})_\alpha{}^\beta \epsilon_\mu p_\nu \tag{9.29}$$

the SUSY algebra $\{Q_\alpha, Q_\beta\} = 2(\gamma^\mu C)_{\alpha\beta} P_\mu$ in ten dimensions closes up to linearized gauge transformations (i.e. changes in $\epsilon^\lambda \sim p^\lambda$ in the second term of the second line)

$$\begin{aligned}
[\delta_{\eta_1}, \delta_{\eta_2}]\chi^\alpha &= 2(\eta_1\gamma^\mu C\eta_2) p_\mu \chi^\alpha \\
[\delta_{\eta_1}, \delta_{\eta_2}]\epsilon^\lambda &= 2(\eta_1\gamma^\mu C\eta_2) p_\mu \epsilon^\lambda - 2(\eta_1\gamma^\mu C\eta_2)\epsilon_\mu p^\lambda
\end{aligned} \tag{9.30}$$

9.3 Closed superstrings and higher-dimensional supersymmetry

The goal of this subsection is to

- construct ten-dimensional type-IIA/IIB supergravity multiplets from massless closed-string states
- recover $\mathcal{N}=8$ supergravity upon dimensional reduction to $D=4$

In the double-copy construction of closed-string states, one can choose independent chiralities of the spin fields S_A and $\bar{S}_B$ in the GSO projection of left- and right-movers, respectively:

- the underlying left-moving OPEs $\psi^\mu(z) S_A(w)$ and $S_A(z) S_B(w)$ decouple from those among the right-movers $\bar{\psi}^\mu, \bar{S}_A$
- only the relative chirality matters for the spectrum

$$\begin{array}{llll} \text{type IIB} : & \text{“(left-handed)}^{\otimes 2}\text{”}, & S_A \to S_\alpha\,, & \bar{S}_B \to \bar{S}_\beta \\ \text{type IIA} : & \text{“(left)}\otimes\text{(right)”}, & S_A \to S_\alpha\,, & \bar{S}_B \to \bar{S}^{\dot{\beta}} \end{array} \tag{9.31}$$

As will be elaborated below, one can decompose the spacetime fermions in the massless closed-string spectrum into irreducibles of $SO(1,9)$. This is similar to the spacetime bosons in the NS-NS sector

$$\epsilon^\mu \otimes \bar{\epsilon}^\nu \to \underbrace{\zeta^{\mu\nu}}_{35 \text{ d.o.f.}} \oplus \underbrace{B^{\mu\nu}}_{28 \text{ d.o.f.}} \oplus \underbrace{\eta^{\mu\nu}_{SO(8)}\Phi}_{1 \text{ d.o.f.}} \tag{9.32}$$

- R-NS sector: by analogy with the NS-NS sector, the $SO(1,9)$ decomposition is of schematic form

$$\chi^\alpha \otimes \bar{\epsilon}_\mu \quad \longrightarrow \quad \text{“}\gamma\text{-traceless”} \oplus \text{“}\gamma\text{-trace”} \tag{9.33}$$

  whose exact form is (note that $\gamma^\mu \gamma_\mu = 10$ in ten dimensions)

$$\chi^\alpha \bar{\epsilon}_\mu = \underbrace{\chi^\alpha \bar{\epsilon}_\mu - \frac{1}{10}\bar{\epsilon}_\lambda (\chi \gamma^\lambda \gamma_\mu)^\alpha}_{\substack{\text{gravitino, spin 3/2-rep. of} \\ SO(1,9) \text{ with } 8(8-1) \text{ d.o.f.}}} + \underbrace{\frac{1}{10}\bar{\epsilon}_\lambda (\chi \gamma^\lambda \gamma_\mu)^\alpha}_{\substack{\text{write as } \Lambda_{\dot{\beta}} \gamma_\mu^{\dot{\beta}\alpha} \text{ with } \Lambda_{\dot{\beta}} \text{ a} \\ \text{right-handed dilatino (8 d.o.f.)}}} \tag{9.34}$$

  the 8 d.o.f. of the dilatino $\Lambda_{\dot{\beta}} = \frac{1}{10}\bar{\epsilon}_\lambda (\chi \gamma^\lambda)_{\dot{\beta}}$ can be traced back to the massless Dirac equation $\Lambda_{\dot{\beta}} \gamma_\mu^{\dot{\beta}\alpha} p^\mu = 0$ (which follows from that of χ and transversality of $\bar{\epsilon}_\lambda$)

- NS-R sector:

$$\begin{array}{lll} \text{type IIB} : & \epsilon^\mu \otimes \bar{\chi}^\alpha & \Longrightarrow \text{ gravitino / dilatino of same chirality as in R-NS sector} \\ \text{type IIA} : & \epsilon^\mu \otimes \bar{\chi}_{\dot{\beta}} & \Longrightarrow \text{ gravitino / dilatino of opposite chirality as in R-NS sector} \end{array} \tag{9.35}$$

- the two gravitini in type II theories necessitate two local spacetime SUSYs (they are needed as gauge-symmetries to remove negative-norm states), i.e. $\mathcal{N} = 2$ in ten dimensions; construct the left- and right-moving supercharges in direct analogy with open strings

$$\begin{array}{lll} \text{type IIB} : & Q_\alpha^{(-1/2)} \sim \oint : e^{-\phi/2} : S_\alpha\,, & \bar{Q}_\beta^{(-1/2)} \sim \oint : e^{-\phi/2} : \bar{S}_\beta \\ \text{type IIA} : & Q_\alpha^{(-1/2)} \sim \oint : e^{-\phi/2} : S_\alpha\,, & \bar{Q}^{(-1/2)\dot{\beta}} \sim \oint : e^{-\phi/2} : \bar{S}^{\dot{\beta}} \end{array} \tag{9.36}$$

In order to understand the Lorentz irreducibles in the R-R sector, decompose bi-spinors of $SO(1,9)$ into antisymmetric tensors, so-called “form-fields” $A^{(k)}_{[\mu_1\mu_2\ldots\mu_k]}$.

- for 16-component Weyl spinors in $D = 10$, which products of 16×16 gamma matrices γ^μ exhaust the (16×16)-dimensional space of bispinors?

  $\longrightarrow$ by $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$, it is sufficient to study antisymmetrized products

$$\gamma^{\mu_1\mu_2\ldots\mu_k} = \frac{1}{k!}\big[\underbrace{\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_k} \pm \text{antisymm}(1,2,\ldots,k)}_{k! \text{ terms}}\big] \tag{9.37}$$

- in $D = 10$, duality w.r.t. $\varepsilon^{\mu_1\mu_2\ldots\mu_{10}}$ relates k-forms $\gamma^{\mu_1\mu_2\ldots\mu_k}$ to $(10{-}k)$-forms $\gamma^{\mu_1\mu_2\ldots\mu_{10-k}}$, with (anti-)self-duality at $k = 5$

$$\begin{aligned} (\gamma^{\mu_1\mu_2\ldots\mu_5}C)_{\alpha\beta} &= \frac{1}{5!}\varepsilon^{\mu_1\mu_2\ldots\mu_{10}}(\gamma_{\mu_6\mu_7\ldots\mu_{10}}C)_{\alpha\beta} \\ (\gamma^{\mu_1\mu_2\ldots\mu_5}C)^{\dot{\alpha}\dot{\beta}} &= -\frac{1}{5!}\varepsilon^{\mu_1\mu_2\ldots\mu_{10}}(\gamma_{\mu_6\mu_7\ldots\mu_{10}}C)^{\dot{\alpha}\dot{\beta}} \end{aligned} \tag{9.38}$$

- for Weyl-spinors $\psi, \bar\lambda$ of $SO(1,9)$ with opposite chirality, can decompose via Fierz identities

$$\psi^\alpha \bar\lambda_{\dot\beta} = \frac{1}{16} \underbrace{(C^{-1})^\alpha{}_{\dot\beta}}_{1 \text{ matrix}} (\psi C \bar\lambda) + \frac{1}{16 \cdot 2!} \underbrace{(C^{-1}\gamma^{\mu\nu})^\alpha{}_{\dot\beta}}_{\binom{10}{2}=45 \text{ matrices}} (\psi \gamma_{\nu\mu} C \bar\lambda) + \frac{1}{16 \cdot 4!} \underbrace{(C^{-1}\gamma^{\mu\nu\rho\sigma})^\alpha{}_{\dot\beta}}_{\binom{10}{4}=210 \text{ matrices}} (\psi \gamma_{\sigma\rho\nu\mu} C \bar\lambda) \tag{9.39}$$

- for Weyl-spinors ψ, λ of $SO(1,9)$ with alike chirality, the analogous Fierz identity is

$$\begin{aligned} \psi^\alpha \lambda^\beta &= \frac{1}{16} \underbrace{(C^{-1}\gamma^\mu)^{\beta\alpha}}_{10 \text{ matrices}} (\psi \gamma_\mu C \lambda) + \frac{1}{16 \cdot 3!} \underbrace{(C^{-1}\gamma^{\mu\nu\rho})^{\beta\alpha}}_{\binom{10}{3}=120 \text{ matrices}} (\psi \gamma_{\rho\nu\mu} C \lambda) \\ &\quad + \frac{1}{2 \cdot 16 \cdot 5!} \underbrace{(C^{-1}\gamma^{\mu\nu\rho\sigma\tau})^{\beta\alpha}}_{\frac{1}{2}\binom{10}{5}=126 \text{ matrices}} (\psi \gamma_{\tau\sigma\rho\nu\mu} C \lambda) \end{aligned} \tag{9.40}$$

  where the denominator in the last term is $2 \cdot 16 \cdot 5!$ rather than $16 \cdot 5!$ to avoid overcounting in view of the self-duality of $\gamma^{\mu\nu\rho\sigma\tau}$; note the symmetry properties $(C^{-1}\gamma^\mu)^{\beta\alpha} = (C^{-1}\gamma^\mu)^{\alpha\beta}$ as well as $(C^{-1}\gamma^{\mu\nu\rho})^{\beta\alpha} = -(C^{-1}\gamma^{\mu\nu\rho})^{\alpha\beta}$ and $(C^{-1}\gamma^{\mu\nu\rho\sigma\tau})^{\beta\alpha} = (C^{-1}\gamma^{\mu\nu\rho\sigma\tau})^{\alpha\beta}$

- component counting in the R-R sector: physical closed-string states arise from the double copy of physical gaugino polarizations

$$(8 \text{ solutions to the massless Dirac equation})^{\otimes 2}$$

  which yields 64 d.o.f. in bi-spinors $(\chi \gamma_{\mu_1 \ldots \mu_k} C \bar\chi)$

- need the k-form expansions of $\chi^\alpha \otimes \bar\chi^\beta$ (type IIB) and $\chi^\alpha \otimes \bar\chi_{\dot\beta}$ (type IIA) to respect $\chi^\alpha \gamma^\mu_{\alpha\dot\beta} p_\mu = 0$ (and same for $\bar\chi$)

Engineer bispinors compatible with the Dirac equation via

$$(\chi \gamma_{\mu_1\mu_2\ldots\mu_k} C \bar\chi) \to F^{(k)}_{\mu_1\mu_2\ldots\mu_k} = k p_{[\mu_1} A^{(k-1)}_{\mu_2\ldots\mu_k]} \tag{9.41}$$

with transversal $(k-1)$-form potentials $A^{(k-1)}_{\mu\nu\ldots} = A^{(k-1)}_{[\mu\nu\ldots]}$

$$p^\mu A^{(k-1)}_{\mu\nu\ldots} = 0 \tag{9.42}$$

- respective field strengths $F^{(k)} = dA^{(k-1)}$ enjoy gauge invariance with $(k-2)$-form parameters $\xi^{(k-2)}$

$$\delta A^{(k-1)} = d\xi^{(k-2)} \tag{9.43}$$

- as an example how the Dirac equation relates to the transversality of the potentials, consider the one-form $A^{(1)}_\mu$ in the R-R sector of type IIA superstrings (with field strength $F^{(2)}_{\mu\nu} = p_\mu A^{(1)}_\nu - p_\nu A^{(1)}_\mu$) which is obtained from the double copy

$$\begin{aligned} \chi^\alpha \bar\chi_{\dot\beta} &\to \frac{1}{2} (C^{-1}\gamma^{\mu\nu})^\alpha{}_{\dot\beta} F^{(2)}_{\mu\nu} = (C^{-1}\gamma^{\mu\nu})^\alpha{}_{\dot\beta} p_\mu A^{(1)}_\nu \\ &= (C^{-1}\gamma^\mu\gamma^\nu)^\alpha{}_{\dot\beta} p_\mu A^{(1)}_\nu = -(C^{-1}\gamma^\nu\gamma^\mu)^\alpha{}_{\dot\beta} p_\mu A^{(1)}_\nu \end{aligned} \tag{9.44}$$

  in replacing $\gamma^{\mu\nu}$ by either $\gamma^\mu\gamma^\nu$ or $-\gamma^\nu\gamma^\mu$, we have used that the $\mu \leftrightarrow \nu$ symmetric part yields $\eta^{\mu\nu}$ whose contraction with $p_\mu A^{(1)}_\nu$ vanishes by transversality; the first representation in the second line of (9.44) manifests that the Dirac equation of χ^α is preserved (contract with $p_\lambda(\gamma^\lambda C)_{\delta\alpha}$) while the second representation manifests that of $\bar\chi_{\dot\beta}$ (contract with $p^\lambda \gamma_\lambda^{\dot\beta\delta}$)

- the type IIA solutions along with their d.o.f. are

$$\chi^{\alpha}\bar{\chi}_{\dot{\beta}} \to \begin{cases} (C^{-1}\gamma^{\mu\nu})^{\alpha}{}_{\dot{\beta}} p_{\mu} A^{(1)}_{\nu} & : \ 8 \text{ d.o.f.} \\ (C^{-1}\gamma^{\mu\nu\rho\sigma})^{\alpha}{}_{\dot{\beta}} p_{\mu} A^{(3)}_{\nu\rho\sigma} & : \ 56 = \binom{8}{3} \text{ d.o.f.} \end{cases} \tag{9.45}$$

  by transversality of the $A^{(k)}_{\mu_1 \ldots \mu_k}$ and their gauge freedom under $d\xi^{(k-1)}$, we can treat the indices μ_j as effectively taking 8 values in the degree-of-freedom counting

- the type IIB solutions along with their d.o.f. are

$$\chi^{\alpha}\bar{\chi}^{\beta} \to \begin{cases} (C^{-1}\gamma^{\mu})^{\alpha\beta} p_{\mu} A^{(0)} & : \ 1 \text{ scalar "axion"} \\ (C^{-1}\gamma^{\mu\nu\rho})^{\alpha\beta} p_{\mu} A^{(2)}_{\nu\rho} & : \ 28 = \binom{8}{2} \text{ d.o.f.} \\ (C^{-1}\gamma^{\mu\nu\rho\sigma\tau})^{\alpha\beta} p_{\mu} A^{(4)}_{\nu\rho\sigma\tau} & : \ 35 = \frac{1}{2}\binom{8}{4} \text{ d.o.f.} \end{cases} \tag{9.46}$$

  where the factor of $\frac{1}{2}$ in the d.o.f. of $A^{(4)}$ stems from the self-duality of its five-form field strength.

- in the same way as the $B_{\mu\nu}$ in the NS-NS sector couples to strings through the worldsheet action $\int_{\Sigma} dX^{\mu} \wedge dX^{\nu} B_{\mu\nu}$, the R-R form fields $A^{(p+1)}$ couple to Dp branes with (p+1)-dimensional worldvolumes Σ_{p+1} via

$$S_{p\text{-form}} \sim \int_{\Sigma_{p+1}} dX^{\mu_1} \wedge dX^{\mu_2} \wedge \ldots \wedge dX^{\mu_{p+1}} A^{(p+1)}_{\mu_1 \mu_2 \ldots \mu_{p+1}} \tag{9.47}$$

- conversely, the R-R sector states single out the stable types of Dp branes in the respective theory

$$\begin{aligned} &\text{type IIA} \ : \quad D0 \leftrightarrow A^{(1)} \, , \quad D2 \leftrightarrow A^{(3)} \\ &\text{type IIB} \ : \quad D(-1) \leftrightarrow A^{(0)} \, , \quad D1 \leftrightarrow A^{(2)} \, , \quad D3 \leftrightarrow A^{(4)} \end{aligned} \tag{9.48}$$

  where the D(-1) brane has a zero-dimensional worldvolume, i.e. it covers a point in spacetime and is referred to as an *instanton*

- since T-duality swaps Dp and D(p±1) branes, their R-R fields are swapped as well

$$\Longrightarrow \quad \text{type IIA} \ \stackrel{\text{T-duality}}{\longleftrightarrow} \ \text{type IIB} \tag{9.49}$$

We shall now assemble the ten-dimensional supergravity multiplets from the massless states in the type IIA/IIB theories. In the following table, the number of physical d.o.f. determined from the above discussions is indicated in parenthesis, and the shorthands LH and RH stand for left- and right-handed spacetime fermions

<table>
<tr><th>$\otimes$</th><th colspan="2">NS</th><th colspan="2">R</th></tr>
<tr><td>NS</td><td colspan="2">graviton $h_{\mu\nu}$ (35)
B-field $B_{\mu\nu}$ (28)
dilaton Φ (1)</td><td colspan="2">LH gravitino χ^{α}_{μ} (56)
RH dilatino $\Lambda_{\dot{\beta}}$ (8)</td></tr>
<tr><td rowspan="2">R</td><td>type IIA</td><td>type IIB</td><td>type IIA</td><td>type IIB</td></tr>
<tr><td>RH $\bar{\chi}^{\mu}_{\dot{\beta}}$ (56)
LH $\bar{\Lambda}^{\alpha}$ (8)</td><td>LH $\bar{\chi}^{\alpha}_{\mu}$ (56)
RH $\bar{\Lambda}_{\dot{\beta}}$ (8)</td><td>$A^{(1)}_{\mu}$ (8)
$A^{(3)}_{\mu\nu\rho}$ (56)</td><td>$A^{(0)}$ (1)
$A^{(2)}_{\mu\nu}$ (28)
$A^{(4)}_{\mu\nu\rho\sigma}$ (35)</td></tr>
</table>

We shall now dimensionally reduce the type-IIB multiplet from ten to four spacetime dimensions. In the following set of reductions, $|\lambda\rangle$ refers to a state with helicity or J^{12} eigenvalue λ, and the range of $SO(1,3)$- and $SO(6)$ indices is again $m,n,\ldots=0,1,2,3$ and $i,j,\ldots=4,5,\ldots,9$ for vectors as well as $a,b=1,2,\ \dot a,\dot b=\dot 1,\dot 2$ and $I,J=1,2,3,4$ for Weyl spinors.

$$
\begin{aligned}
\Phi \ \&\ A^{(0)} \ &\longrightarrow \ 2\times|\lambda{=}0\rangle \\
B_{\mu\nu}\ \&\ A^{(2)}_{\mu\nu} \ &\longrightarrow \ \begin{cases} B_{mn}\ \&\ A^{(2)}_{mn} \ \leftrightarrow \ 2\times|\lambda{=}0\rangle \\ B_{mi}\ \&\ A^{(2)}_{mi} \ \leftrightarrow \ 2\times 12\times|\lambda{=}\pm 1\rangle \\ B_{ij}\ \&\ A^{(2)}_{ij} \ \leftrightarrow \ 2\times 15\times|\lambda{=}0\rangle \end{cases} \\
h_{\mu\nu} \ &\longrightarrow \ \begin{cases} h_{mn} \ \leftrightarrow \ 2\times|\lambda{=}\pm 2\rangle \\ h_{mi} \ \leftrightarrow \ 12\times|\lambda{=}\pm 1\rangle \\ h_{ij} \ \leftrightarrow \ 21\times|\lambda{=}0\rangle \end{cases} \\
A^{(4)}_{\mu\nu\rho\sigma} \ &\longrightarrow \ \begin{cases} A^{(4)}_{ijkl} \ \leftrightarrow \ \binom{6}{4}\times|\lambda{=}0\rangle \\ A^{(4)}_{mijk} \ \leftrightarrow \ \binom{6}{3}\times|\lambda{=}\pm 1\rangle \\ \text{no } A^{(4)}_{mnij},\ A^{(4)}_{mnpi},\ A^{(4)}_{mnpq} \end{cases} \\
\chi^\alpha_\mu,\ \bar\chi^\alpha_\mu \ &\longrightarrow \ \begin{cases} \chi^a_{mI}\oplus\chi^J_{m\dot b}\oplus\left(\substack{\text{same with}\\ \chi\to\bar\chi}\right) \ \leftrightarrow \ 2\times 4\times 2\times|\lambda{=}\pm\tfrac{3}{2}\rangle \\ \chi^a_{iI}\oplus\chi^J_{i\dot b}\oplus\left(\substack{\text{same with}\\ \chi\to\bar\chi}\right) \ \leftrightarrow \ 6\times 2\times 4\times 2\times|\lambda{=}\pm\tfrac{1}{2}\rangle \end{cases} \\
\Lambda_{\dot\beta},\ \bar\Lambda_{\dot\beta} \ &\longrightarrow \ \Lambda_{aI}\oplus\Lambda^{bJ}\oplus\left(\substack{\text{same with}\\ \Lambda\to\bar\Lambda}\right) \ \leftrightarrow \ 2\times 4\times 2\times|\lambda{=}\pm\tfrac{1}{2}\rangle
\end{aligned}
\tag{9.50}
$$

For gravitons, the vanishing of the ten-dimensional trace $\eta^{\mu\nu}h_{\mu\nu}$ interlocks the $SO(1,3)$-trace of h_{mn} with the $SO(6)$-trace of h_{ij}, and we effectively treat h_{mn} as traceless in the degree-of-freedom counting. For the dimensional reduction of $A^{(4)}$, the self-duality of $F^{(5)}$ with momentum $p_\mu=(p_m,0,0,0,0,0,0)$ leaves no independent d.o.f. in components with more than one $SO(1,3)$ index.

The above distribution of four-dimensional helicities precisely ties in with the spectrum of $\mathcal{N}=8$ supergravity

- $1\times|\lambda{=}2\rangle$ from $h_{\mu\nu}$
- $8\times|\lambda{=}\frac{3}{2}\rangle$ from χ^α_μ (4), $\bar\chi^\alpha_\mu$ (4)
- $28\times|\lambda{=}1\rangle$ from $B_{\mu\nu}$ (6), $A^{(2)}_{\mu\nu}$ (6), $h_{\mu\nu}$ (6), $A^{(4)}_{\mu\nu\rho\sigma}$ (10)
- $56\times|\lambda{=}\frac{1}{2}\rangle$ from χ^α_μ (24), $\bar\chi^\alpha_\mu$ (24), $\Lambda_{\dot\beta}$ (4), $\bar\Lambda_{\dot\beta}$ (4)
- $70\times|\lambda{=}0\rangle$ from $B_{\mu\nu}$ (16), $A^{(2)}_{\mu\nu}$ (16), $h_{\mu\nu}$ (21), $A^{(4)}_{\mu\nu\rho\sigma}$ (15), Φ (1), $A^{(0)}$ (1)

with the same counting for the negative-helicity states.

As you will explore in exercise G.4, the ten-dimensional type IIA multiplet can also be obtained as a dimensional reduction of the unique $D=11$ supergravity (with an $\mathcal{N}=1$ Dirac spinor of 32 supercharges). With vector indices $M,N,P=0,1,\ldots,10$ and Dirac-spinor index $B=1,2,\ldots,32$ (there is no notion of chirality in odd dimensions), its particle content is

$$
\begin{aligned}
h_{MN} \ &\longleftrightarrow \ \text{graviton with 44 d.o.f.} \\
A^{(3)}_{MNP} \ &\longleftrightarrow \ \text{3-form potential with } \binom{9}{3}=84 \text{ d.o.f.} \\
\chi^B_M \ &\longleftrightarrow \ \text{gravitino with 128 d.o.f.}
\end{aligned}
\tag{9.51}
$$

10 Superstring amplitudes

Scattering amplitudes in superstring theories admit the same kind of topological expansion in terms of two-dimensional surfaces as you know from bosonic strings. The worldsheet topologies contributing to tree-level and one-loop amplitudes of open and closed strings are depicted in figure 25. One-loop amplitudes of open superstrings for instance have contributions from both cylinder and Moebius-strip diagrams, and their interplay is crucial for the cancellation of UV divergences (which are absent in closed-string loop diagrams from the outset)

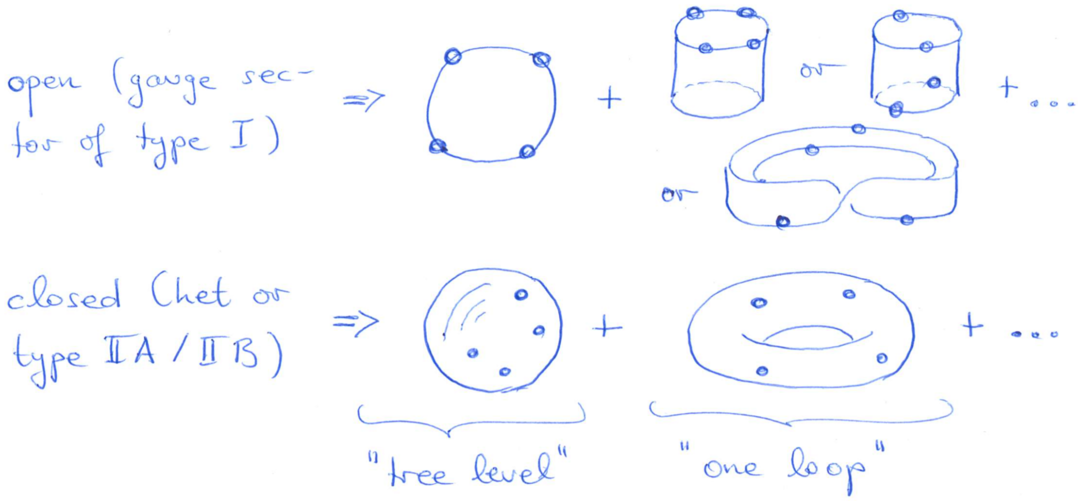


Figure 25: Worldsheet topologies contributing to four-point tree-level and one-loop amplitudes of open and closed superstrings

Still, there are two noteworthy differences to bosonic-string amplitudes:

- the final results for superstring amplitudes are considerably simpler than those in bosonic theories
- strictly speaking, the RNS prescription for superstring amplitudes involves an integral over bosonic *and fermionic* moduli of super-Riemann surfaces (where the ghost pictures address the fermionic moduli at low genus)
- one cannot split super-moduli space at genus $g \geq 5$ in the sense that it is not possible to integrate over the fermionic moduli *first* to then be left with a bosonic integral

10.1 Superstring tree-level amplitudes

For open superstrings, the properties of Chan-Paton charges associated with their endpoints are identical to those of open bosonic strings. Hence, we have the same color-decompositions of open-string tree-level amplitudes in the bosonic and the supersymmetric theory

$$\mathcal{M}^{\rm tree}_{\substack{\rm open \\ \rm super}}(\{\varphi_i, a_i\}; \alpha') = \sum_{\rho \in S_{n-1}} {\rm Tr}(T^{a_{\rho(1)}} T^{a_{\rho(2)}} \ldots T^{a_{\rho(n-1)}} T^{a_n}) \mathcal{A}^{\rm tree}_{\rm super}(\varphi_{\rho(1)}, \varphi_{\rho(2)}, \ldots, \varphi_{\rho(n-1)}, \varphi_n; \alpha') \tag{10.1}$$

where we use collective notation φ_i for polarizations and momenta as well as adjoint indices a_i for the color degrees of freedom.

The color-ordered amplitudes $\mathcal{A}^{\rm tree}_{\rm super}$ are again computed from correlation functions of vertex operators along with c-ghost contributions $\langle c(z_i)c(z_j)c(z_k)\rangle = |z_{ij}z_{ik}z_{jk}|$ for the SL_2-fixed legs at $(z_i,z_j,z_k) = (0,1,\infty)$:

$$\mathcal{A}^{\rm tree}_{\rm super}(\varphi_1,\varphi_2,\ldots,\varphi_n;\alpha') \sim \int\limits_{-\infty<z_1<z_2<\ldots<z_n<\infty} \frac{dz_1\,dz_2\,\ldots\,dz_n}{\mathrm{vol}\,\mathrm{SL}_2(\mathbb{R})} \langle V^{(q_1)}_{\varphi_1}(z_1)V^{(q_2)}_{\varphi_2}(z_2)\ldots V^{(q_n)}_{\varphi_n}(z_n)\rangle \,\big|_{\sum_{j=1}^n q_j=-2} \tag{10.2}$$

- need total superghost charge -2 at tree level (and $2g-2$ at genus g) to saturate the background charge of the β-γ system

$$\left\langle \prod_{j=1}^n :e^{q_j\phi(z_j)}: \right\rangle = \delta\left(2+\sum_{j=1}^n q_j\right) \prod_{1\leq i<j}^n (z_i-z_j)^{-q_iq_j} \tag{10.3}$$

  for each combination of physical states with an even number of spacetime fermions (amplitudes with an odd number of spacetime fermions vanish by Lorentz invariance), one can find a variety of representatives $V^{(q_1)}_{\varphi_1}\ldots V^{(q_n)}_{\varphi_n}$ in admissible ghost pictures

- the result for $\mathcal{A}^{\rm tree}_{\rm super}$ is independent on the distribution of the q_i over the vertex operators (e.g. for n bosons, insert any two legs in the $V^{(-1)}_\varphi$ picture and the remaining $n{-}2$ in the $V^{(0)}_\varphi$ picture); intuitively, this can be understood from the fact that a relocation of the superghost pictures is a worldsheet supersymmetry transformation; the detailed argument can for instance be found in section 5.2.3 of https://edoc.ub.uni-muenchen.de/13381/1/Schlotterer_Oliver.pdf.

10.1.1 Examples of open-string amplitudes

We shall next consider several warm-up examples of open-superstring tree amplitudes. The computation of the relevant correlators will be greatly facilitated by the fact that X^μ, ϕ and the (ψ^μ,S_α)-system furnish three separate CFT sectors (i.e. all correlation functions factorize into those of X^μ, ϕ and (ψ^μ,S_α)).

- for three gluons, recall that $p_i\cdot p_j=0$ in the kinematic phase space of three massless particles such that the Koba-Nielsen factor is 1; without loss of generality, we choose leg 1 to be the only instance of the $V^{(0)}_\epsilon$-picture

$$\begin{aligned}
\mathcal{A}^{\rm tree}_{\rm super}(\epsilon_1,\epsilon_2,\epsilon_3) &\sim |z_{12}z_{13}z_{23}|\langle V^{(0)}_{\epsilon_1}(z_1)V^{(-1)}_{\epsilon_2}(z_2)V^{(-1)}_{\epsilon_3}(z_3)\rangle \\
&= |z_{12}z_{13}z_{23}|\langle :e^{-\phi(z_2)}::e^{-\phi(z_3)}:\rangle \epsilon^1_\mu\epsilon^2_\nu\epsilon^3_\lambda \\
&\quad\times\Big\{ \langle :i\partial_{z_1}X^\mu(z_1)e^{ip_1\cdot X(z_1)}: \prod_{j=2}^3 :e^{ip_j\cdot X(z_j)}:\rangle \,\langle\psi^\nu(z_2)\psi^\lambda(z_3)\rangle \\
&\qquad + 2\alpha'\langle\prod_{j=1}^3 :e^{ip_j\cdot X(z_j)}:\rangle p^1_\rho\langle :\psi^\rho\psi^\mu(z_1): \psi^\nu(z_2)\psi^\lambda(z_3)\rangle\Big\} \\
&\sim \frac{|z_{12}z_{13}z_{23}|}{z_{12}z_{13}z_{23}}\Big\{\underbrace{(p_2\cdot\epsilon_1)(\epsilon_2\cdot\epsilon_3)}_{\text{from }\partial_{z_1}X^\mu} + \underbrace{(\epsilon_1\cdot\epsilon_2)(p_1\cdot\epsilon_3)-(\epsilon_1\cdot\epsilon_3)(p_1\cdot\epsilon_2)}_{\text{from }\psi^\rho\psi^\mu}\Big\} \\
&= \mathcal{A}^{\rm tree}_{\rm SYM}(\epsilon_1,\epsilon_2,\epsilon_3)
\end{aligned} \tag{10.4}$$

  Similar to the correlators of the bosonic string, the net effect of the $i\partial_{z_1}X^\mu(z_1)$ insertion is a factor of $2\alpha'(\frac{\epsilon_1\cdot p_2}{z_{12}}+\frac{\epsilon_1\cdot p_3}{z_{13}}) = 2\alpha'\frac{\epsilon_1\cdot p_2 z_{23}}{z_{12}z_{13}}$ (using transversality and momentum conservation). The remaining correlators have been evaluated via $\langle : e^{-\phi(z_2)} :: e^{-\phi(z_3)} :\rangle = z_{23}^{-1}$ and $\langle : \psi^\rho \psi^\mu(z_1) : \psi^\nu(z_2)\psi^\lambda(z_3)\rangle = (\eta^{\mu\nu}\eta^{\rho\lambda} - \eta^{\mu\lambda}\eta^{\nu\rho})/(z_{12}z_{13})$. Furthermore, $\frac{|z_{12}z_{13}z_{23}|}{z_{12}z_{13}z_{23}} = -1$ is independent on the z_j within the color-ordering of $\mathcal{A}^{\rm tree}_{\rm super}(\epsilon_1, \epsilon_2, \epsilon_3)$ but instead gives $+1$ in case of $\mathcal{A}^{\rm tree}_{\rm super}(\epsilon_1, \epsilon_3, \epsilon_2)$. Accordingly, the color-dressed amplitude depends on the color degrees of freedom via $\text{Tr}(T^{a_1}[T^{a_2}, T^{a_3}]) \sim f^{a_1a_2a_3}$.

Most importantly, the three-gluon amplitude coincides with its field-theory limit, i.e. it is homogeneous in α' and does not involve any analogue of the extra term $2\alpha'(p_i \cdot \epsilon_j)^3$ in the bosonic-string amplitude. This exemplifies a general virtue of superstring amplitudes that the α'-dependence takes a considerably simpler form that for bosonic strings. In the RNS formulation, one can attribute this simplicity to worldsheet supersymmetry, while the amplitude computations in the pure-spinor formalism are simplified by the manifest spacetime supersymmetry.

- for one gluon and two gluinos, it is natural to employ the -1 and $-\frac{1}{2}$ ghost pictures in evaluating

$$\begin{aligned}\mathcal{A}^{\rm tree}_{\rm super}(\epsilon_1, \chi_2, \chi_3) &\sim |z_{12}z_{13}z_{23}| \langle V^{(-1)}_{\epsilon_1}(z_1) V^{(-1/2)}_{\chi_2}(z_2) V^{(-1/2)}_{\chi_3}(z_3)\rangle \\ &= |z_{12}z_{13}z_{23}| \langle : e^{-\phi(z_1)} :: e^{-\phi(z_2)/2} :: e^{-\phi(z_3)/2} :\rangle \langle \prod_{j=1}^3 : e^{ip_j\cdot X(z_j)} :\rangle \\ &\quad \times \epsilon_1^\mu \chi_2^\alpha \chi_3^\beta \, \langle \psi_\mu(z_1) S_\alpha(z_2) S_\beta(z_3)\rangle \\ &\sim \frac{|z_{12}z_{13}z_{23}|}{z_{12}z_{13}z_{23}} \frac{1}{\sqrt{2}} \epsilon_1^\mu (\chi_2 \gamma_\mu C \chi_3) \\ &= \mathcal{A}^{\rm tree}_{\rm SYM}(\epsilon_1, \chi_2, \chi_3)\end{aligned} \tag{10.5}$$

Since we are scattering GSO projected states, all fractional powers of z_{ij} from matter and ghost contributions $\langle : e^{-\phi(z_1)} :: e^{-\phi(z_2)/2} :: e^{-\phi(z_3)/2} :\rangle = (z_{12}z_{13})^{-1/2} z_{23}^{-1/4}$ and $\langle \psi_\mu(z_1) S_\alpha(z_2) S_\beta(z_3)\rangle = \frac{(\gamma_\mu C)_{\alpha\beta}}{\sqrt{2}(z_{12}z_{13})^{1/2} z_{23}^{3/4}}$ conspire to integer ones.

- the results of both three-point examples can be summarized in a spacetime supersymmetric way by

$$\mathcal{A}^{\rm tree}_{\rm super}(1,2,3) = \mathcal{A}^{\rm tree}_{\rm SYM}(1,2,3) \tag{10.6}$$

where $1, 2, 3$ refer to any combination of states in the massless gauge multiplet; the right-hand side refers to the amplitudes of $\mathcal{N} = 1$ in SYM $D = 10$ which can for instance be generated from the Lagrangian

$$S_{\rm SYM}[A, \chi] \sim \int d^{10}X \, \text{Tr}\left\{ \frac{1}{4} F_{\mu\nu}F^{\mu\nu} + \chi^\alpha (\gamma^\mu C)_{\alpha\beta} [D_\mu, \chi^\beta] \right\} \tag{10.7}$$

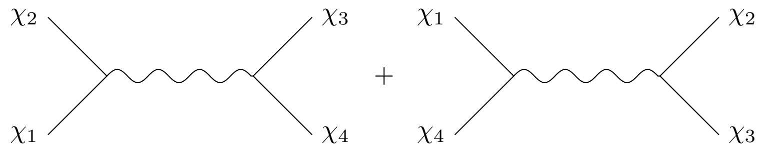
with gauge-covariant derivatives $[D_\mu, \chi] = \partial_\mu \chi - [A_\mu, \chi]$; the bosonic part $\sim F_{\mu\nu}F^{\mu\nu}$ of the Lagrangian is identical to that of pure Yang–Mills theory, that is why the n-gluon tree amplitudes are universal to SYM and pure YM; the Feynman rules of the SYM action consist of the three- and four-gluon vertices known from pure YM and an additional three-point vertex of two gauginos and one gluon, see the above $\mathcal{A}^{\rm tree}_{\rm SYM}(\epsilon_1, \chi_2, \chi_3)$ for its (unique) tensor structure

- among the massless four-point functions, the final result for the four-gluino amplitude takes a particularly compact form (see exercise H.2); with the conventions $s_{ij} = \alpha'(p_i + p_j)^2$ for dimensionless

Mandelstam invariants, we have

$$
\begin{aligned}
\mathcal{A}^{\rm tree}_{\rm super}(\chi_1,\chi_2,\chi_3,\chi_4;\alpha') &\sim \int_0^1 dz_2 \lim_{z_4\to\infty} |z_{13}z_{14}z_{34}| \, \langle V^{(-1/2)}_{\chi_1}(z_1)V^{(-1/2)}_{\chi_2}(z_2)V^{(-1/2)}_{\chi_3}(z_3)V^{(-1/2)}_{\chi_4}(z_4)\rangle \\
&\sim s_{12}\int_0^1 \frac{dz_2}{z_2}|z_2|^{s_{12}}|1{-}z_2|^{s_{23}} \bigg\{ \frac{(\chi_1\gamma_\mu C\chi_2)(\chi_3\gamma^\mu C\chi_4)}{s_{12}} + \frac{(\chi_4\gamma_\mu C\chi_1)(\chi_2\gamma^\mu C\chi_3)}{s_{23}} \bigg\} \\
&\sim \frac{\Gamma(1{+}s_{12})\Gamma(1{+}s_{23})}{\Gamma(1{+}s_{12}{+}s_{23})} \mathcal{A}^{\rm tree}_{\rm SYM}(\chi_1,\chi_2,\chi_3,\chi_4) \qquad (10.8)
\end{aligned}
$$

in the color-ordering under investigation, our choice of SL_2-fixing $(z_1,z_3,z_4)\to(0,1,\infty)$ leads to the integration domain $z_2\in(0,1)$; the four-point SYM amplitude in the curly bracket of the second line is literally the result of the Feynman-diagram computation below



- finally, the four-point result obtained for gauginos uplifts to the entire gauge supermultiplet, i.e. the string tree amplitude for arbitrary combinations of gluons and gluinos takes the universal form

$$
\mathcal{A}^{\rm tree}_{\rm super}(1,2,3,4;\alpha') = \frac{\Gamma(1{+}s_{12})\Gamma(1{+}s_{23})}{\Gamma(1{+}s_{12}{+}s_{23})} \mathcal{A}^{\rm tree}_{\rm SYM}(1,2,3,4) \qquad (10.9)
$$

by the leading term in the α'-expansion $\frac{\Gamma(1+s_{12})\Gamma(1+s_{23})}{\Gamma(1+s_{12}+s_{23})} = 1-\zeta_2 s_{12}s_{23}+\mathcal{O}(\alpha'^3)$, this is consistent with the field-theory limit $\alpha'\to 0$ where any massless $\mathcal{A}^{\rm tree}_{\rm super}$ reduces to the corresponding $\mathcal{A}^{\rm tree}_{\rm SYM}$.

10.1.2 Massless n-point open-string amplitudes

The tight interplay between open-superstring and SYM tree amplitudes generalizes to any number n of external gauge multiplets: As derived in arXiv:1106.2645 from the pure-spinor superstring, such color-ordered n-point amplitudes can be decomposed into

$$
\mathcal{A}^{\rm tree}_{\rm super}(1,2,\ldots,n;\alpha') = \sum_{\rho\in S_{n-3}} F^\rho(s_{ij})\mathcal{A}^{\rm tree}_{\rm SYM}(1,\rho(2,3,\ldots,n{-}2),n{-}1,n) \qquad (10.10)
$$

where all the polarization dependence resides in a $(n{-}3)!$ basis of $\mathcal{A}^{\rm tree}_{\rm SYM}(1,\rho(\ldots),n{-}1,n)$ w.r.t. BCJ relations (the field-theory limits of monodromy relations). The α'-dependence in turn is entirely carried by the basis of disk integrals

$$
F^\rho(s_{ij}) = \int\limits_{0<z_2<z_3<\ldots<z_{n-2}<1} dz_2\, dz_3\, \ldots\, dz_{n-2} \prod_{1\leq i<j}^{n-1} |z_{ij}|^{s_{ij}} \rho\bigg\{ \prod_{k=2}^{n-2}\sum_{m=1}^{k-1} \frac{s_{km}}{z_{km}} \bigg\} \qquad (10.11)
$$

with $(z_1,z_{n-1},z_n)=(0,1,\infty)$, where the permutation $\rho\in S_{n-3}$ acts on the labels of both the z_{km} and s_{km} enclosed by the curly brackets. The α'-expansion of the $(n{-}3)!$ integrals $F^\rho(s_{ij})$ tailored to the disk ordering with $z_i<z_{i+1}$ has leading orders

$$
F^\rho(s_{ij}) = \delta^\rho_{23\ldots n-2} + \mathcal{O}(\alpha'^2) \qquad (10.12)
$$

The field-theory limit for ρ different from the identity permutation is absent such that the above expression for $\mathcal{A}^{\rm tree}_{\rm super}$ consistently reduces to $\mathcal{A}^{\rm tree}_{\rm SYM}$ in the same color-ordering as $\alpha'\to 0$. The first non-vanishing

α'-corrections take the schematic form $\zeta_2 s_{ij}^2 + \zeta_3 s_{ij}^3 + \ldots$, and more generally, the α'^w order features polynomials in s_{ij} accompanied by multiple zeta values $\zeta_{n_1,n_2,\ldots,n_r}$ of weight $w = n_1 + n_2 + \ldots + n_r$, see for instance arXiv:1304.7304. This property is known as *uniform transcendentality* and has also been observed in various instances of field-theory amplitudes and Feynman integrals.

10.1.3 Closed-string tree amplitudes

For tree-level amplitudes of type IIA and type IIB superstrings, the opening line

$$\mathcal{M}^{\text{tree}}_{\text{super}}(\{\Phi_i\};\alpha') \sim \int \frac{d^2 z_1\, d^2 z_2 \ldots d^2 z_n}{\operatorname{vol}\mathrm{SL}_2(\mathbb{C})} \Big\langle \prod_{a=1}^{n} V_{\Phi_a}^{(q_a,\bar q_a)}(z_a) \Big\rangle \Big|^{\sum_{j=1}^n q_j=-2}_{\sum_{j=1}^n \bar q_j=-2} \tag{10.13}$$

involves correlation functions of vertex operators $V_\Phi^{(q,\bar q)}$ with independent left- and right-moving superghost pictures $q, \bar q$. By the background charge of the respective superghost systems, one needs to separately saturate

$$\sum_{j=1}^{n} q_j = -2 = \sum_{j=1}^{n} \bar q_j \tag{10.14}$$

in order to have non-zero correlators for the $: e^{q_j \phi(z_j)} :$ and $: e^{\bar q_j \bar\phi(\bar z_j)} :$ involving chiral bosons ϕ and $\bar\phi$.

The sphere integrals over the resulting correlation functions can be manipulated with the same KLT techniques that you have seen for the closed bosonic string

$$\begin{aligned}\mathcal{M}^{\text{tree}}_{\text{super}}(\{\Phi_1,\Phi_2,\ldots,\Phi_n\};\alpha') \sim \sum_{\rho,\tau\in S_{n-3}} &\mathcal{A}^{\text{tree}}_{\text{super}}(\varphi_1,\rho(\varphi_2,\varphi_3,\ldots,\varphi_{n-2}),\varphi_{n-1},\varphi_n;\tfrac{\alpha'}{4}) \\ &\times S_{\alpha'}(\rho|\tau)\mathcal{A}^{\text{tree}}_{\text{super}}(\bar\varphi_1,\tau(\bar\varphi_2,\bar\varphi_3,\ldots,\bar\varphi_{n-2}),\bar\varphi_n,\bar\varphi_{n-1};\tfrac{\alpha'}{4})\end{aligned} \tag{10.15}$$

The entries of the $(n{-}3)! \times (n{-}3)!$ matrix $S_{\alpha'}(\rho|\tau)$ indexed by permutations ρ and τ are again degree-$(n{-}3)$ polynomials in $\big[\sin(\frac{\pi\alpha'}{2} p_i \cdot p_j)\big]^{n-3}$. This KLT formula is universally valid for any combination of type-IIA or type-IIB states $\Phi_j = \varphi_j \otimes \bar\varphi_j$, and it can be adapted to the heterotic string by replacing one of the $\mathcal{A}^{\text{tree}}_{\text{super}}$ on the right-hand side by $\mathcal{A}^{\text{tree}}_{\text{bos}}$. In other words, the KLT decomposition into products of disk integrals is a universal property of sphere integrals, regardless of the string theory they originate from.

Finally, in case of four massless states (i.e. arbitrary combinations $1,2,3,4$ of states in the type IIA/IIB supergravity multiplets), the trigonometric factor can be used to rearrange the Γ functions of $\mathcal{A}^{\text{tree}}_{\text{super}}(1,2,3,4;\alpha')$ into

$$\begin{aligned}\mathcal{M}^{\text{tree}}_{\text{super}}(\{1,2,3,4\};\alpha') &\sim \sin\left(\frac{\pi\alpha'}{2} p_1\cdot p_2\right)\mathcal{A}^{\text{tree}}_{\text{super}}(1,2,3,4;\alpha')\mathcal{A}^{\text{tree}}_{\text{super}}(\bar 1,\bar 2,\bar 4,\bar 3;\alpha') \\ &\sim \frac{\Gamma(1+\frac{s}{4})\Gamma(1+\frac{t}{4})\Gamma(1+\frac{u}{4})}{\Gamma(1-\frac{s}{4})\Gamma(1-\frac{t}{4})\Gamma(1-\frac{u}{4})}\mathcal{M}^{\text{tree}}_{\text{supergravity}}(\{1,2,3,4\}) \\ &= \Big(1 - \frac{\zeta_3}{32} stu + \mathcal{O}(\alpha'^5)\Big)\mathcal{M}^{\text{tree}}_{\text{supergravity}}(\{1,2,3,4\})\end{aligned} \tag{10.16}$$

On the right-hand side, we have identified the four-point supergravity amplitudes according to the field-theory double copy

$$\mathcal{M}^{\text{tree}}_{\text{supergravity}}(\{1,2,3,4\}) \sim \mathcal{A}^{\text{tree}}_{\text{SYM}}(1,2,3,4)(p_1\cdot p_2)\mathcal{A}^{\text{tree}}_{\text{SYM}}(\bar 1,\bar 2,\bar 4,\bar 3) \tag{10.17}$$

where the type-IIA and type-IIB cases are distinguished by the chirality of the gauginos in the second factor of $\mathcal{A}^{\text{tree}}_{\text{SYM}}$. The first string correction to supergravity occurs at the order of α'^3 and corresponds to an $\alpha'^3\zeta_3\mathcal{R}^4$ interaction together with its $\mathcal{N}=2$ superpartners in $D=10$ dimensions. In other words, the curvature corrections $\alpha'\mathcal{R}^2$ and $\alpha'^2\mathcal{R}^3$ of the bosonic string are absent for type-II superstrings. This can firstly be viewed as a consequence of the dropout of $\alpha' F^3$ interactions from the open superstring. Secondly, the 32 supercharges of type-II superstrings are incompatible with the operators $\alpha'\mathcal{R}^2$ and $\alpha'^2\mathcal{R}^3$ of the closed bosonic string.

Note, however, that the $(n-3)! \times (n-3)!$-permutation sum in the n-point KLT formula does not allow to factor out the analogous $(n{\geq}5)$-point supergravity amplitudes from $\mathcal{M}^{\rm tree}_{\rm super}(\{\Phi_i\};\alpha')$ at $n{\geq}5$.

A Problem set Sept 13$^{\text{th}}$ 2023

A.1 Equations of motion

The purpose of this exercise is to determine the equations of motion (e.o.m.) for the Nambu–Goto and Polyakov actions $S_{\text{NG}}[X]$ and $S_{\text{P}}[X,h]$ introduced in the lecture

(i) derive the following lemmata concerning the induced metric $\gamma_{\alpha\beta} = \frac{\partial X^\mu}{\partial \sigma^\alpha}\frac{\partial X^\nu}{\partial \sigma^\beta}\eta_{\mu\nu}$ on the worldsheet

$$\frac{\partial \gamma_{\beta\delta}}{\partial(\partial_\alpha X_\mu)} = \delta^\alpha_\beta \partial_\delta X^\mu + \delta^\alpha_\delta \partial_\beta X^\mu$$
$$\frac{\partial\sqrt{-\det\gamma}}{\partial \gamma_{\beta\delta}} = \frac{1}{2}\sqrt{-\det\gamma}\,\gamma^{\delta\beta}$$

(ii) show that the e.o.m. derived from the Nambu–Goto action takes the form of a reparametrization invariant wave equation (with tension $T=\frac{1}{2\pi\alpha'}$)

$$\frac{\delta S_{\text{NG}}[X]}{\delta X_\mu} = T\,\partial_\alpha\big(\sqrt{-\det\gamma}\,\gamma^{\alpha\beta}\partial_\beta X^\mu\big)\,.$$

(iii) show that the e.o.m. of the Polyakov action w.r.t. the (inverse) worldsheet metric is given by

$$\frac{\delta S_{\text{P}}[X,h]}{\delta h^{\alpha\beta}} = -\frac{\sqrt{-\det h}}{4\pi\alpha'}\Big\{\partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} - \frac{1}{2}h_{\alpha\beta}(h^{\gamma\delta}\partial_\gamma X^\mu \partial_\delta X^\nu)\eta_{\mu\nu}\Big\}\,.$$

A.2 Warmup on Christoffel symbols

Following your General Relativity course, the Christoffel symbols associated with the metric $h_{\alpha\beta}$ are given by

$$\Gamma^\sigma_{\alpha\beta} = \frac{1}{2}h^{\sigma\rho}(\partial_\alpha h_{\beta\rho} + \partial_\beta h_{\alpha\rho} - \partial_\rho h_{\alpha\beta})\,,$$

and they enter the covariant derivatives of vectors V^β and covectors W_β as follows

$$\nabla_\alpha V^\beta = \partial_\alpha V^\beta + \Gamma^\beta_{\gamma\alpha}V^\gamma\,, \qquad \nabla_\alpha W_\beta = \partial_\alpha W_\beta - \Gamma^\gamma_{\beta\alpha}W_\gamma$$

(i) check that their trace can be written as

$$\Gamma^\sigma_{\sigma\alpha} = \partial_\alpha \log\sqrt{-\det h}\,.$$

(ii) use this to show that the equations of motions $0 = \frac{\delta S_{\text{P}}}{\delta X_\mu}$ can be written as a reparametrization invariant wave equation

$$\partial_\alpha(\sqrt{-\det h}\,h^{\alpha\beta}\partial_\beta X^\mu) = \sqrt{-\det h}\,\nabla_\alpha\nabla^\alpha X^\mu = 0$$

(iii) in the context of diffeomorphism invariance of the Polyakov action, show that the infinitesimal transformation $\delta h_{\alpha\beta} = \eta^\rho\partial_\rho h_{\alpha\beta} + h_{\rho\alpha}\partial_\beta\eta^\rho + h_{\rho\beta}\partial_\alpha\eta^\rho$ can be rewritten as $\delta h_{\alpha\beta} = \nabla_\alpha\eta_\beta + \nabla_\beta\eta_\alpha$.

(iv) given the metric in conformal gauge $h_{\alpha\beta} = e^{2\phi}\eta_{\alpha\beta}$, check that the only non-vanishing Christoffel symbols in lightcone coordinates are

$$\Gamma^+_{++} = 2\partial_+\phi\,, \qquad \Gamma^-_{--} = 2\partial_-\phi\,.$$

(v) compute the resulting components of the Riemann tensor and the Ricci scalar

A.3 Diffeomorphism invariance of the Polyakov action

This exercise helps you to verify the invariance of the Polyakov action under infinitesimal diffeomorphisms

$$\delta X^\mu = \eta^\gamma \partial_\gamma X^\mu \,, \qquad \delta h_{\alpha\beta} = \eta^\gamma \partial_\gamma h_{\alpha\beta} + h_{\gamma\alpha}\partial_\beta \eta^\gamma + h_{\gamma\beta}\partial_\alpha \eta^\gamma \,.$$

(i) show that the above variation of the metric propagates as follows to its inverse and its determinant:

$$\begin{aligned} \delta h^{\alpha\beta} &= \eta^\gamma \partial_\gamma h^{\alpha\beta} - \partial^\alpha \eta^\beta - \partial^\beta \eta^\alpha \\ \delta\sqrt{-\det h} &= \partial_\gamma(\eta^\gamma \sqrt{-\det h}) \end{aligned}$$

(ii) use these lemmata to identify the variation of the Polyakov action as a total derivative

$$\delta S_{\rm P}[X,h] = -\frac{1}{4\pi\alpha'} \int \mathrm{d}^2\sigma \, \partial_\gamma \Big(\eta^\gamma \sqrt{-\det h}\, h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu \Big) \,.$$

A.4 Noether currents for Poincaré symmetry

The purpose of this exercise is to study the Noether currents and conserved charges w.r.t. the global Poincaré symmetry

$$\delta X^\mu = \Lambda^\mu{}_\nu X^\nu + c^\mu \,, \qquad \Lambda^{\mu\nu} = -\Lambda^{\nu\mu}$$

of the Polyakov action.

(i) by writing the variation w.r.t. local transformations in the form

$$\delta S_{\rm P}[X,h] = \int \mathrm{d}^2\sigma \, \Big(P^\alpha_\mu \partial_\alpha c^\mu + \frac{1}{2} J^\alpha_{\mu\nu} \partial_\alpha \Lambda^{\mu\nu} \Big) \,,$$

determine the momentum current P^α_μ and the Lorentz current $J^\alpha_{\mu\nu}$ in terms of a general metric $h_{\alpha\beta}$ (i.e. before gauge fixing)

(ii) show that the associated charges in conformal gauge are given by

$$\begin{aligned} P_\mu &= \int_0^{2\pi} \mathrm{d}\sigma \, P^\tau_\mu = \frac{1}{2\pi\alpha'} \int_0^{2\pi} \mathrm{d}\sigma \, \partial_\tau X_\mu \\ J_{\mu\nu} &= \int_0^{2\pi} \mathrm{d}\sigma \, J^\tau_{\mu\nu} = \frac{1}{2\pi\alpha'} \int_0^{2\pi} \mathrm{d}\sigma \, (X_\nu \partial_\tau X_\mu - X_\mu \partial_\tau X_\nu) \end{aligned}$$

and prove that they are conserved, i.e. that $\partial_\tau P_\mu = \partial_\tau J_{\mu\nu} = 0$, if X^μ obeys the equations of motion

(iii) by inserting the mode expansion

$$X^\mu = x^\mu + \alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{1}{n} (\tilde{\alpha}^\mu_n e^{-in(\tau+\sigma)} + \alpha^\mu_n e^{-in(\tau-\sigma)}) \,,$$

derive the following expressions for the conserved charges

$$\begin{aligned} P^\mu &= p^\mu \\ J^{\mu\nu} &= p^\mu x^\nu - x^\mu p^\nu + i \sum_{n=1}^\infty \frac{1}{n} \big(\alpha^\nu_n \alpha^\mu_{-n} - \alpha^\mu_n \alpha^\nu_{-n} + \tilde{\alpha}^\nu_n \tilde{\alpha}^\mu_{-n} - \tilde{\alpha}^\mu_n \tilde{\alpha}^\nu_{-n} \big) \,. \end{aligned}$$

A.5 Commutation relations

The goal of this exercise is to study the consequences of the equal-time commutators

$$[X^\mu(\tau,\sigma),\Pi_\nu(\tau,\sigma')] = i\delta^\mu_\nu\delta(\sigma-\sigma')$$
$$[X^\mu(\tau,\sigma),X^\nu(\tau,\sigma')] = [\Pi_\mu(\tau,\sigma),\Pi_\nu(\tau,\sigma')] = 0$$

and to derive the Hamiltonian in the canonical quantization of the bosonic string.

(i) use the mode expansion of the canonical momentum $\Pi^\mu = \frac{1}{2\pi\alpha'}\dot{X}^\mu$ to derive ($k\in\mathbb{Z}$)

$$\int_0^{2\pi} \mathrm{d}\sigma\, \Pi^\mu(\tau,\sigma) = p^\mu\,, \qquad \int_0^{2\pi} \mathrm{d}\sigma\, e^{ik\sigma}\Pi^\mu(\tau,\sigma) = \frac{1}{\sqrt{2\alpha'}}\big(\tilde{\alpha}^\mu_k e^{-ik\tau} + \alpha^\mu_{-k}e^{ik\tau}\big)$$

(ii) show that the Hamiltonian takes the following form in terms of the oscillator modes

$$\begin{aligned} H &= \frac{1}{4\pi\alpha'}\int_0^{2\pi}\mathrm{d}\sigma\,\big(\dot{X}^2(\tau,\sigma) + X'^2(\tau,\sigma)\big) \\ &= \frac{\alpha'}{2}p^2 + \frac{1}{2}\sum_{n=1}^{\infty}\big(\alpha_{-n}\cdot\alpha_n + \alpha_n\cdot\alpha_{-n} + \tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n + \tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\big) \end{aligned}$$

and, using the commutation relations in (iii), that normal ordering formally leads to the alternative expression in D spacetime dimensions

$$H = \frac{\alpha'}{2}p^2 + \sum_{n=1}^{\infty}\big(\alpha_{-n}\cdot\alpha_n + \tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n\big) + D\sum_{n=1}^{\infty} n\,.$$

(iii) **OPTIONAL** (no need to submit a solution for this part): use the results of (i) to show that the above equal-time commutators lead to the following commutators among x^μ, p_ν & the oscillators $\alpha^\mu_n, \tilde{\alpha}^\nu_k$ with $n,k\neq 0$,

$$[x^\mu,p_\nu] = i\delta^\mu_\nu\,, \qquad [\alpha^\mu_n,\alpha^\nu_k] = n\,\eta^{\mu\nu}\delta_{n+k,0}\,, \qquad [\tilde{\alpha}^\mu_n,\tilde{\alpha}^\nu_k] = n\,\eta^{\mu\nu}\delta_{n+k,0}$$
$$[x^\mu,x^\nu] = [p_\mu,p_\nu] = [x^\mu,\alpha^\nu_n] = [x^\mu,\tilde{\alpha}^\nu_n] = [p_\mu,\alpha^\nu_n] = [p_\mu,\tilde{\alpha}^\nu_n] = [\alpha^\mu_n,\tilde{\alpha}^\nu_k] = 0$$

To be submitted on Sept 28$^{\text{th}}$ 2023 by 6pm

B Problem set Sept 28th 2023

B.1 Hamiltonian constraints and Polyakov action

The purpose of this exercise is to derive the Polyakov action using Dirac's method of Hamiltonian constraints. We start from the Nambu-Goto action

$$S_{\rm NG}[X] = -T\int \mathrm{d}^2\sigma\,\sqrt{-\det\gamma}\,,$$

where $\gamma_{\alpha\beta} = \partial_\alpha X\cdot\partial_\beta X$ as well as $\dot X_\mu = \partial_\tau X_\mu$ and $X'_\mu = \partial_\sigma X_\mu$.

(i) Compute the canonical momenta $P_\mu = \frac{\delta S_{\rm NG}}{\delta \dot X^\mu}$ and prove the results to obey the constraints

$$P^2 + (TX')^2 = 0\,, \qquad P\cdot X' = 0\,.$$

(ii) Given (i), we can define an alternative action which is equivalent to $S_{\rm NG}$,

$$S_{\rm alt}[X,P,e,u] = \int \mathrm{d}^2\sigma\,\Big\{\dot X\cdot P - \frac{1}{2}e(P^2+(TX')^2) - uP\cdot X'\Big\}\,.$$

In $S_{\rm alt}$ the constraints on the momenta are imposed by the Lagrange multipliers e and u, and we will see that this leads to the same equations of motion as $S_{\rm NG}$. Use the equation $\frac{\delta S_{\rm alt}}{\delta P_\mu} = 0$ to solve for the momenta.

(iii) By substituting the result of (ii) into the action, reduce it to the following form:

$$S_{\rm alt}[X,P,e,u] = -\frac{T}{2}\int \mathrm{d}^2\sigma\,\widetilde h^{\alpha\beta}\partial_\alpha X\cdot\partial_\beta X,$$

$$\widetilde h^{\alpha\beta} = \frac{1}{Te}\begin{pmatrix} -1 & u \\ u & (Te)^2-u^2\end{pmatrix}.$$

Comment: This can be rewritten in the form of the familiar Polyakov action

$$S_{\rm P}[X,h] = -\frac{T}{2}\int \mathrm{d}^2\sigma\,\sqrt{-\det h}\,h^{\alpha\beta}\partial_\alpha X\cdot\partial_\beta X\,,$$

where $h^{\alpha\beta} = \Omega\widetilde h^{\alpha\beta}$ is now an unconstrained worldsheet metric (the dependence on the scaling variable Ω drops out). Since $S_{\rm P}[X,h]$ is known to yield the same equations of motion as $S_{\rm NG}[X]$, we conclude that also $S_{\rm alt}[X,P,e,u]$ does.

B.2 Lightcone quantization

This exercise aims to fill in some intermediate steps and examples from the discussion of lightcone quantization in the lecture.

(i) Check that, in closed-string normalization of p^μ,

$$\partial_+ X_L^- = \frac{1}{\alpha' p^+}\sum_{i=1}^{D-2}\partial_+X^i\partial_+X^i\,, \qquad \partial_- X_R^- = \frac{1}{\alpha' p^+}\sum_{i=1}^{D-2}\partial_-X^i\partial_-X^i$$

imply the following at the level of the mode expansion

$$\alpha' p^- = \frac{1}{p^+} \sum_{i=1}^{D-2} \Big(\frac{\alpha'}{2} p^i p^i + \sum_{n\neq 0} \alpha^i_n \alpha^i_{-n} \Big)$$

$$\alpha^-_n = \sqrt{\frac{1}{2\alpha'}} \frac{1}{p^+} \sum_{k\in\mathbb{Z}} \sum_{i=1}^{D-2} \alpha^i_{n-k} \alpha^i_k \,, \qquad n \neq 0$$

and the same equations with $\alpha^i_n \leftrightarrow \tilde{\alpha}^i_n$.

(ii) Show that there are $\frac{1}{6}(D-2)(D-1)(D+6)$ states with eigenvalue $+3$ under the oscillator number $\sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \alpha^i_{-n} \alpha^i_n$. Explain that this matches the degree-of-freedom counting for a symmetric traceless 3-tensor and an antisymmetric 2-tensor of the little group $SO(D-1)$. You can disregard the $\tilde{\alpha}_n$ oscillators in this exercise (unless you would like to square the above numbers).

B.3 Virasoro algebra

Upon quantization of the closed bosonic string, the energy-momentum modes are defined with the following ordering prescription of the oscillators α^μ_n

$$L_0 = \frac{1}{2}\alpha_0^2 + \sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n \,, \qquad L_{k\neq 0} = \frac{1}{2} \sum_{n\in\mathbb{Z}} \alpha_{k-n} \cdot \alpha_n$$

with analogous definitions for $\tilde{L}_k$ with $\alpha^\mu_n \to \tilde{\alpha}^\mu_n$. The goal of this exercise is to derive their commutation relations – you will find a special case of the Virasoro algebra, where the central charge c equals the number of spacetime dimensions D.

(i) use the commutator identity $[AB, C] = A[B, C] + [A, C]B$ for arbitrary operators A, B, C to derive the lemma

$$[L_k, \alpha^\mu_m] = -m \alpha^\mu_{m+k} \,, \qquad m, k \in \mathbb{Z} \,.$$

(ii) use the lemma to verify that (note the essential restriction $m+n \neq 0$)

$$[L_m, L_n] = (m-n) L_{m+n} \,, \qquad m, n \in \mathbb{Z},\ m+n \neq 0 \,.$$

(iii) relax the restriction $m+n \neq 0$ by showing that

$$[L_m, L_{-m}] = 2m L_0 + \frac{D}{12}(m^3 - m) \,.$$

(you will need to pay particular attention to the operator ordering in L_0)

(iv) convince yourself that the above results imply the Virasoro algebra

$$[L_m, L_n] = (m-n) L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0} \,, \qquad m, n \in \mathbb{Z}$$

at $c = D$ and check that it is consistent with the Jacobi relations

$$[[L_m, L_n], L_p] + [[L_n, L_p], L_m] + [[L_p, L_m], L_n] = 0 \,, \qquad m, n, p \in \mathbb{Z} \,.$$

(v) show that the linear part of the central term $\sim c(m^3 - m)$ can be removed by a redefinition $\hat{L}_m = L_m + \alpha \delta_{m,0}$ with $\alpha \in \mathbb{C}$, i.e. verify that

$$[\hat{L}_m, \hat{L}_n] = (m-n)\hat{L}_{m+n} + \delta_{m+n,0} \Big[\frac{Dm^3}{12} - m\Big(\frac{D}{12} + 2\alpha\Big) \Big] \,, \qquad m, n \in \mathbb{Z} \,.$$

B.4 Zeta function regularization

This exercise aims to strengthen your confidence in the regularized value for the formally divergent sum

$$\sum_{n=1}^{\infty} n = -\frac{1}{12}$$

in the computation of the normal-ordering constant. We will relate it to a special value of the Riemann zeta function $\zeta(s)$ and the Dirichlet eta function $\eta(s)$ which are defined by infinite sums

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} \qquad = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \frac{1}{5^s} + \ldots\,, \qquad \mathrm{Re}(s) > 1 \tag{B.1}$$

$$\eta(s) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^s} = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \frac{1}{5^s} - \ldots\,, \qquad \mathrm{Re}(s) > 0 \tag{B.2}$$

which converge in the specified ranges of $s \in \mathbb{C}$.

(i) show that the eta and zeta functions in their domain of convergence $\mathrm{Re}(s) > 1$ are related by

$$(1 - 2^{1-s})\,\zeta(s) = \eta(s)\,. \tag{B.3}$$

hint: write $2^{-s}\zeta(s)$ as $\frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} + \ldots$ and relate this to the even terms in (B.1) and (B.2)

(ii) check that $\eta(-1) = \frac{1}{4}$ can be obtained from a one-sided limit $x \uparrow 1$ of

$$\frac{1}{(1+x)^2} = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - \ldots$$

which converges as $x \in \mathbb{R}$ approaches 1 from below.

(iii) One defines the analytic continuation of $\zeta(s)$ to regions with $\mathrm{Re}(s) \leq 1$ by insisting that the functional relation (B.3) holds even if the defining series diverges. Use this to infer the regularized value $\sum_{n=1}^{\infty} n = -\frac{1}{12}$ from $\eta(-1) = \frac{1}{4}$.

(iv) Repeat the renormalization procedure introduced in the lecture to obtain

$$\sum_{n=0}^{\infty}(n+q) = -\frac{q^2}{2} + \frac{q}{2} - \frac{1}{12}$$

from the coefficient of ϵ^0 in the Laurent expansion of $-\frac{\partial}{\partial \epsilon}\sum_{n=0}^{\infty} e^{-\epsilon(n+q)}$ around $\epsilon = 0$.

B.5 Mixing Neumann and Dirichlet boundary conditions

The goal of this exercise is to derive the mass spectrum for open strings subject to mixed Neumann and Dirichlet boundary conditions in some of the spacetime directions. Let d_N and d_D denote the number of dimensions with Neumann and Dirichlet boundary conditions for both endpoints, then impose

$$\partial_\sigma X^a(\tau, \sigma = 0) = 0\,, \qquad X^a(\tau, \sigma = \pi) = c^a\,, \qquad d_N + d_D \leq a \leq D-1$$

corresponding to Neumann conditions at $\sigma = 0$ and Dirichlet conditions at $\sigma = \pi$.

(i) start from an ansatz in the usual lightcone coordinates $\sigma^\pm = \tau \pm \sigma$ on the worldsheet

$$X_L^a(\sigma^+) = \frac{x^a}{2} + \alpha' p_L^a \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{r \in \Xi} \frac{1}{r} \tilde{\alpha}_r e^{-ir\sigma^+}$$

$$X_R^a(\sigma^-) = \frac{x^a}{2} + \alpha' p_R^a \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{r \in \Xi} \frac{1}{r} \alpha_r e^{-ir\sigma^-}$$

with some a priori unknown summation range Ξ for r and show that the boundary conditions enforce

$$x^a = c^a\,, \qquad p_L^a = p_R^a = 0\,, \qquad \tilde{\alpha}_r^a = \alpha_r^a\,, \qquad \Xi = \mathbb{Z}-\tfrac{1}{2} = \big\{n-\tfrac{1}{2}\ :\ n\in\mathbb{Z}\big\}$$

(ii) following the open-string conventions of the lecture, pick lightcone gauge $\partial_\pm X^+ = \alpha' p^+$ with $X^\pm = \sqrt{\frac{1}{2}}(X^0 \pm X^{d_N-1})$ and impose boundary conditions

$$\begin{aligned} \partial_\sigma X^i(\tau,\sigma=0) &= 0\,, \qquad \partial_\sigma X^i(\tau,\sigma=\pi) = 0\,, \qquad 0\leq i\leq d_N-1 \\ X^I(\tau,\sigma=0) &= c^I\,, \qquad X^I(\tau,\sigma=\pi) = d^I\,, \qquad d_N\leq I\leq d_N+d_D-1\,. \end{aligned}$$

show that the energy-momentum constraints yield

$$\begin{aligned} \alpha' p^- = \frac{1}{2\alpha' p^+}\bigg\{ &\alpha'^2 \sum_{i=1}^{d_N-2} p^i p^i + \frac{1}{4\pi^2}\sum_{I=d_N}^{d_N+d_D-1}(c^I-d^I)^2 \\ &+ \frac{\alpha'}{2}\sum_{\substack{n\in\mathbb{Z}\\ n\neq 0}}\Big(\sum_{i=1}^{d_N-2}\alpha_n^i\alpha_{-n}^i + \sum_{I=d_N}^{d_N+d_D-1}\alpha_n^I\alpha_{-n}^I\Big) + \frac{\alpha'}{2}\sum_{r\in\mathbb{Z}-\frac{1}{2}}\ \sum_{a=d_N+d_D}^{D-1}\alpha_r^a\alpha_{-r}^a\bigg\} \end{aligned}$$

(iii) bring the oscillators into normal-ordered form and use the result of exercise B.4 to derive the mass spectrum

$$\begin{aligned} \alpha' m^2 = &\frac{1}{4\pi^2\alpha'}\sum_{I=d_N}^{d_N+d_D-1}(c^I-d^I)^2 + \frac{2-D}{24} + \frac{D-d_N-d_D}{16} \\ &+ \sum_{n=1}^{\infty}\Big(\sum_{i=1}^{d_N-2}\alpha_{-n}^i\alpha_n^i + \sum_{I=d_N}^{d_N+d_D-1}\alpha_{-n}^I\alpha_n^I\Big) + \sum_{r\in\mathbb{N}-\frac{1}{2}}\ \sum_{a=d_N+d_D}^{D-1}\alpha_{-r}^a\alpha_r^a\,. \end{aligned}$$

Note the difference between the summation ranges $r\in\mathbb{N}-\frac{1}{2} = \{\frac{1}{2},\frac{3}{2},\frac{5}{2},\ldots\}$ in the normal ordered expression for $\alpha' m^2$ and $r\in\mathbb{Z}-\frac{1}{2}$ in the earlier result for $\alpha' p^-$.

To be submitted on Oct 16$^{\text{th}}$ 2023 by 6pm

C Problem set Oct 17$^{\rm th}$ 2023

C.1 Conformal algebra in general dimensions d

The conformal group in d spacetime dimensions is generated by translations P_μ, Lorentz rotations $J_{\mu\nu}$, dilatations D and special conformal transformations K_μ with $\mu,\nu = 0,1,2,\ldots,d{-}1$ and $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,\ldots,1)$. They give rise to the following group action on scalar functions:

$$P_\mu = -i\partial_\mu \, , \qquad K_\mu = -ix^2\partial_\mu + 2ix_\mu x\cdot\partial$$
$$J_{\mu\nu} = i(x_\mu\partial_\nu - x_\nu\partial_\mu) \, , \qquad D = -ix^\mu\partial_\mu$$

(i) Compute all commutators among $\{P_\mu, K_\mu, J_{\mu\nu}, D\}$.

(ii) In order to bring these commutators into a simpler form, define

$$\mathcal{J}_{\mu\nu} = J_{\mu\nu} \, , \qquad \mathcal{J}_{-2,-1} = D \, , \qquad \mathcal{J}_{-2,\mu} = \frac{1}{2}(P_\mu + K_\mu) \, , \qquad \mathcal{J}_{-1,\mu} = \frac{1}{2}(P_\mu - K_\mu)$$

with $\mathcal{J}_{ab} = -\mathcal{J}_{ba}$ and $a,b = -2,-1,0,1,\ldots,d{-}1$. Show that the Lie algebra spanned by these $\mathcal{J}_{ab}$ is isomorphic to that of $SO(2,d)$, i.e. that the metric η_{ab} in

$$[\mathcal{J}_{ab},\mathcal{J}_{cd}] = i(\eta_{bc}\mathcal{J}_{ad} - \eta_{ac}\mathcal{J}_{bd} - \eta_{bd}\mathcal{J}_{ac} + \eta_{ad}\mathcal{J}_{bc})$$

has two negative entries.

(iii) Consider the Maxwell action in a d-dimensional spacetime with metric $g_{\mu\nu}$,

$$\mathcal{S}_{\rm Max}[A,g] = -\frac{1}{4}\int \mathrm{d}^d x \, \sqrt{-\det g}\, F^{\mu\nu}\, F_{\mu\nu} \, .$$

Show that conformal transformations $g_{\mu\nu}(x) \to \Omega^2(x) g_{\mu\nu}(x)$ leave $\mathcal{S}_{\rm Max}[A,g]$ invariant if the number of spacetime dimensions is chosen to be $d=4$. The Maxwell potential A_μ and its field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ are understood to be unaffected by the conformal transformation.

(iv) Compute the energy-momentum tensor associated with $\mathcal{S}_{\rm Max}[A,g]$ and check that its trace vanishes if and only if $d=4$.

C.2 The free fermion

Consider a free fermion $\psi(z)$ which transforms like a primary of weight $(h,\bar h) = (\frac{1}{2},0)$ under conformal transformations and is taken to anticommute in the classical theory in the sense that

$$\psi(z)\psi(w) = -\psi(w)\psi(z) \, , \qquad \psi(z)\partial_w\psi(w) = -(\partial_w\psi(w))\psi(z) \, , \qquad \text{classically} \, .$$

(i) In passing to the quantum theory in the operator formalism, check that the general prescription for mode expansions given in the lecture yields a sum over $\psi_r\, z^{-r-1/2}$. Explain the consequences for the conformal field theory on the cylinder and the plane if the summation range for r is adjusted to be over half-odd integers,

$$\psi(z) = \sum_{r\in\mathbb{Z}-1/2} \psi_r\, z^{-r-1/2} \, .$$

(ii) Use this mode expansion with $r \in \mathbb{Z} - \frac{1}{2}$ to simplify the state-correspondent $\lim_{z\to 0}\psi(z)|0\rangle$ and explain why this limit is finite.

(iii) You will see in later lectures on the superstring that canonical quantization of the fermionic world-sheet action leads to anticommutation relations

$$\{\psi_r, \psi_s\} = \delta_{r+s,0}\,, \qquad r, s \in \mathbb{Z} - \frac{1}{2}\,.$$

Use this and $\psi_r^\dagger = \psi_{-r}$ to derive the two-point function

$$\langle \psi(z)\psi(w)\rangle = \langle 0|\psi(z)\psi(w)|0\rangle = \frac{1}{z-w}$$

and explain why radial ordering is crucial in this computation.

(iv) Use the resulting OPE for $\psi(z)\psi(w)$ to compute the singular terms in the OPE of $\psi(w)$ with the energy-momentum tensor

$$T(z) = \frac{1}{2} : (\partial_z \psi(z))\psi(z) :$$

and explain why the result is in lines with the conformal weight of $\psi(z)$. Note that the application of the Wick theorem to fermionic fields introduces a minus sign whenever a contraction $\psi(z)\ldots\psi(w) \to \frac{1}{z-w}\ldots$ is performed over an odd number of fermions in the ellipsis, e.g. that

$$\begin{aligned} :\psi_1\psi_2::\psi_3\psi_4: \sim\ & \underbrace{\psi_2\psi_3}\underbrace{\psi_1\psi_4} - \underbrace{\psi_1\psi_3}\underbrace{\psi_2\psi_4} + :\psi_2\psi_3:\underbrace{\psi_1\psi_4} + :\psi_1\psi_4:\underbrace{\psi_2\psi_3} \\ & - :\psi_1\psi_3:\underbrace{\psi_2\psi_4} - :\psi_2\psi_4:\underbrace{\psi_1\psi_3}\,, \end{aligned}$$

where the contraction $\underbrace{\psi_i\psi_j}$ instructs to replace $\psi_i\psi_j$ by their two-point function.

(v) Determine the singular terms in the OPE $T(z)T(w)$ and deduce that the central charge of the free fermion is $c = \frac{1}{2}$.

C.3 OPEs versus (anti-)commutation relations

This exercise guides you to through additional examples of the equivalence of (anti-)commutation relations with OPEs.

(i) Use the contour-integral techniques introduced in the lecture to derive the anticommutation relations $\{\psi_r, \psi_s\} = \delta_{r+s,0}$ of the free-fermion modes $\psi_r = \oint_{B_R(0)} \frac{\mathrm{d}z}{2\pi i} z^{r-1/2}\psi(z)$ from the OPE

$$\psi(z)\psi(w) \sim \frac{1}{z-w} + \ldots\,.$$

Note that by the anticommuting nature of fermions, one identifies $\psi(w)\psi(z)$ at $|w| > |z|$ with *minus* the radially ordered product $\psi(z)\psi(w)$.

(ii) Use the same strategy to reproduce the Virasoro algebra of the energy-momentum-tensor modes $L_m = \oint_{B_R(0)} \frac{\mathrm{d}z}{2\pi i} z^{m+1} T(z)$ in a generic CFT from the OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \ldots\,.$$

C.4 Vertex operators

This exercise aims to familiarize you with vertex operators, conformal primary fields built from normal ordered exponentials of the free boson. As you know from the lecture, the simplest example of a vertex operator

$$V_p(z,\bar{z}) \;=: e^{ip\cdot X(z,\bar{z})} :$$

is the field-correspondent of the oscillator ground state $|0;p\rangle$ with momentum eigenvalue p^μ.

(i) Show that the singular terms in the OPE $\partial X^\mu(z)\, V_p(w,\bar{w})$ are

$$\partial X^\mu(z)\, V_p(w,\bar{w}) \sim -\frac{i}{2}\,\frac{\alpha' p^\mu}{z-w} V_p(w,\bar{w}) + \dots$$

by series-expanding the exponential in intermediate steps.

(ii) Use the same reasoning to determine the singular terms in the OPE $T(z)\, V_p(w,\bar{w})$ and deduce that $V_p(w,\bar{w})$ is a conformal primary of weight $(h,\bar{h}) = (\frac{\alpha'}{4}p^2, \frac{\alpha'}{4}p^2)$.

(iii) As another example of a vertex operator, consider the field

$$V_{p,\zeta}(z,\bar{z}) = \zeta_{\mu\nu} : \partial_z X^\mu(z) \partial_{\bar{z}} X^\nu(\bar{z}) e^{ip\cdot X(z,\bar{z})} :$$

characterized by a momentum vector p^μ and a polarization tensor $\zeta_{\mu\nu}$. Under which conditions on p^μ and $\zeta_{\mu\nu}$ does $V_{p,\zeta}(z,\bar{z})$ furnish a conformal primary field of weight $(h,\bar{h}) = (1,1)$?

C.5 Correlations functions and OPEs

This exercise aims to familiarize you with the techniques to reduce correlation functions involving descendant fields to those of the associated conformal primaries.

(i) Use the results derived in the lecture to show that the descendant fields $\phi^{(-1)}_{h_i}(w) = \partial_w \phi_{h_i}(w)$ and $\phi^{(-k)}_{h_i}(w) = \oint_{B_\epsilon(w)} \frac{\mathrm{d}z}{2\pi i}(z-w)^{1-k} T(z)\phi_{h_i}(w),\ k \geq 1$ of primaries $\phi_{h_i}(w)$ give rise to the two-point functions (with shorthand $z_{ab} = z_a - z_b$)

$$\langle \phi^{(-1)}_{h_i}(z_1)\phi_{h_j}(z_2)\rangle = -\frac{2h_i d_{ij}}{z_{12}^{2h_i+1}}\delta_{h_i,h_j}\,, \qquad \langle \phi^{(-2)}_{h_i}(z_1)\phi_{h_j}(z_2)\rangle = \frac{3h_i d_{ij}}{z_{12}^{2h_i+2}}\delta_{h_i,h_j}$$

as well as the three-point function

$$\begin{aligned}\langle \phi_{h_i}(z_1)\phi_{h_j}(z_2)\phi^{(-2)}_{h_k}(z_3)\rangle &= \frac{C_{ijk}}{z_{12}^{h_i+h_j-h_k} z_{23}^{h_j+h_k-h_i} z_{13}^{h_i+h_k-h_j}} \\ &\times \left\{ \frac{2h_i - h_j + h_k}{z_{13}^2} + \frac{2h_j - h_i + h_k}{z_{23}^2} + \frac{h_k - h_i - h_j}{z_{13}z_{23}} \right\},\end{aligned}$$

where d_{ij} and C_{ijk} are the normalizations of the two- and three-point functions among the primaries.

(ii) Given the generic form of the OPE among primary fields

$$\begin{aligned}\phi_{h_i}(z_1)\phi_{h_j}(z_2) \sim \sum_{\text{primary } k} \frac{C_{ij}{}^k}{z_{12}^{h_i+h_j-h_k}} \Big\{ &\phi_{h_k}(z_2) + z_{12}\beta^{(-1)}_{ijk}\partial_{z_2}\phi_{h_k}(z_2) \\ &+ z_{12}^2 \beta^{(-1,-1)}_{ijk}\partial^2_{z_2}\phi_{h_k}(z_2) + z_{12}^2\beta^{(-2)}_{ijk}\phi^{(-2)}_{h_k}(z_2) + \mathcal{O}(z_{12}^3)\Big\}\,,\end{aligned}$$

show that the Laurent expansion of the three-point function $\langle \phi_{h_i}(z_1)\phi_{h_j}(z_2)\phi_{h_k}(z_3)\rangle$ around $z_1 = z_2$ imposes

$$\beta_{ijk}^{(-1)} = \frac{h_i + h_k - h_j}{2h_k}$$

and constrains the coefficients at the second level to obey

$$2h_k(2h_k+1)\beta_{ijk}^{(-1,-1)} + 3h_k\beta_{ijk}^{(-2)} = \frac{1}{2}(h_i + h_k - h_j)(h_i + h_k - h_j + 1)\,.$$

(iii) Explain without calculation why you would need the three-point function $\langle \phi_{h_i}(z_1)\phi_{h_j}(z_2)\phi_{h_k}^{(-2)}(z_3)\rangle$ (or some other correlator involving a higher descendant of ϕ_{h_k}) in order to determine $\beta_{ijk}^{(-1,-1)}$ and $\beta_{ijk}^{(-2)}$ individually. Why does this imply that $\beta_{ijk}^{(-1,-1)}$ and $\beta_{ijk}^{(-2)}$ depend on the central charge c on top of the conformal weights h_i, h_j, h_k?

Hint: Can you distinguish $\partial_{z_2}^2\phi_{h_k}(z_2)$ from a multiple of $\phi_{h_k}^{(-2)}(z_2)$ through the respective two-point function with $\phi_{h_\ell}(z_3)$ or $\partial_{z_3}^{N\geq 1}\phi_{h_\ell}(z_3)$?

To be submitted on Nov 17th 2023 by 6pm

D Problem set Nov 17th 2023

D.1 The delta function in complex coordinates

This exercise validates the representation

$$\partial_{\bar z}\partial_z \log|z|^2 = \partial_{\bar z}\frac{1}{z} = \pi\delta^2(z,\bar z)$$

of the delta function $\delta^2(z,\bar z) = \delta(x)\delta(y)$ for $z = x+iy$ with $x,y\in\mathbb{R}$ in two different ways.

(i) Use the volume form $\mathrm{d}^2z = \mathrm{d}x\wedge\mathrm{d}y = \frac{1}{2i}\mathrm{d}\bar z\wedge\mathrm{d}z$ and Stokes' theorem for p-forms ω

$$\int_\Omega \mathrm{d}\omega = \oint_{\partial\Omega}\omega$$

to show that $\partial_{\bar z}\frac{1}{z} = \pi\delta^2(z,\bar z)$ is consistent with integrating $\partial_{\bar z}\partial_z\log|z|^2$ over any open region $\Omega\subset\mathbb{C}$ that contains the origin.

(ii) As an alternative validation, study the deformation

$$\partial_z\frac{1}{\bar z} = \partial_z\lim_{\epsilon\to 0}\frac{z}{|z|^2+\epsilon^2}\,.$$

Explain why the sequence of distributions

$$\Delta_\epsilon(z,\bar z) = \frac{\epsilon^2}{(|z|^2+\epsilon^2)^2}$$

approaches $\pi\delta^2(z,\bar z)$ as $\epsilon\to 0$ and conclude that this implies the desired relation $\partial_z\frac{1}{\bar z} = \pi\delta^2(z,\bar z)$. More precisely, show that pointwise, $\lim_{\epsilon\to 0}\Delta_\epsilon(z,\bar z) = 0\ \forall\ z\neq 0$ and study the integral of $\Delta_\epsilon(z,\bar z)$ over disks $0\leq|z|\leq R$ in spherical coordinates $\mathrm{d}^2z = r\,\mathrm{d}r\,\mathrm{d}\varphi$ for $z = re^{i\varphi}$ with $r\in\mathbb{R}_+,\ \varphi\in[0,2\pi)$.

D.2 Anomalous current and two-point function of the (b,c)-ghost system

In this exercise, you are asked to verify certain fingerprints of the background charge of the (b,c)-ghosts.

(i) Using the expressions

$$T(z) = 2:\big(\partial_z c(z)\big)b(z): + :c(z)\partial_z b(z):\,,\qquad j(z) = -:b(z)c(z):$$

for the energy-momentum tensor $T(z)$ and the current $j(z)$ of the (b,c)-system, derive the OPE

$$T(z)j(w) \sim \frac{-3}{(z-w)^3} + \frac{j(w)}{(z-w)^2} + \frac{\partial_w j(w)}{z-w} + \ldots\,,$$

where the triple-pole prevents $j(z)$ from being a conformal primary field.

(ii) Using the canonical anticommutation relations of the b- and c-modes

$$\{b_m,b_n\} = \{c_m,c_n\} = 0\,,\qquad \{b_m,c_n\} = \delta_{m+n,0}\,,$$

derive the two-point function

$$\langle 0|c_{-1}c_0c_1 b(z)c(w)|0\rangle = \frac{1}{z-w}$$

in the operator formalism. In other words, insert the mode expansion

$$b(z)=\sum_{n\in\mathbb{Z}} b_n z^{-n-2}\,, \qquad c(z)=\sum_{n\in\mathbb{Z}} c_n z^{-n+1}$$

and note the following properties of the vacuum (including its 3 units of background ghost charge):

$$b_{n\geq -1}|0\rangle = c_{n\geq 2}|0\rangle = 0\,, \qquad \langle 0|b_{n\leq 1} = \langle 0|c_{n\leq -2} = 0\,, \qquad \langle 0|c_{-1}c_0c_1|0\rangle = 1$$

(iii) Show that the OPEs of $b(z)c(w)$ as well as $j(z)c(w)$ and $j(z)b(w)$ are reproduced by the "bosonized" representations

$$j(z)=\partial_z\phi(z)\,, \qquad b(z)=:e^{-\phi(z)}:\,, \qquad c(z)=:e^{\phi(z)}:\,,$$

where $\phi(z)$ is a chiral boson defined by short-distance singularities $\phi(z)\phi(w)\sim\log(z-w)+\ldots$.

(iv) Determine the constants $u_1,u_2\in\mathbb{Q}$ in the bosonized representation of the energy-momentum tensor of the ghost system

$$T(z)=u_1\ :\big(\partial_z\phi(z)\big)^2:+u_2\,\partial_z^2\phi(z)\,.$$

(v) Show that $:e^{q\phi(z)}:$ generate eigenstates $|q\rangle=\lim_{z\to 0}:e^{q\phi(z)}:|0\rangle$ of the zero mode j_0 of the anomalous ghost current $j(z)$ as specified in the lecture.

(vi) Show that $:e^{q\phi(z)}:$ are conformal primaries and determine their conformal weight.

D.3 Massless open-string states from BRST quantization

The goal of this exercise is to find the physical states of the massless open bosonic string in the framework of BRST quantization, following the methods introduced in the tutorial on November 22$^{\mathrm{nd}}$. We will see that the number of degrees of freedom matches the expectation from lightcone quantization. You will find the conformal weights $(h,\bar h)=(2,0)$ & $(-1,0)$ for b & c and the following singular terms in the OPEs useful

$$b(z)c(w)\sim\frac{1}{z-w}\,.$$

In addition, the BRST operator Q takes the form (with $T(z)=-\frac{1}{\alpha'}:\partial_zX^\mu(z)\partial_zX_\mu(z):$ excluding the ghost contributions to the energy-momentum tensor)

$$Q=\oint\frac{\mathrm{d}z}{2\pi i}(cT+bc\partial c)\,.$$

(i) Write down the most general ansatz for an unintegrated massless vertex operator compatible with conformal weight zero and built from ∂_zX,c,b and possibly their derivatives.

<u>Hint</u>: You should arrive at two vectors and four scalars that parametrize the ansatz.

(ii) Constrain this state to be annihilated by b_0, where

$$b_0=\oint\frac{\mathrm{d}z}{2\pi i}zb\,.$$

(iii) Identify the BRST closed states in the above.

(iv) Subtract all terms that are BRST exact, and explain why your result is compatible with the d.o.f. counting in lightcone quantization.

D.4 Plane-wave correlators in presence of derivatives

You have seen in the lecture how path-integral methods can be used to determine the plane-wave correlator (also known as Koba–Nielsen factor)

$$\Big\langle \prod_{j=1}^{n} : e^{ip_j\cdot X(z_j)} : \Big\rangle = \delta^D\Big(\sum_{j=1}^{n} p_j^\mu\Big) \prod_{1\leq i<j}^{n} |z_{ij}|^{\alpha' p_i\cdot p_j} \,,$$

where $z_{ij} = z_i - z_j$, and the momentum-conserving delta function will be suppressed in the following. The goal of this exercise is to extend these techniques to correlators involving derivatives $\partial X^\mu(z_j) := \partial_{z_j} X^\mu(z_j)$ that are ubiquitous in vertex operators. The idea is to evaluate the generating function

$$\mathcal{G}_n(p_j,\xi_j,z_j) := \Big\langle \prod_{j=1}^{n} : e^{ip_j\cdot X(z_j)+i\xi_j\cdot\partial X(z_j)} : \Big\rangle$$

of such correlators, where the vectors ξ_j^μ with $j=1,2,\ldots,n$ are formal bookkeeping variables.

(i) Use the strategy of the lecture to rewrite the exponentials as a source term in the worldsheet action,

$$e^{ip_j\cdot X(z_j)+i\xi_j\cdot\partial X(z_j)} = \exp\Big(i\int \mathrm{d}^2z\, X_\mu(z)\big[p_j^\mu\delta^2(z-z_j) - \xi_j^\mu\partial_z\delta^2(z-z_j)\big]\Big)\,,$$

to show that

$$\mathcal{G}_n(p_j,\xi_j,z_j) = \prod_{1\leq i<j}^{n} |z_{ij}|^{\alpha' p_i\cdot p_j} \exp\Big(\frac{\alpha'}{2}\sum_{1\leq i<j}^{n}\Big[\frac{\xi_i\cdot p_j}{z_{ij}} + \frac{\xi_j\cdot p_i}{z_{ji}} + \frac{\xi_i\cdot\xi_j}{z_{ij}^2}\Big]\Big)\,.$$

Hint: make use of the Gaussian integral

$$\begin{aligned}\frac{1}{Z}\int \mathcal{D}[X]\, \exp\Big(\int \mathrm{d}^2z\,\Big[\frac{1}{\pi\alpha'}X_\mu(z)\partial_z\partial_{\bar z}X^\mu(z) + iX_\mu(z)J^\mu(z)\Big]\Big)\\ = \exp\Big(\frac{\alpha'}{4}\int\mathrm{d}^2z\int\mathrm{d}^2w\, J^\mu(z)\log|z-w|^2 J_\mu(w)\Big)\end{aligned}$$

and exploit that contractions $p_i\cdot\xi_j$ and $\xi_i\cdot\xi_j$ with $i=j$ are excluded by normal ordering.

(ii) Explain why the "component correlators"

$$\Big\langle \prod_{j=1}^{n} : \big[\xi_j\cdot i\partial X(z_j)\big]^{m_j} e^{ip_j\cdot X(z_j)} : \Big\rangle, \qquad m_1,m_2,\ldots,m_n \in \mathbb{N}_0$$

can be obtained by isolating the multilinear terms $\sim (\xi_1)^{m_1}(\xi_2)^{m_2}\ldots(\xi_n)^{m_n}$ in the ξ-expansion of the exponentials in $\mathcal{G}_n(p_j,\xi_j,z_j)$ and multiplying by $m_1!m_2!\ldots m_n!$. Furthermore explain why these component correlators with insertions $:\big[\xi\cdot i\partial X(z)\big]^m e^{ip\cdot X(z)}:$ are sufficient to infer correlators of $:\zeta_{\mu_1\mu_2\ldots\mu_m} i\partial X^{\mu_1}(z) i\partial X^{\mu_2}(z)\ldots i\partial X^{\mu_m}(z) e^{ip\cdot X(z)}:$ with arbitrary polarization tensors $\zeta_{\mu_1\mu_2\ldots\mu_m}$.

(iii) Check that $\mathcal{G}_n(p_j,\xi_j,z_j)$ obeys the differential equation

$$\frac{\partial}{\partial(\xi_\ell)_\mu}\mathcal{G}_n(p_j,\xi_j,z_j) = \frac{\alpha'}{2}\sum_{\substack{k=1\\k\neq\ell}}^{n}\Big(\frac{p_k^\mu}{z_{\ell k}} + \frac{\xi_k^\mu}{z_{\ell k}^2}\Big)\mathcal{G}_n(p_j,\xi_j,z_j)$$

with $\ell=1,2,\ldots,n$, and show that it implies the recursion or Ward identity for component correlators

$$\begin{aligned}&\Big\langle :i\partial X^\mu(z_\ell)e^{ip_\ell\cdot X(z_\ell)}: \prod_{\substack{j=1\\j\neq\ell}}^{n} :\big[\xi_j\cdot i\partial X(z_j)\big]^{m_j} e^{ip_j\cdot X(z_j)}:\Big\rangle\\
&= \frac{\alpha'}{2}\sum_{\substack{k=1\\k\neq\ell}}^{n}\Big\{\frac{p_k^\mu}{z_{\ell k}}\Big\langle :e^{ip_\ell\cdot X(z_\ell)}: \prod_{\substack{j=1\\j\neq\ell}}^{n} :\big[\xi_j\cdot i\partial X(z_j)\big]^{m_j} e^{ip_j\cdot X(z_j)}:\Big\rangle\\
&\quad + \frac{m_k\xi_k^\mu}{z_{\ell k}^2}\Big\langle :e^{ip_\ell\cdot X(z_\ell)}: :\big[\xi_k\cdot i\partial X(z_k)\big]^{m_k-1} e^{ip_k\cdot X(z_k)}: \prod_{\substack{j=1\\j\neq k,\ell}}^{n} :\big[\xi_j\cdot i\partial X(z_j)\big]^{m_j} e^{ip_j\cdot X(z_j)}:\Big\rangle\Big\}\,.\end{aligned}$$

Note that this resembles the Ward identity for correlators involving energy-momentum insertions and otherwise conformal primaries: The recursion instructs to replace the insertion of $i\partial X^\mu(z_\ell)$ within a correlator by a sum over all its OPE singularities $i\partial X^\mu(z_\ell){:}e^{ip_k\cdot X(z_k)}{:} \sim \frac{\alpha'}{2}\frac{p^\mu}{z_{\ell k}}{:}e^{ip_k\cdot X(z_k)}{:}+\ldots$ and $i\partial X^\mu(z_\ell) i\partial X^\nu(z_k) \sim \frac{\alpha'}{2}\frac{\eta^{\mu\nu}}{z_{\ell k}^2}+\ldots$ with the remaining fields. Like this, all the correlators on the right-hand side have fewer insertions of $i\partial X^\mu(z)$.

(iv) Determine the three-point correlators

$$\Big\langle \prod_{j=1}^{3} : e^{ip_j\cdot X(z_j)} : \Big\rangle , \qquad \Big\langle \Big(\prod_{j=1}^{2} : e^{ip_j\cdot X(z_j)} : \Big) : \xi_3 \cdot i\partial X(z_3) e^{ip_3\cdot X(z_3)} : \Big\rangle ,$$

$$\Big\langle : e^{ip_1\cdot X(z_1)} : \prod_{j=2}^{3} : \xi_j \cdot i\partial X(z_j) e^{ip_j\cdot X(z_j)} : \Big\rangle , \qquad \Big\langle \prod_{j=1}^{3} : \xi_j \cdot i\partial X(z_j) e^{ip_j\cdot X(z_j)} : \Big\rangle .$$

Hint: You can cross-check your results by testing the above Ward identity with $\ell = n = 3$ (though you are not expected to include this check into your solution).

(v) Generalize the generating functions to correlators involving $\bar\partial X^\mu(z_j) = \partial_{\bar z_j} X^\mu(z_j)$, i.e. show that

$$\mathcal{H}_n(p_j,\xi_j,z_j) := \Big\langle \prod_{j=1}^{n} : e^{ip_j\cdot X(z_j) + i\xi_j\cdot\partial X(z_j) + i\bar\xi_j\cdot\bar\partial X(z_j)} : \Big\rangle$$

with additional bookkeeping variables $\bar\xi_j^\mu, \ j=1,2,\ldots,n$ can be written in the factorized form

$$\mathcal{H}_n(p_j,\xi_j,z_j) = \prod_{1\le i<j}^{n} |z_{ij}|^{\alpha' p_i\cdot p_j} \exp\Big(\frac{\alpha'}{2} \sum_{1\le i<j}^{n} \Big[\frac{\xi_i\cdot p_j}{z_{ij}} + \frac{\xi_j\cdot p_i}{z_{ji}} + \frac{\xi_i\cdot\xi_j}{z_{ij}^2} \Big] \Big)$$
$$\times \exp\Big(\frac{\alpha'}{2} \sum_{1\le i<j}^{n} \Big[\frac{\bar\xi_i\cdot p_j}{\bar z_{ij}} + \frac{\bar\xi_j\cdot p_i}{\bar z_{ji}} + \frac{\bar\xi_i\cdot\bar\xi_j}{\bar z_{ij}^2} \Big] \Big) .$$

Hint: You will need to exploit that $|z_{ij}|^{\alpha' p_i\cdot p_j}$ vanishes as $z_i \to z_j$ (upon analytic continuation in $\alpha' p_i\cdot p_j$). Note that the exponentials $: e^{ip_j\cdot X(z_j)+i\xi_j\cdot\partial X(z_j)} :$ and $: e^{ip_j\cdot X(z_j)+i\xi_j\cdot\partial X(z_j)+i\bar\xi_j\cdot\bar\partial X(z_j)} :$ are generating functions for certain open- and closed-string vertex operators. Hence, the factorized form of $\mathcal{H}_n(p_j,\xi_j,z_j)$ implies that these closed-string correlators are products of open-string correlators (up to the Koba–Nielsen factor and the rescaling $\alpha' \to 4\alpha'$ as will be mentioned in the lecture).

(vi) **OPTIONAL** (no need to submit a solution for this part): Comment on how you would modify the above strategy to also compute correlators involving higher derivatives $\partial^N X^\mu(z_j)$ and $\bar\partial^N X^\mu(z_j)$ with $N \ge 2$. Explain why their generating functions will still take a factorized form.

To be submitted on Dec 11$^{\text{th}}$ 2023 by 6pm

E Problem set Dec 12^{th} 2023

E.1 Three tachyons and one gluon or graviton

On this sheet, you will need the vertex operators of the tachyon and the non-abelian gauge boson of the open bosonic string

$$V_T(z) \sim :e^{ip\cdot X(z)}: , \qquad\qquad p^2 = \frac{1}{\alpha'}$$

$$V_\epsilon(z) \sim \epsilon^\mu :i\partial X_\mu(z) e^{ip\cdot X(z)}: , \qquad\qquad p^2 = \epsilon\cdot p = 0$$

and the Euler Beta function

$$\int_0^1 \mathrm{d}z\, z^{a-1}(1-z)^{b-1} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \, .$$

(i) Use the above vertex operators and the SL$_2$-frame $(z_1,z_3,z_4)\to(0,1,\infty)$ to bring the color-ordered four-point amplitude

$$A^{\text{tree}}_{\text{bos}}(\epsilon_1,T_2,T_3,T_4;\alpha') \sim \int_0^1 \mathrm{d}z_2\, \langle cV_{\epsilon_1}(z_1)V_{T_2}(z_2)cV_{T_3}(z_3)cV_{T_4}(z_4)\rangle$$

into the following form ($s=\alpha'(p_1+p_2)^2$ and $t=\alpha'(p_2+p_3)^2$):

$$A^{\text{tree}}_{\text{bos}}(\epsilon_1,T_2,T_3,T_4;\alpha') \sim -2\alpha'\Big[\epsilon_1\cdot p_2\,\frac{\Gamma(s-1)\Gamma(t-1)}{\Gamma(s+t-2)} + \epsilon_1\cdot p_3\,\frac{\Gamma(s)\Gamma(t-1)}{\Gamma(s+t-1)}\Big]$$

(ii) Use Γ-function identities to show that this expression enjoys linearized gauge invariance, i.e. that it vanishes when setting $\epsilon_1\to p_1$.

(iii) Show that your result is antisymmetric under exchange of the second and fourth tachyon,

$$A^{\text{tree}}_{\text{bos}}(\epsilon_1,T_2,T_3,T_4;\alpha') = -A^{\text{tree}}_{\text{bos}}(\epsilon_1,T_4,T_3,T_2;\alpha') \, ,$$

i.e. show that your result changes by a sign when swapping $p_2\leftrightarrow p_4$.

Hint: Eliminate $(\epsilon_1\cdot p_3)$ via momentum conservation and transversality.

E.2 Four-gluon amplitude

This exercise gives you a glimpse of the four-gluon amplitude of the open bosonic string, based on the SL$_2$-frame where $(z_1,z_3,z_4)\to(0,1,\infty)$,

$$A^{\text{tree}}_{\text{bos}}(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4;\alpha') \sim \int_0^1 \mathrm{d}z_2\, \langle cV_{\epsilon_1}(z_1)V_{\epsilon_2}(z_2)cV_{\epsilon_3}(z_3)cV_{\epsilon_4}(z_4)\rangle \, .$$

(i) Show that the contribution of the tensor structure $(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot\epsilon_4)$ to the correlator (excluding ghosts) is proportional to

$$\prod_{1\le i<j}^4 |z_{ij}|^{2\alpha' p_i\cdot p_j} \times \frac{1}{z_{12}^2 z_{34}^2} \, .$$

(ii) Perform the integral over z_2 in terms of Γ functions and use

$$\frac{\Gamma(1+s)\Gamma(1+t)}{\Gamma(1+s+t)} = 1 - \zeta_2\, st - \zeta_3\, stu + \mathcal{O}(\alpha'^4)$$

to expand your result to the third subleading order in α'.

(iii) Explain that your results signal an effective interaction of the form $\text{Tr}(F^3)$, two independent interactions of the form $\text{Tr}(F^4)$ and three independent interactions of the form $\text{Tr}(D^2F^4)$, where $F^{\mu\nu}$ is the non-abelian gluon field strength and D_μ is the gauge-covariant derivative.

(iv) Compute the contribution of the tensor structure $(\epsilon_1 \cdot \epsilon_2)(\epsilon_3 \cdot \epsilon_4)$ to the alternative color-ordered amplitudes $A^{\text{tree}}_{\text{bos}}(\epsilon_2, \epsilon_1, \epsilon_3, \epsilon_4; \alpha')$ and $A^{\text{tree}}_{\text{bos}}(\epsilon_1, \epsilon_3, \epsilon_2, \epsilon_4; \alpha')$ by integrating over $z_2 \in (-\infty, 0)$ and $z_2 \in (1, \infty)$, respectively.

<u>Hint</u>: Use variable transformations $z_2 = \frac{y}{y-1}$ and $z_2 = \frac{1}{x}$ to map the integration regions to $x, y \in (0, 1)$.

(v) Assuming that the remaining tensor structures exhibit the same cancellations as $(\epsilon_1 \cdot \epsilon_2)(\epsilon_3 \cdot \epsilon_4)$, deduce that only one of the $\text{Tr}(F^4)$ and one of the $\text{Tr}(D^2F^4)$ interactions have a non-vanishing abelian limit when the Chan-Paton factors are specialized to $T^{a_j} \to 1$.

E.3 Gamma function versus Riemann zeta function

The Γ function can be alternatively defined through one of the following infinite products

$$\Gamma(z) = \frac{1}{z} e^{-\gamma z} \prod_{n=1}^{\infty} e^{z/n} \Big(1 + \frac{z}{n}\Big)^{-1} = \frac{1}{z} \prod_{n=1}^{\infty} \frac{\big(1 + \frac{1}{n}\big)^z}{\big(1 + \frac{z}{n}\big)} \, ,$$

where γ denotes the Euler–Mascheroni constant

$$\gamma = \sum_{n=1}^{\infty} \Big\{ \frac{1}{n} - \log\Big(1 + \frac{1}{n}\Big) \Big\} = \lim_{N \to \infty} \Big(\sum_{n=1}^{N} \frac{1}{n} - \log N \Big) \approx 0.5772\ldots$$

that will drop out from any application to string tree-level amplitudes.

(i) Show that the logarithmic derivative of the Γ function can be written as

$$\partial_z \log \Gamma(1+z) = -\gamma + \sum_{n=1}^{\infty} \Big(\frac{1}{n} - \frac{1}{n+z} \Big) \, .$$

<u>Hint</u>: Use $z\Gamma(z) = \Gamma(z+1)$ in the first step.

(ii) By integrating the previous identity from 0 to z, derive the following series representation

$$\log \Gamma(1+z) = -\gamma z + \sum_{k=2}^{\infty} \frac{(-z)^k}{k} \zeta_k \, ,$$

where the Riemann zeta values are defined by

$$\zeta_k = \sum_{n=1}^{\infty} \frac{1}{n^k} \, , \qquad k \geq 2 \, .$$

<u>Hint</u>: Perform a geometric-series expansion of $\frac{1}{n+z} = \frac{1}{n} \sum_{k=0}^{\infty} (-\frac{z}{n})^k$ in the integrand.

(iii) The sine function can be written in terms of the Γ functions

$$\sin(\pi z) = \frac{\pi z}{\Gamma(1+z)\Gamma(1-z)} \, .$$

Explain how the zeros and the periodicity $\sin(\pi(z \pm 1)) = -\sin(\pi z)$ of the sine function can be derived from this representation.

(iv) Using the Γ-function representation of the sine function and the above series expansion of $\log \Gamma(1+z)$, show that

$$\sin(\pi z) = \pi z \exp\Big(-\sum_{n=1}^{\infty} \frac{\zeta_{2n}}{n} z^{2n}\Big) .$$

(v) By matching this representation with the more conventional Taylor expansion of the sine function $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7)$, deduce that

$$\zeta_2 = \frac{\pi^2}{6} , \qquad \zeta_4 = \frac{\pi^4}{90} .$$

Note: This method implies that any even zeta value ζ_{2k} is a rational multiple of π^{2k}. However, one cannot extract any information on odd zeta values ζ_{2k+1} from these expansions of the sine function, and all of $\frac{\zeta_3}{\pi^3}$, $\frac{\zeta_5}{\pi^5}, \ldots$ are strongly expected (though not proven) to be irrational.

E.4 Monodromy relations among color-ordered open-string amplitudes

This exercise is dedicated to the monodromy relations among color-ordered open-string amplitudes $A_*^{\rm tree}(\varphi_1, \varphi_2, \ldots, \varphi_n; \alpha')$. The subscript $_*$ is a placeholder for either bosonic strings or superstrings, and the external states φ_j are taken to be massless such that $s = 2\alpha' p_1 \cdot p_2$ and $t = 2\alpha' p_2 \cdot p_3$. The goal is to relate different color orderings or integration regions for the punctures $z_2, z_3, \ldots, z_{n-2}$ in an SL_2-frame where $(z_1, z_{n-1}, z_n) = (0, 1, \infty)$. This will be achieved by exploiting that the underlying correlator among vertex operators is meromorphic except for the universal Koba–Nielsen factor $\sim |z_{ij}|^{2\alpha' p_i \cdot p_j}$.

At four points, the above SL_2-frame leads to the color-ordered amplitudes

$$\begin{aligned}
A_*^{\rm tree}(\varphi_2, \varphi_1, \varphi_3, \varphi_4; \alpha') &\longleftrightarrow \int_{-\infty}^{0} \mathrm{d}z_2 \, |z_2|^s \, |1 - z_2|^t \, f_*(z_2) \\
A_*^{\rm tree}(\varphi_1, \varphi_2, \varphi_3, \varphi_4; \alpha') &\longleftrightarrow \int_{0}^{1} \mathrm{d}z_2 \, |z_2|^s \, |1 - z_2|^t \, f_*(z_2) \\
A_*^{\rm tree}(\varphi_1, \varphi_3, \varphi_2, \varphi_4; \alpha') &\longleftrightarrow \int_{1}^{\infty} \mathrm{d}z_2 \, |z_2|^s \, |1 - z_2|^t \, f_*(z_2) ,
\end{aligned}$$

where the polarization dependent rational function $f_*(z_2)$ is determined by the contractions among the vertex operators, and its detailed form will not be needed in the following.

You will be guided through a proof of the monodromy relations that relies on applying Cauchy's theorem to the meromorphic function $F_*(z_2) = (z_2)^s (1 - z_2)^t f_*(z_2)$ with rational $f_*(z_2)$:

$$\oint_{\mathcal{C}} \mathrm{d}z_2 \, F_*(z_2) = 0 .$$

The integration contour $\mathcal{C}$ is depicted in figure E.1 below and consists of a semi-circle $\mathcal{C}_\infty$ at infinity as well as the real axis with the following shorthands

$$\mathcal{C}_{2134} = (-\infty, 0) , \qquad \mathcal{C}_{1234} = (0, 1) , \qquad \mathcal{C}_{1324} = (1, \infty) .$$

(Strictly speaking, the $\mathcal{C}_{ijkl}$ should be infinitesimally displaced from the real axis by some positive imaginary part $i\epsilon$ to avoid tentative poles of $f_*(z_2)$ at $z_2 = 0$ and $z_2 = 1$.)

(i) Assuming that the semi-circle $\mathcal{C}_\infty$ does not contribute to $\oint_{\mathcal{C}} \mathrm{d}z_2 \, F_*(z_2)$, explain why the integration can be performed using the representation

$$F_*(z_2) = \begin{cases} e^{i\pi s} \, |z_2|^s \, |1 - z_2|^t \, f_*(z_2) & : \; z_2 \in \mathcal{C}_{2134} \\ |z_2|^s \, |1 - z_2|^t \, f_*(z_2) & : \; z_2 \in \mathcal{C}_{1234} \\ e^{-i\pi t} \, |z_2|^s \, |1 - z_2|^t \, f_*(z_2) & : \; z_2 \in \mathcal{C}_{1324} \end{cases}$$

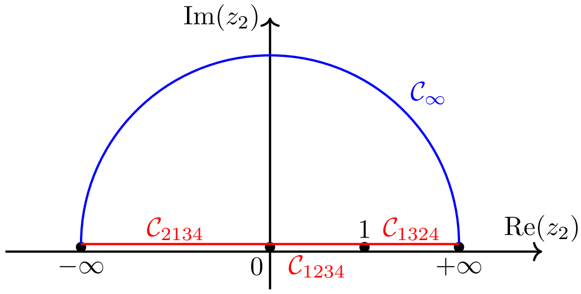


Figure E.1: The contour $\mathcal{C}$ consists of the semicircle $\mathcal{C}_\infty$ drawn in blue and subsets $\mathcal{C}_{2134}$, $\mathcal{C}_{1234}$, $\mathcal{C}_{1324}$ of the real line drawn in red.

of the above integrand $F_*(z_2)$, and explain why Cauchy's theorem implies that

$$e^{i\pi s}\int_{-\infty}^{0} \mathrm{d}z_2 \, |z_2|^s \, |1{-}z_2|^t \, f_*(z_2) + \int_0^1 \mathrm{d}z_2 \, |z_2|^s \, |1{-}z_2|^t \, f_*(z_2) + e^{-i\pi t}\int_1^{\infty} \mathrm{d}z_2 \, |z_2|^s \, |1{-}z_2|^t \, f_*(z_2) = 0 \, .$$

(ii) Assuming that color-ordered amplitudes $A^{\rm tree}_*(\varphi_i,\varphi_j,\varphi_k,\varphi_l;\alpha')$ are real, deduce that

$$\begin{aligned}\cos(\pi s) A^{\rm tree}_*(\varphi_2,\varphi_1,\varphi_3,\varphi_4;\alpha') + A^{\rm tree}_*(\varphi_1,\varphi_2,\varphi_3,\varphi_4;\alpha') + \cos(\pi t) A^{\rm tree}_*(\varphi_1,\varphi_3,\varphi_2,\varphi_4;\alpha') &= 0 \\ \sin(\pi s) A^{\rm tree}_*(\varphi_2,\varphi_1,\varphi_3,\varphi_4;\alpha') - \sin(\pi t) A^{\rm tree}_*(\varphi_1,\varphi_3,\varphi_2,\varphi_4;\alpha') &= 0 \, .\end{aligned}$$

(iii) By isolating the leading order in α', conclude that color-ordered tree amplitudes among non-abelian gauge bosons satisfy the Kleiss–Kuijf relations

$$A^{\rm tree}_{\rm YM}(\varphi_2,\varphi_1,\varphi_3,\varphi_4) + A^{\rm tree}_{\rm YM}(\varphi_1,\varphi_2,\varphi_3,\varphi_4) + A^{\rm tree}_{\rm YM}(\varphi_1,\varphi_3,\varphi_2,\varphi_4) = 0$$

and the Bern–Carrasco–Johansson (BCJ) relations

$$(p_1+p_2)^2 A^{\rm tree}_{\rm YM}(\varphi_2,\varphi_1,\varphi_3,\varphi_4) - (p_2+p_3)^2 A^{\rm tree}_{\rm YM}(\varphi_1,\varphi_3,\varphi_2,\varphi_4) = 0 \, .$$

(iv) **OPTIONAL** (no need to submit a solution for this part): Generalize the integration contour $\mathcal{C}$ and the integrand $F_*(z_2) \rightarrow F_*(z_2,z_3,\ldots,z_{n-2})$ to an n-point setting to derive the monodromy relation

$$\begin{aligned} 0 &= e^{2\pi i\alpha' p_1\cdot p_2} A^{\rm tree}_*(\varphi_2,\varphi_1,\varphi_3,\ldots,\varphi_n;\alpha') + A^{\rm tree}_*(\varphi_1,\varphi_2,\varphi_3,\ldots,\varphi_n;\alpha') \\ &\quad + e^{-2\pi i\alpha' p_2\cdot p_3} A^{\rm tree}_*(\varphi_1,\varphi_3,\varphi_2,\varphi_4,\ldots,\varphi_n;\alpha') + e^{-2\pi i\alpha' p_2\cdot(p_3+p_4)} A^{\rm tree}_*(\varphi_1,\varphi_3,\varphi_4,\varphi_2,\ldots,\varphi_n;\alpha') \\ &\quad + \ldots + e^{-2\pi i\alpha' p_2\cdot(p_3+p_4+\ldots+p_{n-1})} A^{\rm tree}_*(\varphi_1,\varphi_3,\varphi_4,\ldots,\varphi_{n-1},\varphi_2,\varphi_n;\alpha') \end{aligned}$$

and deduce the n-point BCJ relations

$$p_1\cdot p_2 A^{\rm tree}_{\rm YM}(\varphi_2,\varphi_1,\varphi_3,\ldots,\varphi_n) = \sum_{j=3}^{n-1} p_2\cdot(p_3{+}p_4{+}\ldots{+}p_j) A^{\rm tree}_{\rm YM}(\varphi_1,\varphi_3,\ldots,\varphi_j,\varphi_2,\varphi_{j+1},\ldots,\varphi_n) \, .$$

Note: By combining all permutations of the monodromy- and BCJ relations, color-ordered n-point tree amplitudes in string and gauge theory can be reduced to a basis of dimension $(n{-}3)!$.

To be submitted on Jan 12th 2024 by 6pm

F Problem set Jan 11$^{\text{th}}$ 2023

F.1 From string frame to Einstein frame

You have seen in the lecture that the σ-model action for closed strings in background fields $\{G_{\mu\nu}, B_{\mu\nu}, \Phi\}$ only retains conformal invariance if the following beta functions vanish to first order in α':

$$
\begin{aligned}
\beta_{\mu\nu}(G) &= \alpha' \mathcal{R}_{\mu\nu} + 2\alpha' \nabla_\mu \nabla_\nu \Phi - \frac{\alpha'}{4} H_{(\mu}{}^{\lambda\rho} H_{\nu)\lambda\rho} + \mathcal{O}(\alpha'^2) \\
\beta_{\mu\nu}(B) &= -\frac{\alpha'}{2} \nabla^\lambda H_{\lambda\mu\nu} + \alpha' (\nabla^\lambda \Phi) H_{\lambda\mu\nu} + \mathcal{O}(\alpha'^2) \\
\beta(\Phi) &= \frac{D-26}{6} - \frac{\alpha'}{2} \nabla^2 \Phi + \alpha' \nabla_\mu \Phi \nabla^\mu \Phi - \frac{\alpha'}{24} H_{\mu\nu\rho} H^{\mu\nu\rho} + \mathcal{O}(\alpha'^2)
\end{aligned}
$$

(i) Show that the vanishing of these beta functions follows from the equations of motion of the following effective action in D spacetime dimensions to first order in α'

$$
S_{\text{eff}}[G,B,\Phi] = \frac{1}{2\kappa_0^2} \int \mathrm{d}^D X \sqrt{-\det G}\, e^{-2\Phi} \Big\{ R - \frac{1}{12} H_{\mu\nu\lambda} H^{\mu\nu\lambda} + 4 \partial_\mu \Phi \partial^\mu \Phi - \frac{2(D-26)}{3\alpha'} \Big\}
$$

The variables $G_{\mu\nu}$ and Φ are those appearing in the σ-model action, that is why the above expression for $S_{\text{eff}}[G,B,\Phi]$ is said to be in "string frame".

(ii) We shall next transform the string-frame effective action into the so-called "Einstein frame", i.e. introduce the variables $\tilde{G}_{\mu\nu}$ appearing in the Einstein–Hilbert action. Perform the field redefinition

$$
\tilde{G}_{\mu\nu} = e^{2\omega} G_{\mu\nu} \, , \qquad \tilde{\Phi} = \Phi - \Phi_0 \, , \qquad \omega = -\frac{2\tilde{\Phi}}{D-2}
$$

to check that $S_{\text{eff}}[G,B,\Phi]$ is equal to the following Einstein-frame action with $\kappa^2 = \kappa_0^2 e^{2\Phi_0}$,

$$
\begin{aligned}
S_{\text{eff}}[\tilde{G},B,\tilde{\Phi}] = \frac{1}{2\kappa^2} \int \mathrm{d}^D X \sqrt{-\det \tilde{G}} \Big\{ &\tilde{R} - \frac{1}{12} e^{-\frac{8\tilde{\Phi}}{D-2}} H_{\mu\nu\lambda} H^{\mu\nu\lambda} \\
&- \frac{4}{D-2} \partial_\mu \tilde{\Phi} \partial^\mu \tilde{\Phi} + \frac{4(D-1)}{D-2} \nabla^2 \tilde{\Phi} - e^{\frac{4\tilde{\Phi}}{D-2}} \frac{2(D-26)}{3\alpha'} \Big\} \, .
\end{aligned}
$$

You can exploit that the Ricci-scalar $\tilde{R}$ of the transformed metric $\tilde{G}_{\mu\nu}$ is related as follows to the untransformed one

$$
\tilde{R} = e^{-2\omega} \Big(R - 2(D-1) \nabla^2 \omega - (D-2)(D-1) \partial_\mu \omega \partial^\mu \omega \Big) \, .
$$

Note that the indices $\mu, \nu, \lambda, \ldots$ in the Einstein-frame action $S_{\text{eff}}[\tilde{G},B,\tilde{\Phi}]$ are contracted with $\tilde{G}_{\mu\nu}$ rather than $G_{\mu\nu}$.

(iii) Explain why the term $\sim \nabla^2 \tilde{\Phi}$ in the second line of $S_{\text{eff}}[\tilde{G},B,\tilde{\Phi}]$ integrates to zero.

F.2 Born-Infeld equations of motion

The goal of this exercise is to extract the field equations from the Born–Infeld action on a Dp-brane

$$
S_{\text{BI}}[A] = -T_p \int \mathrm{d}^{p+1} X \sqrt{\det(\delta_{ab} + F_{ab})} \, ,
$$

where T_p denotes the brane tension, and the coordinates on the brane have been lined up with the spacetime directions $X^0, X^1, \ldots, X^p$. In order to avoid technical distractions, we are setting $\alpha' = \frac{1}{2\pi}$ within this exercise and consider the Euclidean version of the expression $\sim \sqrt{-\det(\eta_{ab} + 2\pi\alpha' F_{ab})}$ introduced in the lecture.

(i) Use the matrix-identity $\det M = \exp(\mathrm{Tr}\log(M))$ and antisymmetry $F_{ab} = -F_{ba}$ to show that

$$\sqrt{\det(\delta_{ab} + F_{ab})} = \exp\Big(\frac{1}{4}\,\mathrm{Tr}\big(\log(1-F^2)\big)\Big)\,.$$

Hint: Which powers of F_{ab} are traceless?

(ii) Using the series-expansion $\log(1-x) = -\sum_{n=1}^{\infty}\frac{x^n}{n}$, expand the result of (i) to the order of F^6 and check that the F^4-order has the relative factor $\mathrm{Tr}(F^4) - \frac{1}{4}(\mathrm{Tr}(F^2))^2$ between the single- and double-traces.

(iii) Demonstrate at the level of the respective series expansions that

$$\partial_a\Big[\mathrm{Tr}\big(\log(1-F^2)\big)\Big] = -2\partial_a F_{bc}\Big(\frac{F}{1-F^2}\Big)^{cb} = -4\partial_b F_{ac}\Big(\frac{F}{1-F^2}\Big)^{cb}\,,$$

where the second identity follows from the Bianchi identity $\partial_{[a}F_{bc]} = 0$.

(iv) Demonstrate that

$$\partial_a\Big(\frac{F}{1-F^2}\Big)^{ab} = \Big(\frac{F}{1-F^2}\Big)^{ac}\partial_a F_{cd}\Big(\frac{F}{1-F^2}\Big)^{db} + \Big(\frac{1}{1-F^2}\Big)^{ac}\partial_a F_{cd}\Big(\frac{1}{1-F^2}\Big)^{db}\,.$$

Hint: Organize the series expansion of both sides according to the matrix products $(F^M(\partial_a F)F^N)$ with $M, N \geq 0$. It is convenient to check separately that the terms on both sides agree in all the four cases where (M,N) are (even, even), (odd, odd), (even, odd) and (odd, even), respectively.

(v) Since the Born–Infeld action only depends on the photon A_b through its field strength F_{ab} (and no derivatives $\partial_c\partial_d\ldots F_{ab}$), explain why one can simplify the variation w.r.t. A_b to

$$\frac{\delta}{\delta A_b}S_{\mathrm{BI}}[A] = -2\partial_a\Big(\frac{\delta}{\delta F_{ab}}S_{\mathrm{BI}}[A]\Big)\,.$$

(vi) Follow the manipulation of (iii) to show that

$$\frac{\delta}{\delta F_{ab}}S_{\mathrm{BI}}[A] \sim \Big(\frac{F}{1-F^2}\Big)^{ab}\exp\Big(\frac{1}{4}\,\mathrm{Tr}\big(\log(1-F^2)\big)\Big)$$

(vii) Combine the results of (iii) to (vi) to demonstrate that the equations of motion following from the Born–Infeld action are

$$\begin{aligned}0 &= \partial_a\Big[\Big(\frac{F}{1-F^2}\Big)^{ab}\exp\Big(\frac{1}{4}\,\mathrm{Tr}\big(\log(1-F^2)\big)\Big)\Big]\\ &= \Big(\frac{1}{1-F^2}\Big)^{ac}\partial_a F_{cd}\Big(\frac{1}{1-F^2}\Big)^{db}\exp\Big(\frac{1}{4}\,\mathrm{Tr}\big(\log(1-F^2)\big)\Big)\,,\end{aligned}$$

i.e. please justify both equalities. You have now completed the derivation of the Born–Infeld equations of motion introduced in the lecture since the above identity is equivalent to

$$\Big(\frac{1}{1-F^2}\Big)^{ac}\partial_a F_{cd} = 0\,.$$

F.3 Newton gravity in four versus five spacetime dimensions

This exercises guides you to explore Newton's gravitational potential in presence of circular or large extra dimensions and to see the fate of the D-dimensional Newton constant $G^{(D)}$ upon compactification.

(i) Consider the dimension-agnostic Poisson equation

$$\nabla^2 V^{(D)}(x) = 4\pi G^{(D)} \rho_M(x)$$

relating the spatial derivatives $\nabla^2 = \sum_{j=1}^{D-1} \frac{\partial^2}{\partial x_j^2}$ of the D-dimensional gravitational potential $V^{(D)}(x)$ to the density $\rho_M(x) = M\delta^{D-1}(x)$ of a point mass at the origin. Use the divergence theorem to extract the dependence of the spherically symmetric potential on $R = (\sum_{j=1}^{D-1} x_j^2)^{1/2}$ from the integral of the Poisson equation over a $(D-1)$-dimensional ball. More specifically, using the volume

$$\mathrm{vol}\, S^n(R) = \frac{2\pi^{(n+1)/2} R^n}{\Gamma(\frac{1}{2}(n+1))}$$

of an n-dimensional sphere of radius R, show that the D-dimensional potential is given by

$$V^{(D)}(R) = -\frac{2\pi^{(3-D)/2}\Gamma(\frac{1}{2}(D-1)) M G^{(D)}}{(D-3)R^{D-3}}\,, \qquad D \geq 4\,.$$

(ii) Now specialize to $D = 5$ spacetime dimensions with spatial coordinates (x, y, z, w). Compactify the w-direction on a circle of radius a, leading to identification $w \cong w + 2\pi a$. Show that the potential $V_a^{(D=5)}(x, y, z, w{=}0)$ in the compactified setting as a function of $r = \sqrt{x^2 + y^2 + z^2}$ is given by

$$V_a^{(D=5)}(x, y, z, w{=}0) = -\frac{G^{(5)} M}{\pi r^2} \sum_{n=-\infty}^{\infty} \frac{1}{1 + \left(\frac{2\pi n a}{r}\right)^2}\,.$$

Hint: sum over appropriate images of the uncompactified potential $V^{(D=5)}$.

(iii) By evaluating the infinite sum as

$$\sum_{n=-\infty}^{\infty} \frac{1}{1 + (\pi n x)^2} = \frac{1}{x} \coth\left(\frac{1}{x}\right)$$

show that the limit $r \gg a$ of $V_a^{(D=5)}$ reduces to the four-dimensional potential $V^{(D=4)}$ and that the four- and five dimensional Newton constants are related by

$$G^{(4)} = \frac{G^{(5)}}{2\pi a}$$

as argued in the lecture by dimensional reduction of the Einstein–Hilbert action.

(iv) Now investigate the opposite limit and perform a Laurent expansion of $V_a^{(D=5)}$ for small $r \ll a$. Determine the terms of order r^{-2}, r^0 and r^2 and explain how the term $\sim r^{-2}$ is related to earlier quantities seen in this exercise.

F.4 $SU(2)$ covariant approach to massless states

In this exercise, the massless states at the self-dual radius $r = \sqrt{\alpha'}$ are reorganized such that the enhanced $SU(2) \times SU(2)$ gauge symmetry is manifested. A key step is to recover two copies of the $SU(2)$ algebra

$$[t^a, t^b] = i\varepsilon^{abc} t^c\,, \qquad a, b, c \in \{1, 2, 3\}$$

of the triplet of generators $t^a = \frac{1}{2}\sigma^a$ with

$$t^1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad t^2 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad t^3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

at the level of the OPEs. For this purpose, we introduce normalized $(h, \bar{h}) = (1, 0)$ and $(0, 1)$ currents

$$J^3(z) = \frac{i}{\sqrt{\alpha'}}\partial_z X_{D-1}(z), \qquad J^\pm(z) =: e^{\pm \frac{2i}{\sqrt{\alpha'}} X^L_{D-1}(z)} :$$
$$\overline{J}^3(z) = \frac{i}{\sqrt{\alpha'}}\partial_{\bar{z}} X_{D-1}(z), \qquad \overline{J}^\pm(z) =: e^{\pm \frac{2i}{\sqrt{\alpha'}} X^R_{D-1}(z)} :$$

whose OPEs follow from those of the chiral halves of $X_{D-1}(\tau, \sigma) = X^L_{D-1}(\tau{+}\sigma) + X^R_{D-1}(\tau{-}\sigma)$,

$$X^L_{D-1}(z)X^L_{D-1}(w) \sim -\frac{\alpha'}{2}\log(z-w) + \ldots, \qquad X^R_{D-1}(z)X^R_{D-1}(w) \sim -\frac{\alpha'}{2}\log(\bar{z}-\bar{w}) + \ldots$$

Note that there are no mutual singularities between $X^L_{D-1}(z)X^R_{D-1}(w)$ or $J^a(z)\overline{J}^b(w)$.

(i) By first deriving the OPEs among $\{J^3, J^+, J^-\}$ and then converting to

$$J^1 = \frac{J^+ + J^-}{2}, \qquad J^2 = \frac{J^+ - J^-}{2i},$$
$$\overline{J}^1 = \frac{\overline{J}^+ + \overline{J}^-}{2}, \qquad \overline{J}^2 = \frac{\overline{J}^+ - \overline{J}^-}{2i},$$

show that the OPEs can be written in $SU(2)$-covariant[5] form as

$$J^a(z)J^b(w) \sim \frac{\delta^{ab}/2}{(z{-}w)^2} + \frac{i\varepsilon^{abc}J^c(w)}{z{-}w} + \ldots, \qquad \overline{J}^a(z)\overline{J}^b(w) \sim \frac{\delta^{ab}/2}{(\bar{z}{-}\bar{w})^2} + \frac{i\varepsilon^{abc}\overline{J}^c(w)}{\bar{z}{-}\bar{w}} + \ldots.$$

The residue of the simple pole reflects the above $SU(2)$ algebra, and the double-pole term corresponds to its so-called central extension (similar to the central term $\sim c$ in the Virasoro algebra).

(ii) For the two $SU(2)$ triplets of gauge-boson vertex operators

$$V^{a|\mathbb{1}}_\epsilon(z) = \epsilon_\mu J^a(z) : i\partial_{\bar{z}} X^\mu(z) e^{ip\cdot X}(z) :, \qquad V^{\mathbb{1}|a}_\epsilon(z) = \epsilon_\mu \overline{J}^a(z) : i\partial_z X^\mu(z) e^{ip\cdot X}(z) : \tag{F.1}$$

work out the linear combinations of their components $a = 1, 2, 3$ which reproduce the vertex operators $V_\pm, V^L_\pm, V^R_\pm$ given in the lecture. The Lorentz index of ϵ_μ and the dot product $p \cdot X$ in the exponent exclude the $(D{-}1)$-direction.

(iii) Determine the three-point function of the currents $J^a(z)$ and use this to compute the three-point amplitude of three gauge bosons $V^{a|\mathbb{1}}_\epsilon(z)$ in $SU(2)$-covariant form.

Hint: see the three-gluon amplitude given in the lecture for the polarization dependence.

(iv) The massless closed-string spectrum at the self-dual radius contains 9 scalars with vertex operators

$$V^{a|b}(z) = J^a(z)\overline{J}^b(z) : e^{ip\cdot X}(z) :$$

that transform in the bi-adjoint representation of $SU(2) \times SU(2)$. Determine the momentum- and winding numbers m, n in $(p_L, p_R) = (\frac{n}{r} + \frac{mr}{\alpha'}, \frac{n}{r} - \frac{mr}{\alpha'})$ of their 3×3 components and give their three-point amplitude in $(SU(2) \times SU(2))$-covariant form.

To be submitted on Jan 31st

[5] Covariance means that the indices a, b, c on the right-hand side are attached to the invariant tensors δ and ε of $SU(2)$.

G Problem set Jan 27th 2023

G.1 The ground-state energy of the Ramond sector

The goal of this exercise is to derive the ground-state energy in the Ramond sector of the worldsheet fermions ψ^μ which amounts to the conformal weight $h = \frac{D}{16}$ of the spin fields $S_A(z)$. The Ramond sector is defined by antiperiodic boundary conditions $\psi^\mu(e^{2\pi i}z) = -\psi^\mu(z)$ on the complex plane due to mode expansions

$$\psi^\mu(z) = \sum_{r\in\mathbb{Z}} \psi^\mu_r z^{-r-1/2}$$

with integer r as opposed to the half-odd integer r in the Neveu-Schwarz sector. The modes ψ^μ_r obey the canonical anticommutation relations (equivalent to the OPE $\psi^\mu(z)\psi^\nu(w) \sim \eta^{\mu\nu}(z-w)^{-1} + \ldots$)

$$\{\psi^\mu_r, \psi^\nu_s\} = \eta^{\mu\nu}\delta_{r+s,0}\,, \qquad r,s\in\mathbb{Z}\,.$$

(i) Show that the modes of the energy-momentum tensor in the Ramond sector

$$T_\psi(z) = \frac{1}{2}:\!\big(\partial_z\psi^\mu(z)\big)\psi_\mu(z)\!: = \sum_{m\in\mathbb{Z}} L_m z^{-m-2}$$

are given by

$$L_{m\neq 0} = \frac{1}{2}\sum_{r\in\mathbb{Z}} r\,\psi^\mu_{m-r}\,\psi^\nu_r\eta_{\mu\nu}\,, \qquad L_0 = h + \sum_{r=1}^{\infty} r\,\psi^\mu_{-r}\psi^\nu_r\eta_{\mu\nu}\,,$$

where you are supposed to ignore terms $\psi^\mu_0\psi^\nu_0\eta_{\mu\nu}$ for the moment. The expression for L_0 features a commuting offset $h\in\mathbb{Q}$ which captures the contributions of the zero modes ψ^μ_0 and whose value will be determined below. The normal-ordering prescription $:\ldots:$ in the definition of $T_\psi(z)$ moves annihilation operators $\psi^\mu_{r>0}$ to the right of creation operators $\psi^\mu_{r<0}$,

$$:\!\psi^\mu_r\psi^\nu_{-r}\!: = \begin{cases} \psi^\mu_r\psi^\nu_{-r} & : \ r<0\,, \\ -\psi^\nu_{-r}\psi^\mu_r & : \ r>0\,. \end{cases}$$

Since the normal-ordering prescription for zero modes $:\!\psi^\mu_0\psi^\nu_0\!:$ is ambiguous from the viewpoint of creation and annihilation operators, one introduces an a priori undetermined offset h into the definition of L_0.

(ii) Show that the canonical anticommutators of the ψ^μ modes imply

$$[L_m, \psi^\mu_s] = -\Big(\frac{m}{2}+s\Big)\psi^\mu_{m+s}\,.$$

Hint: You may find the lemma $[AB,C] = A\{B,C\} - \{A,C\}B$ useful.

(iii) Use the lemma in (ii) to verify the following commutator among energy-momentum modes

$$[L_m, L_{-m}] = \frac{Dm^3}{24} + \frac{Dm}{12} + 2m\sum_{r=1}^{\infty} r\,\psi^\mu_{-r}\psi^\nu_r\eta_{\mu\nu}\,.$$

Hint: Use the corollary $\eta_{\mu\nu}\psi^\mu_0\psi^\nu_0 = \frac{D}{2}$ of the canonical anticommutators and be careful about non-vanishing anticommutators in bringing bilinears of ψ^μ modes into normal-ordered form (cf. exercise B.3). You can also use the following sums seen in earlier exercises:

$$\sum_{s=1}^{m} s = \frac{m(m+1)}{2}\,, \qquad \sum_{s=1}^{m} s^2 = \frac{m(m+1)(2m+1)}{6}$$

(iv) Show that the commutator in (iii) can only line up with the Virasoro algebra

$$[L_m, L_{-m}] = \frac{c}{12}(m^3 - m) + 2mL_0$$

at central charge $c = \frac{D}{2}$ if the offset in the definition of L_0 is chosen as $h = \frac{D}{16}$. Note that this identifies the conformal weight of the spin fields which generate the Ramond-sector ground states.

G.2 NS vertex operators at the first mass level

In the NS-sector of the open-superstring spectrum, the first mass level compatible with GSO projection occurs at $m^2 = -p^2 = \frac{1}{\alpha'}$. One can construct its vertex operators from an ansatz

$$V^{(-1)}_{m^2=1/\alpha'} = :\Big(B_{\mu\nu} i\partial_z X^\mu(z)\psi^\nu(z) + E_{\mu\nu\lambda}\psi^\mu\psi^\nu\psi^\lambda(z) + H_\mu \partial_z \psi^\mu(z)\Big) e^{-\phi(z)} e^{ip\cdot X(z)}:$$

involving polarization tensors $B_{\mu\nu}, E_{\mu\nu\lambda}$ and H_μ. As explained in the lecture, the matter part (after peeling off the superghost contribution $:e^{-\phi(z)}:$) must be the lower component of a superconformal primary, i.e. one cannot have any double poles in its OPE with the supercurrent

$$G(z) = \frac{1}{\sqrt{2\alpha'}} i\partial_z X^\mu(z)\psi_\mu(z)\,.$$

(i) Show that the superconformal-primary condition constrains the polarizations to obey

$$\begin{aligned} \eta^{\mu\nu}B_{\mu\nu} + p^\mu H_\mu &= 0 \\ \frac{1}{2}(B_{\mu\nu} - B_{\nu\mu}) + 3p^\lambda E_{\lambda\mu\nu} &= 0 \\ 2\alpha' p^\nu B_{\mu\nu} + H_\mu &= 0\,. \end{aligned}$$

(ii) One can show (for instance using BRST methods which have not been covered in the lecture) that the following vertex operators in $D = 10$ dimensions

$$\begin{aligned} V^{(-1)}_{\text{sp A}} &= :\Big((\eta_{\mu\nu} + 4\alpha' p_\mu p_\nu) i\partial_z X^\mu(z)\psi^\nu(z) + 6\alpha' p_\mu \partial_z\psi^\mu(z)\Big) e^{-\phi(z)} e^{ip\cdot X(z)}: \\ V^{(-1)}_{\text{sp B}} &= :\Big(\Sigma_{[\mu\nu]} i\partial_z X^\mu(z)\psi^\nu(z) + \alpha' \Sigma_{[\mu\nu} p_{\lambda]}\psi^\mu\psi^\nu\psi^\lambda(z)\Big) e^{-\phi(z)} e^{ip\cdot X(z)}: \\ V^{(-1)}_{\text{sp C}} &= :\Big((p_\mu \pi_\nu + p_\nu \pi_\mu) i\partial_z X^\mu(z)\psi^\nu(z) + 2\pi_\mu \partial_z\psi^\mu(z)\Big) e^{-\phi(z)} e^{ip\cdot X(z)}: \end{aligned}$$

with $p^\mu \Sigma_{\mu\nu} = p^\mu \pi_\mu = 0$ describe spurious states (similar to the gluon at $\epsilon_\mu \to p_\mu$) that decouple from amplitudes. Verify that they are compatible with the superconformal-primary conditions.

(iii) By their decoupling from ampliudes, the spurious states in (ii) can be identified with zero. Use this to show that the vertex operators describing physical states can be brought into the form

$$\begin{aligned} V^{(-1)}_{\text{spin 2}} &= B_{\mu\nu} : i\partial_z X^\mu(z)\psi^\nu(z) e^{-\phi(z)} e^{ip\cdot X(z)} : \\ V^{(-1)}_{\text{3-form}} &= E_{\mu\nu\lambda} : \psi^\mu\psi^\nu\psi^\lambda(z) e^{-\phi(z)} e^{ip\cdot X(z)} :, \end{aligned}$$

where the polarizations satisfy

$$B_{\mu\nu} = B_{\nu\mu}\,, \qquad \eta^{\mu\nu}B_{\mu\nu} = p^\mu B_{\mu\nu} = 0\,, \qquad p^\lambda E_{\lambda\mu\nu} = 0\,.$$

(iv) Explain why the constraints on $B_{\mu\nu}$ & $E_{\mu\nu\lambda}$ leave 44 & 84 physical degrees of freedom, respectively.

(v) Show that the representatives of the spin-2 and the 3-form vertex operators in the zero ghost picture are proportional to

$$
\begin{aligned}
V^{(0)}_{\text{spin }2} &\sim B_{\mu\nu}{:}\Big(i\partial_z X^\mu i\partial_z X^\nu(z) + 2\alpha'(p\cdot\psi) i\partial_z X^\mu \psi^\nu(z) + 2\alpha'(\partial_z\psi^\mu)\psi^\nu(z)\Big)e^{ip\cdot X(z)}{:}\\
V^{(0)}_{\text{3-form}} &\sim E_{\mu\nu\lambda}{:}\Big(3i\partial_z X^\mu \psi^\nu\psi^\lambda(z) + 2\alpha'(p\cdot\psi)\psi^\mu\psi^\nu\psi^\lambda(z)\Big)e^{ip\cdot X(z)}{:}\,.
\end{aligned}
$$

G.3 Supersymmetry transformations

This exercise aims to familiarize you with fermion vertex operators and supersymmetry transformations in ten dimensions (using the shorthand $\gamma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$ for antisymmetrized products below).

(i) Demonstrate that the supercharge (to be contracted with anticommuting Weyl spinors η^α of $SO(1,9)$)

$$
\mathcal{Q}^{(-1/2)}_\alpha \sim \oint \frac{\mathrm{d}z}{2\pi i} S_\alpha(z){:}e^{-\phi(z)/2}{:}
$$

transforms massless fermionic vertex operators $V^{(-1/2)}_\chi(z) \sim \chi^\alpha S_\alpha(z){:}e^{-\phi(z)/2}e^{ip\cdot X(z)}{:}$ to bosonic vertex operators with polarization $\epsilon^\mu = (\eta\gamma^\mu C\chi)$.

(ii) Use the identities $(\gamma_\mu C)_{\alpha\beta}\gamma^\mu_{\delta\dot\omega} + (\gamma_\mu C)_{\beta\delta}\gamma^\mu_{\alpha\dot\omega} + (\gamma_\mu C)_{\delta\alpha}\gamma^\mu_{\beta\dot\omega} = 0$ and $(\gamma_\mu C)_{\alpha\beta} = (\gamma_\mu C)_{\beta\alpha}$ to show that

$$
\big[\delta_{\eta_1}, \delta_{\eta_2}\big]\chi^\alpha = 2(\eta_1\gamma_\mu C\eta_2)p^\mu\chi^\alpha\,,
$$

where the supersymmetry transformations of gluinos and gluons are encoded through the shorthands

$$
\delta_\eta\epsilon^\mu = (\eta\gamma^\mu C\chi)\,, \qquad \delta_\eta\chi^\beta = \eta^\alpha(\gamma^{\mu\nu})_\alpha{}^\beta\epsilon_\mu p_\nu\,.
$$

(iii) Use the identity $(\gamma^{\lambda\rho}\gamma^\mu C)_{\alpha\beta} + (\gamma^{\lambda\rho}\gamma^\mu C)_{\beta\alpha} = 2\eta^{\mu\rho}(\gamma^\lambda C)_{\alpha\beta} - 2\eta^{\mu\lambda}(\gamma^\rho C)_{\alpha\beta}$ to show that

$$
\big[\delta_{\eta_1}, \delta_{\eta_2}\big]\epsilon^\mu = 2p^\lambda(\eta_1\gamma_\lambda C\eta_2)\epsilon^\mu - 2\epsilon^\lambda(\eta_1\gamma_\lambda C\eta_2)p^\mu\,,
$$

where the second term amounts to a linearized gauge transformation and does not affect amplitudes.

To be submitted on Feb 12$^{\text{th}}$

H Problem set February 9th 2023

H.1 Spacetime supersymmetry of the superstring spectrum

The purpose of this exercise is to verify the matching of bosonic and fermionic d.o.f. in the superstring spectrum including massive modes, a necessary condition for spacetime supersymmetry. This will be done via generating functions of the form

$$Z(q) = \sum_k c_k q^k$$

with a formal bookkeeping variable $q \in \mathbb{C}$, where the expansion coefficients $c_k \in \mathbb{N}_0$ track the number of states at mass level k. We shall furthermore rely on the superstring analogue of lightcone quantization where only transverse oscillators α^i_n, ψ^i_r with vector indices $i = 1, 2, \ldots, 8$ of $SO(8)$ are taken into account. Similar to the discussion for the bosonic string, restriction to transverse oscillators automatically incorporates all the physical-state constraints from the OPE with the supercurrent. Moreover, all the spurious states (such as linearized gauge transformations $\epsilon_\mu \to k_\mu$ of the gluon polarization vector) are automatically subtracted.

Just like in the manifestly $SO(1,9)$ covariant approach of the lectures, the modes of ψ^i_r have $r \in \mathbb{Z} - \frac{1}{2}$ in the NS sector and $r \in \mathbb{Z}$ in the R sector. However, the Ramond-sector ground states $|\hat{a}\rangle_{\rm R}$ are now labelled by a Weyl-spinor index $\hat{a} = 1, 2, \ldots, 8$ of $SO(8)$ since the Clifford algebra $\{\psi^i_0, \psi^j_0\} = \delta^{ij}$ of the zero modes is now adapted to a smaller range $i, j = 1, 2, \ldots, 8$. As before, GSO projection only admits an odd number of $\psi^i_{r \in \mathbb{Z} - \frac{1}{2}}$ oscillators acting on the NS ground state $|0\rangle_{\rm NS}$ while the R-sector states may involve any number of $\psi^i_{r \in \mathbb{Z}}$ acting on a left-handed Weyl-spinor ground state $|\hat{a}\rangle_{\rm R}$.

At the massless level, for instance, the 8 physical gluon d.o.f. are encoded in $\psi^i_{-1/2}|0\rangle_{\rm NS}$ with $i = 1, 2, \ldots, 8$ while the 8 physical gluino d.o.f. stem from $|\hat{a}\rangle_{\rm R}$ with $\hat{a} = 1, 2, \ldots, 8$ (the transversality constraint and the massless Dirac equation are automatically incorporated). Here and below, the spacetime momenta p^μ characterizing the ground states $\lim_{z\to 0} {:}e^{ip\cdot X(z)}{:}|0\rangle_{\rm NS}$ and $\lim_{z\to 0} {:}e^{ip\cdot X(z)}{:}|\hat{a}\rangle_{\rm R}$ are left implicit.

(i) Show that the counting of transverse oscillators at the first mass level of the open superstring (see exercise G.2) leads to 128 states in both the NS and the R sector.

(ii) Consider a fixed spacetime direction i of the bosonic and fermionic oscillators. Explain why the $k^{\rm th}$-order coefficients in the following q-expansions count the number of states $\alpha^i_{m_1} \ldots \alpha^i_{m_a} \psi^i_{r_1} \ldots \psi^i_{r_b}$ with overall oscillator number $-\sum_{j=1}^a m_j - \sum_{j=1}^b r_j = k$

- only bosonic oscillators $\alpha^i_{n<0}$ $\implies$ $Z_{\rm bos}(q) = \displaystyle\prod_{m=1}^\infty \frac{1}{1-q^m}$
- only fermionic NS-sector oscillators $\psi^i_{r<0}$ $\implies$ $Z_{\rm NS}(q) = \displaystyle\prod_{m=1}^\infty (1 + q^{m-1/2})$
- only fermionic R-sector oscillators $\psi^i_{r<0}$ $\implies$ $Z_{\rm R}(q) = \displaystyle\prod_{m=1}^\infty (1 + q^m)$

You can assume the oscillators to act on a reference state $|0\rangle$ which is annihilated by all $\alpha^i_{n>0}, \psi^i_{r>0}$ and ignore the zero modes in the R sector for the moment.

(iii) For the complete R sector of the superstring with eight values of i, eight ground-state components $|\hat{a}\rangle_{\rm R}$ and the possibility to combine α^i_n and ψ^i_n oscillators, explain that the generating function for the number of states at different mass levels is

$$Z^{\rm super}_{\rm R}(q) = 8\left(\prod_{m=1}^{\infty} \frac{1+q^m}{1-q^m}\right)^8$$

and determine the number of spacetime fermions at the 2nd mass level of the open-superstring spectrum.

For your information: The same logic leads to the generating function

$$Z^{\not{\rm G}{\rm SO}}_{\rm NS}(q) = q^{-1/2}\left(\prod_{m=1}^{\infty} \frac{1+q^{m-1/2}}{1-q^m}\right)^8$$

for the complete NS sector of the superstring **before** GSO projection (with the prefactor $q^{-1/2}$ to account for the fact that the state without any α^i_n or ψ^i_n would be a tachyon at mass level $-\frac{1}{2}$).

(iv) In order to implement GSO projection at the level of the NS-sector generating function, it is convenient to introduce the building block (again for ψ^i_r with a fixed value of $i = 1, 2, \ldots, 8$)

$$\overline{Z}_{\rm NS}(q) = \prod_{m=1}^{\infty}(1-q^{m-1/2})$$

whose minus sign tracks the number of ψ^i_r-oscillators in a NS-sector state (with $(-1)^\ell$ for ℓ fermionic oscillators). Explain why the coefficients of q^n in

$$Z^{\rm super}_{\rm NS}(q) = q^{-1/2}\left(\prod_{m=1}^{\infty} \frac{1}{1-q^m}\right)^8 \frac{1}{2}\Big(Z_{\rm NS}(q)^8 - \overline{Z}_{\rm NS}(q)^8\Big)$$

generate the number of GSO projected NS-sector states at level n of the open superstring.

(v) Verify up to and including the second mass level that the numbers of spacetime bosons and spacetime fermions in the open-superstring spectrum match (which is a necessary condition for spacetime supersymmetry).

(vi) Express $Z^{\rm super}_{\rm NS}(q)$ and $Z^{\rm super}_{\rm R}(q)$ in terms of the Dedekind eta function $\eta(q)$ as well as the Jacobi theta functions $\theta_j(q),\ j = 2, 3, 4$ defined by

$$\begin{aligned}
\eta(q) &= q^{1/24}\prod_{m=1}^{\infty}(1-q^m)\\
\theta_2(q) &= 2q^{1/8}\prod_{m=1}^{\infty}(1-q^m)(1+q^m)^2\\
\theta_3(q) &= \prod_{m=1}^{\infty}(1-q^m)(1+q^{m-1/2})^2\\
\theta_4(q) &= \prod_{m=1}^{\infty}(1-q^m)(1-q^{m-1/2})^2
\end{aligned}$$

and explain why spacetime supersymmetry of the entire string spectrum relies on Jacobi's abstruse identity (from the year 1829)

$$\big(\theta_3(q)\big)^4 - \big(\theta_2(q)\big)^4 - \big(\theta_4(q)\big)^4 = 0\,.$$

(vii) **OPTIONAL** (no need to submit a solution for this part): Prove Jacobi's abstruse identity.

H.2 Four-fermion amplitude

The goal of this exercise is to derive the following expression for the color-ordered four-gaugino amplitude

$$\mathcal{A}^{\rm tree}_{\rm super}(\chi_1,\chi_2,\chi_3,\chi_4) = \mathcal{A}^{\rm tree}_{\rm SYM}(\chi_1,\chi_2,\chi_3,\chi_4)\frac{\Gamma(1+s)\Gamma(1+t)}{\Gamma(1+s+t)}$$

with wave functions $\chi^\alpha_{j=1,2,3,4}$ and the following field-theory amplitude in ten-dimensional SYM

$$\mathcal{A}^{\rm tree}_{\rm SYM}(\chi_1,\chi_2,\chi_3,\chi_4) = \frac{(\chi_1\gamma^\mu C\chi_2)(\chi_3\gamma_\mu C\chi_4)}{(p_1{+}p_2)^2} + \frac{(\chi_4\gamma^\mu C\chi_1)(\chi_2\gamma_\mu C\chi_3)}{(p_2{+}p_3)^2}\,.$$

(i) Revisit the three-point functions of (ψ^μ, S_α) and the superghost system to derive the OPE

$$:\!S_\alpha e^{-\phi/2}\!:\!(z)\ :\!S_\beta e^{-\phi/2}\!:\!(w) \sim \frac{(\gamma^\mu C)_{\alpha\beta}}{\sqrt{2}(z{-}w)}:\!\psi_\mu e^{-\phi}\!:\!(w) + \ldots$$

and the three-point function

$$\langle :\!\psi_\mu e^{-\phi}\!:\!(z_2)\ :\!S_\gamma e^{-\phi/2}\!:\!(z_3)\ :\!S_\delta e^{-\phi/2}\!:\!(z_4)\rangle = \frac{(\gamma_\mu C)_{\gamma\delta}}{\sqrt{2}z_{23}z_{24}z_{34}}\,.$$

(ii) Show that the following expression for a four-point spin-field correlator

$$\begin{aligned}&\langle :\!S_\alpha e^{-\phi/2}\!:\!(z_1)\ :\!S_\beta e^{-\phi/2}\!:\!(z_2)\ :\!S_\gamma e^{-\phi/2}\!:\!(z_3)\ :\!S_\delta e^{-\phi/2}\!:\!(z_4)\rangle \\ &= \frac{1}{2(z_{13}z_{24})}\left(\frac{(\gamma_\mu C)_{\alpha\beta}(\gamma^\mu C)_{\gamma\delta}}{z_{12}z_{34}} - \frac{(\gamma_\mu C)_{\alpha\delta}(\gamma^\mu C)_{\gamma\beta}}{z_{14}z_{23}}\right)\end{aligned} \tag{H.1}$$

is consistent with the OPE from (i) in the limits $z_1 \to z_{2,3,4}$. Note that $(\gamma_\mu C)_{\alpha\beta} = (\gamma_\mu C)_{\beta\alpha}$ and $(\gamma_\mu C)_{\alpha\beta}(\gamma^\mu C)_{\gamma\delta}{+}\text{cyc}(\alpha,\beta,\gamma) = 0$ by representation theory of $SO(1,9)$. Moreover, the OPE relevant to $z_1 \to z_3$ needs to be weighted with a minus sign from swapping the positions of $:\!S_\beta e^{-\phi/2}\!:\!(z_2)$ and $:\!S_\gamma e^{-\phi/2}\!:\!(z_3)$.

For your information: Once you have checked consistency of (H.1) with the limits $z_1 \to z_{2,3,4}$, you have in fact succeeded to prove it for any $z_j \in \mathbb{C}$: You have shown that the difference between the left-hand side and right-hand side of (H.1) does not have any poles in z_1. Since this difference is holomorphic $\forall\ z_1 \in \mathbb{C}$ and bounded by the absence of poles (it cannot blow up as $z_1 \to \infty$), it must be constant in z_1 by Liouville's theorem. Finally, the constant difference between the left- and right-hand side of (H.1) must be zero since the falloff of the desired correlator of $h = 1$ conformal primaries is $\sim z_1^{-2}$ as $z_1 \to \infty$.

(iii) Use the prescription for open-superstring tree amplitudes given in the lecture to show that

$$\mathcal{A}^{\rm tree}_{\rm super}(\chi_1,\chi_2,\chi_3,\chi_4) \sim \int_0^1 \mathrm{d}z_2\,|z_2|^s|1-z_2|^t\left(\frac{(\chi_1\gamma^\mu C\chi_2)(\chi_3\gamma_\mu C\chi_4)}{z_2} + \frac{(\chi_4\gamma^\mu C\chi_1)(\chi_2\gamma_\mu C\chi_3)}{1-z_2}\right).$$

Note that $\chi^\alpha_i\chi^\beta_j = -\chi^\beta_j\chi^\alpha_i$ are anticommuting variables.

(iv) Evaluate the disk integrals in terms of Gamma functions to arrive at the expression for the four-point function stated at the beginning of the exercise.

H.3 Massive three-point amplitude

This exercise is dedicated to the three-point tree amplitude

$$\mathcal{A}^{\rm tree}_{\rm super}(B,\epsilon_2,\epsilon_3) = |z_{12}z_{13}z_{23}| \cdot \langle V^{(q_1)}_{\rm spin\,2}(z_1) V^{(q_2)}_{\epsilon_2}(z_2) V^{(q_3)}_{\epsilon_3}(z_3)\rangle \tag{H.2}$$

involving two gluons with transverse polarizations ϵ_2, ϵ_3 and one massive spin-two state from the first mass level of the open superstring with $p_1^2 = -\frac{1}{\alpha'}$: The spin-two state is characterized by a transverse & symmetric-traceless polarization tensor $B_{\mu\nu}$, and its vertex operators $V^{(q_1)}_{\rm spin\,2}(z_1)$ in the ghost pictures $q_1 = 0, -1$ can be found in exercise G.2.

(i) Show that the open-string Koba–Nielsen factor for the above state configuration is given by

$$\left\langle \prod_{j=1}^{3} :e^{ip_j\cdot X(z_j)}: \right\rangle = \left| \frac{z_{12}z_{13}}{z_{23}} \right| .$$

(ii) Explain why the color-dressed amplitude is proportional to $\mathrm{Tr}(T^{a_1}T^{a_2}T^{a_3} + T^{a_1}T^{a_3}T^{a_2})$.

(iii) Evaluate the color-ordered three-point amplitude (H.2) in the ghost pictures $(q_1,q_2,q_3) = (-1,-1,0)$.

(iv) Manifest linearized gauge invariance by expressing the results in terms of the linearized field strengths $f_j^{\mu\nu} = p_j^\mu \epsilon_j^\nu - p_j^\nu \epsilon_j^\mu$ of the gluons.

(v) Derive the three-point correlator

$$B_{\mu\nu}\langle :(\partial_{z_1}\psi^\mu)\psi^\nu:(z_1)\psi^\lambda(z_2)\psi^\rho(z_3)\rangle = \frac{z_{23}B^{\lambda\rho}}{z_{12}^2 z_{13}^2}$$

(vi) Use (v) to reproduce the result of (iv) from the ghost pictures $(q_1,q_2,q_3) = (0,-1,-1)$ in (H.2).

H.4 Dimensional reduction and type-IIA supergravity

OPTIONAL (no need to submit a solution for this exercise, the required material has not been covered in the lecture but can be looked up in the lecture notes)

As you have seen in the lecture, the massless spectrum of the type-IIA superstring in $D = 10$ comprises

- NS-NS sector: graviton $h_{\mu\nu}$, B-field $B_{\mu\nu}$, dilaton Φ
- R-NS sector: left-handed gravitino χ^α_μ, right-handed dilatino $\Lambda_{\dot\beta}$
- NS-R sector: right-handed gravitino $\bar\chi^\mu_{\dot\beta}$, left-handed dilatino $\bar\Lambda^\alpha$
- R-R sector: one-form potential $A^{(1)}_\mu$ and three-form potential $A^{(3)}_{\mu\nu\lambda}$

(i) Dimensionally reduce the type-IIA multiplet to $D = 4$ and verify that the helicities follow the counting of $\mathcal{N} = 8$ supergravity.

(ii) Show that the type-IIA multiplet in ten dimensions follows from dimensional reduction of the $D = 11$ supergravity multiplet which comprises a graviton h_{MN}, a three-form potential $A^{(3)}_{MNP}$ and a gravitino χ^B_M (with vector indices $M,N,P = 0,1,\ldots,10$). The $D = 11$ gravitino is a 32-component Dirac spinor with index $B = 1,2,\ldots,32$ (odd dimensions do not admit a notion of chirality). You can use that Dirac spinors in $D = 11$ decompose into one left-handed and one right-handed Weyl spinor in $D = 10$ dimensions.

To be submitted on Feb 28$^{\rm th}$